ETS-88™
System Manual

Applies to the Following Tester Models:
ETS-88™

DOCP1052 Rev. 9 – May '12

For use with software release EV 2012A / MST 2011B

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1 Introduction

Welcome to the ETS-88™ Test System. These test systems offer you cost-effective, high voltage component test solutions. Eagle Test Systems, Inc. (ETS) wants you to be successful with this equipment. Our application department exists to serve you. Please call with any questions you have (see the end of this chapter for contact information).

Training courses are held on a regular basis at our Illinois office. The ETS Training School provides system training as well as "hands-on" test writing and debugging experience. ETS training courses focus on lab work to bring up frequently asked questions. These questions can be answered quickly in a classroom situation as opposed to the inconvenience of remote communications.

1.1 System Documentation

This manual is intended to be a reference to assist you in developing and running test programs on the ETS-88™ Test System. It contains information on all hardware and software for the standard configuration of the tester.

Please see the Table of Contents for an overview of this manual, and use the Index to locate information on specific topics. For information on the system computer, peripherals, operating system and the Visual C++® test development environment, refer to the ETS Software Help File, or DOCP0419 – The Eagle Vision Software Suite Manual. For information on maintaining or servicing your system, please refer to DOCP1053 – The ETS-88™ Service Manual. For preventative maintenance procedures, see DOCP1031 – The ETS Preventative Maintenance Guide.

1.1.1 Computer Operation Materials

The ETS-88™ main console is an Intel processor based computer. Documentation associated with the ETS-88™ main console includes the following:

- Visual C++® OnLine Help
- Printer operation manual
- Monitor operation manual

Any other manuals or material you receive with your system are the result of purchasing some other option.

1.2 System Overview

1.2.1 Introduction

The dual test heads of the ETS-88™ tester are integrated into the mainframe cabinet. These test heads include an IEEE-488/GPIB bus interface, which allows them to be connected to and communicate with GPIB instruments. The test software provides high-level control of the hardware, and a great deal of hardware flexibility is available at the application board.

1.2.2 System Installation Requirements

Please refer to the ETS-88™ Site Guide (available on the ETS Website) or DOCP1053 – The ETS-88™ Service Manual for full system installation requirements, including tester footprint diagrams.

1.2.2.1 Electrical Ratings

The ETS-88™ system can be operated with the following single phase AC(∼) power line conditions:

Table 1-1 – ETS-88 Power Line Requirements

Select One	A	B	C	D	E
Single Phase AC Line Voltage (50/60 Hz)	200 V	210 V (208 V)	220 V	230 V	240 V
Current Required	40 A	40 A	40 A	40 A	40 A

1.2.2.2 Plugs

The ETS-88™ is designed to accept a facility hard-wire connection. See section for hard-wiring instructions. For convenience, a flexible 11 ft. (3.4 m), 3-conductor, 8 AWG (3.26 mm) cord is available for mains power connection. For US customers, a NEMA 14-50 P connector wired as shown in Figure 1-1 below is also available.

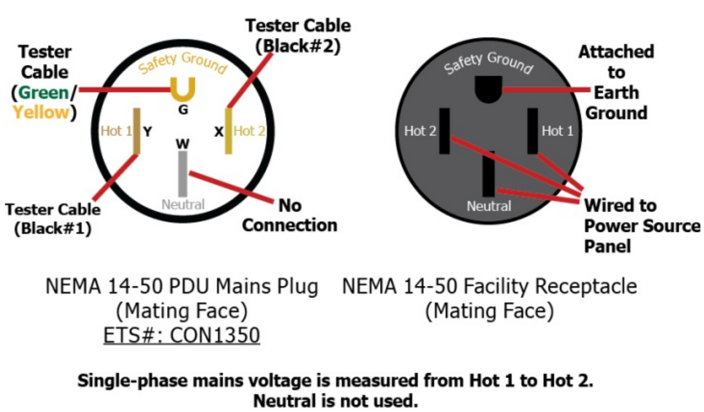

Figure 1-1 – 8 KVA PDU Plug Definition

1.2.3 Architecture Overview

1.2.3.1 Computer Console

Referring to Figure 1-2, note that the standard configuration of the system computer consists of an Intel processor based computer with hard disk, DVD drive, keyboard, and color monitor. Other options are available upon request.

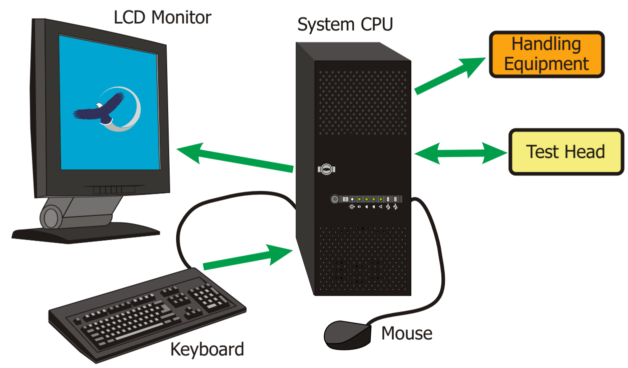

Figure 1-2 – System Computer Architecture

1.2.3.2 Mainframe

The mainframe cabinet is the main component in ETS-88™ Test Systems. The mainframe contains all the power supplies for the system, plus the dual test heads, which contain the system resources. Please refer to Figure 1-3 on the following page for a diagram of a typical ETS-88™ mainframe configuration.

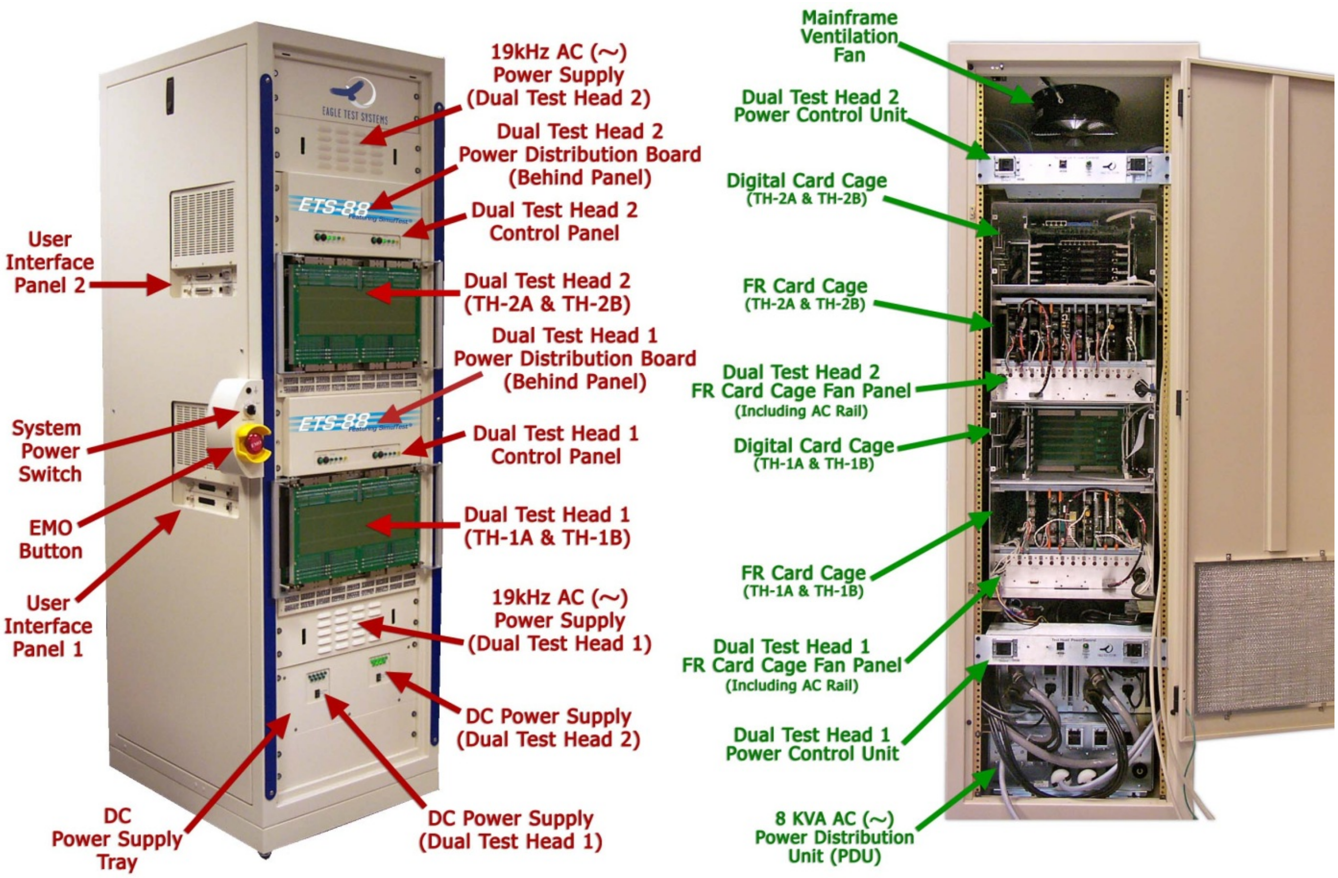

Figure 1-3 – ETS-88™ System Components (2M Cabinet)

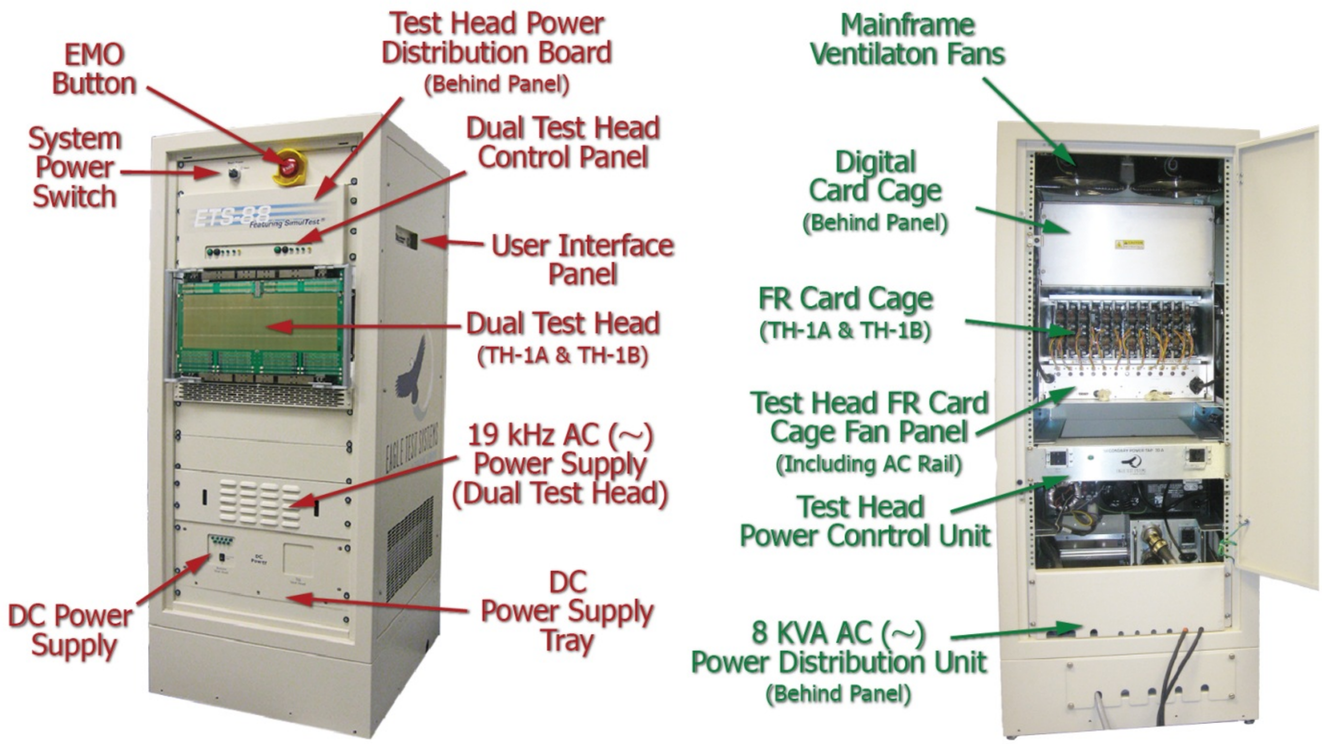

Figure 1-4 – ETS-88™ System Components (1.5M Cabinet)

Functional Testers (Dual Test Head Segments)

The mainframe cabinet can physically accommodate two dual test heads, each of which includes two segments. Each of these segments can function as an individual tester, with its own dedicated set of resources, or they can be bridged to work as one. These "functional testers" are referenced using the TH-xA and TH-xB convention in this manual, where 'x' is the number of dual test head in question. The bottom dual test head – or the sole dual test head in systems with only one dual test head – is Dual Test Head 1 (TH-1A and TH-1B). In systems with two dual test heads, the top dual test head is Dual Test Head 2 (TH-2A and TH-2B – see Figure 1-3 on the previous page).

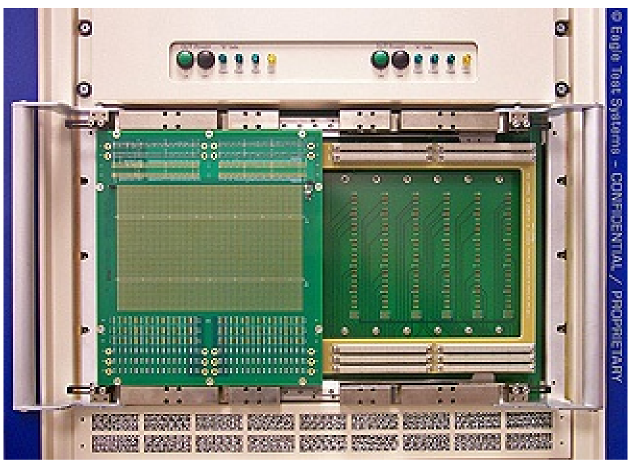

Figure 1-5 – Single Standard Application Board on an ETS-88™ Dual Test Head

Each "functional tester" (TH-1A, TH-1B, TH-2A, TH-2B) has its own dedicated set of resources that can operate independently from each other. These resources are contained in the digital and floating card cages within each dual test head. Each dual test head has its own set of power supplies, with an independent power switch for each set.

For applications that require a larger set of resources, the "functional testers" (TH-xA and TH-xB) can be "bridged" together to operate as a single tester, instead of independently. When operating in bridged mode, the application board is twice as large and covers the entire width of the dual test head instead of just half (see Figure 1-5 and Figure 1-6).

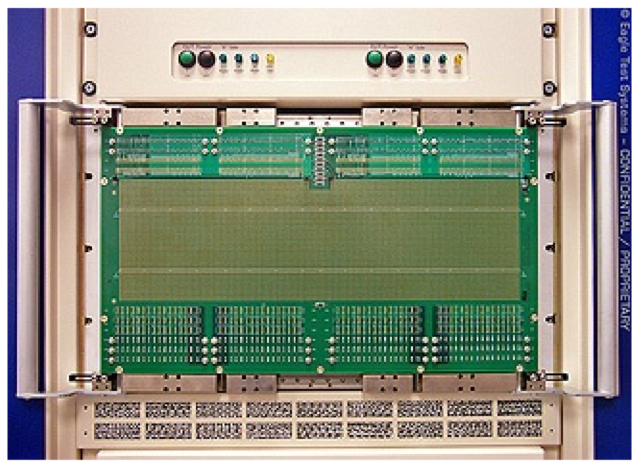

Figure 1-6 – "Bridged" Application Board on an ETS-88™ Dual Test Head

Figure 1-7 on the following page shows the test head segmentation as viewed from the rear of the mainframe.

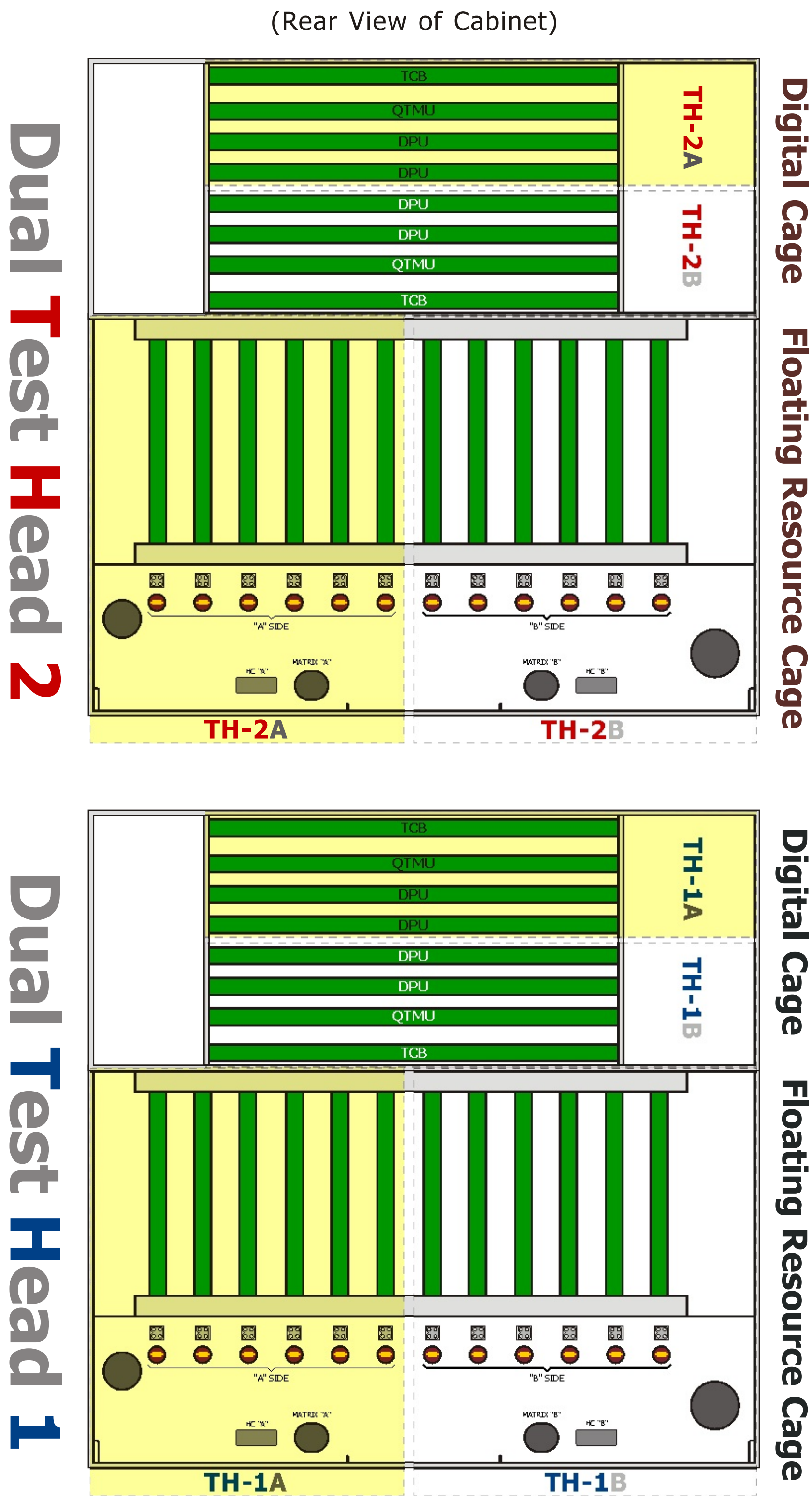

Figure 1-7 – ETS-88™ Dual Test Head Segmentation

Each half of a dual test head requires a Test Head Control Board (TCB), which provides communication between the various components of the system. The TCB includes a Programmable Control Bits (C-Bits) module, a PC Interface for communicating with the system PC, an Op Box interface for communicating with the Operator Box, a MS Handler interface for communicating with handling equipment, and an Isolated Communications interface, for communicating with ISO-COMM-based resources. The TCB is the only required resource in the system. Please refer to Chapter 2 for further details on the TCB and the other resources used by the "functional testers" of the ETS-88™ system.

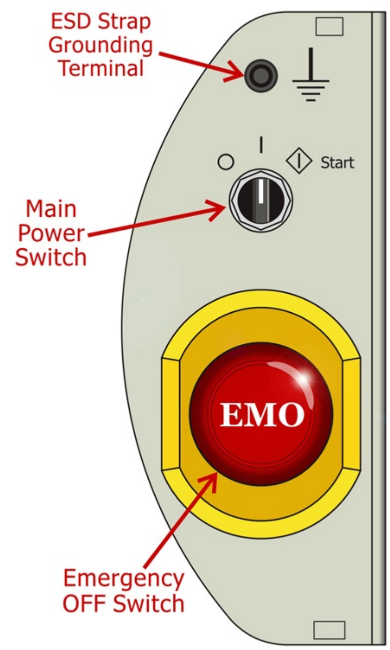

Figure 1-8 – System Power Control Panel (2M Mainframe Cabinet)

1.2.3.3 Power Switch / EMO Panel

This small, crescent-shaped panel that extends off the front corner of the mainframe is the main power control for the system. The panel (Figure 1-8) includes the main power switch, emergency shut-off button (EMO), and an ESD grounding strap terminal (which should be used whenever handling any boards used in the system).

Main Power Switch

Use this switch to enable power to the system. The switch does not power-up individual components in the system, but must be ON before the supplies for components such as the dual test heads and system PC can be turned on.

Use the following sequence to enable system power:

1.) Connect the mains power cord to the proper AC line voltage (facility power).

2.) Close the mains and output circuit breakers (on the 8 KVA PDU).

3.) Close the rear mainframe door interlock switch.

4.) Make sure the **EMO** switch is not depressed.

5.) Turn the main power switch to the ⬦ (Momentary Start) position, then release it to the | (ON) position.

Once all these conditions are met, the power will turn on. The actuator of the main power switch remains at the | (ON) position to maintain power.

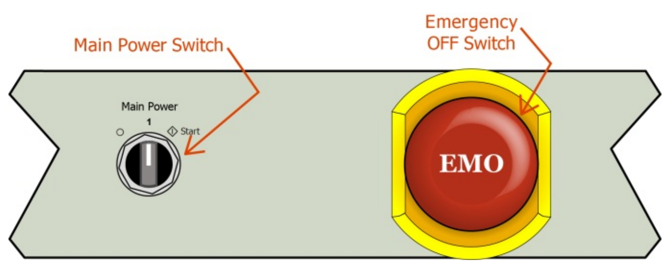

Figure 1-9 – System Power Control Panel (1.5M Mainframe Cabinet)

The standard power-down operation is to rotate the main power switch to the ○ (OFF) position. This action disables all power from the AC Power Conditioner or 8 KVA PDU. From this state, power can be restored by rotating the main power switch to the | (ON) position.

EMO Button

The ETS-88™ includes an Emergency Off (EMO) button in case of emergency. The button is palm-sized, mushroom-shaped, red in color in front of a yellow background, and clearly labeled "EMO."

Press the **EMO** button to activate the EMO circuit. Activating the EMO circuit disables all power from the 8 KVA PDU. Opening the back door of the standard mainframe cabinet also activates the EMO circuit.

The mainframe power and remote control circuitry are accessed through the J7 round three-position connector of the 8 KVA PDU. This connector must be connected to enable any power output from 8 KVA PDU.

1.2.3.4 Operator Box

The OP Box output is a RJ-45 cable connector that connects to the remote Operator Box. The Operator Box is a small (approx. 6 in by 4 in), aluminum box that includes controls for remotely starting/stopping testing manually, or activating continuous testing. The Operator Box also includes a pair of LEDs that allow the box to serve as a remote status indicator. The other controls on the box are disabled by default and may be programmed though the test executive. The box includes a magnetic strip on the bottom panel that allows it to be placed on a vertical surface for convenience.

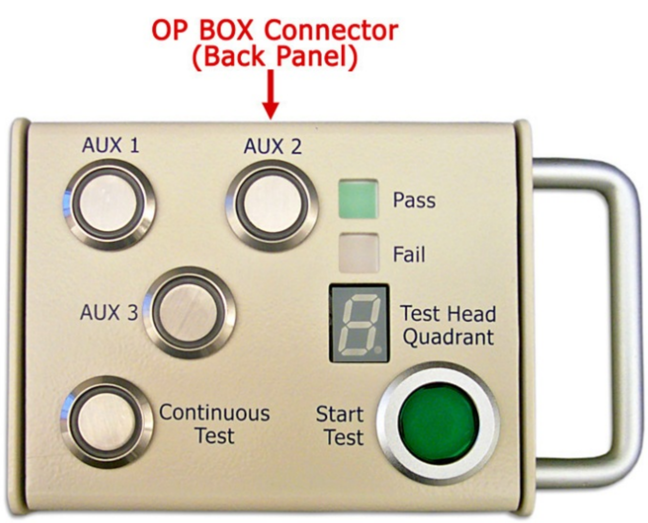

Figure 1-10 – The Operator Box

The Operator Box RJ-45 connector signal definitions and pin assignments are shown in Figure 1-11 and Table 1-2 below.

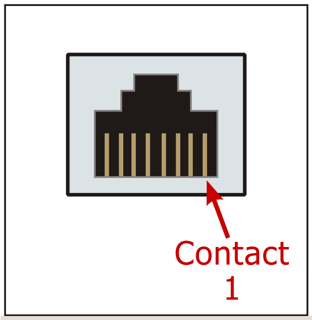

Figure 1-11 – OP BOX RJ-45 Connector Pin

Table 1-2 – OP BOX RJ-45 Connector Signal Definitions

Contact	Signal
1	+5 V
2	+5 V
3	PASS
4	FAIL
5	Spare
6	Start Of Test
7	AUX Switches

1.2.4 Programming Overview

1.2.4.1 General

The software environment in the ETS-88™ system is based on Windows® XP (English).

All of the software developed by ETS has been developed under Windows® XP (English). This software includes utilities that give you direct control over the test head via the "C++" programming language. The ETS utilities look and act like a natural part of the language.

1.2.4.2 Language

C++ is a compiled programming language and is the standard test programming software provided with the ETS family of testers. The ETS utilities interface to the language as external functions. Developing a test program is a four step process:

Step	Description	Notes	File Usage
1	Edit a text file	Use the editor	File.CPP
2	Compile the text file	Compile "File.CPP" from the editor	File.OBJ
3	Link the object file	LINK "File.OBJ" from PWB	File.DLL
4	Execute the file	Run file from the Shell	c:\ets\bin\shell\testexecutive.exe

Visual C++®, in conjunction with Eagle's Shell, combines the steps required to create an executable program. In Step 1, you use the text editor to edit a skeleton file created by Eagle's Shell. In Steps 2 & 3, Visual C++® compiles your file and links it with other files to create an executable program. The other files that are linked to your test program include the ETS utilities interface and any reusable modules of code. A program can be linked together with another module to take advantage of pre-existing routines. In Step 4, Eagle's Shell invokes or executes the compiled program.

The Eagle Shell system incorporates the entire edit/compile/link/run process into a series of menu selections. Refer to the ETS Software Help File or Eagle Vision Software Suite Manual for more information.

1.2.4.3 Utilities

Standard languages such as C, Pascal, and Basic do not support IC testing. For this reason, the ETS Family utilities are necessary to allow the software to interface with the test head hardware. As mentioned earlier, the ETS utilities are linked with the test program. These utilities allow you to set power supplies, measure voltages, measure currents, load patterns into the AWG, measure time intervals, etc. The utilities are documented in The ETS Software Help File.

1.2.4.4 Debuggers

The ETS Family offers a two-level debugging system. This system is based on the Visual C++® Source Code Debugger and Eagle's RAIDE environment (discussed in Chapter 3 of this manual). The Visual C++® Source Code Debugger lets you set break points, watch and change variables, single step through a program, and view program output. This debugger focuses strictly on the program itself.

The RAIDE environment is hardware-oriented. It gives you a direct link to the test hardware, allowing you to check or change the status of any tester resource at any time. By using the RAIDE environment from within the Visual C++® Source Code Debugger, you can pause at any point in your test and see the interaction between the software, hardware and DUT.

Together, the Visual C++® Source Code Debugger and RAIDE give you a powerful tool for test development, which makes your work easier and decreases your development time.

1.2.5 Test Development Overview

1.2.5.1 General

Some aspects of a test development sequence are the same on any piece of test equipment. The programmer must understand the device, the parameters to be tested, and the hardware available for testing to maximize efficiency and accuracy. We will now discuss the steps necessary to create test programs on the ETS Family system.

1.2.5.2 Test Specification

Creating a test specification is usually a time consuming task; however, it makes sense to have a test specification for the device being tested. A good test specification includes all the critical parameters and functions to be tested. The more completely you specify the test, the more assured you are of a properly tested device.

1.2.5.3 Program Control Specification

A program control specification defines exactly what the program itself must do. Although the test may be defined, the flow of the test program must also be defined. For example, a customer who is trying to fill an incoming inspection requirement may be interested solely in go/no-go testing. A manufacturer may be interested in characterization testing where data output is vital.

The program control specification also defines how the various tests will be run and what will be done with the generated data. A program flow chart can speed test development, and is very helpful to describe the test to others or as a future reference.

1.2.5.4 Software Development

After you complete the hardware interface design, the software development begins. Certain ETS utilities allow you to use the test and program control specifications to create the software for testing the device. You will use the four step process of editing, compiling, linking, and executing the program several times as you prepare the program. ETS recommends writing the program one step at a time, stopping at each step to debug the new code. This allows you to catch any mistakes and correct them early in the test development sequence.

1.2.5.5 Test Debugging

Test debugging should be part of the ongoing process of program writing. As mentioned earlier, the Visual C++® Source Code Debugger and RAIDE provide powerful tools for diagnosing and correcting problems. Programmers should never assume that the test is working until every portion of the test performs as expected. Once you complete the program, thoroughly evaluate it using as many different test devices as possible. Any unexpected test results should be recorded and bench tested for correlation.

1.3 Getting Started

1.3.1 Introduction

This section of the manual explains some of the fundamental properties of the system. The following procedures are covered here:

- Setting up the system
- Booting up the system
- Learning more

1.3.2 Setting Up the System

1.3.2.1 General

NOTE: Normally, your ETS-88™ Test System is installed by Eagle factory personnel; however, it is useful to understand some of the aspects of the system setup. If you ever need to execute some portion of the setup procedure, it is provided here.

The following items come pre-installed in the ETS-88™ mainframe:

- 8 KVA AC Power Distribution Unit (PDU)
- Test Head Power Control Units (TPCs)
- Digital and Floating Resource Card Cages (DCC and FRC)
- Test Head Fan Panel and AC Rail
- Mainframe Ventilation Fan

These items will still need to be connected using the proper cabling. This cabling is described in the next section.

1.3.2.2 System Power Components

With the PDU and TPCs pre-installed, the next step is to install the units that convert and supply the power from these main sources.

1.) Inspect the jumpers on the rear of the PDU to verify that it has been properly tapped for your facility's power (refer to the silkscreen on the front of the unit for guidance).

2.) Connect the 8 KVA PDU to facility power via its main power cord.

3.) Connect the TPC(s) to the PDU via the TPC's main power cord(s).

4.) Install the DC Power Supply (or Supplies) in the DC Supply Tray.

5.) Install the 19 kHz Power Supply (or Supplies) and associated vent panel(s).

6.) Connect the DC and 19 kHz supplies to their appropriate switched outlets on the TPCs.

7.) Connect the high and low current cables between the DC Power Supply and the Test Head Power Distribution Board

The connectors on these cables are keyed so that they only connect to the one correct power jack on the rear of the test head. The current ratings are as follows for the ASM5116 DC Power Supply:

Table 1-3 – ETS-88 DC Supply Current Ratings

Nominal	Low Limit	High Limit	Usable DUT Current	Current Capacities
+3.3 V⎓	+3.25 V⎓	+3.35 V⎓	0 A	60 A
+5 V⎓	+4.8 V⎓	+5.2 V⎓	10 A*	60 A
-5.2 V⎓	-5.24 V⎓	-5.16 V⎓	2 A*	10 A
+12 V⎓	+11.7 V⎓	+12.3 V⎓	6 A*	17 A
-12 V⎓	-12.3 V⎓	-11.7 V⎓	1 A*	10 A
+15 V⎓	+14.6 V⎓	+15.4 V⎓	2 A*	8 A
-15 V⎓	-15.4 V⎓	-14.6 V⎓	2 A*	4 A
+24 V⎓	+23.4 V⎓	+24.6 V⎓	1.6 A*	4 A

8.) Connect the Test Head Power Switch interlock cable between each TPC and DC Power Supply.

9.) Connect the Test Head Interlock cable between the 19 kHz Supply and the Test Head Fan Panel.

10.) Connect the Fan Power cable between the unswitched outlet on the TPC, the test head fans, and the mainframe ventilation fan.

11.) Secure all cabling to the mainframe strain relief.

12.) Perform a system power supply check (contact ETS for details).

1.3.2.3 Computer Console

NOTE: This subsection describes how the test system computer has been configured specifically for use in the ETS-88™ system. For more specific information regarding the main console computer, refer to its manuals, which are included with the system.

The computer console consists of a computer, keyboard, and monitor. The computer is freestanding and housed in a mini-tower case. The keyboard has a coiled cable that plugs directly into the back of the computer. The monitor plugs into the back of the computer by way of a fifteen pin connector.

The basic setup procedure for the system computer is as follows:

1.) Connect the monitor, keyboard and mouse to the appropriate connections on the rear of the computer.

2.) Install the dongle onto the printer parallel port or USB connection on the rear of the computer (unless using FLEXnet® software licensing – see the "Options" section below).

3.) Plug the computer and monitor's power cords into the unswitched outlets on the TPC (recommended for surge protection), or any facility AC outlets.

Options

Because computers have a vast array of available options, we cannot cover them all here. If you have requirements beyond your present system's capabilities, please contact ETS to discuss available options and how they will affect your test system.

Licensing of the ETS Software on the system computer can either be controlled with a physical software key (dongle), or through FLEXnet® software licensing. This licensing will be established prior to shipment of the tester. If using a dongle, you will simply need to plug it in as described above.

Other

ETS strongly recommends backing up all the supplied master software disks at least twice. The originals should be stored in a safe location and only used in an emergency. Be aware that Microsoft software carries license agreements that are binding directly between Microsoft and your company.

Registration cards are provided for each of the Microsoft packages. You should immediately fill out these registration cards and send them to Microsoft. If this is done, you will be informed of the latest versions of the Microsoft software. ETS will assume that you have sent in your registration cards and are receiving update information from Microsoft, and will not provide customers with this update information.

When you receive new or updated software from Microsoft, call your ETS sales administrator before installing it. ETS must verify that the new software does not affect the operation of the system adversely. Because the evaluation of new software takes time, you may wish to contact ETS before the purchase of any new revisions to see when (or if) it will be supported.

1.3.2.4 Dual Test Heads

The setup procedure for the test heads is as follows:

1.) Install the resources (seat all boards firmly against the backplane, and connect each floating resource's power cord to the AC Rail).
2.) Connect the High Current cables (if any MPUs or HPUs are installed).
3.) Connect the QTMU cables between the QTMU and DPU-16 resources.
4.) Connect the TCB(s) to the PC Interface Board(s) in the system computer.
5.) Connect the MS Handler cables.
6.) Connect the Operator Box.
7.) Connect the ISO-COMM cables.
8.) Connect the CIB to the QPLU (if one is installed).
9.) Install the Digital Card Cage cover(s).

When a test head is being set up for the first time after shipment, the boards must be installed in the card cages, and seated properly against the backplane before power is applied. Access the card cages by opening the mainframe's rear door, and removing the cage cover(s).

Figure 1-12 and Figure 1-13 (on the following page) illustrate the location of the resources and cables within a dual test head for a typical configuration. Install the TCB(s) before installing the other boards.

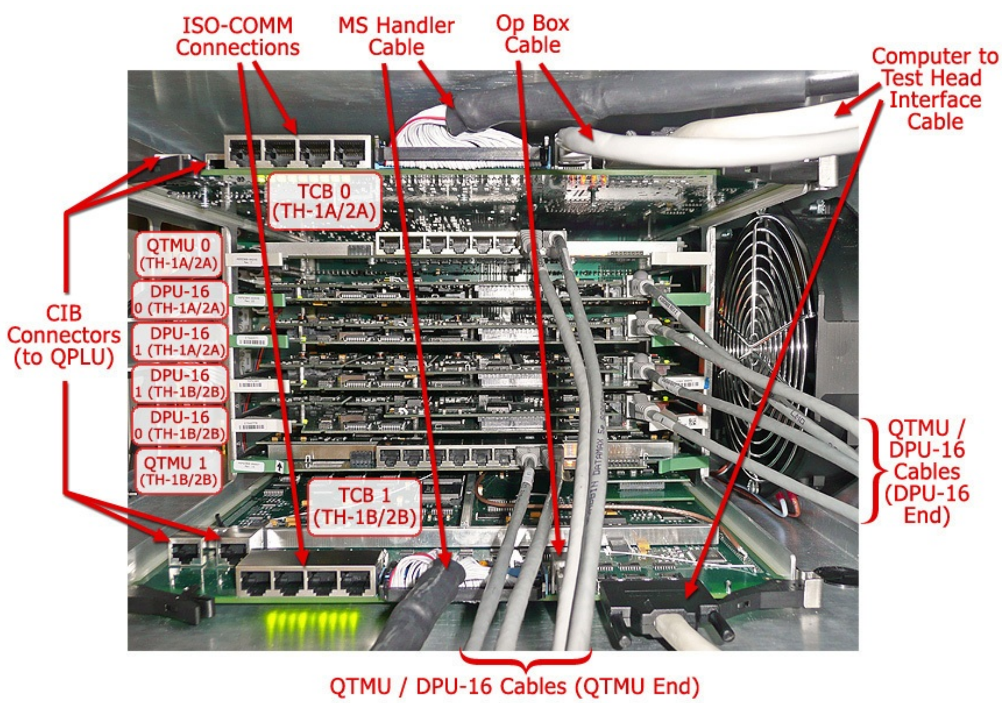

Figure 1-12 – Digital Card Cage Contents

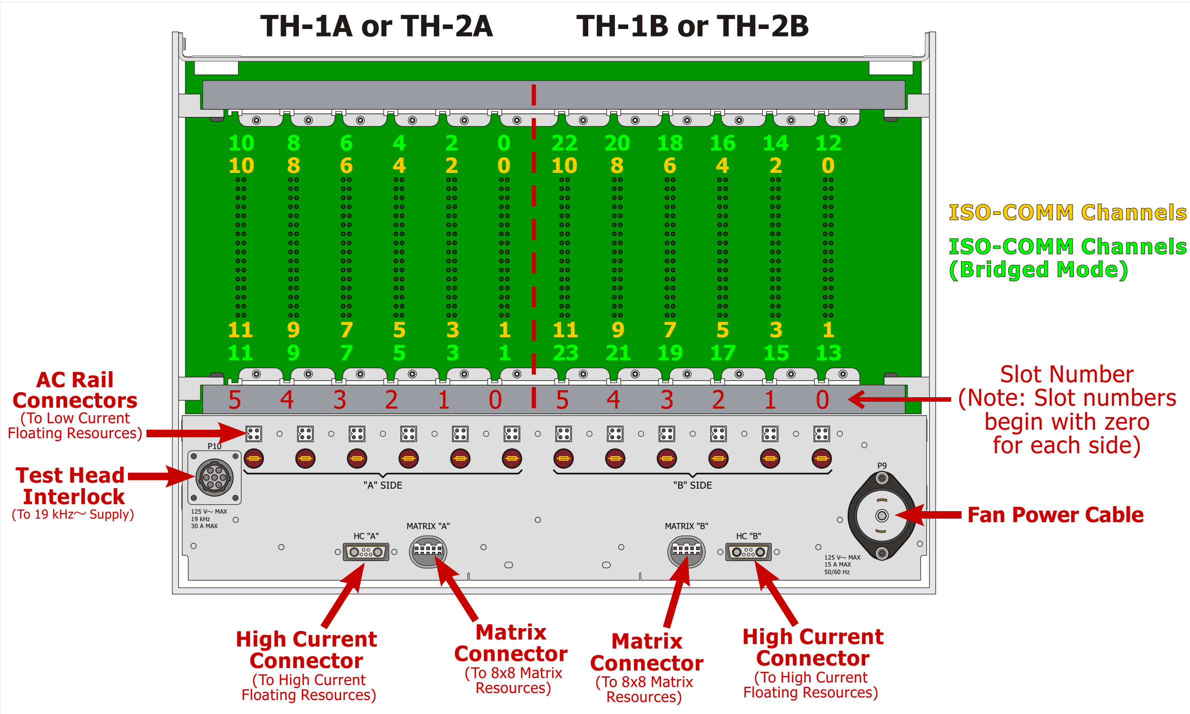

Figure 1-13 – FR Card Cage Overview

1.3.2.5 Miscellaneous

If the system will be connected to a handler, connect it via the MS Handler connector on the User Interface panel on the side of the mainframe.

At this time, the operator box connections can be made. The operator box is controlled through a 25 pin D connector which plugs into the back of the test head. The AC line cable may now be installed between the power supply box and the power strip at the bottom of the mainframe cabinet. The AC input cable to the power strip should be routed through the bottom of the cabinet and plugged into the nearest available outlet (110 VAC, 60 Hz, 30 A).

If the system ever needs to be shipped to another location, it is recommended that all of the boards be removed from the backplane and individually packed to protect them. If the system is ever to be relocated, consult ETS in order to protect your system warranty. After the test head is properly set up with power applied, run the test head diagnostics to verify the operation of the system. Once the system passes the diagnostics, the set up operation is complete.

1.3.3 Booting the System

When applying power to the test head for the first time, or after installing a new or replacement resource, it is recommended that the following power up sequence is followed:

1.) Verify that all 19 kHz power supply, DC power supply, and mainframe power switches are in their ○ (OFF) position.

2.) Verify that the mains circuit breaker is in its | (ON) position, and the mainframe back door is closed.

3.) Switch and hold the mainframe power switch to its ⌽ (START) position for approximately one second (the test head and mainframe fans should operate, but none of the power supply lights should power up). After approximately one second, let the mainframe power switch spring back to the | (ON) position.

4.) Apply power to the dual test head(s) using the Test Head Power Switch(es) on the DC Power

Supply (all of the LEDs should immediately begin to glow steadily).

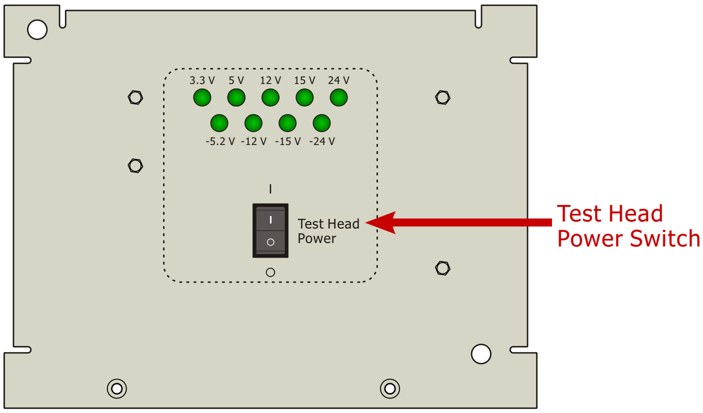

Figure 1-14 – Test Head Power Switch Location on the DC Power Supply

5.) Switch the 19 kHz power supply ON (the protect light should come on and, after an initial delay, the Power indicator should light).

6.) The test head is now powered up and ready for use.

Whenever power is interrupted either by switching the mainframe power switch to ○, engaging the EMO switch, disengaging the rear door interlock switch, or by a power loss at the source, you must switch and hold the mainframe power switch to its ◇ (START) position for approximately one second to power-up again. After approximately one second, let the mainframe power switch spring to the | (ON) position.

When powering up the tester for the first time after a shipment, pay particular attention to the LEDs on the DC supplies. These LEDs are directly connected to the various DC supply voltages within the system. If any of the LEDs fail to light, the tester should be powered down immediately. A good procedural habit is to check these LEDs every time the system is powered up.

Once the power is applied to the system, the system PC can be booted up by pressing its power switches. Booting the system PC runs a startup file that initializes certain portions of the computer, allowing it to work properly with the ETS Software.

When booting is finished, you will be in the Eagle Shell operating environment. See the Eagle Vision Software Suite Manual for information on finding your way through the menus and screens available under the Shell.

1.3.4 General Safety and Operation Considerations

Hazardous Voltages are present inside the mainframe cabinet when powered.

Observe precautions for handling static-sensitive devices when working with boards used within the system. Use the ESD strap grounding point on the Power Switch / EMO Panel.

CAUTION: Certain parts of the ETS-88™ system exceed two-person lift capacity. If you are unsure of an item's weight, use a proper lifting device (which meets national and local safety standards) to be safe if transport is necessary.

1.3.5 Learning More

The rest of this manual describes the various components of the system – the resources (force/measure devices, matrices, etc.), the software, signals running through the system, connections, etc. For more information on getting started with your system, refer to the ETS Software help files, the tutorials on the Documentation CD that shipped with your system software CD, and visit the ETS Web site (www.eagletest.com) to access application notes and other useful documentation.

If you still have questions after reading through the available documentation, please contact ETS/Teradyne Global Support by any of the following means:

Eagle Test Systems, Inc. a Teradyne Company
2200 Millbrook Drive
Buffalo Grove, IL 60089
World Wide Web: http://www.eagletest.com
E-Mail: Support@eagletest.com
Phone: (847) 367-8282
Fax: (847) 367-8640

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2 System Resources

2.1 Introduction

This chapter discusses the ETS-88™'s hardware resources. The resources are described as follows:

1.) Features and functionality

2.) Interfacing with the application board

3.) Hardware specifications.

Each description includes a programmer's block diagram and implementation notes for the practical application and use of each resource. Figures 2-1 and 2-2 show simplified block diagrams of the overall ETS-88™ tester, to illustrate how all the resources are related from a system perspective.

NOTE: Resource availability is dependent on the system software. Refer to each resource's "Software Support" section to determine compatibility with your system. For programming details, see the applicable software's help file.

2.2 Test Head Resources

2.2.1 General

This section explains the overall configuration of the resources available in the ETS-88™. Each functional block of the system is explained individually in subsequent sections.

The Test Head Interface Board, commonly known as the THIB, is located on the TCB (Testhead Control Board) as opposed to a stand-alone board that is found on an ETS-300/364/600. It is responsible for controlling the activities of the entire test head. The THIB controls test head operation, and also handles the various test head interfaces:

- The Operator Box Interface
- The Handler/Prober Interface
- The IEEE-488 Interface

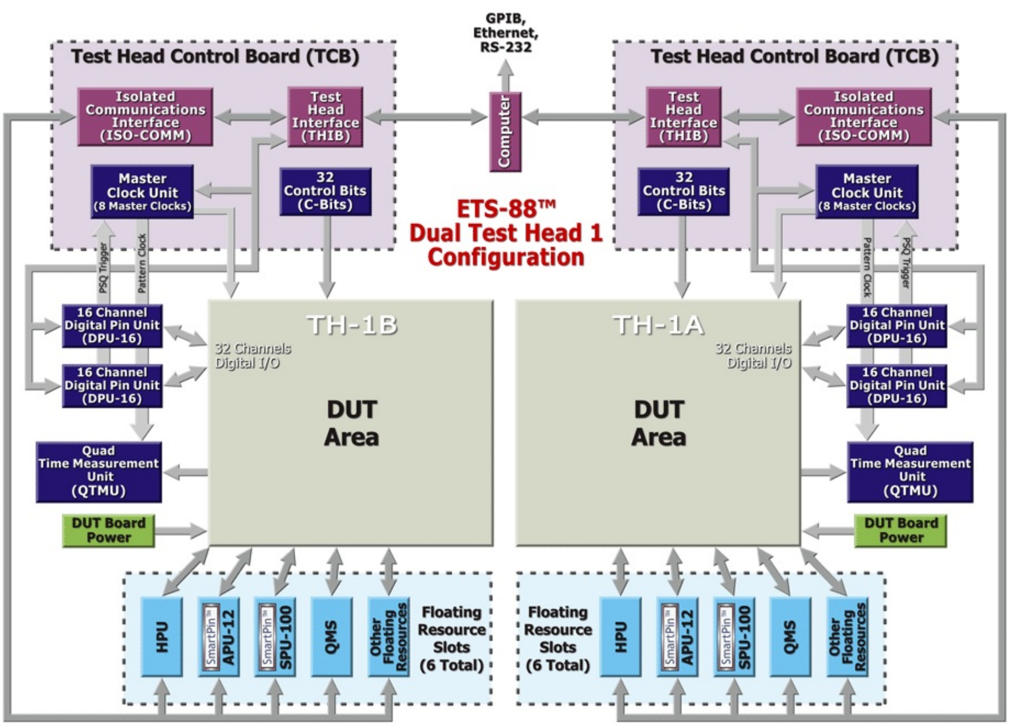


Figure 2-1 – Example ETS-88™ System Simplified Block Diagram (Dual Test Head 1 Configuration)

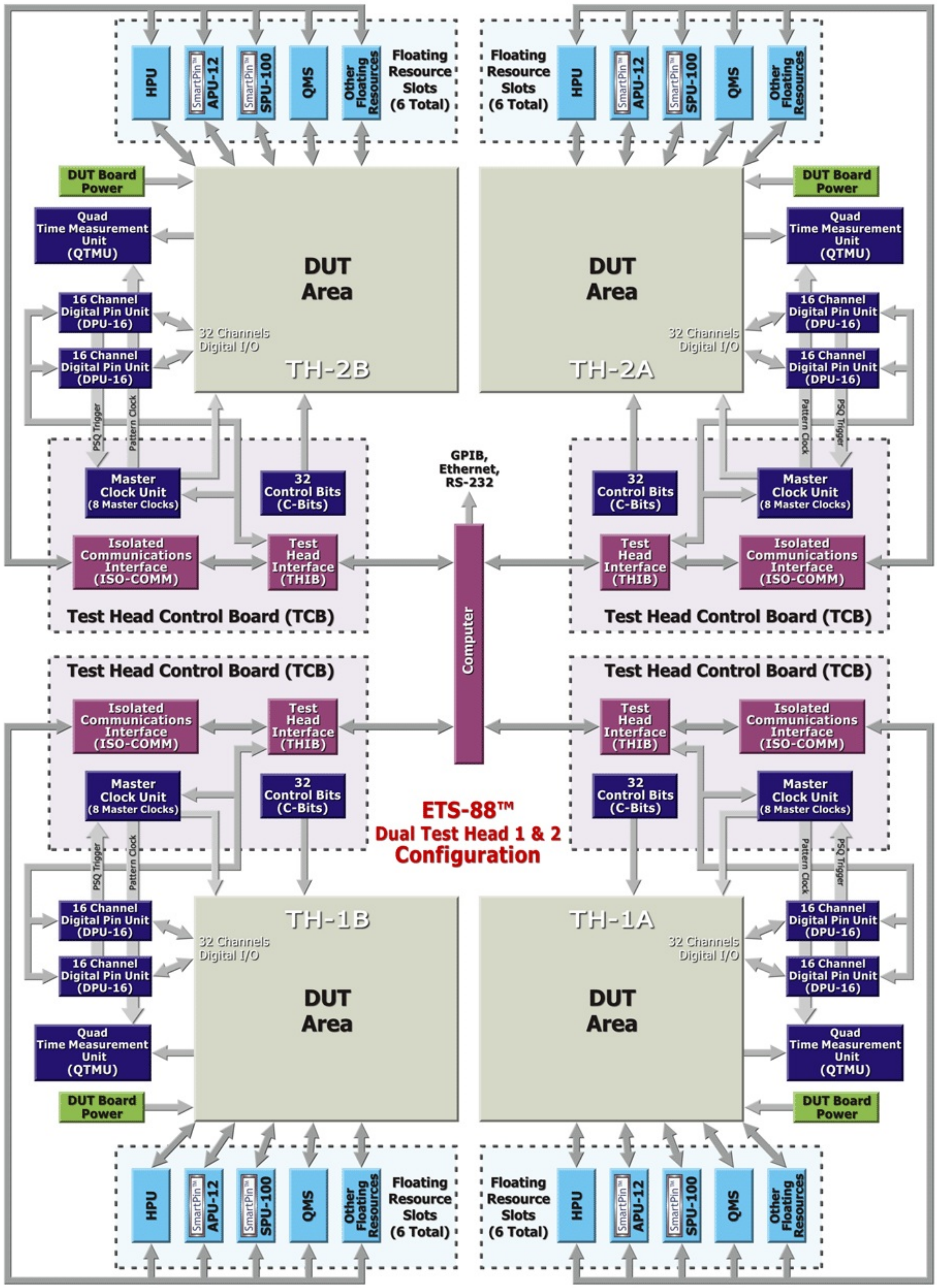


Figure 2-2 – Example ETS-88™ System Simplified Block Diagram (Dual Test Head 1 & 2 Configuration)

2.2.2 User Interface

A test program for a DUT usually does more than exercise the device and determines whether it passes or fails. Particularly in a production environment, the program must also recognize start-of-test signals, output an end-of-test signal, and communicate information to an operator, automatic handler, or external instrument. The ETS-88™ features several ports that facilitate this exchange of information.

All communication over the system ports is accomplished through calls to ETS-88™ utilities. The ports are listed below, accompanied by the names of the applicable utilities:

- Operator Box – `sot()`, `bin()`, `lbin()`, `aux()`
- MS Handler Port – `mshsotset()`, `mshhandler()`, `mshinit()`, `bin()`, `lbin()`
- GPIB Port – `gpib()`

See Appendix A for cable pinout diagrams, and The ETS Software Help File for descriptions of these utilities.

2.2.3 Implementation Notes

A typical test program waits for a start signal to determine when to begin the next test. Three sources can start a test:

- The "Start Test" switch on the Operator Box
- The handler test control line on the MS Handler port
- The console keyboard

The auxiliary switches on the operator box can be used to control the mode of a test program. For example, an auxiliary switch could be used to switch between "stop on fail" and "continue after fail" operation.

2.3 8 x 8 Matrix

2.3.1 Features

- Eight (8) two-wire force/sense output channels
- Eight (8) two-wire force/sense resource input channels
- Output channel isolation: 1000 V
- Input channel isolation:
    - Channels 0 – 5: 500 V
    - Channels 6 – 7: 1000 V
- >20 MHz bandwidth
- Fast channel closure response time: <2 ms

2.3.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]

2.3.3 Theory of Operation

The 8x8 Matrix is a high voltage force/sense crosspoint matrix. Any input channel can be connected to any output channel. Force/Sense lines are closed with the same command. All outputs are rated for 1000 V. Inputs 0 – 5 are rated at 500 V and inputs 6 & 7 are rated at 1000 V. Inputs 6 and 7 are internally cabled to the external inputs on the ETS-88™ application board per your tester's configuration.

The most common use for the 8x8 Matrix is connecting a single tester resource to several DUT pins. This is done by connecting the matrix input lines to the resource via the application board, and connecting the DUT pins to the matrix outputs again via the application board. The 8x8 Matrix may also be used to stack floating resources.

Because its inputs are not dedicated, and are available at the application board, the 8x8 Matrix is extremely flexible and its use is determined on an application-to-application basis rather than being hardwired in the test system.

2.3.4 Block Diagram

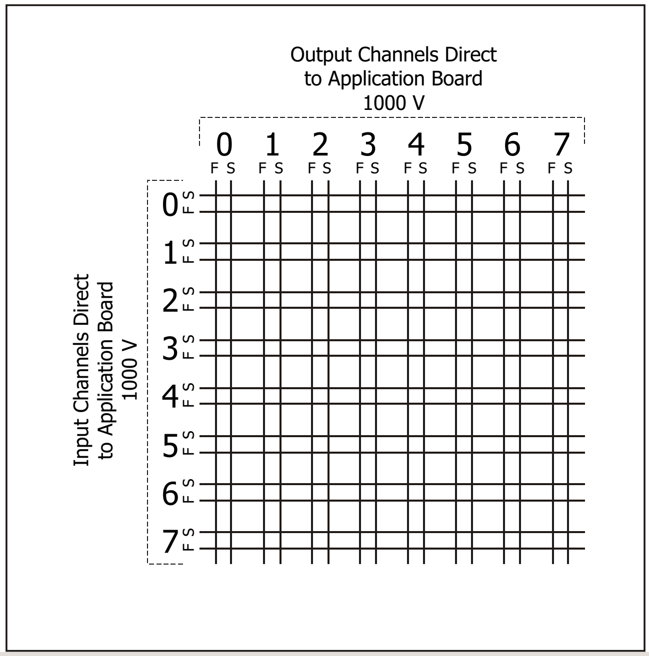


Figure 2-3 – 8x8 Matrix Block Diagram

2.3.5 Specifications*

Number of Output Channels	8
Number of Input Channels	8
Maximum V	1000 V
Maximum Continuous I	1 A
Maximum Pulsed I	2 A
Bandwidth	>20 MHz
Capacitance (Closed Contacts)	<100 pf/Channel

*Specifications subject to change without notice.

2.3.6 User Interface

2.3.6.1 Software

All functions of the 8x8 Matrix are programmed using Matrix Utility (`matxxx()`) calls from your C test program. These utilities and their syntax and usage are described in the ETS software help file.

2.3.6.2 Hardware

The 8x8 Matrix resides in the Floating Resource Card Cage. It contains 64 double-pole relays configured as eight dual-path (force and sense) input channels, and eight dual-path (force and sense) output channels. The ETS-88™ test system can support up to four 8x8 Matrix cards (one per side of each dual test head).

There are four 8x8 Matrix cables that route channels 6 and 7 to the application board. The cables are connected to the P501, P502, P503, and P504 external input connectors on the alignment board. Figure 2-4 below shows example connections for an 8x8 Matrix in slot 0. Please refer to Chapter 4 for details on 8x8 Matrix connections to the Application Board.

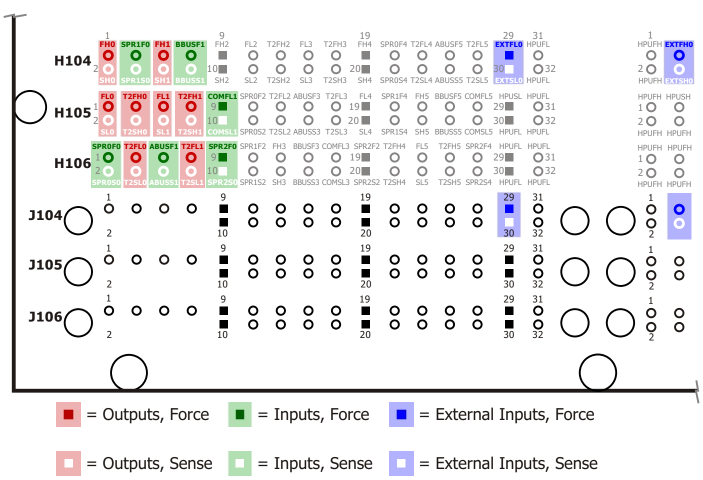


Figure 2-4 – 8x8 Matrix 0 Application Board Connections

2.4 Analog Pin Unit 10 µA (APU-10)

2.4.1 Features

- Eight (8) force/sense channels per board
- Two (2) force voltage ranges:
  ±30 V, 10 V (16 bit resolution per pin)
- Two (2) measure voltage ranges:
  ±30 V, 10 V (16 bit resolution, shared ADC)
- Three (3) force current ranges:
  ±100 mA, 10 mA, 1 mA
  (16 bit resolution per pin)
- Five (5) measure current ranges:
  ±100 mA, 10 mA, 1 mA, 100 µA, 10 µA
  (16 bit resolution)
- Hardware current clamps
  (limit current to 110% of range)
- Hardware voltage clamps
  (limit voltage to 110% of range)
- 100 KSPS AWG programmable to any or all eight channels (16 bit resolution)
- 100 KSPS digitizer multiplexed to eight channels (16 bit resolution, one per board)
- 2x8 matrix available per board
  - Maximum Voltage / Current: 200 V / 1A
- Fully floating (board isolation to ±30 V)
- Software measurement/test limit comparison functions (for ultra fast continuity testing)

2.4.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]

2.4.3 Theory of Operation

The Analog Pin Unit (APU-10) resource provides general purpose per-pin force and measure capability covering a range of ±30 V and up to ±100 mA. With eight channels per board, the APU-10 gives you a great deal of functionality at a modest cost.

Key features include 16 bit force and measure capability combined with fast MUX software compare measurements. Additionally, the APU-10 has a number of features which are typically not found in similar resources in the industry. A module-based design strategy has made it possible to offer synchronized AWG and digitizer capabilities on a per-board basis.

The APU-10 is capable of low leakage, fast settling measurements, unlike other design approaches. In the ETS system architecture, APU-10s are placed in the test head, where cable length and capacitance is minimal. Relative to other design approaches, the APU-10 has greatly reduced leakage.

APU-10s provide great flexibility in continuity and other parallel/multisite DC testing situations. Measurements are made via a 10 µs 16 bit shared ADC. The APU-10s may also be used in situations where an audio-based signal must be synthesized or digitized. The shared AWG and Digitizer on board are linked to the master clock, which allows coherent synchronization of analog and digital events. All of this with ±30 V / 100 mA, four quadrant capability.

2.4.4 V/I Quadrant Diagram

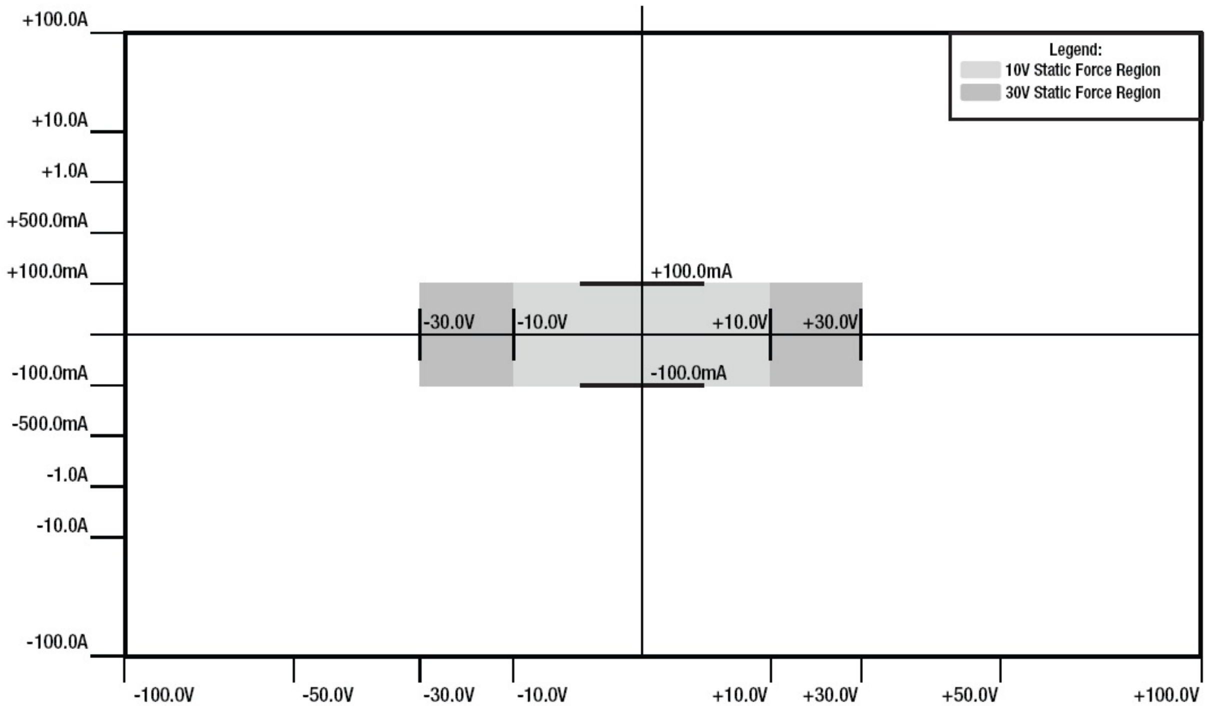


Figure 2-5 – APU-10 V/I Quadrants

2.4.5 Block Diagram

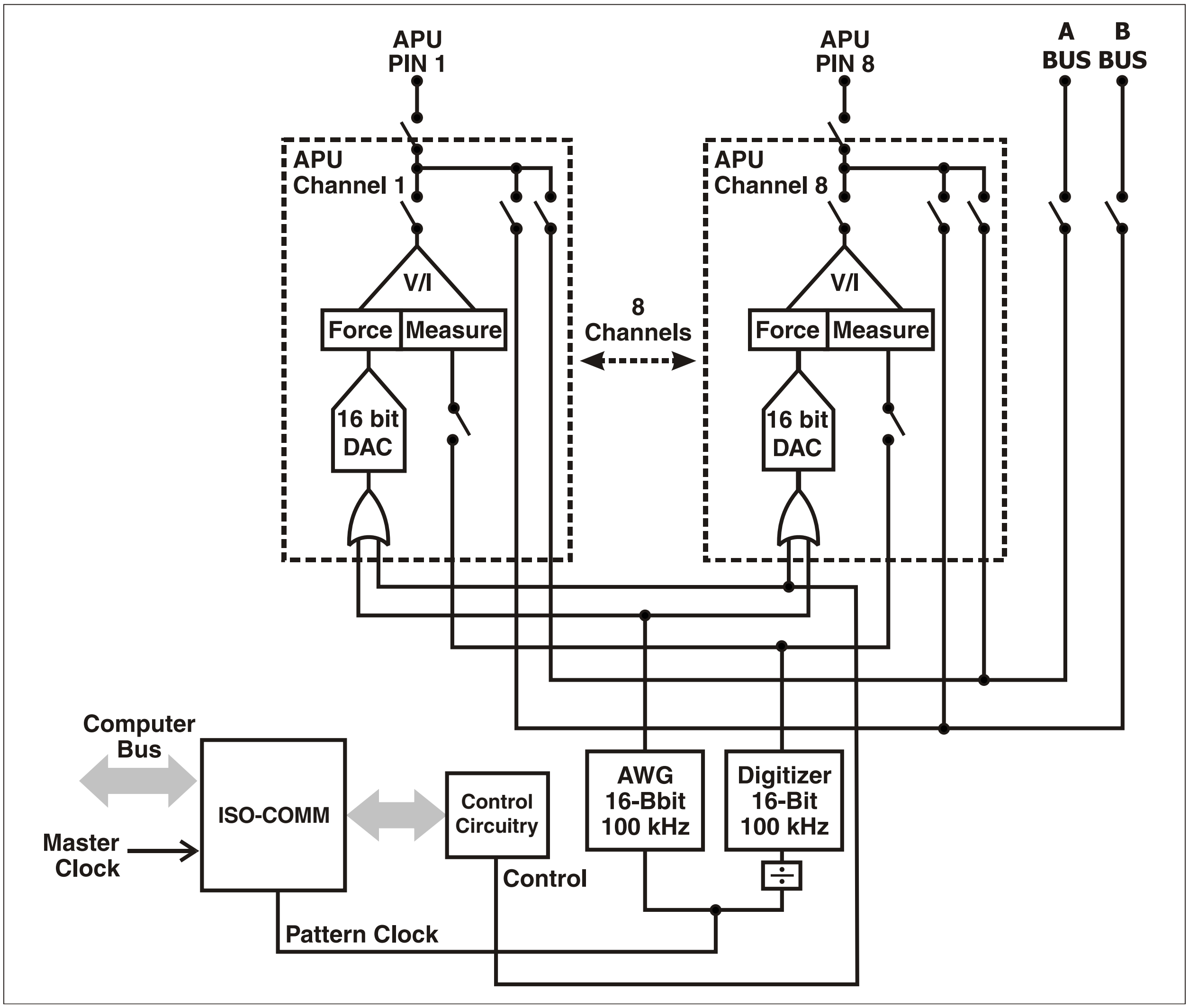


Figure 2-6 – APU-10 Block Diagram

2.4.6 Specifications

2.4.6.1 Voltage Force/Measure

Range	Resolution (16 Bit)	Accuracy
±10 V	0.305 mV	±(1.3 mV + 0.05% Rdg)
±30 V	0.915 mV	±(4 mV + 0.05% Rdg)

2.4.6.2 Current Force/Measure

Range	Resolution (16 bit)	Accuracy
±10 μA	.305 nA	±(8 nA + 0.1% Rdg + 0.2 nA/V)*
±100 μA	3.05 nA	±(20 nA + 0.1% Rdg + 0.8 nA/V)*
±1 mA	30.5 nA	±(125 nA + 0.1% Rdg + 8 nA/V)
± 10 mA	305 nA	±(1.25 μA + 0.1% Rdg + 80 nA/V)
±100 mA	3.05 μA	±(12.5 μA + 0.1% Rdg + 0.8 μA/V)

* Measure only

Waveform Digitizer

- One per eight pins
- Resolution: 16 bit
- Maximum Digitizer Sample Rate: 100 kHz (10 μsec)
- Digitizer Capture Memory: 4K

Arbitrary Waveform Generator

- Programmable to any combination of eight channels
- Resolution: 16 bit
- Maximum Clock Rate: 100 kHz
- Pattern Depth: 4K
- Pattern Looping allows continuous operation

2.4.7 User Interface

2.4.7.1 Software

All functions of the APU-10 are programmed using utility function calls from your C test program. These utilities and their syntax and usage are described in The ETS Help File.

NOTE: The TCB must be set up and started in order to clock the AWG and/or the digitizer. Please refer to the Test Head Control Board section (Section 2.17 on page 2-113) or the APU and MCB utility descriptions (in The ETS Help File) for further information.

2.4.7.2 Hardware

There are eight APU pins on an APU-10 board and they are housed in the Floating Resource Card Cage (FR Cage).

See Chapter 4 for the connections and pinouts for APU-10s.

2.5 Analog Pin Unit, 12 Channel (APU-12)

2.5.1 Features

- Twelve (12) Force / Sense Channels Per Board
  - Dual Bank Architecture
  - Six (6) Force Hi / Sense Hi Connections Per Bank
  - One (1) Isolated Force Lo / Sense Lo Connection Per Bank
- Three (3) Force / Measure Voltage Ranges: 30 V, 10 V, 3.6 V
  - 16-Bit Resolution
- Six (6) Force / Measure Current Ranges: 200 mA, 100 mA, 10 mA, 1 mA, 100 µA, 10 µA
  - 16-Bit Resolution
  - 2.4 Amp Capability Per Board
- 100 KSPS AWG Programmable to Any or All Channels Per Bank
  - 16-Bit Resolution
  - 256K Depth
  - Multiple Loop Capability
  - Independent Per-Pin AWG Patterns
- 100 KSPS Digitizer Multiplexed to All Channels Per Bank
  - 16-Bit Resolution
  - 32K Depth in ADC Mode (4K Depth for MI or MV)
  - Real-Time Hardware Measurement Accumulator – Instant Results Averaging
- Fixed Hardware Current Clamps (Limit Current: 120% to 150% of Range)
- Fixed Hardware Voltage Clamps (Limit Voltage: Up to 110% of Range)
- 2 x 6 Matrix Available Per Bank
  - 2 x 12 Matrix Available in Combined Mode
  - Maximum Voltage / Current: 200 V / 1 A
- Fully Floating (Board Isolation to ±30 V from Ground)
- High Side Kelvin Detect
  - Software Measurement / Test Limit Comparison Functions
- Ultra Fast Continuity Testing
- Fully Compatible with APU, APU-10
  - Eight (8) Force / Sense Channels + 2 x 8 Matrix Per Board
  - Common Force Lo / Sense Lo Per Board

2.5.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]
- Eagle Vision MST [Rev. 2011.1.1002.17 and up]

2.5.3 Theory of Operation

The Analog Pin Unit-12 (APU-12) is a single-slot, twelve channel, ±30 V, four quadrant V/I with six current ranges ranging from 200 mA to 10 µA.

The resource is organized in two banks of six channels. Each bank contains its own independent low side connections, ADC, and AWG to provide true single-board multisite capability, and also to allow coherent synchronization of analog and digital events.

In addition, two internal busses allow up to three channels to be "ganged" together for higher current capability. For example, an APU-12 could be configured so that four groups of three pins provide 600 mA from each group. The "ganged" channels must be in the same bank. Alternatively, one bank can "float" on the other bank, providing voltages up to ±60 V on each channel of the floating bank.

The V/I is stable with almost any combination of inductive and/or capacitive loads. Bandwidth and settling time are optimized to maximize measurement speed. Analog switches are used extensively for high reliability and fast switching speeds.

Other key features of the APU-12 include 16-bit force and measure capability combined with fast, MUX-based measurement comparisons. Additionally, the APU-12 has a number of features not typically found in similar products in the industry. In contrast to other design approaches, Eagle's architecture places the APU-12 in the test head, substantially reducing cable length and capacitance. One benefit of this architecture is the ability to make low leakage, fast settling measurements.

The APU-12 provides flexibility in continuity and other parallel/multisite DC testing situations. Measurements are made via a 10 µs 16-bit shared ADC. APU-12s can also be used in situations where an audio-based signal must be synthesized or digitized. The shared on-board AWG and Digitizer are linked to the master clock, which allows coherent synchronization of analog and digital events. All of this with ±30 V / 200 mA, four quadrant capability.

2.5.3.1 Waveform Digitizer

There are two ADCs on each APU-12. Channels 0 – 5 share ADC 1, and channels 6 – 11 share ADC 2. The 32K RAM of each ADC gives the user a powerful measurement tool. The clock coming into the APU-12 may be divided down to a sampling rate of 1 Hz – 100 kHz. This separate clock divider for the digitizer lets you measure at one sampling rate and force data with the AWG at a higher frequency. The digitizer can run concurrently with any forcing function, whether it is an AWG pattern or a DC forced voltage/current.

2.5.3.2 Arbitrary Waveform Generator (AWG)

The 16-bit AWG gives the APU-12 tremendous forcing capability, allowing you to reproduce any waveform from a sine wave to simulating the output of a digital driver into the DUT. The AWG has a maximum clock rate of 100 kHz, with 256K of RAM pattern depth behind each pin, which provides the ability to force either voltage or current (depending on the mode selected). The software lets you load concatenated patterns into the AWG, and then run these patterns individually and/or nonsequentially. The APU-12 will allow individual pins to output different AWG patterns, in differing modes and ranges, concurrently.

2.5.4 V/I Quadrant Diagram

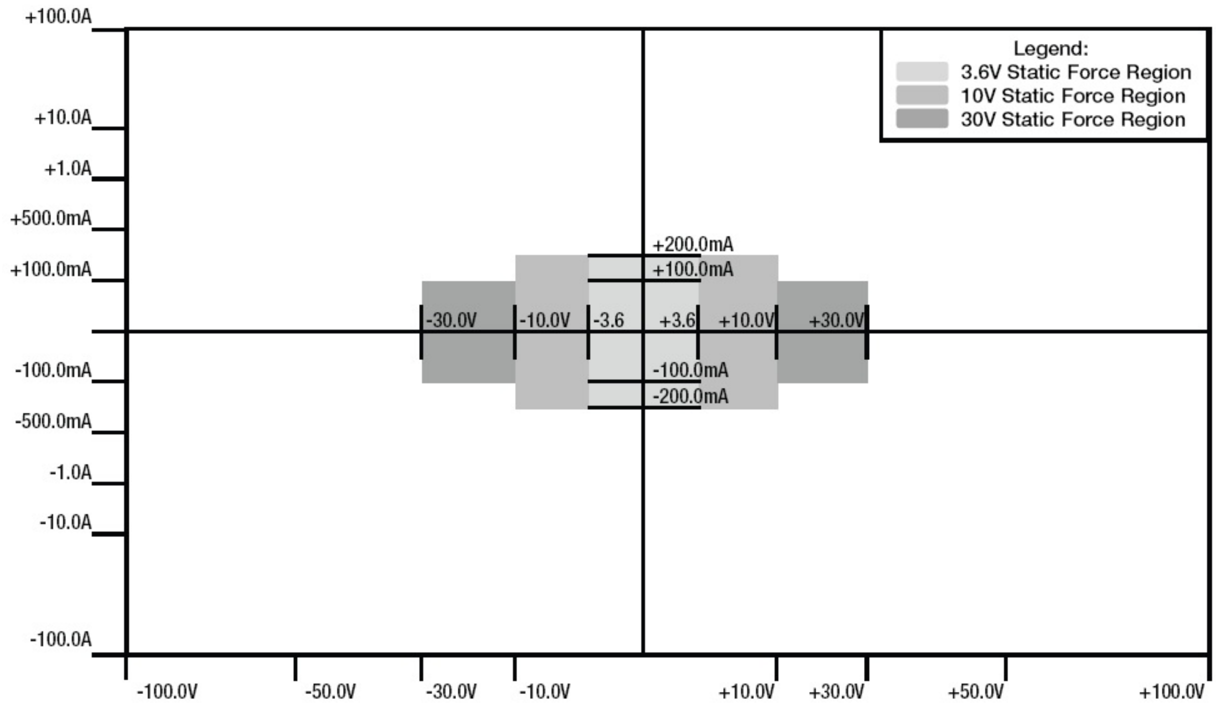


Figure 2-7 – APU-12 V/I Quadrants

2.5.5 Block Diagram

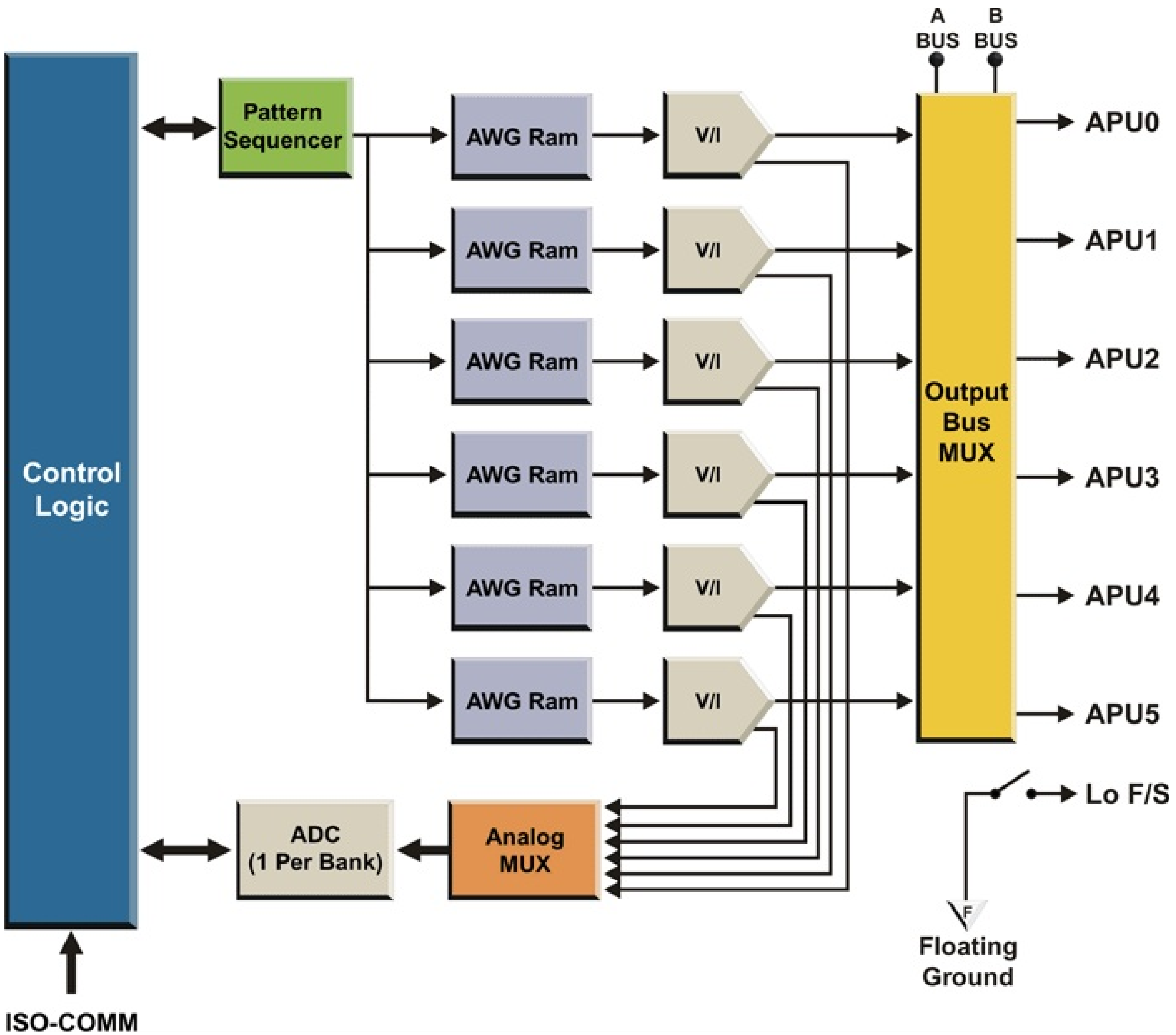


Figure 2-8 – APU-12 Internal Architecture (½ Board)

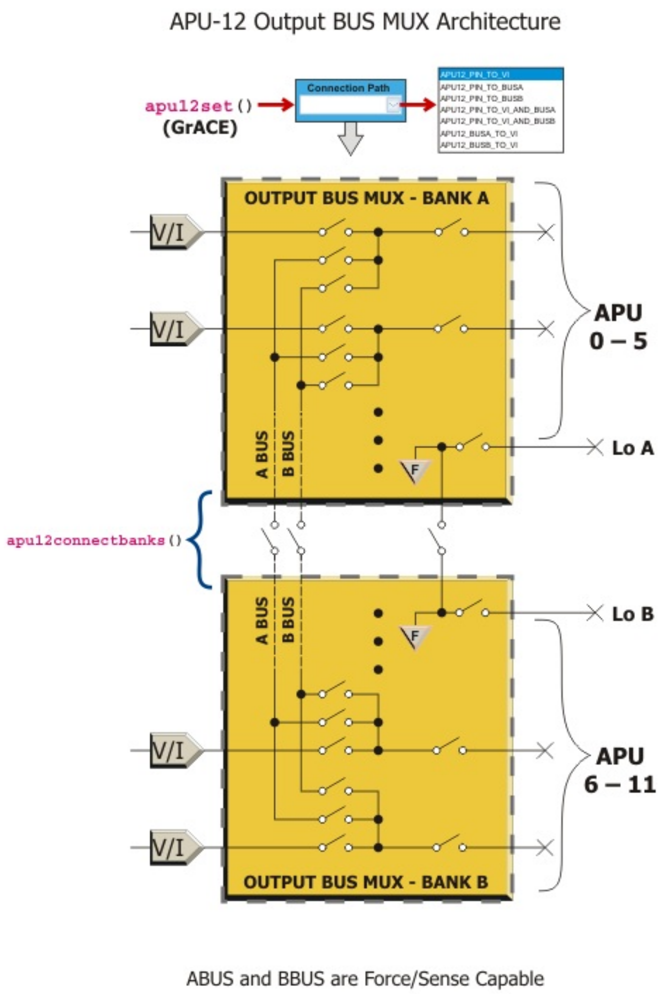


Figure 2-9 – APU-12 Output MUX Diagram

2.5.6 APU-10 Emulation Mode

The APU-12 can also be configured to emulate the 8-channel APU-10. To accomplish this, eight of the APU-12's output channels (force Hi and Sense Hi) are used. The force/sense lo output for APU-12 channels 0-5 acts as the common force/sense lo output to the eight APU-12 channels. In order to emulate the ABUS and BBUS capability of the APU-10, two of the APU-12s output channels are converted to ABUS and BBUS connections while in this emulation mode.

This emulation mode will allow customers purchase APU-12s for additional analog pins plus the improved capability in Pattern-Based testing and in speed of measurements, while still providing backward compatibility for test programs written for APU-10's, thus allowing users to improve tester performance for newer applications and still maintain compatibility and correlation with older test programs.

APU-10 Emulation Mode
Architecture

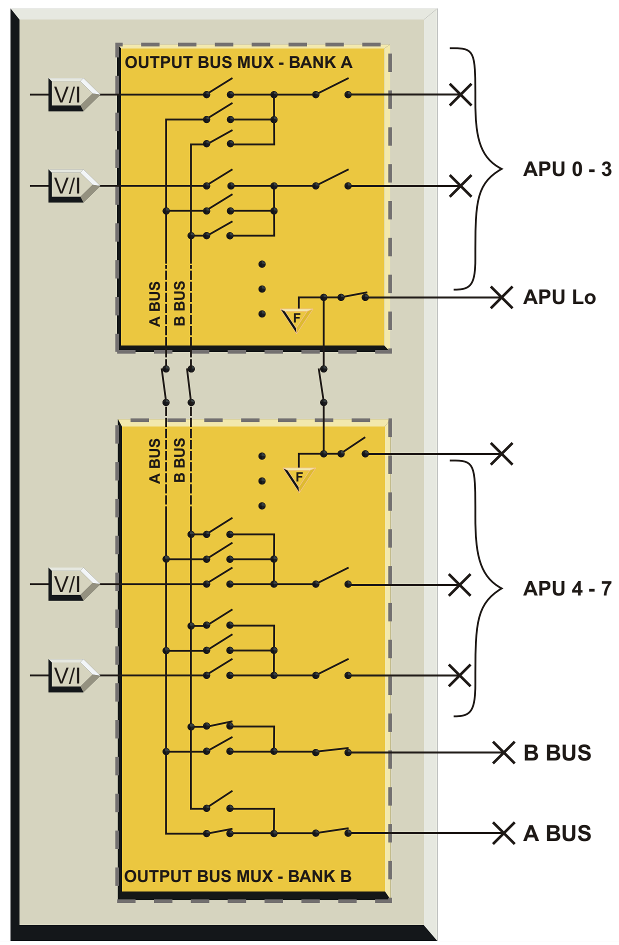


Figure 2-10 – APU-12 Emulation Mode Diagram

2.5.7 Specifications

Voltage Force/Measure

Range	Resolution (16-Bit)	Accuracy
±3.6 V	122 µV	±(.45 mV + .025% Rdg)
±10 V	305 µV	±(1.3 mV + .025% Rdg)
±30 V	915 µV	±(4.0 mV + .025% Rdg)

Current Force/Measure (x1 amplifier)

Range	Resolution (16-Bit)	Accuracy
±10 µA	0.305 nA	±(8 nA + .05% Rdg + 0.2 nA/V)
±100 µA	3.05 nA	±(20 nA + .05% Rdg + 0.8 nA/V)
±1 mA	30.5 nA	±(125 nA + .05% Rdg + 8 nA/V)
±10 mA	305 nA	±(1.25 µA + .05% Rdg + 80 nA/V)
±100 mA	3.05 µA	±(12.5 µA + .05% Rdg + 0.8 µA/V)
±200 mA*	6.10 µA	±(25.0 µA + .05% Rdg + 1.6 µA/V)

* 10V & 3.6V ranges only.

Current Measure (x10)

Current Force Range	Resolution (16-Bit)	Accuracy
±10 µA	Not available	Not available
±100 µA	0.305 nA	±(12 nA + .05% Rdg + .4 nA/V)
±1 mA	3.05 nA	±(60 nA + .05% Rdg + 4 nA/V)
±10 mA	30.5 nA	±(600 nA + .05% Rdg + 40 nA/V)
±100 mA	305 nA	±(6 µA + .05% Rdg + .4 µA/V)
±200 mA*	610 nA	±(12 µA + .05% Rdg + .8 µA/V)

* 10V & 3.6V ranges only.

Hardware Clamping Limits

Mode	Range	Typical Limit
Current	±200 mA	120% of Range
	±100 mA	130% of Range
	±10 mA, ±1 mA, ±100 µA, ±10 µA	150% of Range
Voltage	±30 V, ±10 V	110% of Range
	±3.6 V	110% of 10 V Range

2.5.8 User Interface

2.5.8.1 Software

All functions of the APU-12 are programmed using utility function calls from your C++ test program. These utilities and their syntax and usage are described in Eagle Vision software help file.

NOTE: The TCB must be set up and started in order to clock the AWG and/or the digitizer. Please refer to the Test Head Control Board section (Section 2.17 on page 2-113) or the APU-12 and MCLK utility descriptions (in the Eagle Vision software help file) for further information.

2.5.8.2 Hardware

There are 12 APU pins on APU-12 boards, which are housed in the Floating Resource Card Cage (FR Cage). See Chapter 4 for the connections and pinouts for APU-12s.

2.6 Digital Pin Unit, 16 Channel (DPU-16, DPU-16/8M)

NOTE: In this document, 'DPU-16' refers to both the standard and enhanced (8 Meg) versions of the resource unless stated otherwise.

Also NOTE: Standard and enhanced DPU-16 resources cannot co-exist in a system.

2.6.1 Features

Pin Electronics

- 16 complete I/O channels per board
- Per-pin drive levels (-1.0 to +7.0 V; 16 bit resolution)
- 50 Ohm output impedance
- Per-pin voltage level window compare (-1.0 to +7.0 V; 16 bit resolution)
- >150 MHz receive bandwidth
- Per-pin selectable TMU input ranges (-1.0 V to +7.0 V; -3 V to +21 V)
- Integrated TMU start/stop multiplexing

Vector Speed and Timing

- 66 MHz vector rate (SDR)
- 132 MHz vector rate (DDR)
- Independent timing per pin
- Robust set of data formats
- Drive timing (±500 psec skew; 50 psec resolution; 3.75 nsec minimum pulse width)
- Receive timing (±500 psec skew; 50 psec resolution)
- Selectable DDS-based DUT clock (available to each pin)
- Serial mode (2 to 16 bit)
- LVDS capable
- Timeset switching

Pattern Memory

- Vector Depth:
  - DDR –
    DPU-16 = 8 Meg DPU-16/8M = 16 Meg
  - SDR –
    DPU-16 = 4 Meg DPU-16/8M = 8 Meg
  - Capture Memory
    (per-vector-step control):
    4 Meg (DDR)
    2 Meg (SDR)
- 8 K fail memory

Pattern Sequencer (PSQ)

- PSQ per board (per 16 pins; supports independent multisite operation)
- Serial mode (2 to 16 bit)
- 16 bit burst/repeat counter
- Nested loops up to 256 deep (16 bit loop counter)
- Nested subroutines up to 256 deep
- Conditional branching (pass/ fail; match mode)
- PSQ-based trigger to start master clock channels
- PSQ-based clock to substitute as a master clock channel for analog resource clocking
- TMU arming trigger
- Global step counter

DCs

- Per-Pin V/I (16 bit force/measure)
    - Output voltage ranges:
      -1 to +7 V, -2 to +8 V
    - Seven current ranges:
      32 mA, 8 mA, 2 mA, 512 µA, 128 µA,
      32 µA, 8 µA
    - Programmable Voltage Clamps
    - Parametric Comparator

2.6.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]
- Eagle Vision MST [Rev. 2011.1.1002.17 and up]

2.6.3 Theory of Operation

Overview

The DPU-16 resource provides 16 full-featured digital I/O channels supporting vector rates up to 132 MHz (DDR mode), with an 8 Meg standard vector depth (Same Cycle I/O Mode: 66 MHz at 4 Meg Vectors). The pin electronics operate over a range of -1.0 V to +7.0 V with independent programmable drive and receive levels per pin. The enhanced version of the resource (DPU-16/8M) includes all the functionality of the standard version, plus twice the pattern memory size (16 Meg for DDR, and 8 Meg for SDR).

Each board also has per-pin resistive loads (terminated to 16 bit DACs), a microcode-based pattern sequencer (designed for multisite operation), a per-pin V/I (for DC measurements), and a fully integrated TMU multiplexer (for time measurements).

Vector Speed and Timing

DDR (double data rate) is an important feature of the DPU-16 that allows all channels to operate at effectively double speed. In this mode, two sets of pattern data are accessed on every pattern step and multiplexed to produce double data rates. Any channel on any step can be configured as I or O at double data rate. Drive and Receive data each have independent timing. Instructions and control bits are executed once per pattern step regardless of mode, and data can be captured at double data rates.

Differential operation is supported at both data rates; however, differential operation consumes two channels. Any combination of channels can be operated at single or double data rates independently.

A wide variety of drive data formats are supported at both single and double data rates (see the data format chart on the following page), and deskew circuitry is provided for deskewing the edges of all formats, including the beginning of the period, the leading edge, and the trailing edge.

The DPU-16 also supports timeset switching on-the-fly. At single data rates (SDR), two timesets are available; however, if a pin is defined to be uni-directional, then up to four timesets are available per pin. At double data rates, two timesets are available only if a pin is defined to be uni-directional. Refer to Figure 2-12 and Table 2-1 on the following pages for more information on the DPU-16's on-the-fly timeset switching.

Notes:

1) If a pin is defined as a receive pin, then its driver is always tristated.

2) Receive strobes are active during drive steps so data can be "captured;" however, data comparisons are suppressed.

DC Measurements

The DPU-16 is also capable of making fast, accurate DC measurements with its per-pin V/Is (PPVIs). Each channel has access to a multi-range PPVI for parallel continuity and leakage measurements.

Every pin on the DPU-16 also has an independent window comparator (and DACs) the input of which is an analog voltage generated by the per-pin V/I. This window comparator is normally isolated from the DUT pin during patterns but can be switched in during parametric tests to provide real-time continuity results without using the on-board ADC. The dedicated DACs ensure true "set and forget" capability.

The comparator bits for all pins are routed to a single register so that the status of all comparators can be read in one pass. In addition, you can use the on-board ADC to measure the actual value of the voltage if desired.

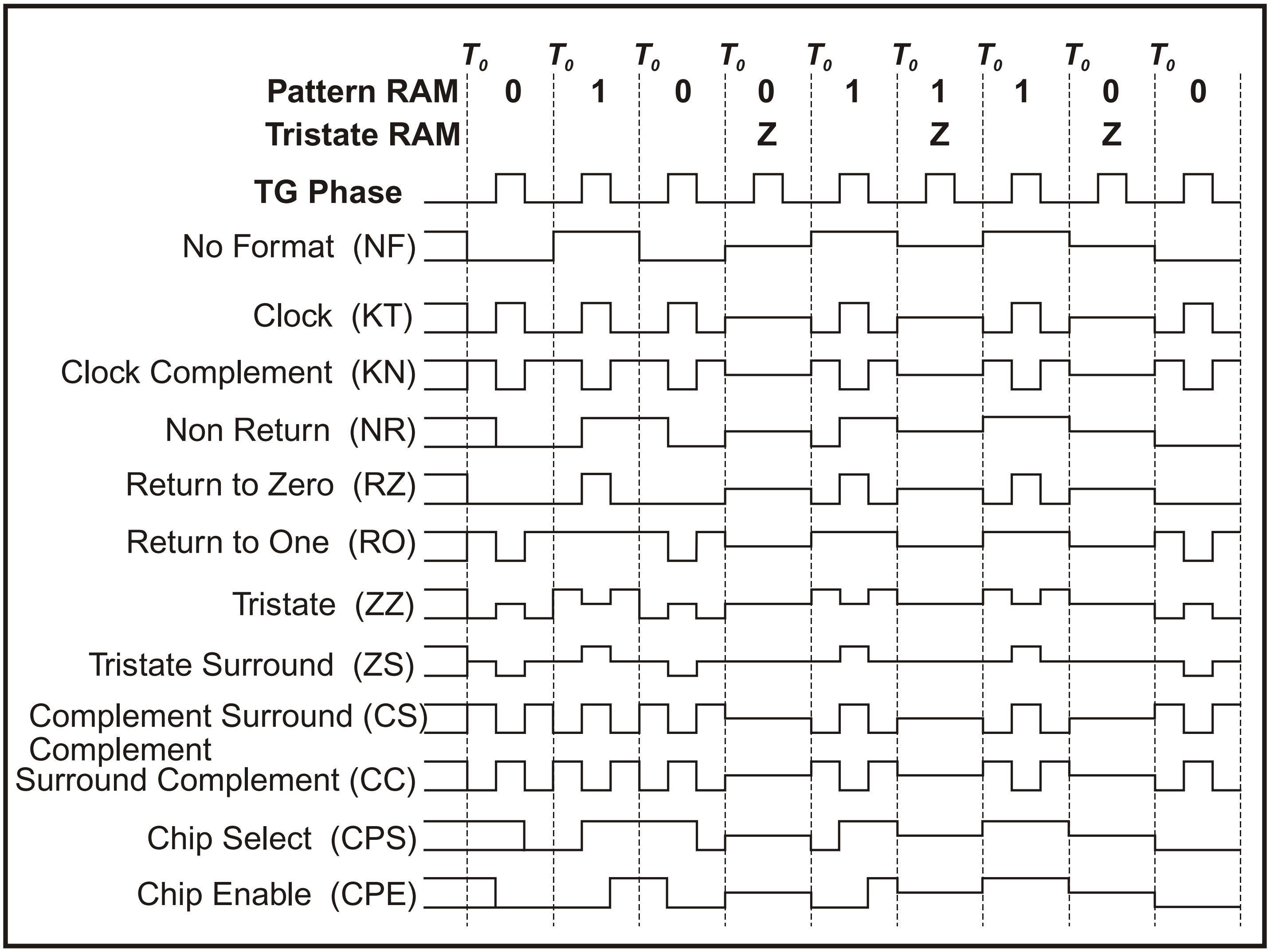


Figure 2-11 – DPU-16 Digital Data Format Chart

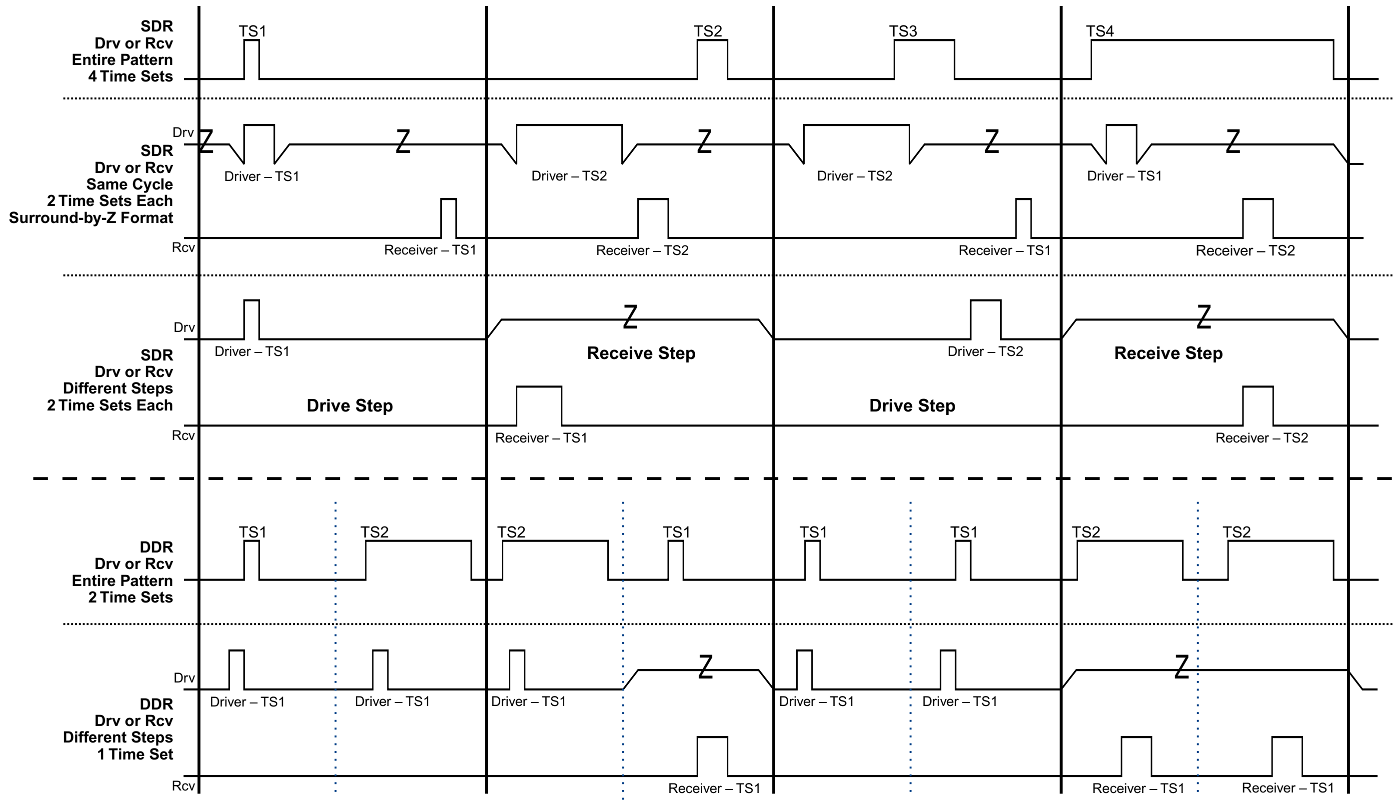


Figure 2-12 – On-the-Fly Timeset-Switching Diagram

Table 2-1 – DPU-16 Timeset Capabilities

Pin Type	SDR # of Timesets	DDR # of Timesets
In	4	2
Out	4	2
In or Out	2	1

Pattern Memory and Sequencing

A microcoded pattern sequencer (PSQ) supports a number of sophisticated pattern sequencing operations, including conditional branching, nested subroutines, nested looping, and match mode. This basic design addresses the various multisite problems associated with mixed-signal device testing.

Each site or device can operate independently as needed for the highest throughput with little or no compromising.

The pattern sequencer commands are shown in the following list:

Instruction Set	Description
`NOP`	The NOP (No-Operation) instruction flushes the instruction pipeline.
`Burst: #`	The Burst instruction causes the current step to execute repeatedly, the number of times specified in the event RAM, before going to the next instruction.
`Halt`	The Halt instruction stops the Pattern Sequencer from running.
`Set Loop: #`	The Set Loop instruction pushes the current loop counter onto the loop counter stack, if the current loop counter is not zero. Then the specified loop counter value is placed in the loop counter register.
`End Loop: label`	The End Loop instruction will jump to the specified label if the loop counter register is not zero. If the jump is taken (i.e. the loop count register is not zero), the loop count register is decremented.If the jump is not taken (i.e. the loop count register is zero), the top entry of the loop stack will be popped back into the loop count register
`Jmp: label`	The Jmp (Jump) instruction will unconditionally jump to the specified label.
`Call: label`	The Call instruction will unconditionally jump to the specified label. The pattern step (address) counter is pushed onto the pattern address stack. When the next Return instruction is executed, the pattern sequencer will pop the stored pattern address and return execution back to the next step after the calling step.
`Return`	The Return instruction will unconditionally jump to the pattern address popped from the top of the pattern address stack. This will return execution back to the next step after the calling step.
`Set Fail`	The Set Fail instruction forces the fail flag to be set.
`Clr Fail`	The Clear Fail instruction forces the fail flag to be cleared.
`If ( Fail ) Jmp: label`	The If (Fail) Jmp: label instruction will conditionally jump to the specified label if the Fail flag is set. If the jump is taken (i.e. the Fail flag is set), the pattern address will be set to the address specified by the label. If the jump is not taken (i.e. the Fail flag is not set), the pattern address is incremented to the next step.
`If ( Fail ) Call: label`	The If (Fail) Call: label instruction will conditionally jump to the specified label if the Fail flag is set. If the jump is taken (i.e. the Fail flag is set), the pattern step (address) counter is pushed onto the pattern address stack. When the next Return instruction is executed, the pattern sequencer will pop the stored pattern address and return execution back to the next step after the calling step. If the jump is not taken (i.e. the Fail flag is not set), the pattern address is incremented to the next step.

Instruction Set	Description
`If ( Fail ) Return`	The If (Fail) Return instruction will conditionally jump to the pattern address popped from the top of the pattern address stack if the fail flag is set. If the jump is taken (i.e. the flag is set), the pattern address will be set to the next step after the address popped off the top of the pattern address stack. If the jump is not taken (i.e. the Fail flag is not set), the pattern address is incremented to the next step.
`If ( ! Fail ) Jmp: label`	The If (! Fail) Jmp: label instruction will conditionally jump to the specified label if the Fail flag is not set. If the jump is taken (i.e. the Fail flag is not set), the pattern address will be set to the address specified by the label. If the jump is not taken (i.e. the Fail flag is set), the pattern address is incremented to the next step.
`If ( ! Fail ) Call: label`	The If (! Fail) Call: label instruction will conditionally jump to the specified label if the Fail flag is not set. If the jump is taken (i.e. the Fail flag is not set), the pattern step (address) counter is pushed onto the pattern address stack. When the next Return instruction is executed, the pattern sequencer will pop the stored pattern address and return execution back to the next step after the calling step. If the jump is not taken (i.e. the Fail flag is set), the pattern address is incremented to the next step.
`If ( ! Fail ) Return`	The If (!Fail) Return instruction will conditionally jump to the pattern address popped from the top of the pattern address stack if the Fail flag is not set. If the jump is taken (i.e. the Fail flag is not set), the pattern address will be set to the next step after the address popped off the top of the pattern address stack. If the jump is not taken (i.e. the Fail flag is set), the pattern address is incremented to the next step.
`Set Match`	The Set Match instruction places the DPU into match mode.
`Clr Match`	The Clear Match instruction releases the DPU from match mode.
`If ( Match ) Jmp: labell`	The If (Match) Jmp: label instruction will conditionally jump to the specified label if the Match flag is set. If the jump is taken (i.e. the Match flag is set), the pattern address will be set to the address specified by the label. If the jump is not taken (i.e. the Match flag is not set), the pattern address is incremented to the next step.
`If ( Match ) Call: label`	The If (Match) Call: label instruction will conditionally jump to the specified label if the Match flag is set. If the jump is taken (i.e. the Match flag is set), the pattern step (address) counter is pushed onto the pattern address stack. When the next Return instruction is executed, the pattern sequencer will pop the stored pattern address and return execution back to the next step after the calling step. If the jump is not taken (i.e. the Match flag is not set), the pattern address is incremented to the next step.
`If ( Match ) Return`	The If (Match) Return instruction will conditionally jump to the pattern address popped from the top of the pattern address stack if the Match flag is set. If the jump is taken (i.e. the Match flag is set), the pattern address will be set to the next step after the address popped off the top of the pattern address stack. If the jump is not taken (i.e. the Match flag is not set), the pattern address is incremented to the next step.
`If ( ! Match ) Jmp: label`	The If (! Match) Jmp: label instruction will conditionally jump to the specified label if the Match flag is not set. If the jump is taken (i.e. the Match flag is not set), the pattern address will be set to the address specified by the label. If the jump is not taken (i.e. the Match flag is set), the pattern address is incremented to the next step.

Instruction Set	Description
`If ( ! Match ) Call: label`	The If (! Match) Call: label instruction will conditionally jump to the specified label if the Match flag is not set. If the jump is taken (i.e. the Match flag is not set), the pattern step (address) counter is pushed onto the pattern address stack. When the next Return instruction is executed, the pattern sequencer will pop the stored pattern address and return execution back to the next step after the calling step. If the jump is not taken (i.e. the Match flag is set), the pattern address is incremented to the next step.
`If ( ! Match ) Return`	The If (! Match) Return instruction will conditionally jump to the pattern address popped from the top of the pattern address stack if the Match flag is not set. If the jump is taken (i.e. the Match flag is not set), the pattern address will be set to the next step after the address popped off the top of the pattern address stack. If the jump is not taken (i.e. the Match flag is set), the pattern address is incremented to the next step.
`Clr Fail Mem`	The Clr Fail Mem (Clear Fail Memory) instruction resets the pointer into failure memory where failures are written when the expected input does not match the actual input.
`Clr Capture Mem`	The Clr Capture Mem (Clear Capture Memory) instruction resets the pointer into capture memory where capture data is written.
`Clr Loop Count`	The Clear Loop Count instruction sets the loop counter to zero. If the loop counter stack is not empty, the top entry of the stack is popped into the loop counter register.
`Pop Loop Count`	The Pop Loop Count instruction takes the entry on the top of the loop counter stack and places it into the loop count register and then discards the entry on the top of the loop count stack.
`Clr Call Stack`	The Clr Call Stack (Clear Call Stack) instruction resets the pattern address counter and event counter stack pointers to the beginning of their respective stacks.
`Dec Call Stack Ptr`	The Dec Call Stack Ptr instruction discards the entry on the top of the pattern address stack.
`Serial`	The Serial instruction causes designated serial pins to shift data on every clock.

Many applications require digital signals to be captured for some type of post-processing. This is especially true for analog-to-digital converters, where an FFT or other algorithm is customary. The PSQ supports capture mode operation, where received data can be transferred to capture memory on each specified vector step. This per-step capture control greatly improves the efficiency of capture RAM utilization. The on-board DSP can be used for real-time analysis of the data once it is captured.

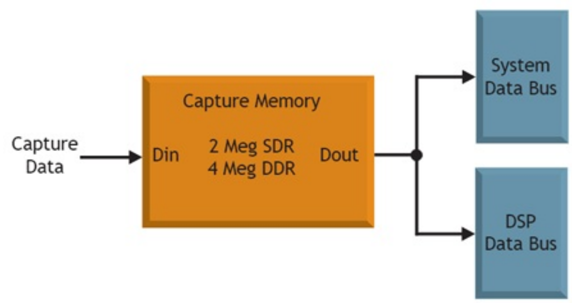


Figure 2-13 – Capture Memory Interface to DSP

Quad TMU Support

Another integrated feature of the DPU-16 is its built-in support of the QTMU (Quad Time Measurement Unit). The QTMU's input structure uses the same pin electronics as the standard digital channels (-1.0 V to +7.0 V). This means that QTMU measurements do not introduce any extra pin-loading to the device under test. In addition, both resources use the same controlled-impedance signal paths.

For higher voltage applications, a special buffered high voltage range is provided (-3 V to +21 V). This specially buffered path has an analog bandwidth of >20 MHz, and is selectable per I/O pin. Each of the channel's outputs is multiplexed to the QTMU resource as required for timing measurements.

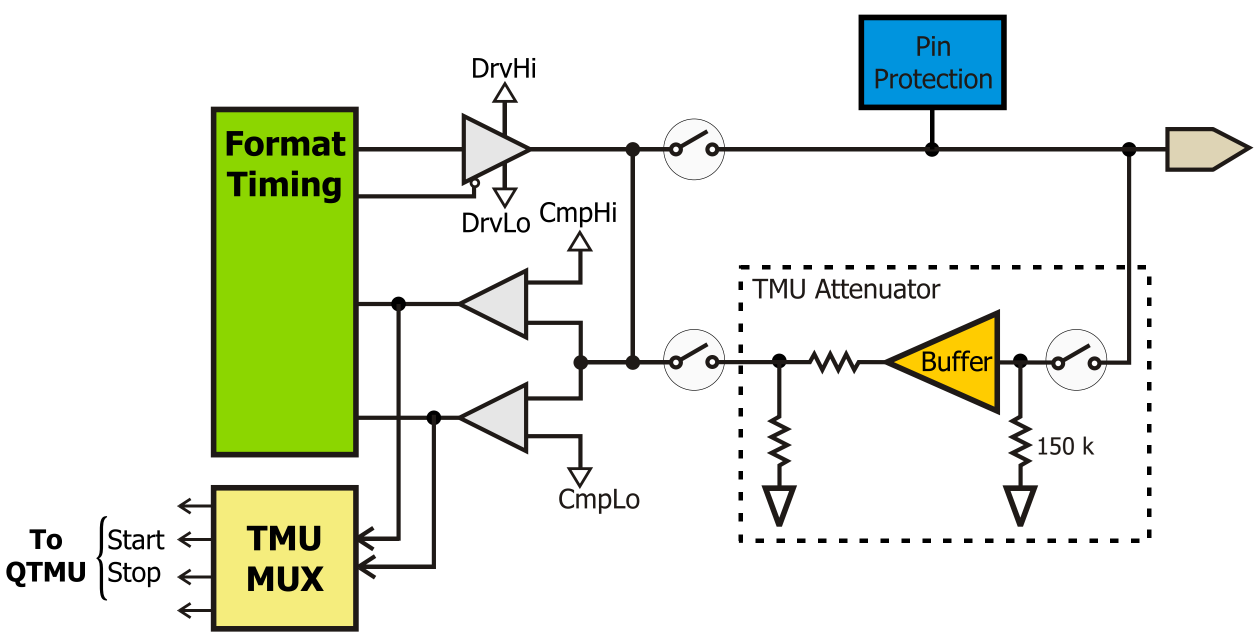


Figure 2-14 – QTMU Digital Channel Input Structure

2.6.4 Block Diagram

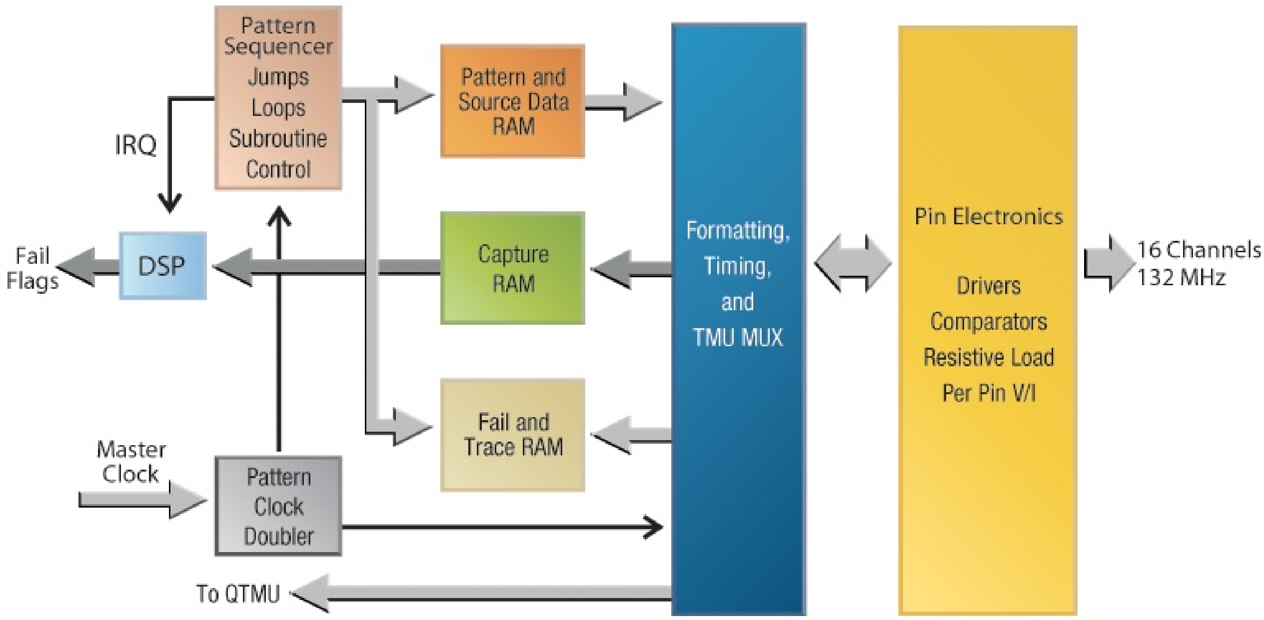


Figure 2-15 – DPU-16 Overall Block Diagram

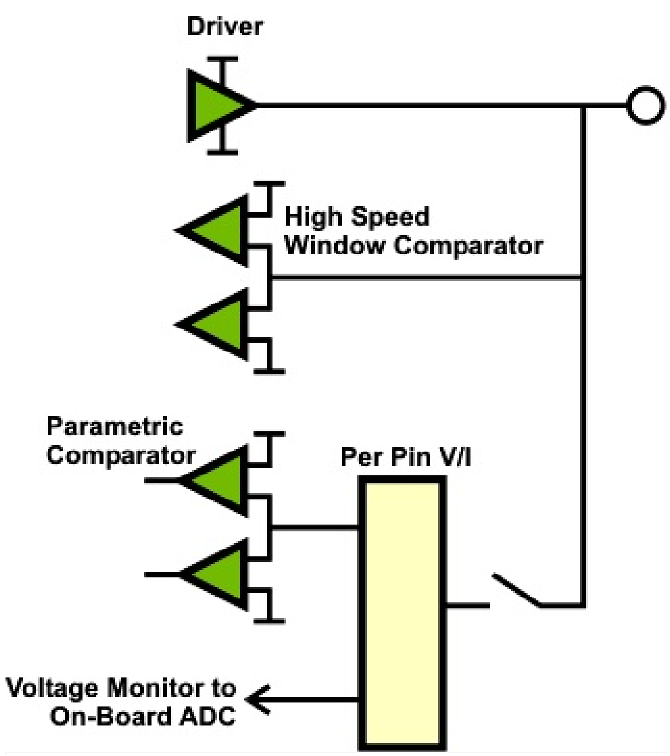


Figure 2-16 – DPU-16 I/O Pin Structure

2.6.5 Specifications

Per Pin V/I Capabilities (DC)

Voltage Force/Measure

Range	Resolution	Accuracy (no load)
-1.0 to +7.0 V	122.0 µV	±13 mV ±.05% Rdg ±3 mV / mA
-2.0 to +8.0 V	152.5 µV	±26 mV ±.05% Rdg ±3 mV / mA

Current Measure

Range	Resolution	Accuracy (1-6)
8 µA	0.24 nA	±20 nA + 0.1% of Rdg + 2 nA / V
32 µA	0.97 nA	±64 nA + 0.1% of Rdg + 8 nA / V (5)
128 µA	3.9 nA	±256 nA + 0.1% of Rdg + 32 nA / V (4)
512 µA	15.6 nA	±2 µA + 0.1% of Rdg + 125 nA / V (4)
2 mA	61.0 nA	±4 µA + 0.1% of Rdg + 500 nA / V (4)
8 mA	244.1 nA	±16 µA + 0.1% of Rdg + 2 µA / V (2,3)
32 mA	976.6 nA	±64 µA + 0.1% of Rdg + 8 µA / V (2,3)

Current Force

Range	Resolution	Accuracy (1-6)
8 µA	0.24 nA	±24 nA + 0.1% of Setting + 13 nA / V
32 µA	0.97 nA	±96 nA + 0.1% of Setting + 50 nA / V (5)
128 µA	3.9 nA	±384 nA + 0.1% of Setting + 200 nA / V (4)
512 µA	15.6 nA	±2 µA + 0.1% of Setting + 0.8 µA / V (4)
2 mA	61.0 nA	±6 µA + 0.1% of Setting + 3.2 µA / V (4)
8 mA	244.1 nA	±24 µA + 0.1% of Setting + 13 µA / V (2,3)
32 mA	976.6 nA	±96 µA + 0.1% of Setting + 50 µA / V (2,3)

NOTES

(1) Accuracy specifications only valid when operated within the maximum current operating range.
See 8 V and 10 V maximum current operating range plots on page 2-.

(2) Above +7.0 V, maximum source current determined by the equation Max Current = (+FS Range Current)*(2.4 - (Output Voltage * 0.2))

(3) Below +0.75 V, maximum sink current determined by the equation Max Current = (-FS Range Current)*(-0.7955 - (Output Voltage * 0.2727))

*DPU-16 Maximum Current Operating Range Plots**

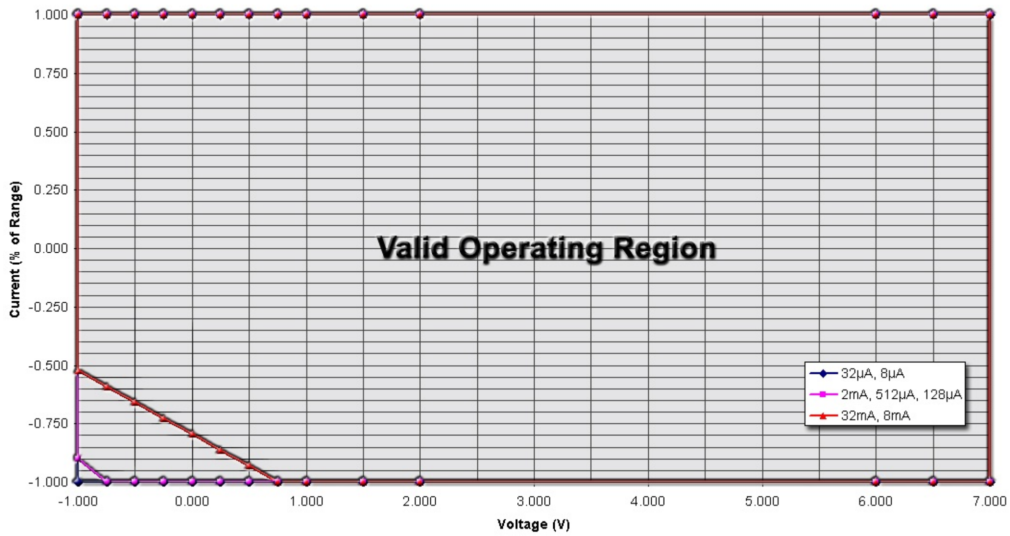


Figure 2-17 – 8 V Maximum Current Operating Range

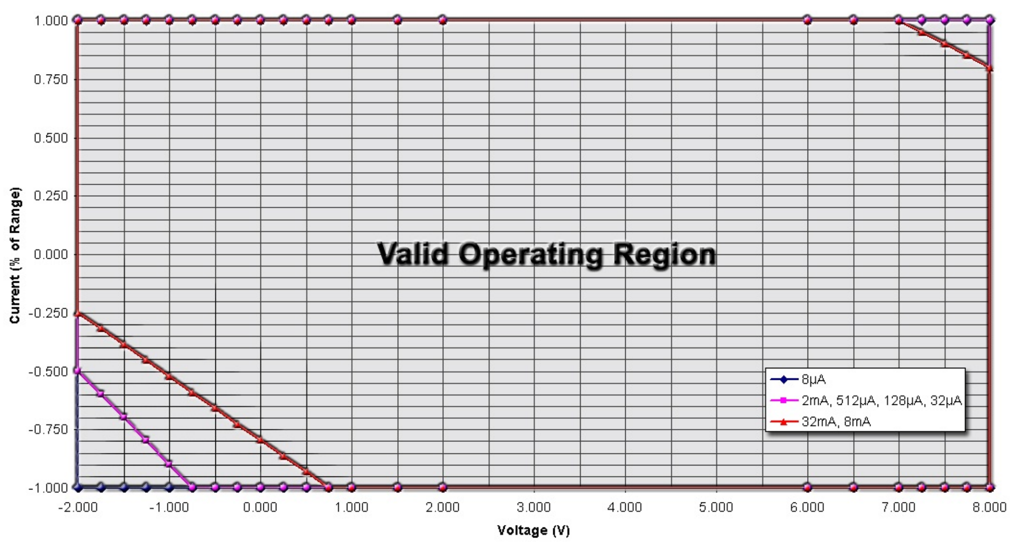


Figure 2-18 – 10 V Maximum Current Operating Range

***Operation only valid within box region defined for current range. Multiply value on Y-axis by the current range to determine maximum currents.**

Pin Electronics (Dynamic Specifications)

Digital Driver Specifications

VOH Range	-1.0 V to + 7.0 V
VOL Range	-1.0 V to + 7.0 V
Resolution	122 µV (16 Bit)
VOH/VOL Level Accuracy	±(25 mV) no load
Impedance	50 Ohms nominal
Skew	± 0.5 nsec all formats
Minimum Pulse Width	3.75 nsec @ 3 V
Rise Time	1.6 nsec (typ)
DC Output Current	± 35 mA
HiZ Leakage	±150 nA max
Off State Isolation Voltage	±24V max

Digital Comparator Specifications

VIH Range	-1.0 V to + 7.0 V
VIL Range	-1.0 V to + 7.0 V
Resolution	122 µV (16 Bit)
VIH/VIL Threshold Accuracy	±(25 mV)
Skew	±0.5 nsec
Minimum Pulse Width	3.0 nsec (typ)
Bandwidth	>150 MHz
TMU Buffer Bandwidth (-3 V to +21 V Range)	> 20 MHz

Timing Accuracy

Overall Timing Accuracy	±1 nsec – All timing, All pins

2.6.6 User Interface

2.6.6.1 Software

All functions of the DPU-16 are programmed using Digital Pin Unit software utilities. These utilities (which begin the prefix "dpin") can be called from your C++ test program or in real time from RAIDE. Their syntax and usage are described in detail in the Digital Pin Unit (DPU-16) Utilities section of the Eagle Vision software help file.

2.6.6.2 Hardware

DPU-16s are located in the Digital Card Cage in the dual test heads.

Each DPU-16 is connected to the QTMU in its respective Digital Card Cage via a single RJ-45 cable, known as a QTMU Cable. These 16 cables are included with the system regardless of tester configuration. As DPU-16 boards are added to the tester, the QTMU cables must be plugged into specific connectors on the QTMU board as shown below.

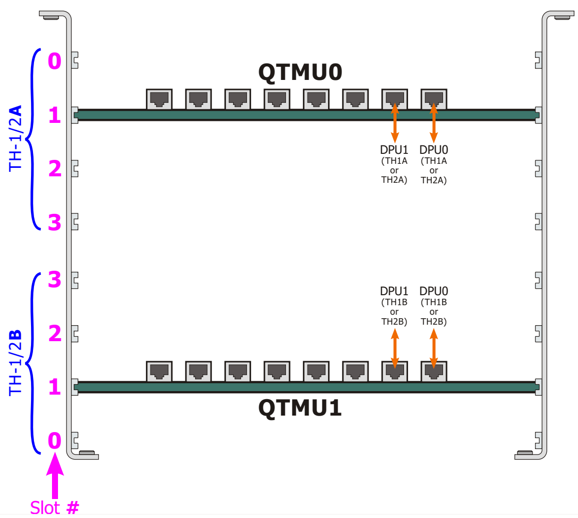


Figure 2-19 – QTMU Wiring Diagram

See Chapter 4 for the connections and pinouts for the DPU-16.

2.7 High Power Unit (HPU-25/100)

2.7.1 Caution to Users

SAFETY NOTICE:
This resource is designed to operate in a test system environment that is designed with the following safety features:

- Access to this resource requires the use of a tool to remove a cover
- Access to Input/Output connections to this resource is blocked by mechanical barriers
- An electrical interlock circuit inhibits the output of this resource

CAUTION: RISK OF SHOCK.

****

Hazardous Voltages Present! This resource generates hazardous voltages and must be operated in a properly designed enclosure with safety features in place. Always turn power off prior to handling this resource. Eagle Test Systems, Inc. accepts no responsibility for harm from handling or misuse of this resource.

****

Use high-voltage-insulated wiring when wiring connections from a HPU to points on the application boards. Teflon- and silicone-insulated wire offer dielectric strengths in hundreds to thousands of volts.

2.7.2 Features

- Fully independent single channel SmartPin™ resource for high current testing (±25 A, or ±100 A with optional power booster)
- Fully floating and stackable (±200 VDC from ground max)
- Three (3) voltage ranges, ten (10) current ranges:
  - LOW Current Ranges (Continuous Current):

    ±100 V @
    ±(500 mA, 100 mA, 10 mA, 1 mA, 100 µA, 10 µA, 1 µA)

    ±30 V @
    ±(1 A, 100 mA, 10 mA, 1 mA, 100 µA, 10 µA, 1 µA)

    ±10 V @
    ±(2 A, 200 mA, 20 mA, 2 mA, 200 µA, 20 µA, 2 µA)
  - HIGH Current Ranges (Pulsed Operation):

    ±75 V @ ±100 A, ±10 A, ±4 A
    (1 A Continuous Current)

    ±30 V @ ±100 A, ±10 A, ±4 A
    (1 A Continuous Current)

    ±10 V @ ±100 A, ±10 A
    (1 A Continuous Current)

    ±10 V @ ± 4 A
    (4 A Continuous Current)
  - Additional 10X and 100X measure gain settings are available in most voltage and current ranges.
- Two (2) 500 KSPS digitizers to capture both voltage and current simultaneously
- Independent high/low programmable voltage/current clamps with alarms
- Kelvin error detect and measurement full-scale alarms
- Driver/signal generator mode (LOW current range only)
  - High speed AWG:
    16 bit, 25 MSPS, up to 5 MHz Sine
    (±30 V / 10 V ranges only)
  - High Resolution AWG:
    18 bit, 350 KSPS, up to 50 kHz Sine
    (±30 V / 10 V ranges only)
  - Audio Mode:
    18 bit, 350 KSPS,
    better than -96 dB THD @ 1 kHz
    (±10 V / 1 V ranges only)

- Enhanced voltage measurement ranges
    - Pedestal mode:
      ±1.1 V around 0 V, 1 V, 3 V, 5 V
- Volt meter mode:
  ±(1000 V, 100 V, 30 V, 10 V)
- On-board DSP with robust math library for rapid evaluation of complex results
- Real-time measurement accumulator for instant results averaging
- Change V/I settings under AWG pattern control (on-the-fly)
    - Force voltage/current mode
    - Current ranges
    - ADC gain and filtering
    - ADC sample clock gate on/off
- Results accumulator supports up to 32 sample sets per pattern
- Interlocks provided for operator safety
- Hardware and software designed for multisite applications
- Digitizer self-trigger mode for asynchronous signals (includes pre-trigger sampling)

2.7.3 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]
- Eagle Vision MST [Rev. 2011.1.1002.17 and up]

2.7.4 Theory of Operation

The High Power Unit (HPU-25/100) is a single slot, single channel, ±100V SmartPin™ resource with 10 current ranges. The SmartPin™ architecture incorporates an AWG and a dual digitizer within a conventional four quadrant V/I. This resource includes all the standard capabilities of full-featured V/I's (programmable clamps, Kelvin detect, alarms, etc.) plus advanced characteristics such as pattern-based range changing and sample clock control.

The V/I is stable with almost any combination of inductive and/or capacitive loads. The programmable clamps cross over from voltage to current or vice-versa with minimal overshoot or instability.

Bandwidth and settling time are optimized to maximize measurement speed. The HPU-25/100 design uses analog switches extensively for excellent reliability and switching speeds.

The 18-bit AWG makes it possible to generate arbitrary voltage and current-based signals that are synchronized to the other digital and analog resources in the test system. This synchronization enables test engineers to create dynamic test conditions that can quickly locate analog thresholds and other complex parameters.

For waveform generation, three special driver modes (18 bit 350 KSPS, 16 bit 25 MSPS and a special audio mode) support waveform generation by providing direct access to the buffered AWG output. This proves extremely useful for general-purpose applications requiring AC signals in and above the audio range. These signals may be synchronized to the other analog and digital resources of the system.

In the audio mode, a specialized differential line driver is switched into the output force lines to provide a high quality audio signal for THD and noise testing. Both differential and single-ended configurations are supported.

The high current output stage accommodates pulsed currents on three ranges up to ±25 A. For currents greater than 25A, the optional HPU Booster Board can provide ±100 A capability to four HPU-25/100 units simultaneously. The high current output stage draws its power from a capacitor bank that is charged continuously. Using pattern-based programming techniques, the HPU-25/100 can output precise current and/or voltage pulses of any amplitude and duration up to the limits of the selected range. This can greatly reduce test time and avoid excessive die heating.

The HPU-25/100 also incorporates integrated dual digitizers for parallel measurements of both voltage and current. Under pattern-based control, the digitizer can be switched on and off and the ADC filter and gain settings can be changed. Sample results are stored in on-board memory. The on-board DSP with its robust math library can be used to evaluate complex test results thereby avoiding time-consuming data transfer operations.

Capturing sporadic asynchronous signals can be a difficult operation for a digitizer. The HPU-25/100 includes a self-trigger mode specifically for this task. With this mode, you can trigger on the incoming signal, based on a programmable trigger threshold setting. The self-trigger mode also includes the ability to specify a certain number of pre- and post-trigger samples. Using these samples lets you capture and use the entire waveform of interest. The self-trigger mode facilitates capturing sporadic signals with a high sample rate, without using a large amount of capture memory.

NOTE: When the HPU and HPU Booster Board are placed in the Smart Power Expansion Chassis (SPEC), the measure voltage is limited to ±100 V, the forced or measured current operates on current ranges of 1 mA and above, and audio mode is disabled.

2.7.5 Applications

The HPU-25/100 is useful for advanced measurement applications as well as for general purpose V/I tasks. The various voltage and current ranges make it possible to address a wide variety of test applications. For static force/measure applications, the real-time measurement accumulator reduces measurement times with built-in hardware results averaging. Combining these features with 18 bit force, and 16 bit measurement resolution supplements the HPU-25/100's performance with high precision.

SmartPin™ resources such as the HPU-25/100 are well suited for testing various devices, because they make it possible to initiate a wide range of test conditions in rapid sequence. The use of a pattern-based V/I makes it possible to change force conditions on-the-fly (under hardware control).

The pattern RAM contains the V/I force values and a number of synchronized control bits that make it possible to change the operating state of the V/I on-the-fly and to enable/disable the on-board digitizer to capture the desired test results at selectable pattern locations. The digitizer is also capable of on-the-fly averaging, where the average value of each sample set is stored in RAM along with each set of sample values. For pattern-based DC tests, this mode of averaging greatly reduces data transfer time during post-processing by reducing the amount of data returned to the host computer.

With this type of hardware available, it is possible to string together many test conditions, while simultaneously storing the measured results. After the pattern runs, the system controller will typically read back the results for test limit comparison purposes.

Due to the real-time averaging of sample sets (Results Accumulator), reading the results from the hardware is a fast process. If a more complex evaluation method is required, the on-board DSP can be used as needed, or the sampled data can be transferred to controller memory for further mathematical evaluation.

Threshold searches are a common application problem that the HPU-25/100 can simplify. Often, test engineers must use either successive approximation techniques or design specialized application circuitry to speed up these normally time-consuming tests; however, this does not have to be the case. The HPU-25/100 makes it possible to locate current-based threshold points and the associated threshold hysteresis levels with a single up/down ramp pattern. Because the ramp signal is

AWG driven, the resolution and speed of the ramp can be optimized for the best trade-off in test speed vs. measurement accuracy and repeatability. The on-board DSP can quickly perform the search operations to keep test time to a minimum.

For static DC force/measure operations, which are still heavily used in many applications, the real-time measurement accumulator provides instant averaging of measured results. This means that the samples are summed mathematically in real-time. When sampling is complete, the answer is read directly from the resource, avoiding any further data transfers. The full data is also still available for plotting or other engineering purposes, offering the best of both worlds.

The pedestal measurement mode of the HPU-25/100 provides 16 bit resolution in a ±1.1 V measurement range, which can be offset by the following voltages: 0 V, 1 V, 3 V, 5 V. The pedestal measurement mode is operational during the normal V/I forcing mode. This feature provides much higher voltage measurement accuracy for testing voltage regulators, power FETs, and the like.

2.7.6 HPU Booster Board

The HPU Booster Board is an optional resource that enables the HPU-25/100 to deliver high current pulses up to 100 Amps (10 V, >5 mS pulse). Each channel is fully isolated from the other channels.

An integrated high current booster cable connects one booster board channel to one HPU-25/100 resource. Each booster board is capable of supplying high current pulse power to four HPU boards simultaneously.

The HPU Booster Board occupies a single slot in the Floating Resource card cage (FR Cage) or in the SPEC. The booster board must reside in the same cabinet with the HPU boards.

Two continuously charged capacitor banks (for positive and negative currents) are available per channel. Each bank begins recharging immediately during the current pulse and continues afterwards, if necessary, until fully charged. See the Pulse Width Curves for maximum pulse width available versus current output.

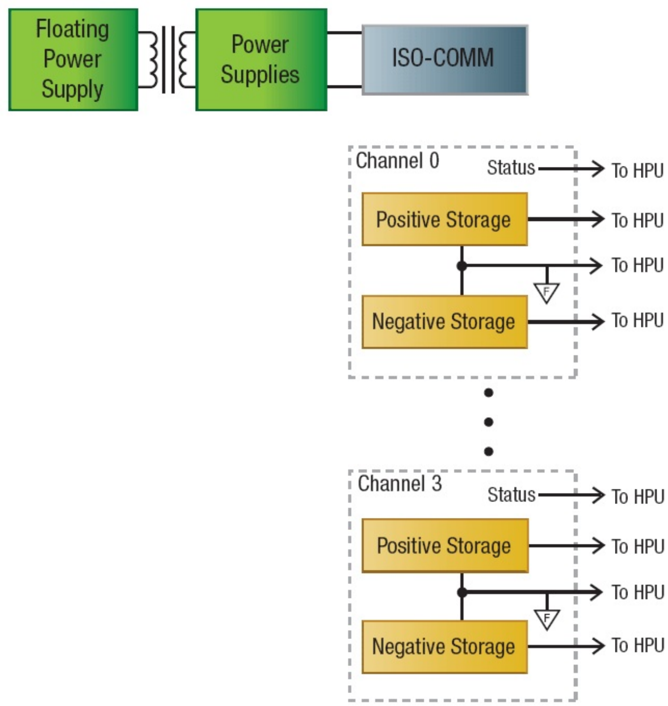


Figure 2-20 – HPU Booster Board Block Diagram

2.7.7 Specifications

The following pages list the specifications for the HPU-25/100. These specifications are subject to change at ETS's discretion.

2.7.7.1 Volt Meter Mode

Measure	Range	Resolution	Accuracy	Bandwidth	Input R
Voltage	1000 V	30 mV	±(125 mV + .025%)	25 kHz	5 MΩ
	100 V	3 mV	±(12.5 mV + .025%)	25 kHz	>200 MΩ
	30 V	900 µV	±(4 mV + .025%)	50 kHz	>200 MΩ
	10 V	300 µV	±(1.2 mV + .025%)	50 kHz	>200 MΩ

2.7.7.2 Driver Mode (Low Current Path Only)

Force	Range	Resolution	Typical Distortion @ Frequency
High Speed (16-bit, 25 MSPS)	±30 V	900 µV	< -75 dB @ 100 kHz
	±10 V	300 µV	< -80 dB @ 100 kHz
High Resolution (18-bit, 350 KSPS)	±30 V	225 µV	< -75 dB @ 10 kHz
	±10 V	80 µV	< -80 dB @ 10 kHz
Audio * (18-bit, 350 KSPS) (Differential)	10 V (pk – pk)	50 µV	Better than -96 dB @ 1 kHz
	1 V (pk – pk)	5 µV	Better than -96 dB @ 1 kHz
* Not supported when located in the remote card cage (mainframe)			

2.7.7.3 10 V Range (Low Current & High Current Path)

Force	Range	Resolution	Accuracy
Voltage	10 V	80 µV	±(0.8 mV + .025% of setting)
Current	2 A	16 µA	±(250 µA + .05%)
	200 mA	1.6 µA	±(25 µA + .05%)
	20 mA	160 nA	±(2.5 µA + .05%)
	2 mA	16 nA	±(250 nA + .05%)
	200 µA	1.6 nA	±(20 nA + .05%)
	20 µA	160 pA	±(7 nA + .1%)
	2 µA	16 pA	±(5 nA + .1%)

HPU 10 V Range (Low Current & High Current Path) Specifications (continued)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±10 V	1X	±10 V	300 µV	±(800 µV + .025%)
	±10 V	10X	±1 V	30 µV	±(600 µV + .025%)
	±10 V	100X	±100 mV	3 µV	±(300 µV + .025%)
Current	±2 A	1X	±2 A	60 µA	±(250 µA + .05%)
	±2 A	10X	±200 mA	6 µA	±(125 µA + .05%)
	±2 A	100X	±20 mA	600 nA	±(60 µA + .05%)
	±200 mA	1X	±200 mA	6 µA	±(25 µA + .05%)
	±200 mA	10X	±20 mA	600 nA	±(12 µA + .05%)
	±200 mA	100X	±2 mA	60 nA	±(6 µA + .05%)
	±20 mA	1X	±20 mA	600 nA	±(2.5 µA + .05%)
	±20 mA	10X	±2 mA	60 nA	±(1.2 µA + .05%)
	±20 mA	100X	±200 µA	6 nA	±(600 nA + .05%)
	±2 mA	1X	±2 mA	60 nA	±(250 nA + .05%)
	±2 mA *	10X	±200 µA	6 nA	±(125 nA + .05%)
	±2 mA *	100X	±20 µA	600 pA	±(60 nA + .1%)
	±200 µA *	1X	±200 µA	6 nA	±(25 nA + .05%)
	±200 µA *	10X	±20 µA	600 pA	±(15 nA + .1%)
	±200 µA *	100X	±2 µA	60 pA	±(12 nA + .1%)
	±20 µA *	1X	±20 µA	600 pA	±(15 nA + .05%)
	±20 µA *	10X	±2 µA	60 pA	±(10 nA + .1%)
	±20 µA *	100X	±200 nA	6 pA	±(7 nA + .1%)
	±2 µA *	1X	±2 µA	60 pA	±(12 nA + .05%) **
	±2 µA *	10X	±200 nA	6 pA	±(6 nA + .1%)
	±2 µA *	100X	±20 nA	600 fA	±(4 nA + .1%)

*** Not supported when located in the remote card cage (mainframe)**
**** Accuracy improvement with auto-zero tare: ±2 µA Range: (±2 nA + .05%)**

2.7.7.4 30 V Range (Low Current & High Current Path)

Force	Range	Resolution	Accuracy
Voltage	30 V	225 µV	±(2.5 mV + .025% of setting)
Current	1 A	8 µA	±(125 µA + .05%)
	100 mA	800 µA	±(12.5 µA + .05%)
	10 mA	80 nA	±(1.25 µA + .05%)
	1 mA	8 nA	±(125 nA + .05%)
	100 µA	800 pA	±(25 nA + .05%)
	10 µA	80 pA	±(10 nA + .1%)
	1 µA	8 pA	±(5 nA + .1%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±30 V	1X	±30 V	900 µV	±(2.5 mV + .025%)
	±30 V	10X	±3 V	90 µV	±(2 mV + .025%)
	±30 V	100X	±300 mV	9 µV	±(1 mV + .025%)
Current	±1 A	1X	±1 A	30 µA	±(125 µA + .05%)
	±1 A	10X	±100 mA	3 µA	±(60 µA + .05%)
	±1 A	100X	±10 mA	300 nA	±(30 µA + .05%)
	±100 mA	1X	±100 mA	3 µA	±(12.5 µA + .05%)
	±100 mA	10X	±10 mA	300 nA	±(6 µA + .05%)
	±100 mA	100X	±1 mA	30 nA	±(3 µA + .05%)
	±10 mA	1X	±10 mA	300 nA	±(1.25 µA + .05%)
	±10 mA	10X	±1 mA	30 nA	±(600 nA + .05%)
	±10 mA	100X	±100 µA	3 nA	±(300 nA + .05%)
	±1 mA	1X	±1 mA	30 nA	±(125 nA + .05%)
	±1 mA *	10X	±100 µA	3 nA	±(60 nA + .05%)
	±1 mA *	100X	±10 µA	300 pA	±(30 nA + .1%)
	±100 µA *	1X	±100 µA	3 nA	±(20 nA + .05%)
	±100 µA *	10X	±10 µA	300 pA	±(12 nA + .1%)
	±100 µA *	100X	±1 µA	30 pA	±(10 nA + .1%)
	±10 µA *	1X	±10 µA	300 pA	±(12 nA + .05%)
	±10 µA *	10X	±1 µA	30 pA	±(8 nA + .1%)
	±10 µA *	100X	±100 nA	3 pA	±(6 nA + .1%)
	±1 µA *	1X	±1 µA	30 pA	±(8 nA + .05%) **
	±1 µA *	10X	±100 nA	3 pA	±(4 nA + .1%)

*** Not supported when located in the remote card cage (mainframe)**
**** Accuracy improvement with auto-zero tare: ±2 µA Range: (±2 nA + .05%)**

2.7.7.5 100 V Range (Low Current & High Current Path)

Force	Range	Resolution	Accuracy
Voltage	100 V	800 µV	±(12.5 mV + .025% of setting)
Current	500 mA	8 µA	±(125 µA + .05%)
	100 mA	800 nA	±(12.5 µA + .05%)
	10 mA	80 nA	±(1.25 µA + .05%)
	1 mA	8 nA	±(125 nA + .05%)
	100 µA	800 pA	±(25 nA + .05%)
	10 µA	80 pA	±(10 nA + .1%)
	1 µA	8 pA	±(5 nA + .1%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±100 V	1X	±100 V	3 mV	±(12.5 mV + .025%)
	±100 V	10X	±10 V	300 µV	±(6 mV + .025%)
	±100 V	100X	±1 V	30 µV	±(3 mV + .025%)
Current	±500 mA	1X	±500 mA	15 µA	±(125 µA + .05%)
	±500 mA	10X	±50 mA	1.5 µA	±(60 µA + .05%)
	±500 mA	100X	±5 mA	150 nA	±(30 µA + .05%)
	±100 mA	1X	±100 mA	3 µA	±(12.5 µA + .05%)
	±100 mA	10X	±10 mA	300 nA	±(6 µA + .05%)
	±100 mA	100X	±1 mA	30 nA	±(3 µA + .05%)
	±10 mA	1X	±10 mA	300 nA	±(1.25 µA + .05%)
	±10 mA	10X	±1 mA	30 nA	±(600 nA + .05%)
	±10 mA	100X	±100 µA	3 nA	±(300 nA + .05%)
	±1 mA	1X	±1 mA	30 nA	±(125 nA + .05%)
	±1 mA *	10X	±100 µA	3 nA	±(60 nA + .05%)
	±1 mA *	100X	±10 µA	300 pA	±(30 nA + .1%)
	±100 µA *	1X	±100 µA	3 nA	±(50 nA + .05%)
	±100 µA *	10X	±10 µA	300 pA	±(25 nA + .1%)
	±100 µA *	100X	±1 µA	30 pA	±(20 nA + .1%)
	±10 µA *	1X	±10 µA	300 pA	±(25 nA + .05%)
	±10 µA *	10X	±1 nA	30 pA	±(12.5 nA + .1%)
	±1 µA *	1X	±1 µA	30 pA	±(12.5 nA + .05%)

*** Not supported when located in the remote card cage (mainframe)**

2.7.7.6 10 V Range (Pulsed High Current Path Only)

Force	Range	Continuous Current	Resolution	Accuracy
Voltage	10 V	N/A	80 µV	±(0.8 mV + .025% of setting)
Current	±100 A	1 A	800 µA	±(30 mA + 0.5%)
	±10 A	1 A	80 µA	±(3.0 mA + 0.5%)
	±4 A	1 A	30 µA	±(1.25 mA + 0.2%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Current	±100 A	1X	±100 A	3 mA	±(30 mA + 0.5%)
	±100 A	10X	±10 A	300 µA	±(15 mA + 0.5%)
	±100 A	100X	±1 A	30 µA	±(7.5 mA + 0.5%)
	±10 A	1X	±10 A	300 µA	±(3.0 mA + 0.5%)
	±10 A	10X	±1 A	30 µA	±(1.5 mA + 0.5%)
	±10 A	100X	±100 mA	3 µA	±(750 µA + 0.5%)
	±4 A	1X	±4 A	120 µA	±(1.25 mA + 0.2%)
	±4 A	10X	±400 mA	12 µA	±(625 µA + 0.2%)
	±4 A	100X	±40 mA	1.2 µA	±(312 µA + 0.2%)

2.7.7.7 30 V Range (Pulsed High Current Path Only)

Force	Range	Continuous Current	Resolution	Accuracy
Voltage	30 V	N/A	225 µV	±(2.5 mV + .025% of setting)
Current	±100 A	1 A	800 µA	±(30 mA + 0.5%)
	±10 A	1 A	80 µA	±(3.0 mA + 0.5%)
	±4 A	1 A	30 µA	±(1.25 mA + 0.2%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Current	±100 A	1X	±100 A	3 mA	±(30 mA + 0.5%)
	±100 A	10X	±10 A	300 µA	±(15 mA + 0.5%)
	±100 A	100X	±1 A	30 µA	±(7.5 mA + 0.5%)
	±10 A	1X	±10 A	300 µA	±(3.0 mA + 0.5%)
	±10 A	10X	±1 A	30 µA	±(1.5 mA + 0.5%)
	±10 A	100X	±100 mA	3 µA	±(750 µA + 0.5%)
	±4 A	1X	±4 A	120 µA	±(1.25 mA + 0.2%)
	±4 A	10X	±400 mA	12 µA	±(625 µA + 0.2%)
	±4 A	100X	±40 mA	1.2 µA	±(312 µA + 0.2%)

2.7.7.8 75 V Range (Pulsed High Current Path Only)

Force	Range	Continuous Current	Resolution	Accuracy
Voltage	75 V	N/A	800 µV	±(12.5 mV + .025% of setting)
Current	±100 A	1 A	800 µA	±(30 mA + 0.5%)
	±10 A	1 A	80 µA	±(3.0 mA + 0.5%)
	±4 A	1 A	30 µA	±(1.25 mA + 0.2%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Current	±100 A	1X	±100 A	3 mA	±(30 mA + 0.5%)
	±100 A	10X	±10 A	300 µA	±(15 mA + 0.5%)
	±100 A	100X	±1 A	30 µA	±(7.5 mA + 0.5%)
	±10 A	1X	±10 A	300 µA	±(3.0 mA + 0.5%)
	±10 A	10X	±1 A	30 µA	±(1.5 mA + 0.5%)
	±10 A	100X	±100 mA	3 µA	±(750 µA + 0.5%)
	±4 A	1X	±4 A	120 µA	±(1.25 mA + 0.2%)
	±4 A	10X	±400 mA	12 µA	±(625 µA + 0.2%)
	±4 A	100X	±40 mA	1.2 µA	±(312 µA + 0.2%)

2.7.7.9 Pedestal Voltage Measurement Mode (10 V Range Only)

Force	Pedestal Voltage	Continuous Current	Resolution (16-Bit)	Accuracy
Voltage	0 V	-1.1 V to +1.1 V	35 µV	±(250 µV + 0.01% of Reading)
Current	1 V	-0.1 V to +2.1 V	35 µV	±(250 µV + 0.01% of (Reading - 1 V))
	3 V	+1.9 V to +4.1 V	35 µV	±(250 µV + 0.01% of (Reading - 3 V))
	5 V	+3.9 V to +6.1 V	35 µV	±(250 µV + 0.01% of (Reading - 5 V))

2.7.8 Pulse Duration Curves

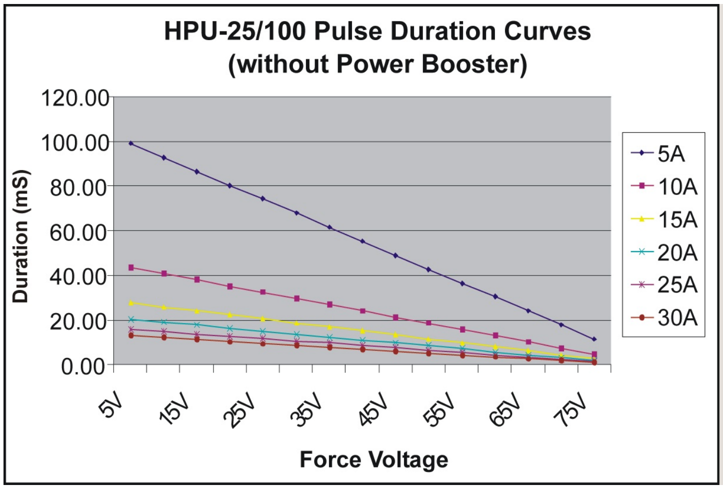


Figure 2-21 – HPU-25/100 Pulse Duration Curves (without Power Booster)

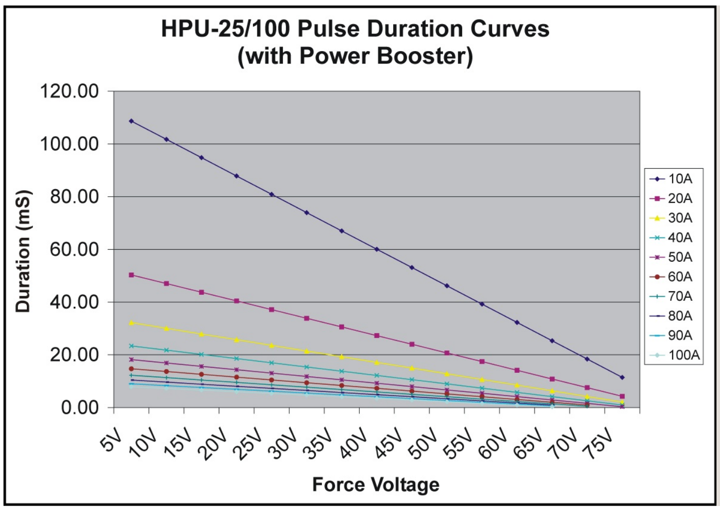


Figure 2-22 – HPU-25/100 Pulse Duration Curves (with Power Booster)

2.7.9 V/I Quadrant Diagram

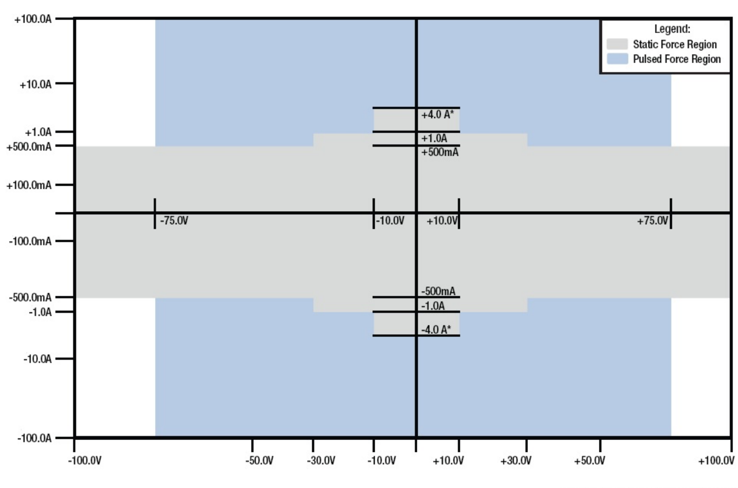


Figure 2-23 – HPU-25/100 V/I Quadrants

2.7.10 Block Diagram

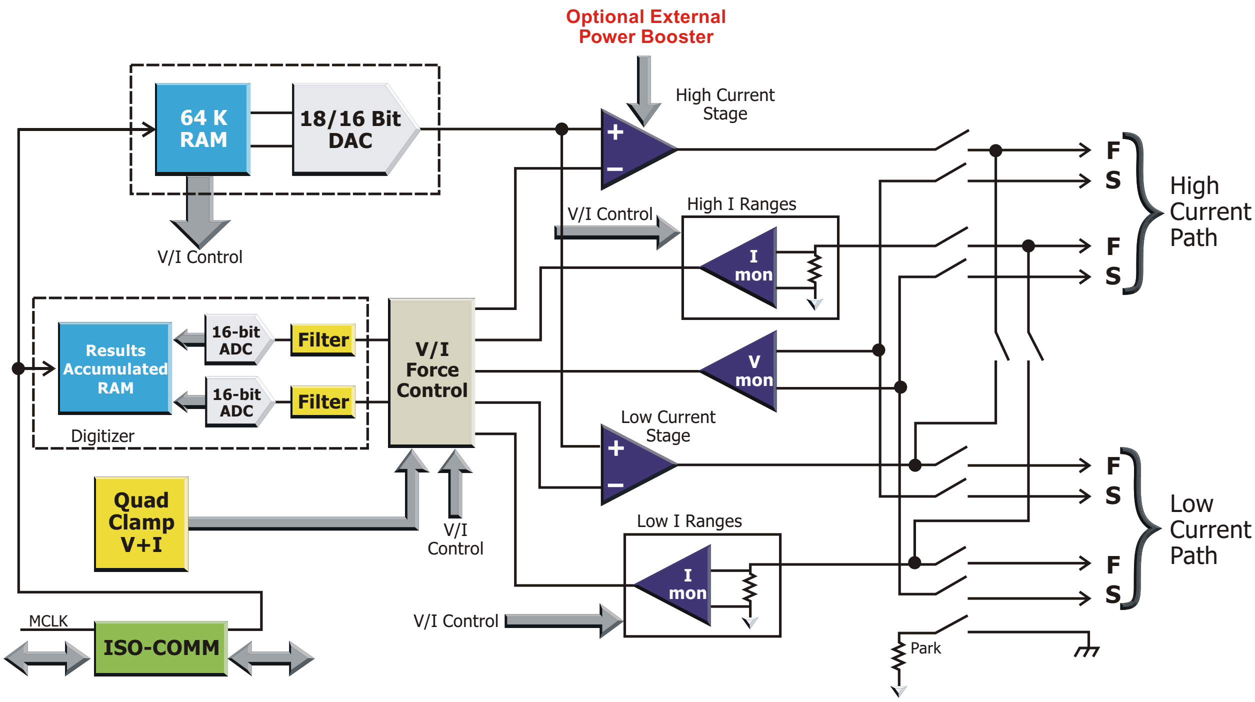


Figure 2-24 – HPU-25/100 Block Diagram

2.7.11 User Interface

2.7.11.1 Software

All functions of the HPU are programmed using utility function calls from your C test program. These utilities and their syntax and usage are described in The ETS Software Help File.

NOTE: The Master Clock portion of the TCB must be set up and started in order to clock the AWG and/or the digitizer. Please refer to the Test Head Control Board section (Section 2.17 on page 2-113) or the HPU and MCB utility descriptions (in The ETS Software Help File) for further information.

2.7.11.2 Hardware

There is a single SmartPin™ per HPU card, and they are housed in the Floating Resource Card Cage (FR Cage). HPU signals can be routed to the DUT on a low current or high current output path. The paths are described below.

Low Current Path

The low current output is designated for DC current less than 1 A, and pulsed current less than 4 A. Figure 2-26 shows the Application Board connections when using HPU 0.

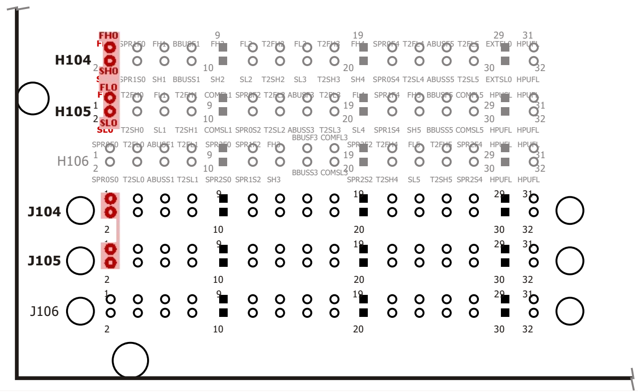


Figure 2-26 – ETS-88™ HPU-25/100 Application Board Connections (Low Current Path)

High Current Path

The second type of output from the HPU is for high current. This path can be used for all values of current that the HPU can provide.

There is one high current connection per side of a dual test head, for up to four HPUs in a system. The high current connectors are located on the test head fan panel, and are labeled 'HC "A"' and 'HC "B"'. The corresponding connections on the Application Board are labeled using the HPUFL_X / HPUSL_X / HPUFH_X / HPUSH_X convention, where X represents the side of the dual test head (A or B).

See Chapter 4 for specific connections based on the HPU's slot position in the Floating Resource Card Cage.

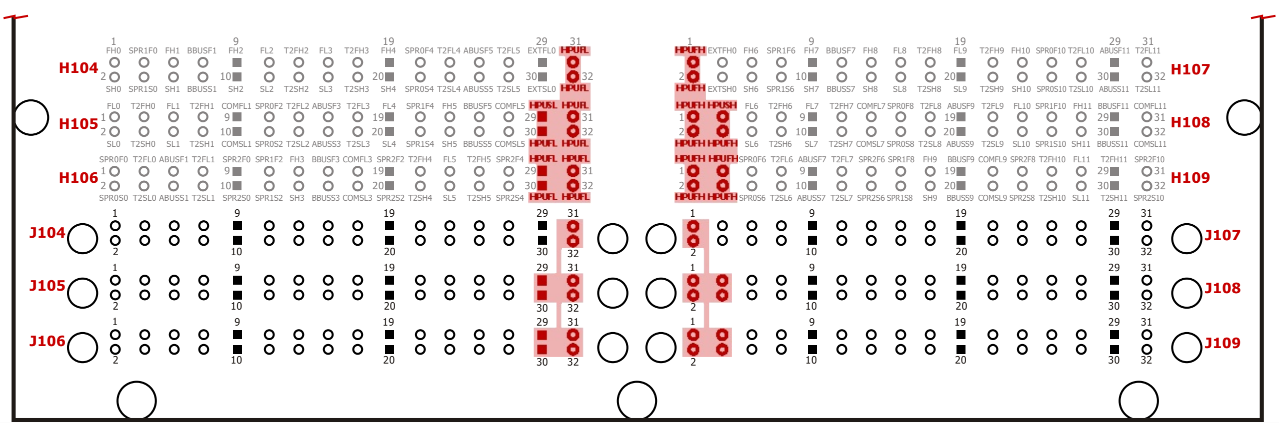


Figure 2-25 – ETS-88™ HPU-25/100 Application Board Connections (High Current Path)

2.7.12 Safety Considerations

For safety purposes, the HPU-25/100's output is inhibited if the DUT board interlock is not satisfied. Take great care to make sure that potentially hazardous voltages are not accessible to operators or any other personnel who may come in contact with the test apparatus. All electrical surfaces that can be energized to a potential above ±48VDC must be adequately covered to eliminate possible electrical contact with humans.

WARNING: RISK OF SHOCK.

****

Hazardous Voltages Present. Due to the nature of this resource and its use, the user must assume the burden of protecting operators and other personnel from possible shock hazard. Eagle Test Systems, Inc. accepts no responsibility for any possible harm this resource may cause to personnel.

NOTE: There are two LEDs on the edge of the HPU board that illuminate – indicating the discharge of the resource's capacitive charge – when the test system is turned off. DO NOT touch the board after powering-down the system until BOTH of these LEDs have turned off!

****

Figure 2-27 – HPU Cap. Discharge LEDs

2.8 Medium Power Unit (MPU)

2.8.1 Features

- Fully floating operation up to 1200 V
- Floating V/I has four-wire remote sensing, six current ranges, three voltage ranges, current and voltage measurement (single or digitized)
- Floating 16-bit Waveform Digitizer with 4K memory and variable sampling rate up to 100 kHz
- Floating 16-bit Arbitrary Waveform Generator (AWG) with 4K (64K optional) pattern RAM and variable sampling rate up to 1 MHz

2.8.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]

2.8.3 Theory of Operation

The basic modes of operation are: V/I Mode (Force (DC or AWG) and Measure (single or digitized)). This one instrument combines the AWG, Waveform Digitizer, DC Supply (or DC Force/Measure).

A typical application for such an instrument would be $RDS_{(ON)}$, $V_{FORWARD}$, $VCE_{(SAT)}$ and similar tests that require high currents at voltages up to 120 V. Since the MPU is a floating resource, multiple MPUs in a configuration can be stacked to obtain higher voltages, or paralleled to obtain higher currents.

2.8.3.1 V/I Mode

The MPU has a full four quadrant V/I with three voltage ranges and six current ranges. Remote sensing is accomplished with a 4-wire output (High Force/Sense, Low Force/Sense). Sense must be connected or force and measure voltage will not function. The MPU is fully floating and can be used to force and measure signals within 1200 V with respect to Chassis GND.

NOTE: The user must connect force and sense at the application board, in order for the force voltage and measure voltage functions to work properly.

Force and measure functions both have 16-bit resolution. As a standard part of the software, measurement averaging is supported to provide flexibility to the user in obtaining the best trade off between measurement repeatability and test time.

Use the utilities `mpumi()` and `mpumv()` to measure DC current and voltage, respectively. DC voltage and current measurements can be obtained without the use of the waveform digitizer.

2.8.3.2 Voltage and Current Clamps

Each voltage and current range has programmable upper and lower clamps. These clamps essentially define a "window" of allowable voltage or current. Current clamps and voltage clamps can be programmed simultaneously. In the FV mode, the current clamps are enabled, while in the FI mode, the voltage clamps are enabled.

Use the utility `mpuset()` to program the clamps, as well as other parameters.

Clamping Guidelines

Use the following guidelines to set current and voltage clamps correctly in applications.

Clamp Resolution:

The following tables provide a breakdown of the clamping ranges for the MPU, including the resolution for each range:

ASM1911 MPU Current Clamps		
Voltage Ranges	Current Range	Clamp Resolution
10V, 40V, 120V	400 µA	102 nA
10V, 40V, 120V	4 mA	1.02 µA
10V, 40V, 120V	40 mA	10.4 µA
10V, 40V, 120V	40 mA	104 µA
10V, 40V, 120V	4 A	1.04 mA
10V, 40V, 120V	40 A	10.4 mA

ASM1911 MPU Voltage Clamps		
Voltage Ranges	Current Ranges	Clamp Resolution
10V	400 µA, 4 mA, 40 mA, 400 mA, 4 A, 40 A	2.6 mV
40V	400 µA, 4 mA, 40 mA, 400 mA, 4 A, 40 A	10.4 mV
120V	400 µA, 4 mA, 40 mA, 400 mA, 4 A, 40 A	31.2 mV

Clamping Tips

The following list describes commonly overlooked factors related to setting the MPU's clamps, and provides tips for avoiding problems due to each factor.

1.) The clamps have approximately ±5 counts of offset error during set-up.
Tip: Write your program accordingly, factoring-in this offset error.

2.) The programmable voltage and current clamps are 12 bits, and these clamps are not calibrated.
Tip: Take precautions when setting these clamps to avoid problems due to assumed trip points. Test the clamps to determine exactly where they trip; this will help you determine precisely where to set each clamp.

3.) Setting the clamps too close together can produce non-uniform clamp levels when comparing negative and positive clamp response because of clamp overlap.
Tip: Set the negative and positive clamps at least 10 counts of resolution apart to prevent clamp overlap.

4.) Setting the current and voltage clamps to 0 will always cause the clamps to trigger.
Tip: Be sure the clamps in your application are set to a value greater than zero. Refer to the tables in the "Clamp Resolution" section above to determine the proper clamp value for the conditions of your test.

5.) Setting either the upper or lower current clamp to or near 0 can prevent the MPU from sourcing or sinking current, resulting in unexpected operation such as the appearance of a "railed" condition.
Tip: Always set upper and lower current clamps to values appropriate for the conditions of the test. Refer to the tables in the "Clamp Resolution" section above for valid clamp values according to range.

6.) When forcing voltage, the force-voltage feedback loop is in control. If a current clamp condition occurs, the feedback loop changes to a current feedback loop (switching modes during a clamp condition). When this happens, and the output has no load, the MPU is forcing current into an open, causing the voltage to rail.
Tip: Implement protection for such situations where they are most likely, or have historically occurred.

2.8.3.3 Arbitrary Waveform Generator (AWG)

The 16-bit AWG allows the user to reproduce any waveform from a sine wave to simulating the output of a digital driver into the DUT. The AWG has a maximum clock rate of 1 MHz and 4K (64K optional) RAM pattern depth. The user can force either current or voltage, depending on the mode selected. The software allows the user to load concatenated patterns with one utility call into the AWG, and then run these patterns in any order.

Use the utility `mpuawg()` to load an array of values, in volts or amps, into AWG memory. Use the utility `mpuawgstartstep()` to specify an AWG start address other than 0. The MCB furnishes the signal that clocks data out of the AWG. You select an MCB channel and connect it to the MPU, then program the channel with the necessary timing sequence(s). Refer to the Waveform Digitizer discussion below for an example of programming the AWG.

2.8.3.4 Waveform Digitizer

The 4K RAM Waveform Digitizer provides another powerful measurement tool. The clock coming into the MPU may be divided down to a sampling rate between 1 Hz and 100 kHz. Having a separate clock divider for the digitizer lets you measure at one frequency, and force data with the AWG at a higher frequency.

The digitizer may run concurrently with any forcing function, such as the AWG described above or a DC voltage/current. As with the AWG, the MCB furnishes the signal that clocks data into the digitizer. You select an MCB channel and connect it to the MPU, then program the channel with the necessary timing sequence(s).

EXAMPLE: Clock 1000 voltage values at 100 kHz out of the AWG of MPU8 using MCB channel 2. Set digitizer to digitize the resultant current at 50 kHz. Voltage values have already been loaded with `mpuawg()`.

```
/* Set MCB clock to 10 MHz */
mclkset(10.0);

/* 10 MHz/100 = 100 kHz */
mclkmode(2, MCLK_CLK, 100, 0);

/* Connect Channel 2 to MPU8 */
mclkchannel(2, "MPU8");

/* 100 kHz/2 = 50 kHz */
mpuadcmode(8, MPU_MI, 2);

/* Create Sequence 0 */
mclksequence(2, "CLEAR, 1000 ON");

/* Run Sequence 0 */
mclkstart(0);

mclkstop();
```

After the clock stops, use ETS data analysis utilities to retrieve digitized values, plot, perform mathematical operations, etc.

2.8.4 Block Diagram

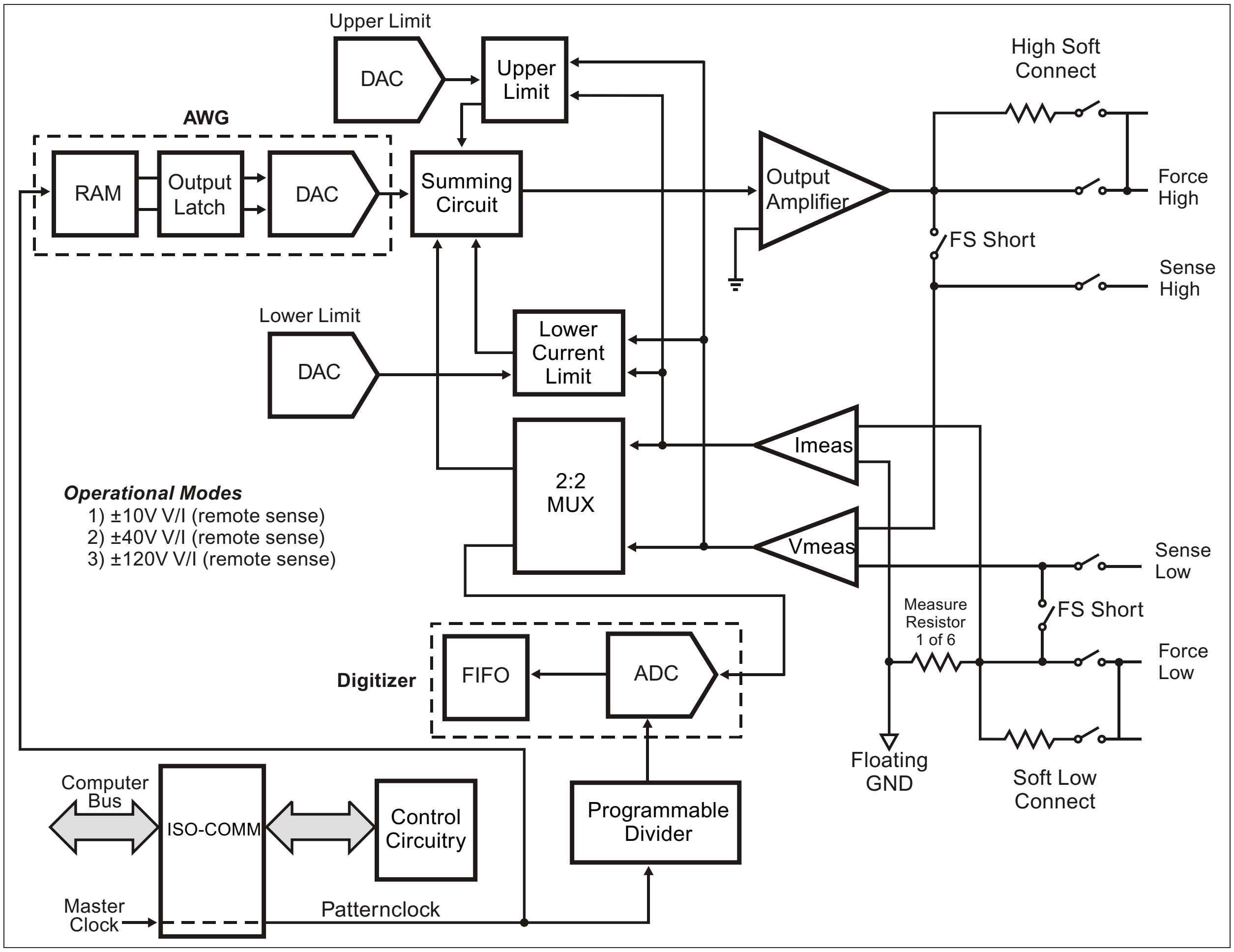


Figure 2-28 – MPU Block Diagram

2.8.5 Specifications

Voltage Ranges	Current Ranges
±10 V	±40 A
±40 V	±4 A
±120 V	±400 mA
	±40 mA
	±4 mA
	±400 µA

2.8.6 V/I Quadrants

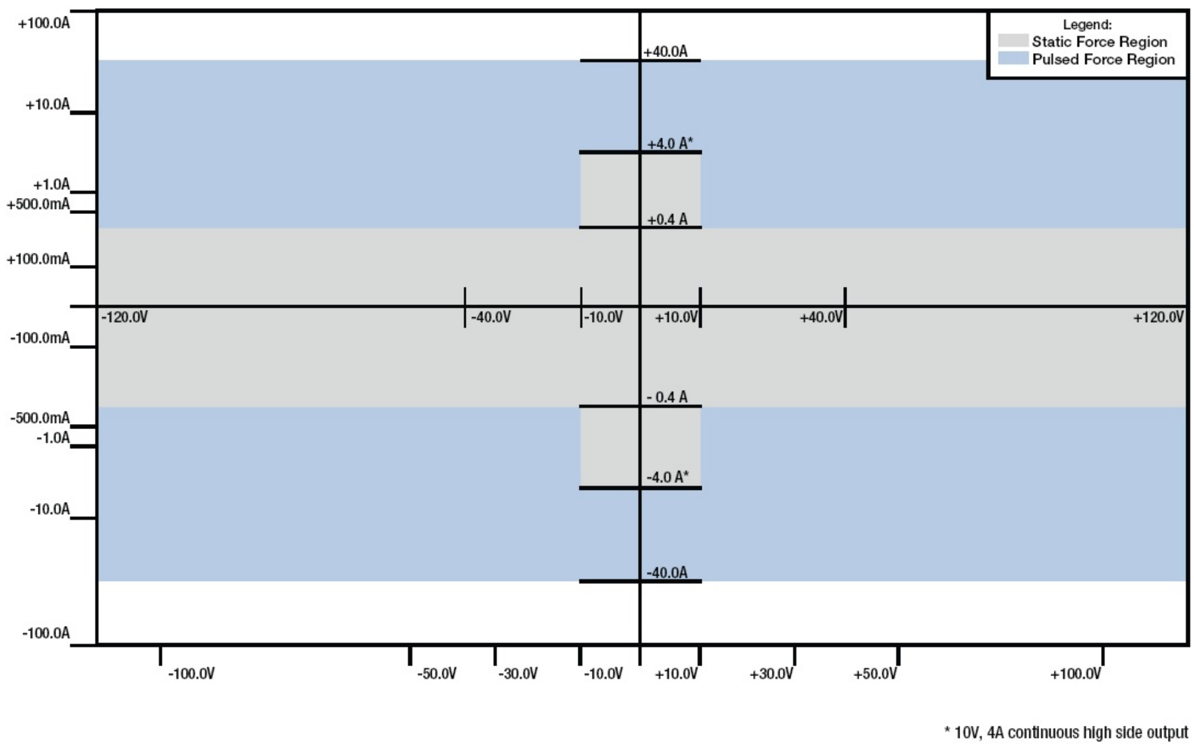


Figure 2-29 – MPU V/I Quadrant Diagram

2.8.7 User Interface

2.8.7.1 Software

All functions of the MPU are programmed using MPU software utilities. These utilities can be called from a test program, or in real time from RAIDE. Their syntax and usage are described in detail in The ETS Software Help File.

To use the AWG and Digitizer of the MPU, it is also necessary to program the Master Clock Board. The MCLK utilities are described in The ETS Software Help File. On-line help is available for all ETS Utilities while in the programming environment or RAIDE.

2.8.7.2 Hardware

There is a single SmartPin™ per MPU card, and they are housed in the Floating Resource Card Cage (FR Cage). MPU signals can be routed to the DUT on a low current or high current output path. The paths are described below.

Low Current Path

The low current output is designated for DC current less than 1 A, and pulsed current less than 4 A. Figure 2-31 shows the Application Board connections when using MPU 0.

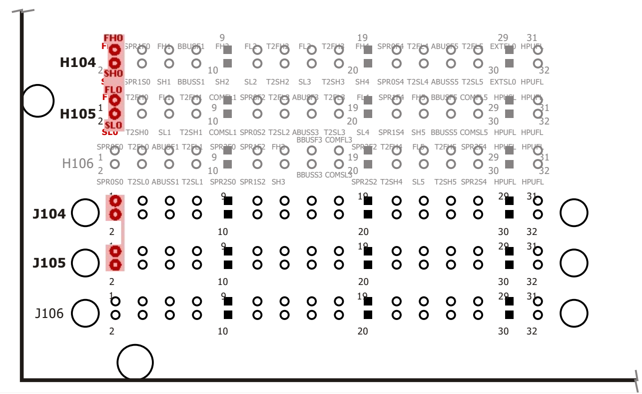


Figure 2-31 – ETS-88™ MPU Application Board Connections (Low Current Path)

High Current Path

The second type of output from the MPU is for high current. This path can be used for all values of current that the MPU can provide.

There is one high current connection per side of a dual test head, for up to four MPUs in a system. The high current connectors are located on the test head fan panel, and are labeled 'HC "A"' and 'HC "B"'. The corresponding connections on the Application Board are labeled using the HPUFL_X / HPUSL_X / HPUFH_X / HPUSH_X convention, where X represents the side of the dual test head (A or B).

See Chapter 4 for specific connections based on the MPU's slot position in the Floating Resource Card Cage.

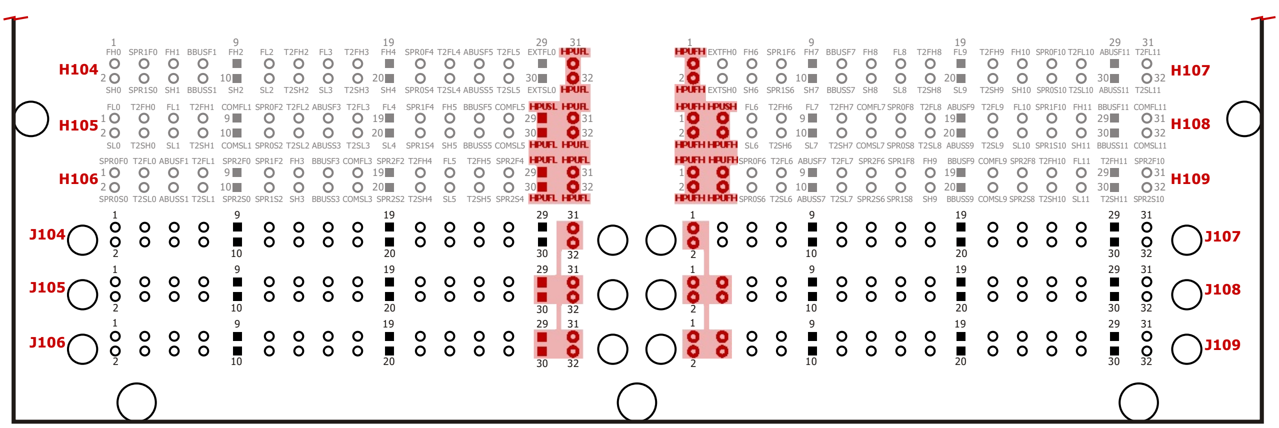


Figure 2-30 – ETS-88™ MPU Application Board Connections (High Current Path)

2.8.7.3 DC Current Capability

The MPU's DC forcing capabilities at each of its voltage ranges are as follows:

- 120 V Range:
  400 mA MAX
- 40 V Range:
  1 A MAX
- 10 V Range:
  Low I Channel = 400 mA MAX
  High I Channel = 4 A MAX

Either the low- or high-current output can be used for DC current values less than 1 A. For values greater than or equal to 1 A, the high current output must be used. See the `mpuset()` utility description in ETS Software help files for details on specific voltage/current range combinations.

Use the utility `mpuset()` to program the MPU to force DC voltages and currents. Use the utilities `mpumv()` and `mpumi()` to measure DC voltage and current values, respectively.

2.8.7.4 High Current Pulse Capability

The MPU employs a method of capacitive discharge for pulsing high current values. This method provides the following programmed modes of operation:

- 120 V Range:
  400 mA – 40 A
- 40 V Range:
  1 A – 40 A
- 10 V Range:
  4 A – 40 A

Use the utility `mpuset()` to program the MPU to force voltage and current pulses.

After a pulsing operation executes, a 400 mA current source begins recharging the capacitor banks to full potential. Total recharge time is a function of the voltage range selected, and the duration and magnitude of the current pulse. It ranges from a few msec, up to 250 msec maximum.

This is not meant to imply that the caps have to be charged to full potential before a pulse can be executed. Rather, it is to admonish the programmer to interrogate the SOA and RAIL DROOP alarms during development of a test program that pulses high values of current. The presence of either of these alarms indicates that the recharge time may have to be increased between current pulse operations.

Use the utility `mpualarm()` to determine if you are allowing adequate time between pulsing operations for the MPU to recharge. At higher values of voltage in the selected voltage range, the RAIL DROOP alarm (status Bit 15) is likely to be set if the recharge time is inadequate. Conversely, at lower values of voltage in the selected voltage range, the SOA alarms (status Bits 7 and 8) are likely to be set. For more information, refer to the discussions of Safe Operating Area and MPU Alarms that follow.

On the 40 V and 10 V ranges only, it is possible to halve the maximum recharge time, from 250 msec to 125 msec. Use the utility `mpuhicharge()` to enable this mode for a selected MPU.

NOTE: Only one MPU in the configuration can operate in this mode at a time.

2.8.7.5 MPU Alarms

The utility mpualarm() returns a bit-mapped status word. A non-zero value for status indicates the presence of a fault condition. Descriptions of the meaningful status conditions are discussed below. Bits not expressly described are always LO.

Bit 0

OSC Detect. This bit is meaningful only if the MPU is in a DC force mode. If this bit is set, look for an oscillation condition at the output connection from the MPU. If no such condition is detected, call `mpualarm()` again.

Bit 1

LO Kelvin. A setting of this bit indicates that the Kelvin connection has not been made on the LO side output of the MPU. Check that MPU Force and Sense LO are connected together at the application board.

Bit 2

HI Kelvin. A setting of this bit indicates that the Kelvin connection has not been made on the HI side output of the MPU. Check that MPU Force and Sense HI are connected together at the application board.

Bit 3

HW I Lo Limit. A setting of this bit indicates that the fixed (hardware) lower current clamp was encountered during the last MPU force operation. Run the tester diagnostic if this bit gets set. The current should be limited by the programmable lower clamp (Bit 5).

Bit 4

HW I Hi Limit. A setting of this bit indicates that the fixed (hardware) upper current clamp was encountered during the last MPU force operation. Run the tester diagnostic if this bit gets set. The current should be limited by the programmable upper clamp (Bit 6).

Bit 5

Clamp Lo Limit. A setting of this bit indicates that the lower programmed voltage or current clamp was encountered during the last MPU force operation. Call `mpualarm()` again to see if the condition is still in effect.

Bit 6

Clamp Hi Limit. A setting of this bit indicates that the upper programmed voltage or current clamp was encountered during the last MPU force operation. Call `mpualarm()` again to see if the condition is still in effect.

Bit 7

SOA Lo. A setting of this bit indicates that the safe operating area was exceeded in the negative direction during the last MPU force pulse operation. If this bit is set, the MPU has been shut down. It must be reset before forcing again.

Bit 8

SOA Hi. A setting of this bit indicates that the safe operating area was exceeded in the positive direction during the last MPU force pulse operation. If this bit is set, the MPU has been shut down. It must be reset before forcing again.

Bit 13

Heat Sink Temperature. Run the tester diagnostic if this bit gets set.

Bit 14

Ambient Temperature. Run the tester diagnostic if this bit gets set.

Bit 15

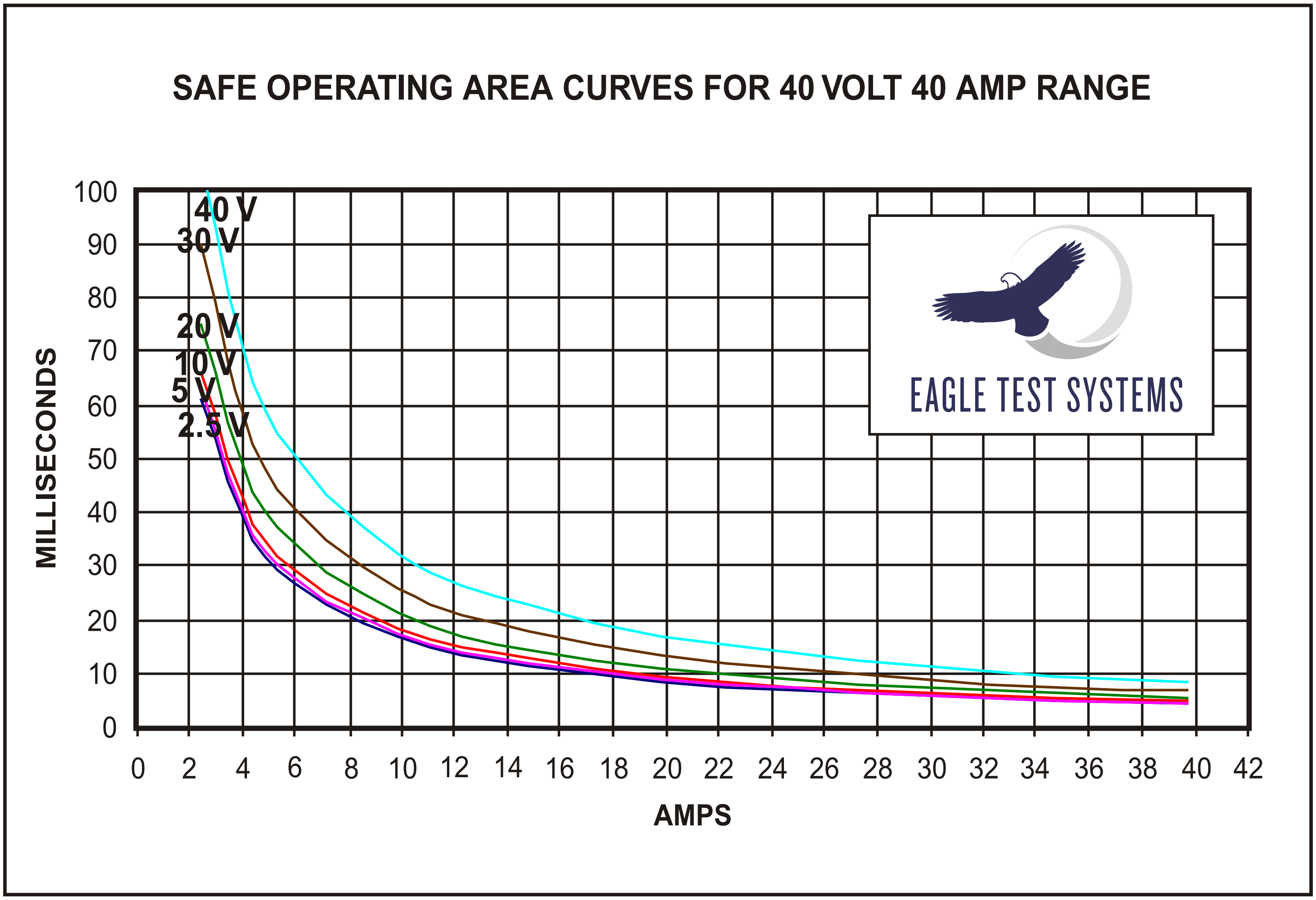
Rail Droop. A setting of this bit indicates the voltage in the discharge circuitry dropped below the specified tolerance during the last MPU force pulse operation.

Bit 31

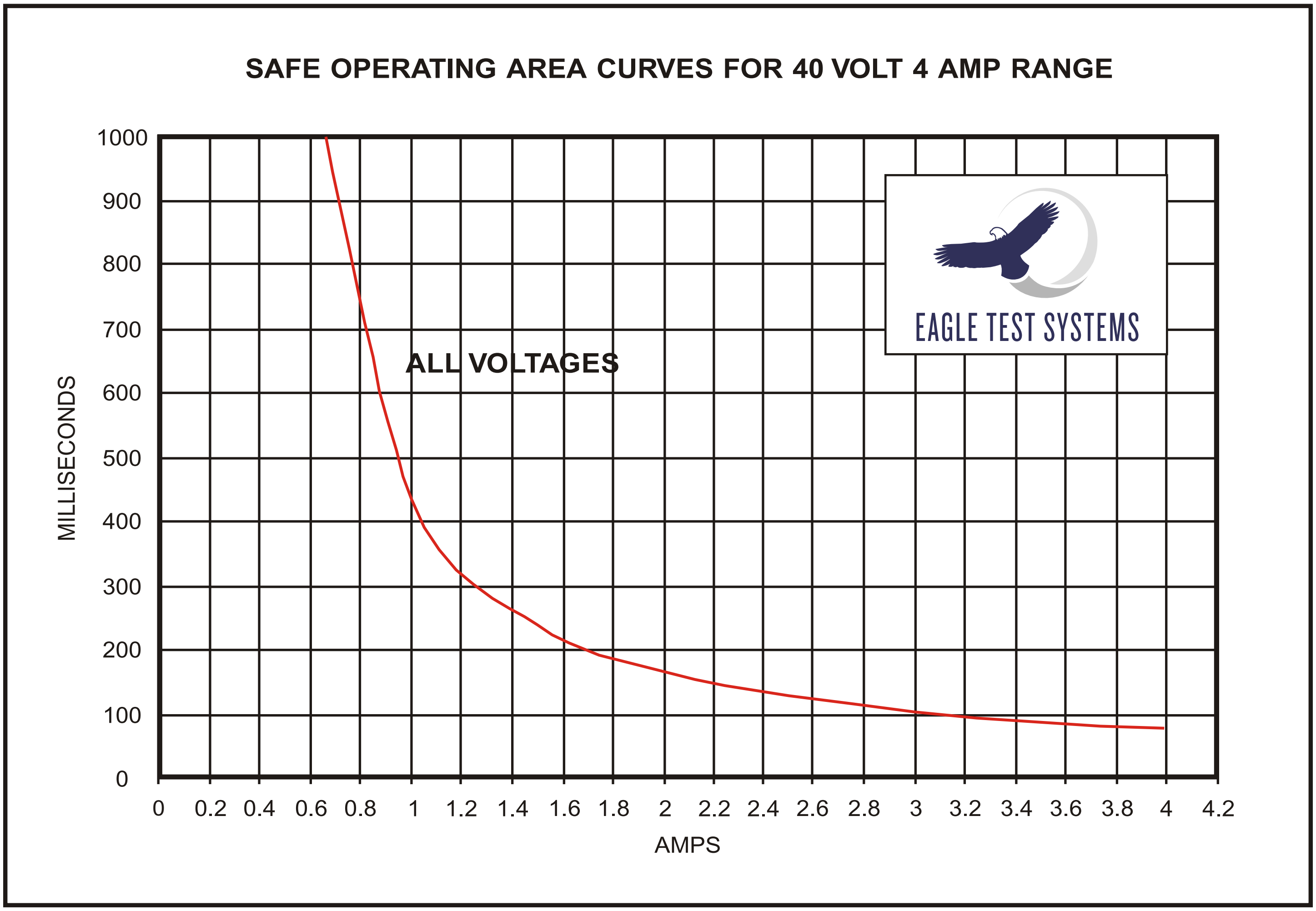
Interlock. This bit monitors the interlock circuitry, which should always be enabled. Run the tester diagnostic if this bit gets set.

2.8.8 MPU Safe Operating Area Curves

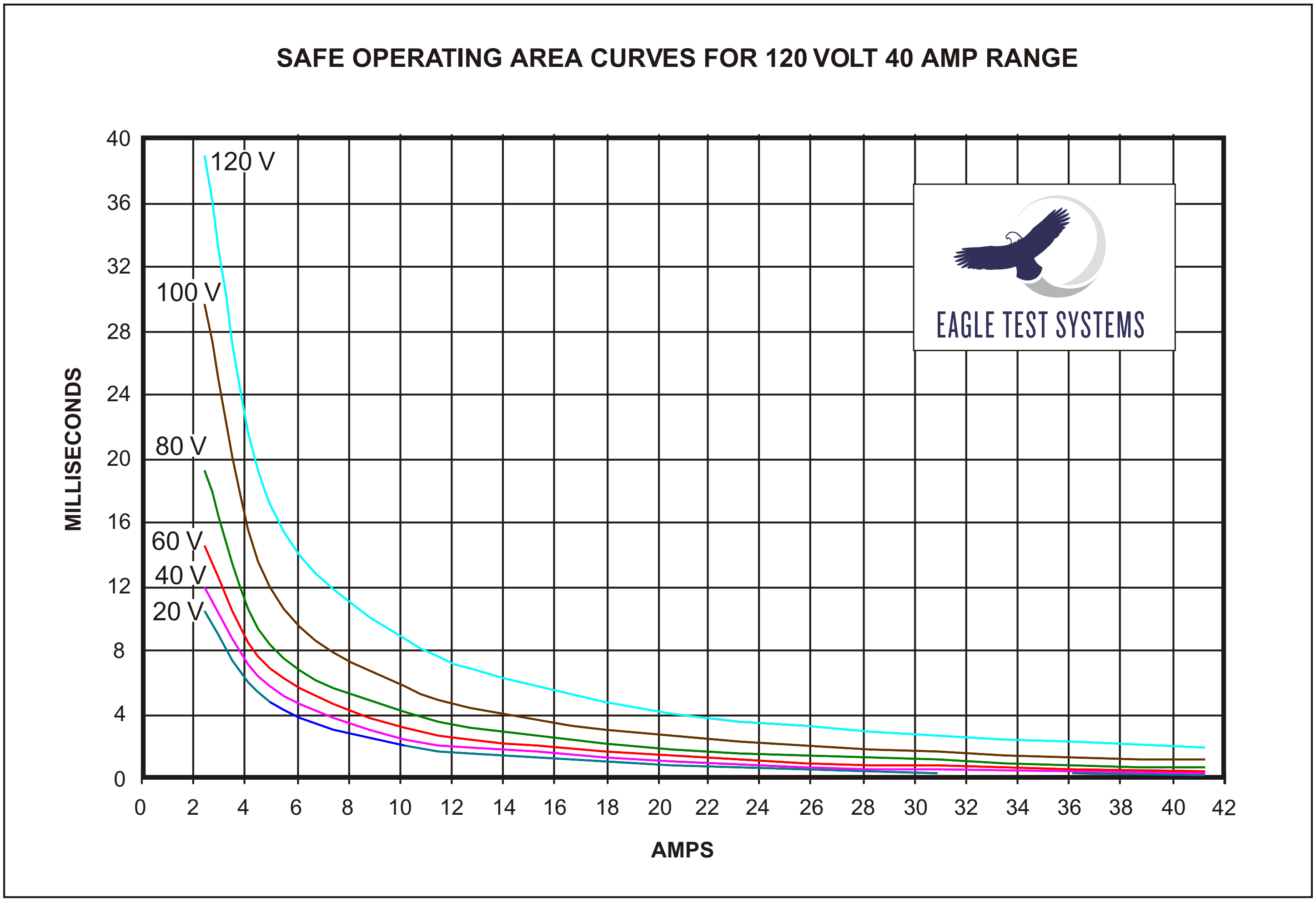
40 V / 40 A Range



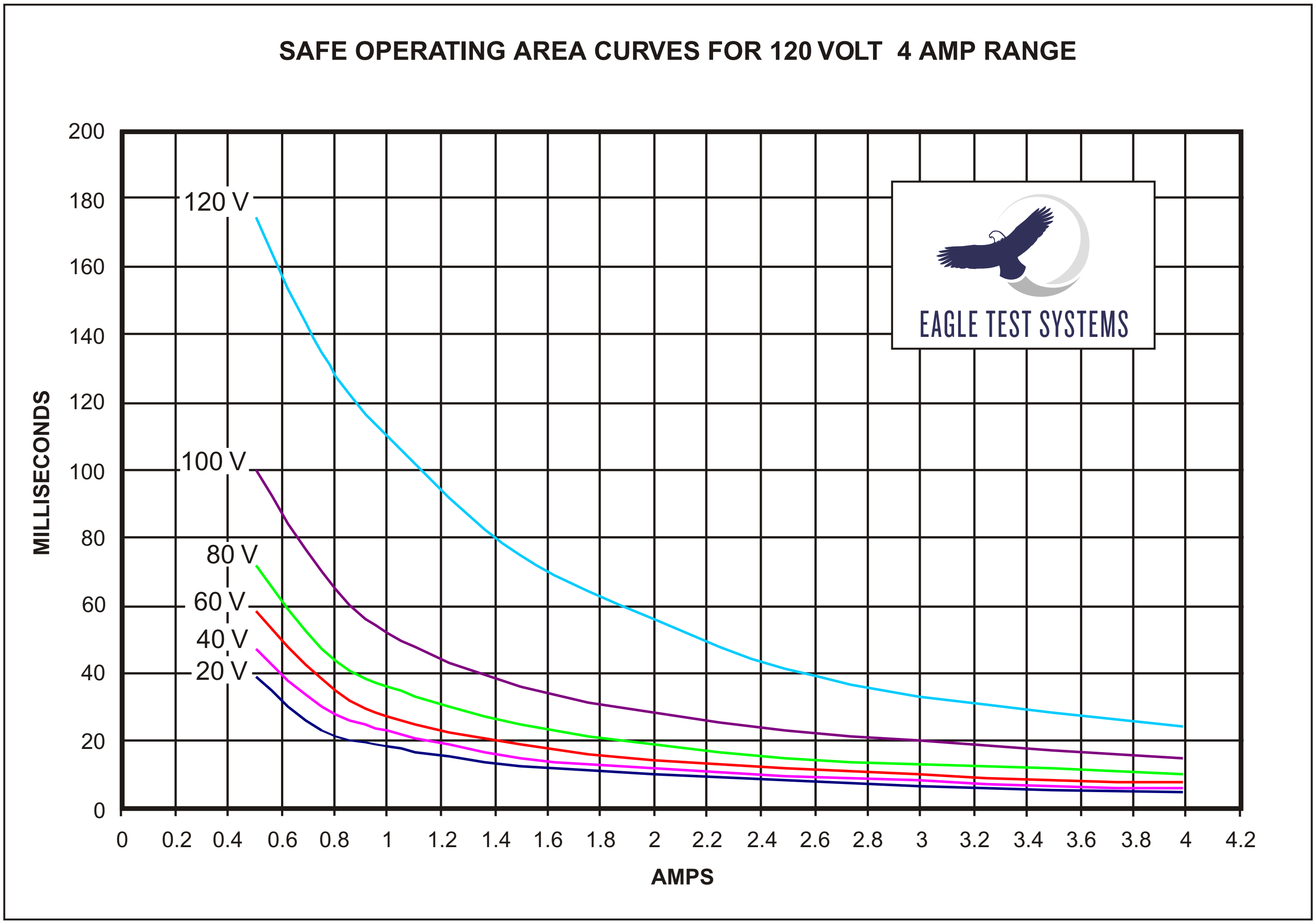
40 V / 4 A Range


SAFE OPERATING AREA CURVES FOR 40 VOLT 4 AMP RANGE
1000
900
800
700
600
500
400
300
200
100
0
MILLISECONDS
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2 2.2 2.4 2.6 2.8 3 3.2 3.4 3.6 3.8 4 4.2
AMPS
ALL VOLTAGES
EAGLE TEST SYSTEMS

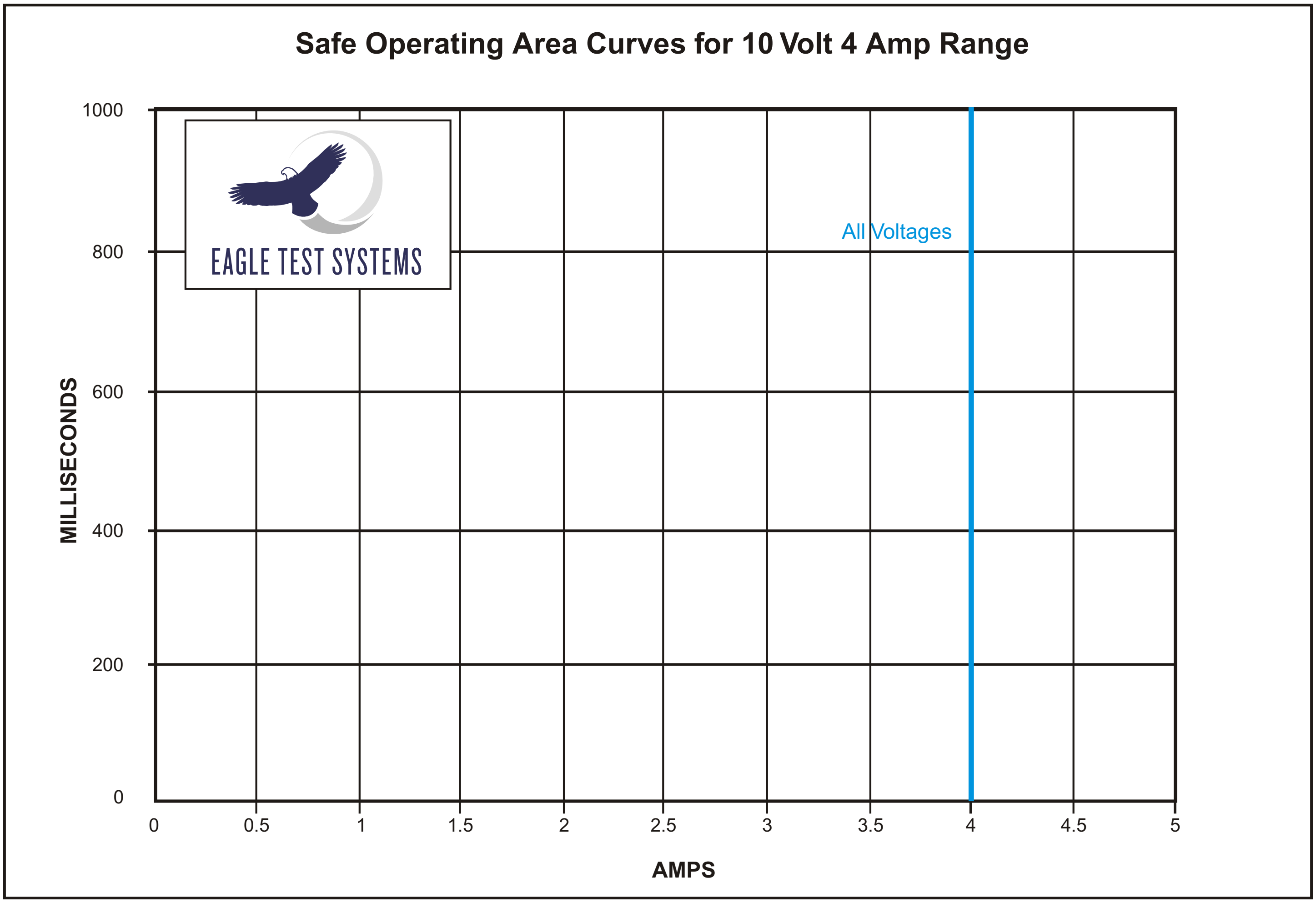
120 V / 40 A Range



120 V / 4 A Range



10 V / 4 A Range


Safe Operating Area Curves for 10 Volt 4 Amp Range
EAGLE TEST SYSTEMS
All Voltages
MILLISECONDS
1000
800
600
400
200
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
AMPS

2.9 Quad High Speed Signal Unit (QHSU)

2.9.1 Features

2.9.1.1 General Features

- Two independent dual-channel source/measure signal analyzers
- Suitable for signals ranging from DC to 50 MHz
- Supports quad-site operation (simultaneous source and measure)
- DSP per analyzer for fast parallel measurements
- Frequency-locked, low jitter clocking for FFT-based measurements
- Self-trigger modes for asynchronous applications

2.9.1.2 Analyzer Source Features

- Two selectable AWG-based signal generators:
    - 14-Bit 160 MSPS

      or

    - 16-Bit 50 MSPS
- 1 Meg pattern RAM
- Micro-coded pattern sequencer (PSQ)
- Selectable attenuation and filtering
- Programmable DC offset
- Single-ended or differential operation

2.9.1.3 Analyzer Measure Features

- Two selectable signal digitizers:
    - 14-Bit 80 MSPS

      or

    - 16-Bit 1 MSPS
- 1 Meg sample memory
- PSQ controlled digitizer gating control
- Selectable input gain and filtering
- Programmable DC offset
- Single-ended or differential operation

2.9.1.4 DSP Operation

- Supports all existing data analysis routines for post-processing data
- Run-time access to all QHSU ports
- DSP shares memory with the digitizers – avoids data transfer time
- Supports AWG pattern amplitude control as a user-initiated feature

2.9.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]

2.9.3 Theory of Operation

2.9.3.1 Overview

The QHSU, designed to test analog and mixed-signal devices, consists of two dual-channel high-speed signal analyzers, which cover the 0 Hz to 50 MHz frequency range. Each dual-channel signal analyzer is fully independent, and can be configured to source and/or measure. Each analyzer has a dedicated DSP, and a micro-coded AWG engine. With each dual channel analyzer, it is possible to source two separate waveforms simultaneously, and measure two incoming signals. Thus, with one resource board it is possible to source and measure four different signals simultaneously providing capability to test up to four sites in parallel.

The QHSU's two channels use the same clock source and can be operated alone, or under pattern sequencer (PSQ) control for simultaneous I/O operation. Sampling under PSQ control allows digitizing to be turned on and off at specific times relative to the outgoing signal. This ability is very useful for synchronized testing scenarios. Each measurement channel has a self-trigger option. This trigger option, similar to an oscilloscope trigger, allows measurements and/or AWG sequences to be triggered by an incoming test signal.

Signal Sourcing

Each source channel has the following features:

- 512 Words of PSQ RAM – One PSQ instruction can define a single contiguous waveform of any size
- 1 Meg of Pattern RAM
- Two AWG DAC Choices: 16-Bit, 50 MSPS and 14-Bit, 160 MSPS
- One Shared-Sine-Wave, 12-Bit, 300 MHz DDS
- 14-Tap Attenuator, 0 to -66 dB in 4 and 6 dB steps
- Six Low-Pass Filter Choices
- 50-Ohm Source Terminated, Differential Output Buffer

Each signal source pin can be driven by a 14-bit AWG (Arbitrary Waveform Generator), or a 16-bit AWG, covering the frequency range of 0 to 40 MHz. In addition, each channel can produce sine wave signals up to 100 MHz with the use of an on-board DDS (one per analyzer). The dual-range AWG is designed for optimal high frequency performance, using a specialized multiplexed 14-bit DAC with sample rates up to 160 MSPS, and a precision 16-bit DAC with sample rates up to 50 MSPS. The sine wave output is provided via a DDS-based 300 MSPS, 12-bit DAC.

The QHSU generates arbitrary waveforms via a micro-coded pattern sequencer (PSQ). One mega-word of memory is available for storing large amounts of signal data. The pattern sequencer effectively expands the available signal memory by allowing loops, jumps and subroutines within the memory space.

A unique case exists when using the 14-bit DAC. As mentioned earlier, the 14-bit DAC is a multiplexed (or double data rate) DAC. Two signal values (28 bits) are clocked into the DAC at half of the output frequency. For example, if the AWG clock is set to 160 MHz, the PSQ and pattern memory will be clocked at 80 MHz. The measure channels will also be clocked at 80 MHz, or some lower rate controlled by the PSQ.

Signal Measurement

Each measure channel has the following features:

- 1 Meg of capture RAM
- Two ADC choices: 16-Bit 1 MSPS, and 14-Bit o 80 MSPS
- Stand-alone and AWG-PSQ controlled measurement capabilities.
- Differential or single-ended signal capture capability
- Three termination selections: high impedance, 50 Ohms to system ground, and 100 Ohms differential
- Self-trigger option for oscilloscope-style triggering

Each measure channel has a 14-bit, 80 MSPS A/D and 16-bit, 1 MSPS A/D (selectable through programming commands), five anti-alias filter selections, and ten gain settings. One mega-word of capture memory is available for storing large amounts of sample data.

The QHSU has two identical signal analyzers. The QHSU block diagram shows the architecture of a single analyzer. Each analyzer has two source channels and two measurement channels. All four channels share a common main clock. Selections for the main clock source are system PCLK (0 – 33 MHz), local DDS 100 kHz – 100 MHz), local PLL (100 MHz – 160 MHz), or external clock (160 MHz maximum).

Measurement samples can be captured using the main clock directly, or under AWG pattern sequencer (AWG-PSQ) control. Each AWG pattern step can be programmed to trigger a measurement. Although the QHSU resides in the Floating Resource Cage, it is permanently parked – there is no option to float this resource.

Signal Processing

Each analyzer contains a dedicated DSP for "true parallel" data processing. This eliminates lengthy data transfer times, and other bottlenecks commonly associated with processing large amounts of data. The Eagle Vision Software Suite provides a robust function library for easy multisite coding.

Target Applications

- Video Filters
- ADSL Front-End Devices
- In-System Programmable Analog Devices
- Analog-to-Digital Converters

2.9.3.2 Operating Modes

Clocking

The QHSU has three basic clocking modes – Pattern Clock (PCLK), DDS Clock, and PLL Clock – which can be used for sourcing, measuring, and/or mixed operation.

In cases where the sample clock is 25 MHz or below, it makes sense to use PCLK as the sample clock. This is a standard operating mode, and assures that every execution will operate the same way each time (phase-reproducible operation).

For clock rates between 25 MHz and 80/100 MHz, use the DDS-based clock. The DDS clock is a programmable clock source that is derived from the 10 MHz clock reference. A local PLL multiplies the 10 MHz reference up to 300 MHz, which is suitable for the DDS sample clock. This means that the DDS is frequency-locked to the MCU-16. When using the DDS clock, the PCLK signal can be used as a start/stop gate.

NOTE: To support asynchronous gating of the digitizer, users must specify which samples went with each "time-slice." You can do this using the AWG Editor. See the "Operation for the QHSU" section of the AWG Editor documentation in the Eagle Vision Software Suite Manual for more information.

In mixed source/measure mode operation, the digitizer clock comes directly from a PSQ control bit. With this hardware approach, PSQ commands can be used to emulate the sample clock being gated on/off, or to make the clock behave as if each sample can be individually specified. The file loader will set the control bit as needed for either case syntax.

One option for sampling is to specify "Sample" on a "Burst: #" step. This holds the AWG at a fixed location, while taking "#" samples. Another way would be to specify "Gate On" once at (or near) the beginning of a pattern, causing the digitizer to sample until some later step where a "Gate Off" instruction is specified.

For AWG operation above 80/100 MSPS, the PLL clock source itself is used directly. You can achieve sample rates up to 160/200 MSPS by using a double speed or multiplexed input DAC. This means that the PSQ and AWG RAM will operate at a maximum speed of 80/100 MHz. The PLL clock will be used for sampling at these higher rates. A PLL/2 clock drives the PSQ and RAM address generator, allowing us to obtain a low jitter clock with 50% duty cycle.

For RF I/Q operation, two synchronized AWG channels provide excellent phase relationship reproducibility. Because I/Q signals are in the RF range, the DDS clock source must be used. The QHSU's dual channel design – with two AWGs running from the same PSQ – guarantees easily reproducible phase control.

Clock Mode Summary

- PCLK (0 to 25 MHz)
    - Gated clocking (PCLK controls turn on/off)
    - Gate on with continuous clocking (PSQ controls turn off)
      - PSQ Digitizer Clock Control
- DDS Clock (25 MHz to 80/100 MHz)
    - PLL multiplies up to 300 MHz for DDS fixed clock source
    - PCLK gate on/off PLL clock output
    - PCLK trigger on only (PSQ controls turn-off after PCLK trigger)
      - PSQ Digitizer Clock Control
- PLL Clock (80/100 MHz to 160/200 MHz)
    - PCLK gate on/off PLL clock output
    - PCLK trigger on only (PSQ controls turn-off after PCLK trigger)
      - PSQ Digitizer Clock Control

AC Source (PCLK, DDS, PLL Clocking)

When using the QHSU as an AC source, patterns run based on the selected clock mode (see above). The patterns may or may not contain microcode depending on your intent. Patterns containing microcode are typically generated in the AWG Editor tool and loaded from a file. This is in contrast to the AWG patterns, which are typically loaded from arrays within the program.

The high speed 160/200 MHz DAC can operate in a double speed mode. In this mode, the PSQ is operating at half the sample rate of the DAC, and the actual DAC sample clock is derived from the PLL. The PSQ and pattern RAM operate at half the actual DAC sample rate.

To source signals from the audio range up to 10 MHz of bandwidth, use the QHSU's 16-Bit, 50 MSPS DAC. This DAC uses the same filter sections as the 160/200 MSPS DAC.

AC Source Mode Summary

- PCLK, DDS, and PLL clock sources
- Microcoded vs Non-microcoded operation
- Two AWG DACs: 16-Bit 50 MSPS, and 14-Bit 160/200 MSPS

AC Measure (PCLK, DDS, PSQ, PLL Clocking)

In the typical digitizer mode of operation, the channel is set to select a given digitizer with range and filter settings. The sample data is stored in a memory area that is shared with the DSP.

AC Source / Measure (PCLK, DDS, PLL, PSQ Clocking)

This mode is driven from the PSQ with microcode and control bit event control. The AWG patterns can loop and subroutine. A control bit clocks the digitizer on and off as needed.

2.9.3.3 EPROM Utilization

The QHSU's on-board EPROM (Flash memory) stores the following information:

- Board serial number
- Hardware revision #
- Software revision # (Calibration rev. #)
- Date of last calibration
- Calibration data.

2.9.4 Block Diagram

Figure 2-29 on the following page shows the overall block diagram of the QHSU resource. Figure 2-30 then shows a detail of a single analyzer within the QHSU.

QHSU Overall Block Diagram

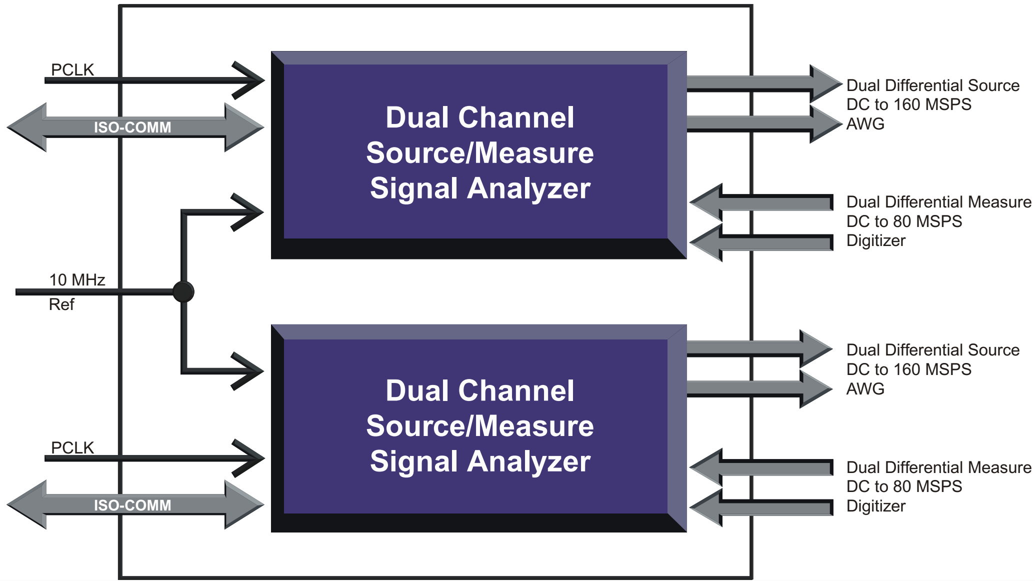


Figure 2-33 – QHSU Overall Block Diagram

Single QHSU Analyzer Block Diagram

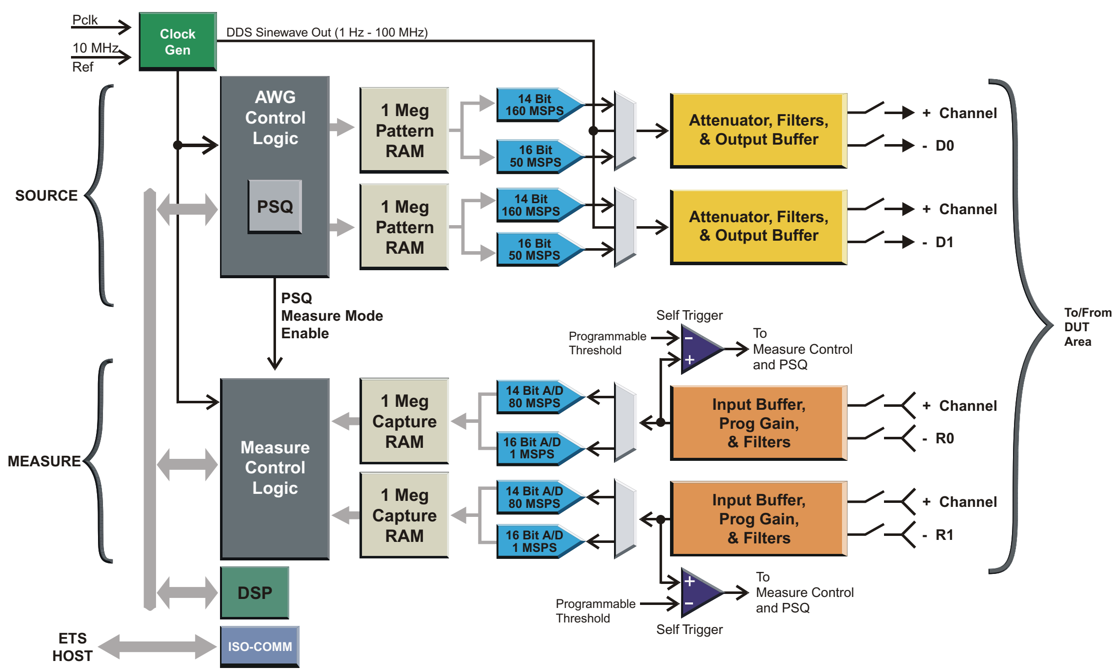


Figure 2-32 – Single QHSU Analyzer

2.9.5 Specifications

2.9.5.1 Source

Analog Output

Parameter	Min	Typical	Max	Units
Max Vout (1)			±4	Volts
Max Output Offset Adjustment(1)	±3.4		±4.042	Volts
Source Termination	45		55	Ohms
DC Accuracy	±(2% of range + 8 mV)			
(1) The magnitude is Volts peak, the connection is single-ended (applies to both plus and minus outputs), and the termination load is 1 MΩ.				

Source Ranges

Source Selection	Range Selection
16-Bit DAC	4 V, 2 V, 1.2 V, 600 mV, 400 mV, 200 mV, 120 mV, 60 mV, 40 mV, 20 mV, 12 mV, 6 mV, 4 mV, 2 mV
14-Bit DAC	2 V, 1 V, 600 mV, 300 mV, 200 mV, 100 mV, 60 mV, 30 mV, 20 mV, 10 mV, 6 mV, 3 mV, 2 mV, 1 mV
DDS Sine Wave	2 V, 1 V, 600 mV, 300 mV, 200 mV, 100 mV, 60 mV, 30 mV, 20 mV, 10 mV, 6 mV, 3 mV, 2 mV, 1 mV

Sample Frequency

Parameter	Min	Typical	Max	Units
16-Bit			50	MHz
14-Bit			160	MHz
12-Bit DDS Sine			300	MHz

(QHSU Source Specifications cont'd)

Bandwidth

Parameter	Min	Typical	Max	Units
16-Bit – 2 V Range, 1 V pk		16		MHz
14-Bit –				
1 V Range, 1 V pk		66		MHz
2 V Range, 1 V pk		42		MHz
12-Bit –				
1 V Range, 0.5 V pk		96		MHz
2 V Range, 2 V pk		52		MHz

Low Pass Filters

Parameter	Min	Typical	Max	Units
60 MHz, 1 V Range, 1 V pk		69		MHz
60 MHz, 2 V Range, 2 V pk		43		MHz
15 MHz, 1 V Range, 1 V pk		14		MHz
15 MHz, 2 V Range, 2 V pk		14		MHz
5 MHz, 1 V Range, 1 V pk		5		MHz
5 MHz, 2 V Range, 2 V pk		5		MHz
2 MHz, 1 V Range, 1 V pk		2		MHz
2 MHz, 2 V Range, 2 V pk		2		MHz
500 kHz, 1 V Range, 1 V pk		500		kHz
500 kHz, 2 V Range, 2 V pk		500		kHz

THD(1)

Parameter	Min	Typical	Max	Units
16-Bit DAC				
500 kHz, 2 V Range, 0.5 V pk		-82		dB
1 MHz, 2 V Range, 1 V pk		-73		dB
14-Bit DAC				
400 kHz, 2.0 V Range		-73		dB
1 MHz, 2.0 V Range		-67		dB
12-Bit DDS Sine Wave				
50 MHz, 1.0 V Range		-57		dB
50 MHz, 2.0 V Range		-45		dB
(1) THD using appropriate low pass filter.				

2.9.5.2 Measure

Analog Input

Parameter	Min	Typical	Max	Units
Max Vin [1]			±4	Volts
Max Input Offset Adjustment [1]			±4	Volts
Input Impedance				
No Termination	100 k			Ohms
Singled Ended Termination	45		55	Ohms
Differential Termination	90		110	Ohms
DC Accuracy	±(2% of range + 8 mV)			

(1) The magnitude is Volts peak, the connection is single-ended (applies to both plus and minus inputs), and the termination load is 1 MΩ.

Measure Ranges

Digitizer Selection	Range Selection
16-Bit ADC	4 V, 2 V, 1.2 V, 600 mV, 400 mV, 200 mV, 120 mV, 60 mV, 40 mV, 20 mV
14-Bit ADC	2 V, 1 V, 600 mV, 300 mV, 200 mV, 100 mV, 60 mV, 30 mV, 20 mV, 10 mV

Sample Frequency

Parameter	Min	Typical	Max	Units
16-Bit ADC	1		80	MHz
14-Bit ADC			1	MHz

Bandwidth

Parameter	Min	Typical	Max	Units
16-Bit ADC – 2.0 V Range		8		MHz
14-Bit ADC				
1.0 V Range, 1 V pk		50		MHz
2.0 V Range, 1 V pk		71		MHz

(QHSU Measure Specifications cont'd)

Low Pass Filters

Parameter	Min	Typical	Max	Units
40 MHz –				
1 V Range, 1 V pk		34		MHz
2 V Range, 2 V pk		38		MHz
20 MHz –				
1 V Range, 1 V pk		20		MHz
2 V Range, 2 V pk		20		MHz
5 MHz –				
1 V Range, 1 V pk		5		MHz
2 V Range, 2 V pk		5		MHz
2 MHz –				
1 V Range, 1 V pk		2		MHz
2 V Range, 2 V pk		2		MHz
500 kHz –				
1 V Range, 1 V pk		500		kHz
2 V Range, 2 V pk		500		kHz

THD(1)

Parameter	Min	Typical	Max	Units
16-Bit ADC				
100 kHz, 4 V Range, 2 V pk		-90		dB
14-Bit ADC				
1 MHz, 2.0 V Range, 1 V pk		-70		dB
5 MHz, 2.0 V Range, 1 V pk		-59		dB
(1) THD/SDFR using appropriate low pass filter.				

SFDR(1)

Parameter	Min	Typical	Max	Units
16-Bit ADC				
100 kHz, 2 V Range, 2 V pk		64		dB
14-Bit ADC				
1 MHz, 2.0 V Range, 1 V pk		70		dB
5 MHz, 2.0 V Range, 1 V pk		60		dB
(1) THD/SDFR using appropriate low pass filter.				

2.9.6 User Interface

2.9.6.1 Software

All functions of the QHSU are programmed using utility function calls from your C test program. These utilities and their syntax and usage are described in The ETS Software Help File.

NOTE: The Master Clock must be set up and started in order to clock the AWG and/or the digitizer. Please refer to the Test Head Control Board section (Section 2.17 on page 2-113) or the QHSU and MCB utility descriptions (in The ETS Software Help File) for further information.

2.9.6.2 Hardware

The QHSU is composed of a motherboard (which includes SMA jacks for connecting to the 10 MHz reference, and RJ45 ISO-COMM connections), and four modules: two converter modules, and two filter modules. The board is housed in the Floating Resource Card Cage, and connects to the AC rail for power via a four-wire cable.

The example in Figure 2-34 below shows the QHSU connections to the Application Board if you are using QHSU0 – Channel 0, 1, 2 and 3. Please see Chapter 4 for details on QHSU connections to the Application Board.

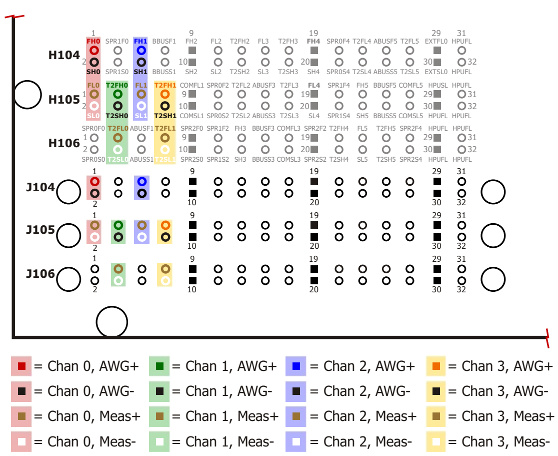


Figure 2-34 – ETS-88™ QHSU Application Board Connections

2.10 Quad Measurement System (QMS)

2.10.1 Features

- Four independent floating measurement systems in a single resource.
- High speed, high precision parallel measurement capability.
- High precision 18 bit resolution ADC mode with 16 bit pedestal DAC.
- Precision 16 bit resolution ADC mode to 200 kHz sample rate.
- High speed 12 bit resolution ADC mode to 10 MHz sample rate.
- Nine AC/DC measure ranges ±200 V to ±0.5 V.
- 16 k sample memory per channel.
- Four high speed "on board DSP" for high speed calculations.
- High precision DC voltage reference output with 16 bit resolution.
- Complete isolation to ±200 VDC.
- QMS-T supports high-speed Turbo Mode operation.

2.10.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]
- Eagle Vision MST [Rev. 2011.1.1002.17 and up]

2.10.3 Theory of Operation

The QMS was designed from start to finish with multisite testing in mind. System architectures with only one precision measurement resource are quite inefficient for multisite tests due to the requirement to make precision measurements serially by site. When DSP is required, having to move all data through a single DSP processor can also become a throughput bottleneck.

The QMS helps eliminate the effects of sequential precision voltage measurement mentioned earlier. The QMS can obtain precision measurements faster than previously available precision voltage measurement resources. It can also process the data on the instrument itself with a DSP processor, returning 'the answer' regardless of the number of data points digitized. All of this can be done on four independent channels in true parallel, yielding lightening-fast test times.

The QMS exhibits excellent precision DC measurement capability for applications such as precision references, E(sat), line regulation, load regulation, V(drop-out) (true differential), and many others.

Each instrument is fully floating on all ranges; ±200 V, ±100 V, ±50 V, ±20 V, ±10 V, ±5 V, ±2 V, ±1 V and ±0.5 V. A special pedestal mode is also available to measure microvolt-level voltage differentials on signals with large DC offsets. For example, a regulator output may be nominally 5.0 V, but may require microvolt measurement resolution to guaranty its ±250 µV line regulation specification. The QMS can make this measurement on four sites simultaneously without any additional application board hardware.

The QMS is also a powerful tool for measuring AC parameters such as; RMS, Pk-Pk, SNR, THD, AVE DC, template comparisons and many others. The user has easy access to a large library of DSP routines.

The low speed path provides access to a 16-bit, 200KSPS digitizer with three anti-aliasing filter settings. Filtering cut-off frequencies are available at 10 kHz, 100 kHz and 400 kHz for each range. A high-speed path is also available, providing access to a higher bandwidth (4 MHz) 12-bit, 10 MSPS digitizer.

The 12-bit digitizer supports the input ranges of ±.5 V, ±1 V, and ±2 V only, and supports pedestal mode on these ranges. The 12-bit digitizer can also be programmed for use on the nine range low speed path (400 kHz maximum BW) for applications where better time resolution is desired.

2.10.3.1 DC

DC measurements will use the selected filtering (10 kHz, 100 kHz or 400 kHz) when using the 16-bit ADC. Run the following utilities to select the filtering for DC measurements:

```
qmsset();

qmsmv();
groupgetresults();
```

The 'precision' mode is a one pass 16-bit ADC digitization which will take the requested number of data points using the desired sampling rate. The 'high precision' mode is an automatic two pass method as follows:

1.) Takes the 16-bit measurement with the ranging and filtering specified.

2.) Sets the pedestal to the same voltage as read in step 1, gains up the output stage amplifier to 50X and digitizes again.

3.) Returns result.

2.10.3.2 Selective DC

Implementing the pedestal mode and selecting an output gain stage other than 0.9X automatically selects the QMS's internal 10 kHz filtering. These higher gains allow flexibility in getting precise ESAT, line regulation and/or load regulation results. Run the following commands to select the QMS's internal 10 kHz filtering:

```
qmsset();

// Initial qms reading
init_val = qmsmv();

// Set pedestal to 'init_val' and
// 50X gain
qmspedestal();

// Final reading
final_val = qmsmv();
```

NOTE: The example above is shown as single site for simplicity but can certainly be done in a multisite scenario.

2.10.3.3 AC

Eagle has gone to great lengths to accommodate the user with much-needed AC performance utilities (i.e. THD, THD+N, SNR, FFT, etc.). The primary utilities for these calculations are:

```
// Give the digitized data in
// the QMS a name
namearray();
```

```
dspthdofspect();
dspthdnofspect();
```

Both of the utilities, `dspthdofspect()` and `dspthdnofspect()`, will take the input waveform, calculate the 2N FFT and return a result in one command. In attempting to foresee the users need to exclude certain Frequency bins these utilities will allow the input of two sets of bin numbers to include. In addition, if a voltage reference is preferred, rather than specifying the fundamental bin, the FFT created will be scaled to the voltage reference specified in the utility. The FFT created by these utilities can be plotted during debug and is represented in V2.

Other AC utilities are available when those specified above do not meet the need. They are as follows:

```
// Give the digitized data in the QMS
// a name
namearray();

//In volts2
dsppsd();

//In dBV, Calculated with dspspect();
dspsnr();

//In Volts
dspspect();

//In dBV
dspspectrum();

//In dBV, Calculated with dsppsd();
dspthd();

//In dBV, Calculated with dsppsd();
dspthdn();
dspfft();
```

2.10.3.4 Data Manipulation

Basic data manipulation is performed in the DSP processor residing on the QMS itself. Each QMS has its own DSP processor to evaluate the data. See the ETS DSP Data Analysis Utilities help files for further information on data manipulation with the QMS. The ability to process this data in parallel will be a tremendous asset in dropping the test times of many devices. Where necessary, it is still possible to bring the data back to the computer memory or into the test program for further manipulation.

2.10.4 Block Diagrams

See the following page for both an overall block diagram of the QMS and a more detailed block diagram of one of the four channels.

Figure 2-35 shows the block diagram of one of the four sections of the QMS.

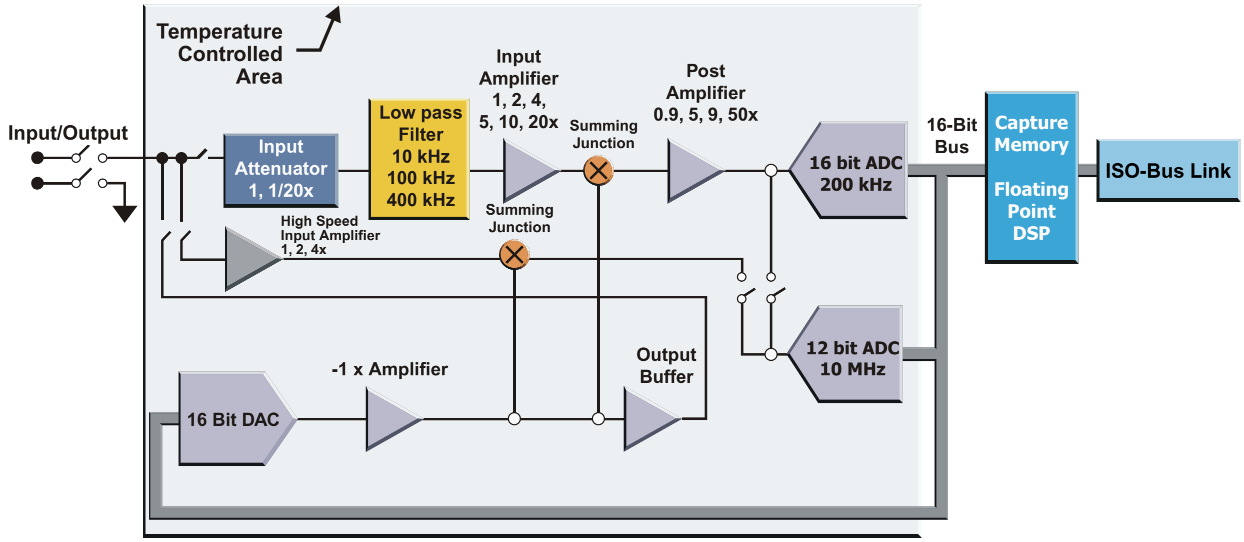


Figure 2-35 – QMS Block Diagram (Section 1 of 4)

Figure 2-36 is the overall block diagram of the QMS.

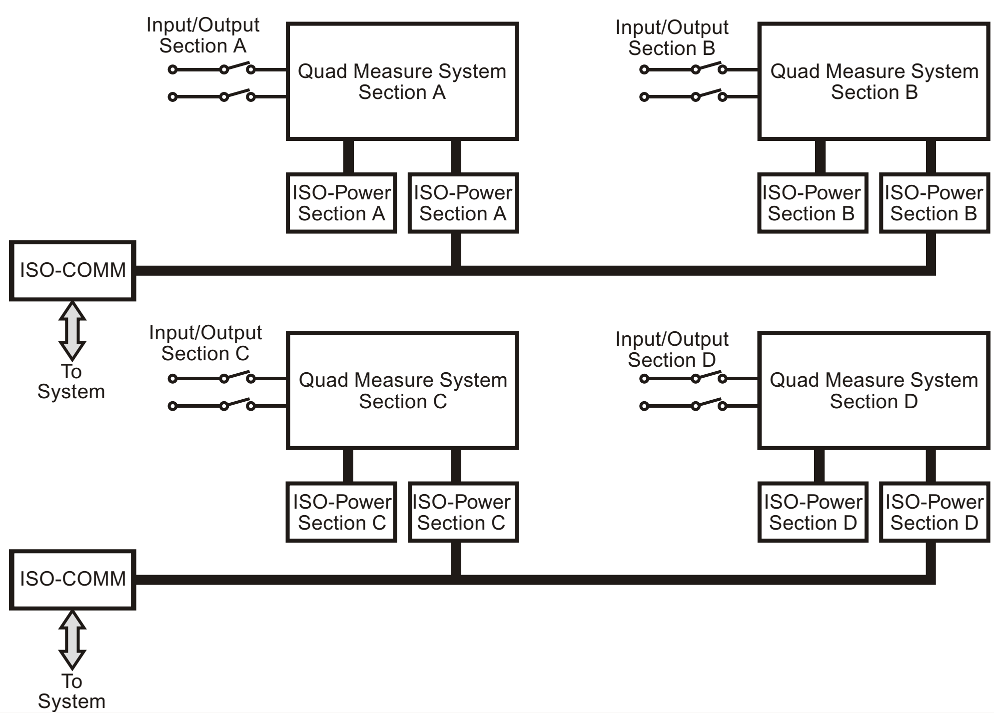


Figure 2-36 – QMS Block Diagram (Section 1 of 4)

2.10.5 Specifications

16 bit ADC Pedestal DAC			High Precision Measure Mode 50 µsec + (Maximum Sample Rate 200 kHz)	
Range	Full Scale	Input Z	Resolution	Accuracy (Offset + % Rdg)
1	±0.5 V	> 20 MΩ	3.8 µV	±(30 µV + 0.002%)
2	±1 V	> 20 MΩ	7.6 µV	±(45 µV + 0.002%)
3	±2 V	> 20 MΩ	15.2 µV	±(60 µV + 0.0015%)
4	±5 V	> 20 MΩ	38 µV	±(100 µV + 0.001%)
5	±10 V	> 20 MΩ	76 µV	±(200 µV + 0.001%)
6	±20 V	1 MΩ	152 µV	±(500 µV + 0.0015%)
7	±50 V	1 MΩ	380 µV	±(1.2 mV + 0.0015%)
8	±100 V	1 MΩ	760 µV	±(2.4 mV + 0.0015%)
9	±200 V	1 MΩ	1.52 mV	±(4.8 mV + 0.0015%)

16 bit ADC			Normal Precision Measure Mode (Maximum Sample Rate 200 kHz)	
Range	Full Scale	Input Z	Resolution	Accuracy (Offset + % Rdg)
1	±0.5 V	> 20 MΩ	15.2 µV	±(100 µV + 0.007%)
2	±1 V	> 20 MΩ	30.5 µV	±(150 µV + 0.006%)
3	±2 V	> 20 MΩ	61 µV	±(200 µV + 0.005%)
4	±5 V	> 20 MΩ	152 µV	±(500 µV + 0.005%)
5	±10 V	> 20 MΩ	305 µV	±(1 mV + 0.005%)
6	±20 V	1 MΩ	610 µV	±(2 mV + 0.007%)
7	±50 V	1 MΩ	1.52 mV	±(5 mV + 0.007%)
8	±100 V	1 MΩ	3.05 mV	±(10 mV + 0.007%)
9	±200 V	1 MΩ	6.1 mV	±(20 mV + 0.007%)

QMS Specifications (cont'd)

16 Bit ADC			AC Performance – Full Scale THD (Maximum Sample Rate = 200 kHz)		
Range	Full Scale	Input Z	-3 dB Frequency	1 kHz Sine	10 kHz Sine
1	±0.5 V	> 20 MΩ	100 kHz	-96 dB	-94 dB
2	±1 V	> 20 MΩ	100 kHz	-96 dB	-94 dB
3	±2 V	> 20 MΩ	100 kHz	-96 dB	-94 dB
4	±5 V	> 20 MΩ	100 kHz	-96 dB	-94 dB
5	±10 V	> 20 MΩ	100 kHz	-96 dB	-94 dB
6	±20 V	1 MΩ	100 kHz	-94 dB	-92 dB

12 Bit ADC Low Speed Path			AC Performance – Full Scale THD (Maximum Sample Rate = 10 MHz)		
Range	Full Scale	Input Z	-3 dB Frequency	10 kHz Sine	100 kHz Sine
1	±0.5 V	> 20 MΩ	400 kHz	-72 dB	-70 dB
2	±1 V	> 20 MΩ	400 kHz	-72 dB	-70 dB
3	±2 V	> 20 MΩ	400 kHz	-72 dB	-70 dB
4	±5 V	> 20 MΩ	400 kHz	-72 dB	-70 dB
5	±10 V	> 20 MΩ	400 kHz	-72 dB	-70 dB

2.10.6 User Interface

2.10.6.1 Software

Utility function calls from your C test program control all functions of the QMS. The ETS Software Help File includes descriptions of the QMS utilities, their syntax, and usage.

NOTE: Setup/starting of the Master Clock board is required in order to clock the digitizer. Please refer to the Test Head Control Board section (Section 2.17 on page 2-113) or to the QMS or Master Clock utility descriptions (in The ETS Software Help File) for further information.

2.10.6.2 Hardware

There are four QMSs per card and they are housed in the Floating Resource Cage. The example in Figure 2-37 below shows the QMS connections to the Application Board if you are using QMS4 – Channel 0, 1, 2 and 3:

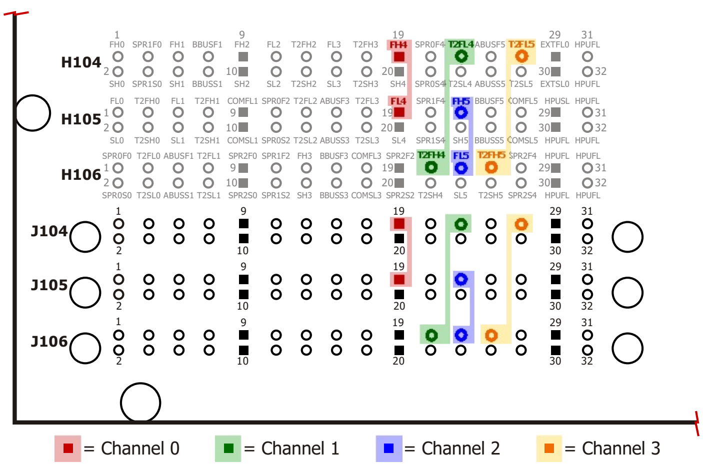


Figure 2-37 – ETS-88™ QMS Application Board Connections

2.11 Quad Precision Linearity Unit (QPLU)

2.11.1 QPLU Features

- Four (4) fully independent channels utilizing a single slot in the Floating Resource Card Cage.
- Supports multisite DC linearity testing of precision ADCs and DACs.
- Each QPLU channel consists of:

Source

- Modes of operation:
  - DC source
  - High precision ramp
  - Servo loop utilizing Rapid Dither™ algorithm*
- ±11.5 V high stability, low noise, output source with up to 25+ bit resolution
- Three dither ranges of ±1.2 V, ±120 mV, ±12 mV
- Single-ended or differential output with programmable common mode voltage
- Extremely fast settling to 20+ bits (< 1 μS typical – Dither DAC change only)

Measure

- Programmable pedestal voltage for canceling DC up to ±11.5V with three voltage measurement ranges:
  - ±10 V,
  - ±1 V,
  - ±100 mV (16-bit resolution)
- Up to 1 MSPS digitization rate
- Results accumulator for on-the-fly averaging
- On-board error amplifier and digitizer for measuring the difference between the pedestal DAC and summed DC output or DUT input
- Internal resource ground reference is remotely sensed at DUT ground
- Two-way communications with DPU-16 DSP processor for fast Pattern-Based Testing™ *
- Dual ±11.5V high stability, low noise, programmable voltage references (16-bit resolution)
  - Share common voltage reference with output source to minimize errors

2.11.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]

2.11.3 Theory of Operation

The QPLU is a low noise, high stability, high precision, analog source and measure resource with short-term accuracy and stability to 1 PPM (20 bits) and resolution of 381nV. A specialized composite Pedestal DAC and Dither DAC architecture are used to create the analog input signal to the DUT. In this design, the Dither DAC fills-in the gaps between pedestal DAC settings. This architecture provides high resolution, and high speed settling, which are both critical for data converter testing. The Pedestal DAC supports a ±11.5 V full-scale. The Dither DAC is summed together with the Pedestal DAC, and is ranged for ±1.2 V, ±120 mV, and ±12 mV with 16-bit resolution.

A common mode DAC is also provided for additional output flexibility, supporting level shifting for differential and single-ended operation. The on-board DACs and ADC use a common voltage reference. Referencing the DUT to the same reference as the pedestal DAC reduces errors caused by reference drift.

Each QPLU channel includes a voltage measure mode that can be used to test high precision devices. The QPLU can make accurate measurements of the analog output voltage of a precision DUT, such as DACs or voltage references. In this mode, an on-board PGA is used to gain-up the difference between the DUT output voltage and the on-board precision source. Selectable gains of 1x, 10x, and 100x are available with resolution as low as 3 µV on a ±11.5V full-scale range.

The QPLU is also useful for testing the DC linearity errors of ADCs using either ramp or servo techniques. Its servo loops facilitate fast code transition voltage measurements for A/D converters using a DSP-based Rapid Dither™ servo algorithm. In this mode, there is a closed loop including the DPU-16, which captures the resulting DUT code, and the QPLU that provides the analog input voltage. This mode involves a fast-acting, two-way communication path between the digital and analog sections of the test system. The Rapid Dither™ servo algorithm, briefly described below, has several configurations and test modes to address the requirements of different types of data converters.

For high precision testing (20-bit accuracy), the on-board error amplifier and 16-bit digitizer are used to measure the difference between the pedestal DAC and the servo output. Each QPLU channel is thermally stabilized to reduce errors associated with temperature drift. This method is used to achieve the utmost thermal stability. At the final stage of the transition search algorithm, an on-board ADC measures the error amplifier output to reduce even this small error.

Because the Rapid Dither™ algorithm is DSP-based and runs in the DPU-16's DSP processor, up to 16-site parallel testing can be supported on the ETS-600® platform. The DSP coding supports many different modes of operation. The DSP-based servo algorithm is used to rapidly process DUT results, avoiding the time penalties associated with data transfer and processing on the host computer.

If required, all the DUT response data is available within the DPU-16 for further processing or debug. In addition, the DSP provides megabytes of code under test (CUT) table memory as well as results memory.

2.11.4 Applications

2.11.4.1 General

- ADC Linearity Testing (Both low and high resolution; supports high accuracy devices)
- DAC Linearity Testing (Both low and high resolution; supports high accuracy devices)
- General Purpose, High Stability, Low Noise, DC Source and Measure Resource

2.11.4.2 Rapid Dither™ Search Algorithm

The Rapid Dither™ transition search algorithm is a specialized servo loop for very efficient testing of linearity errors of ADCs. It is made up of four main stages, transition intercept, fast dither, slow dither, and error sampling. The first three stages progressively reduce the DUT input voltage step size per DUT conversion. The purpose of the last stage is to make an accurate determination of the transition voltage of the ADC under test. Due to the nature of transition noise, the algorithm is tolerant of conversion errors.

2.11.4.3 Operating Modes

Ramp Mode –
Moves from code to code +N after each sample.

Single Code Dither Mode –
Executes Rapid o Dither™ on a single code, then stops.

Multi-Code Dither Mode –
Executes Rapid o Dither™ on the code-under-test table without software intervention.

Continuous Dither Mode –
Executes Rapid o Dither™ on a code, then dithers continuously on slow dither.

- This mode assumes external measurement of -Vin and is typically used for debugging.

2.11.4.4 Rapid Dither™ Error Conditions

There are a small number of error conditions that can occur during linearity testing. If a DUT is not functioning properly, the DSP algorithm must detect and report these possible errors.

- Railed dither DAC
- Sparkle code
- Missing code
- Un-locked condition (failure to locate desired code transition)

2.11.4.5 Calibration Methodology

The QPLU is designed to be a high accuracy / high stability resource; however, the unique demands of ADCs require local buffering of the QPLU signal. Because the local buffer can introduce errors, scaling, or even inversions, the Eagle Vision software includes all the necessary software commands to re-calibrate the QPLU at the DUT board level. These correction factors are stored and applied to all measurements before being returned to the user. This DUT-level calibration data is temporary, and is deleted upon exiting the application.

The calibration time will vary depending on many factors such as table size, and the number of sites. Eagle Vision software has been optimized to calibrate only what is required for each application, creating minimal down time. A quad site application can be calibrated in as little as 1 to 3 minutes.

To maintain traceability, the QPLU requires frequent calibration. The approximately 1 PPM accuracy specification is only guaranteed for a 24-hour period. After this time, a DUT board re-calibration is required to eliminate any component drift within the QPLU. The end user determines the actual re-calibration interval, and the QPLU does not track this time; therefore, care must be taken to ensure a periodic re-calibration. Generally, this DUT-level calibration is performed at application initialization time, and periodically thereafter by creating a countdown timer within the code.

Users should come to expect a DUT-level calibration when programs using a QPLU are initialized.

2.11.5 Specifications

The following pages list the specifications for the QPLU. These specifications are subject to change at ETS's discretion.

2.11.5.1 Pedestal DAC

Voltage Range:	±11.5 V (plus 5% overrange)
Resolution:	16-Bit

2.11.5.2 Dither DAC

Voltage Range:	±1.2 V, ±120 mV, ±12 mV
Resolution:	16-Bit

2.11.5.3 Combined Output

Voltage Range:	±11.5 V (plus 5% overrange)

Dither Range	±1.2 V	±120 mV	±12 mV
Resolution	38.1 µV (~19.3 bits)	3.81 µV (~22.5 bits)	381 nV (~26 bits)
Short Term Absolute Accuracy (<24 Hrs)	±(80 µV ± 0.0002% Value)	±(8 µV ± 0.00005% Value)	±(6 µV ± 0.00005% Value)
Short Term Relative Accuracy (Dither DAC change only, < 1 sec)	±(40 µV ± 0.01% Change)	±(4 µV ± 0.01% Change)	±(2 µV ± 0.01% Change)

Calibrated Absolute Accuracy (for Any Random Voltage)	±(15 µV ± 0.0002% Value)
Stability:	
Over Temperature	0.25 PPM per degree C
Short Term (24 Hour)Stability	1 PPM
Long-Term Stability	0.05 PPM/hr
Noise	<50 nV/root Hz (0.1 Hz to 1 MHz)
Output	Single-ended or Differential

2.11.5.4 Common Mode Output

Voltage Range:	±11.5 V (plus 5% overrange)
Resolution:	366 µV (16 Bit)
Stability:	0.3 PPM per degree C
Long-Term Stability:	0.05 PPM/Hr
Noise:	<25 nV/root Hz (0.1 Hz to 10 kHz)

2.11.5.5 Reference Outputs [2]

Voltage Range:	±11.5 V (plus 5% overrange)
Resolution:	366 µV (16 Bit)
Stability:	0.3 PPM per degree C
Long-Term Stability:	0.05 PPM/Hr
Noise:	<25 nV/root Hz (0.1 Hz to 10 kHz)
Output:	Single-Ended

2.11.5.6 Measurement ADC

Measurement Common Mode	±11.5 V	±11.5 V	±11.5 V
Measurement Range	±10 V (Max input voltage ±14.5 V)	±1.0 V	±100 mV
Resolution (16 bits)	305 µV	30.5 µV	3.05 µV
Absolute Accuracy	±(1 mV + 0.001% Rdg)	±(150 µV + 0.001% Rdg)	±(20 µV + 0.001% Rdg)
Relative Accuracy (Same Range, < 1 Sec)	±(800 µV + 0.01% of Change)	±(80 µV + 0.01% of Change)	±(10 µV + 0.01% of Change)
Stability	1 PPM per degree C		
Long-Term Stability	1 PPM/hr		

2.11.5.7 GND Reference Driver

Bandwidth:	< 100 Hz
Voltage Range from System Ground:	±0.6 V

2.11.6 Block Diagram

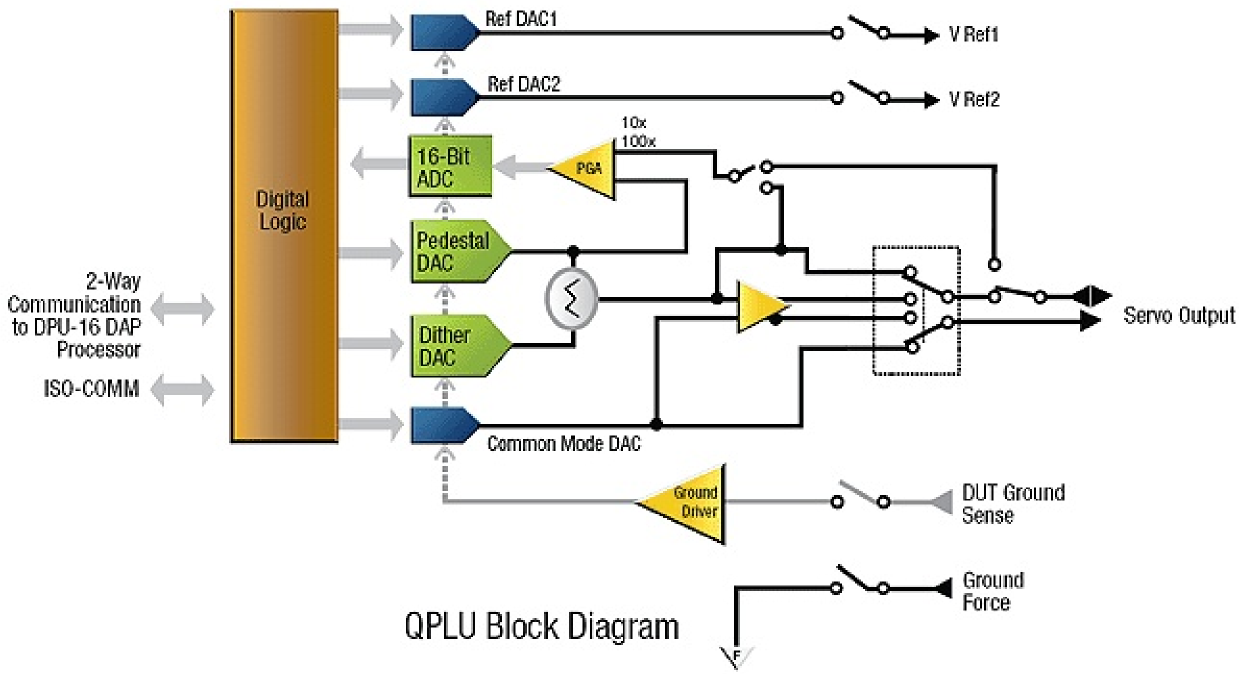


Figure 2-38 – QPLU Block Diagram (Overall)

2.11.7 User Interface

2.11.7.1 Software

All functions of the QPLU are programmed using the ETS Vector Editor and utility function calls from your C test program. These utilities and their syntax and usage are described in The ETS Software Help File.

NOTE: The TCB must be set up and started in order to clock the AWG and/or the digitizer. Please refer to the Test Head Control Board section (Section 2.17 on page 2-113), or the QPLU and MCU utility descriptions (in The ETS Software Help File) for further information.

Communication between the QPLU and the DPU-16 is also controlled via the QPLU Utilities. See the next section for information on how this communication is accomplished.

2.11.7.2 Hardware

There are four channels per QPLU card (28 total connections to the DUT), and the resource is housed in the Floating Resource Card Cage.

The QPLU communicates with the DPU-16 via the Communications Interface Board (CIB), a full duplex, crosspoint matrix. The CIB resides in the Digital Card Cage in the spare slot. The communications path is established from the DPU-16 DSP serial bus, through the backplane to the CIB, and then from the CIB to the QPLU in the FR Cage via CAT 5E style cables. The four QPLU-to-CIB cable receptacles are along the left hand edge on what is the rear of the QPLU board as it sits within the FR Cage.

Please refer to Chapter 4 for details on QPLU connections to the Application Board.

2.12 Quad Time Measurement Unit (QTMU)

2.12.1 Features

- Up to eight independent TMUs multiplexed to the digital pins (start/stop; arm signal per sequencer)
- Measures propagation delays, rise/fall times, frequency, duty cycle, events, etc.
- Three input ranges
  - Standard DPU-16 pin electronics (-1.0 V to +7.0 V; 200 MHz bandwidth)
  - DPU-16 buffered analog input (-3 V to +21 V; 20 MHz bandwidth)
  - TMU directs (±50 V; 2 start/stop inputs per TMU; 20 MHz bandwidth)
- <10 psec time resolution
- No range setting/changing required
- Complex sampling and arming supported
  - Averaging of multiple start/stop events (improves measurement repeatability)
  - Supports multiple arming events and sample sets per digital pattern
  - Self arming (arms on the signal itself)
  - PSQ arming (arming synchronized to a digital pattern)
  - Direct arming (arming based on a DUT board derived signal)
- 4K sample memory
- New timing events can begin every 200 nsec (5 MSPS re-sample rate)
- Frequency vs time analysis mode (direct demodulation of FM signals)
- Start/stop event counters for delayed triggering
- Trigger hold-off modes for noise rejection

2.12.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]
- Eagle Vision MST [Rev. 2011.1.1002.17 and up]

2.12.3 Theory of Operation

The QTMU is an essential resource for a mixed-signal test system. Most test systems offer only one system TMU. A single TMU restricts multisite test throughput by forcing timing measurements to be executed serially. Having access to at least one TMU per site is essential for a modern multisite mixed-signal test system.

The QTMU has a number of advanced operating modes and features. The QTMU is capable of automatically averaging a number of samples per arming event. With the integrated PSQ-based arming, it is possible to take a number of separate sample sets within a single pattern execution. This feature offers to greatly reduce acquisition times by getting more information from a single pattern.

Each QTMU channel has direct start/stop connection paths to the DUT interface board. These ±50 V input paths are also available in combination with the normal digital interface paths (provided through the DPU-16 pin electronics).

A system with up to 256 I/O pins can accept two QTMUs for a total of eight TMUs.

2.12.4 Block Diagram

Figure 2-36 on the following page provides a block diagram of the QTMU resources. Note the lines representing the QTMU cable connection to the DPU-16 resources.

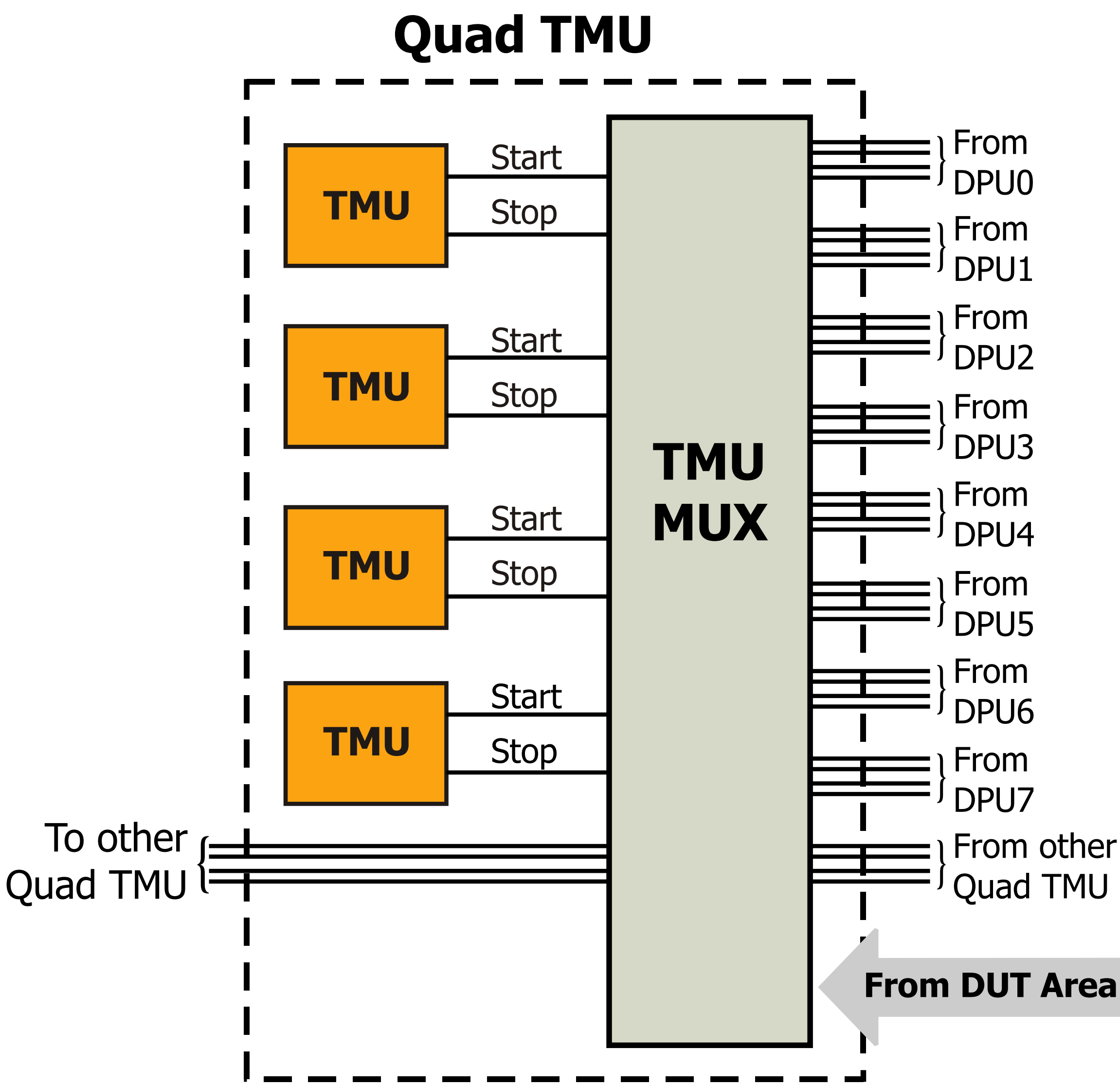


Figure 2-39 – QTMU Block Diagram

2.12.5 Specifications

Digital Input Channel Specifications:

Input Voltage Range	-1.0 V to +7.0 V
Input Analog Bandwidth	>150 MHz
Time Measure Accuracy	±2 nsec (for signals with >200V/µsec slew rate)
Time Measure Resolution	<10 psec
Maximum Time Measurement	13.4 sec
Maximum Input Frequency	50 MHz

Alternate Digital Input Channel Specifications:

Input Voltage Range	-3 V to +21 V
Input Analog Bandwidth	>20 MHz
Time Measure Accuracy	±4 nsec (for signals with >200V/µs slew rate)

Direct Input Channel Specifications:

Input Voltage Range	±50 V (MAX)
Input Voltage Resolution	25 mV
Input Voltage Accuracy	±(100 mV + 0.1%)
Input Analog Bandwidth	>20 MHz
Time Measurement Accuracy	±8 nsec

2.12.6 User Interface

2.12.6.1 Software

All functions of the QTMU are programmed using QTMU software utilities. These utilities (which begin the prefix "qtmu") can be called from your C test program or in real time from RAIDE. Their syntax and usage are described in detail in the Quad Time Measurement Unit (QTMU) Utilities section of the ETS Help File.

2.12.6.2 Hardware

Quad Time Measurement Units are located in the Digital Card Cage in the test head.

See Chapter 4 for the connections and pinouts for the QTMU. Also, refer to the DPU-16 Hardware User Interface section for details on connecting the QTMU to the digital I/O channels.

2.13 Smart Pin Unit 100 V / 2 A (SPU-100)

2.13.1 Features

- Fully independent dual channel SmartPin™ resource
- Each channel fully floating and stackable (±1000 VDC from ground max.)
- Three (3) voltage ranges, Seven (7) current ranges
  - ±100 V @ ±500mA, 100mA, 10mA, 1mA, 100µA, 10µA, 1µA
  - ±30 V @ ±1A, 100mA, 10mA, 1mA, 100µA, 10µA, 1µA
  - ±10 V @ ±2A, 200mA, 20mA, 2mA, 200µA, 20µA, 2µA
- Two (2) 500 KSPS digitizers to capture both voltage and current simultaneously
- Independent high/low programmable voltage/current clamps with alarms
- Kelvin error detect and measurement full-scale alarms
- Driver / signal generator mode
  - High speed AWG (16 Bit; 25 MSPS; up to 1 MHz Sine)
  - High Resolution AWG (18 Bit; 350 KSPS; up to 50 kHz Sine)
  - Audio Mode (18 Bit; 350 KSPS; Better than -96 dB THD @ 1 kHz)
- Volt meter mode: ±(1000 V, 100 V, 30 V, 10 V)
- Real-time measurement accumulator for instant results averaging
- Change V/I settings under AWG pattern control
  - FV/FI selectable on the fly
  - Current ranges selectable on the fly
  - ADC gain and filtering selectable on the fly
  - ADC sample clock gate on/off on the fly
- Results accumulator supports up to 16 sample sets per pattern
- Interlocks provided for operator safety
- Hardware and software designed for multisite applications

2.13.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]
- Eagle Vision MST [Rev. 2011.1.1002.17 and up]

2.13.3 Theory of Operation

The Smart Pin Unit 100 V / 2 A (SPU-100) is a single slot, dual channel, ±100V SmartPin™ resource with seven current ranges. The SPU-100 spans a wide range of voltage and current combinations, making it an extremely versatile instrument. The SmartPin™ architecture incorporates an AWG and a digitizer within a conventional four quadrant V/I. This resource includes all the standard capabilities of full-featured V/Is (programmable clamps, Kelvin detect, alarms, Etc.) plus the advanced characteristics that SmartPin™ users have come to expect.

The V/I is stable with almost any combination of inductive and/or capacitive loads. The programmable clamps are very well behaved, crossing over from voltage to current or visa-versa with minimal overshoot or instability. Bandwidth and settling time are optimized for excellent measurement speed. Analog switches are used extensively for excellent reliability and switching speeds.

The 18 bit AWG makes it possible to generate arbitrary voltage and current based signals that are synchronized to all other digital and analog resources in the test system. This enables test engineers to easily create dynamic test conditions that have previously been impossible in ATE without custom application circuitry.

The three special driver modes (18 bit 350 KSPS, 16 bit 25 MSPS and a special audio mode) support waveform generation by providing direct access to the buffered AWG output. This proves extremely useful for general purpose applications requiring AC signals in and above the audio range. These signals may be synchronized to the other analog and digital resources of the system.

In the audio mode, a specialized differential line driver is switched into the output force lines to provide a high quality audio signal for THD and noise testing. In this mode, both differential and single-ended configurations are supported.

The dual integrated digitizers simplify parallel measurements. As explained in the following section, the ability to digitize both voltage and current simultaneously simplifies and speeds many applications.

2.13.4 Applications

The SPU-100 is useful for advanced measurement applications as well as for use as a general purpose V/I. The three voltage ranges and seven current ranges make it possible to address a wide range of test applications. For static force/measure applications, the real-time measurement accumulator radically reduces measurement times with built-in hardware results averaging. Combining these features with 18 bit force and 16 bit measurement resolution creates a performance standard that is unrivaled in the industry.

SmartPin™ resources are valuable for testing various devices, because they make it possible to initiate a wide range of test conditions in rapid sequence. The use of a pattern-based V/I makes it possible to change force conditions on the fly. The pattern RAM contains the V/I force values and a number of synchronized control bits that make it possible to change the operating state of the V/I on the fly and to enable/disable the on-board digitizer to capture the desired test results at selectable pattern locations.

The digitizer is also capable of on-the-fly averaging, such that the average value of each sample-set is stored in RAM along with each set of sample values. For pattern-based DC tests, this greatly reduces data transfer times during post processing.

With this type of hardware available, it is possible to string together many test conditions, while simultaneously storing the measured results. After the pattern runs, the system controller will typically read back the results for test limit comparison purposes. Due to the real-time averaging of sample-sets (Results Accumulator), very little time is required to read the results from the hardware. If a more complex evaluation method is required, the sampled data can be transferred to controller memory for further mathematical evaluation.

Threshold searches are a common application problem in the industry. Often, test engineers are required to use either successive approximation techniques or design specialized application circuitry to speed up these normally time consuming tests; however, this does not have to be the case. The SPU-100 makes it possible to locate current based threshold points and the associated threshold hysteresis levels with a single up/down ramp pattern. This greatly reduces test time and improves measurement repeatability. Because the ramp signal is AWG driven, the resolution and speed of the ramp can easily be optimized for the best trade-off in test speed vs. measurement accuracy and repeatability.

These Smart-Pin-based test techniques also let you measure parameters such as dynamic PSRR (power supply rejection ratio) or to string together several forcing/loading conditions that are typical in regulator testing. This AWG-pattern-based test approach is quickly becoming the method of choice for regulator manufacturers throughout the industry.

For static DC force/measure applications, which are still heavily used in many applications, the real-time measurement accumulator provides instant averaging of measured results. This means that the samples are summed mathematically in real-time. When sampling is complete, the answer may be read directly from the resource, avoiding any further data transfers. However, the full data is still available for plotting or other engineering purposes, offering the best of both worlds.

2.13.5 Block Diagram

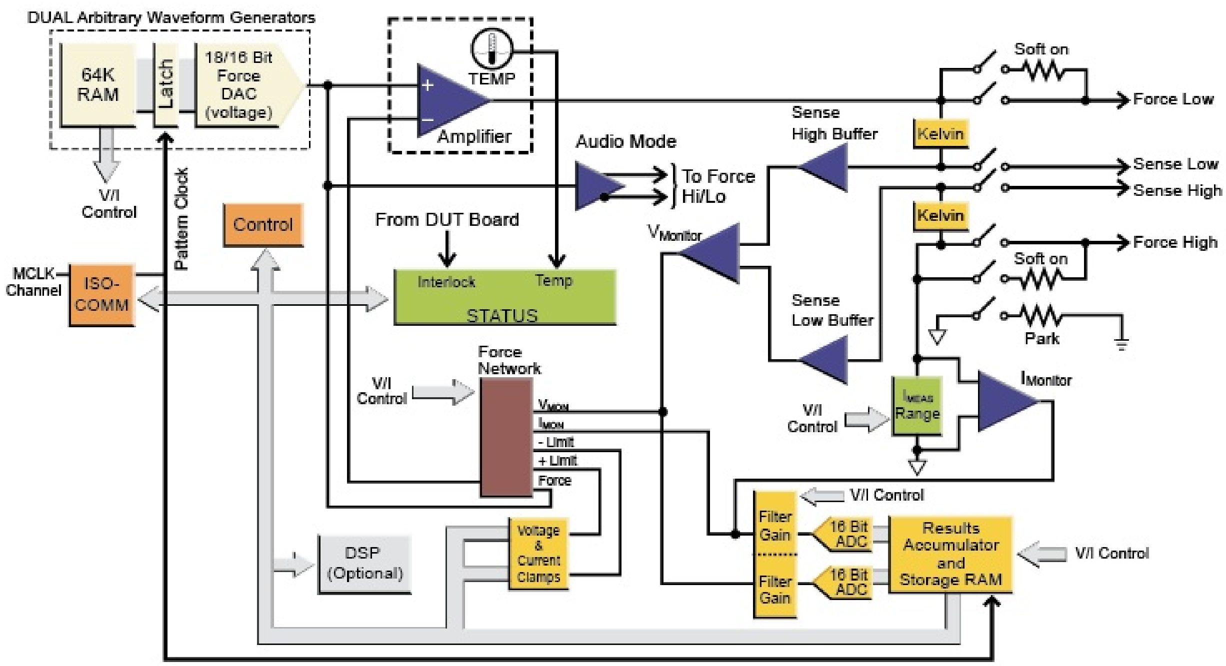


Figure 2-40 – SPU-100 Block Diagram

2.13.6 Specifications

The tables on the following pages provide the detailed voltage and current force and measure specifications for the SPU-100.

2.13.6.1 10 V Range

Force	Range	Resolution	Accuracy
Voltage	10 V	80 µV	±(0.8 mV + .025%)
Current	2 A	16 µA	±(250 µA + .05%)
	200 mA	1.6 µA	±(25 µA + .05%)
	20 mA	160 nA	±(2.5 µA + .05%)
	2 mA	16 nA	±(250 nA + .05%)
	200 µA	1.6 nA	±(20 nA + .05%)
	20 µA	160 pA	±(7 nA + .1%)
	2 µA	16 pA	±(5 nA + .1%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±10 V	1X	±10 V	300 µV	±(800 µV + .025%)
	±10 V	10X	±1 V	30 µV	±(600 µV + .025%)
	±10 V	100X	±100 mV	3 µV	±(600 µV + .025%)
Current	±2 A	1X	±2 A	60 µA	±(250 µA + .05%)
	±2 A	10X	±200 mA	6 µA	±(125 µA + .05%)
	±2 A	100X	±20 mA	600 nA	±(125 µA + .05%)
	±200 mA	1X	±200 mA	6 µA	±(25 µA + .05%)
	±200 mA	10X	±20 mA	600 nA	±(12 µA + .05%)
	±200 mA	100X	± 2 mA	60 nA	±(12 µA + .05%)
	±20 mA	1X	±20 mA	600 nA	±(2.5 µA + .05%)
	±20 mA	10X	±2 mA	60 nA	±(1.2 µA + .05%)
	±20 mA	100X	±200 µA	6 nA	±(1.2 µA + .05%)
	±2 mA	1X	± 2 mA	60 nA	±(250 nA + .05%)
	±2 mA	10X	±200 µA	6 nA	±(125 nA + .05%)
	±2 mA	100X	±20 µA	600 pA	±(125 nA + .1%)
	±200 µA	1X	±200 µA	6 nA	±(25 nA + .05%)
	±200 µA	10X	±20 µA	600 pA	±(15 nA + .1%)
	±200 µA	100X	±2 µA	60 pA	±(15 nA + .1%)
	±20 µA	1X	±20 µA	600 pA	±(15 nA + .05%)
	±20 µA	10X	±2 µA	60 pA	±(10 nA + .1%)
	±20 µA	100X	±200 nA	6 pA	±(10 nA + .1%)
	±2 µA	1X	±2 µA	60pA	±(12 nA + .05%)*
	±2 µA	10X	±200 nA	6 pA	±(6 nA + .1%)
	±2 µA	100X	±20 nA	600 fA	±(6 nA + .1%)

* Accuracy improvement with auto-zero tare: ±2 µA Range: (±2nA + .05%)

2.13.6.2 30 V Range

Force	Range	Resolution	Accuracy
Voltage	30 V	225 µV	±(2.5 mV + .025%)
Current	1 A	8 µA	±(125 µA + .05%)
	100 mA	800 nA	±(12.5 µA + .05%)
	10 mA	80 nA	±(1.25 µA + .05%)
	1 mA	8 nA	±(125 nA + .05%)
	100 µA	800 pA	±(25 nA + .05%)
	10 µA	80 pA	±(10 nA + .1%)
	1 µA	8 pA	±(5 nA + .1%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±30 V	1X	±30 V	900 µV	±(2.5 mV + .025%)
	±30 V	10X	±3 V	90 µV	±(2 mV +. 025%)
	±30 V	100X	±300 mV	9 µV	±(2 mV + .025%)
Current	±1 A	1X	± 1 A	30 µA	±(125 µA + .05%)
	±1 A	10X	±100 mA	3 µA	±(60 µA + .05%)
	±1 A	100X	±10 mA	300 nA	±(60 µA + .05%)
	±100 mA	1X	±100 mA	3 µA	±(12.5 µA + .05%)
	±100 mA	10X	±10 mA	300 nA	±(6 µA + .05%)
	±100 mA	100X	±1 mA	30 nA	±(6 µA + .05%)
	±10 mA	1X	±10 mA	300 nA	±(1.25 µA + .05%)
	±10 mA	10X	±1 mA	30 nA	±(600 nA + .05%)
	±10 mA	100X	±100 µA	3 nA	±(600 nA + .05%)
	±1 mA	1X	±1 mA	30 nA	±(125 nA + .05%)
	±1 mA	10X	±100 µA	3 nA	±(60 nA + .05%)
	±1 mA	100X	±10 µA	300 pA	±(60 nA + .1%)
	±100 µA	1X	±100 µA	3 nA	±(20 nA + .05%)
	±100 µA	10X	±10 µA	300 pA	±(12 nA + .1%)
	±100 µA	100X	±1 µA	30 pA	±(12 nA + .1%)
	±10 µA	1X	±10 µA	300 pA	±(12 nA + .05%)
	±10 µA	10X	±1 µA	30 pA	±(8 nA + .1%)
	±10 µA	100X	±100 nA	3 pA	±(8 nA + .1%)
	±1 µA	1X	±1 µA	30 pA	±(8 nA + .05%)*
	±1 µA	10X	±100 nA	3 pA	±(4 nA + .1%)

* Accuracy improvement with auto-zero tare: ±1 µA Range: (±1.5nA + .05%)

2.13.6.3 100 V Range

Force	Range	Resolution	Accuracy
Voltage	100 V	800 µV	±(12.5 mV + .025%)
Current	500 mA	8.0 µA	±(125 µA + .05%)
	100 mA	800 nA	±(12.5 µA + .05%)
	10 mA	80 nA	±(1.25 µA + .05%)
	1 mA	8 nA	±(125 nA + .05%)
	100 µA	800 pA	±(60 nA + .05%)
	10 µA	80 pA	±(30 nA + .1%)
	1 µA	8 pA	±(20 nA + .1%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±100 V	1X	±100 V	3 mV	±(12.5 mV + .025%)
	±100 V	10X	±10 V	300 µV	±(6 mV + .025%)
	±100 V	100X	±1 V	30 µV	±(6 mV + .025%)
Current	±500 mA	1X	±500 mA	15 µA	±(125 µA + .05%)
	±500 mA	10X	±50 mA	1.5 µA	±(60 µA + .05%)
	±500 mA	100X	±5 mA	150 nA	±(60 µA + .05%)
	±100 mA	1X	±100 mA	3 µA	±(12.5 µA + .05%)
	±100 mA	10X	±10 mA	300 nA	±(6 µA + .05%)
	±100 mA	100X	±1 mA	30 nA	±(6 µA + .05%)
	±10 mA	1X	±10 mA	300 nA	±(1.25 µA + .05%)
	±10 mA	10X	±1 mA	30 nA	±(600 nA + .05%)
	±10 mA	100X	±100 µA	3 nA	±(600 nA + .05%)
	±1 mA	1X	±1 mA	30 nA	±(125 nA + .05%)
	±1 mA	10X	±100 µA	3 nA	±(60 nA + .05%)
	±1 mA	100X	±10 µA	300 pA	±(60 nA + .1%)
	±100 µA	1X	±100 µA	3 nA	±(50 nA + .05%)
	±100 µA	10X	±10 µA	300 pA	±(25 nA + .1%)
	±100 µA	100X	±1 µA	30 pA	±(25 nA + .1%)
	±10 µA	1X	±10 µA	300 pA	±(25 nA + .05%)
	±10 µA	10X	±1 µA	30 pA	±(12.5 nA + .1%)
	±1 µA	1X	±1 µA	30 pA	±(12.5 nA + .05%)

2.13.6.4 Volt Meter Mode

Measure	Range	Resolution	Accuracy	Bandwidth	Input R
Voltage	1000 V	30 mV	±(125 mV + 0.05%)	25 kHz	5 Meg
	100 V	3 mV	±(12.5 mV + .025%)	25 kHz	> 200 Meg
	30 V	900 µV	±(4 mV + .025%)	50 kHz	> 200 Meg
	10 V	300 µV	±(1.2 mV + .025%)	50 kHz	> 200 Meg

2.13.6.5 Driver Mode

Force	Range	Resolution	Typical Distortion @ Frequency
High Speed (16-bit 25 MSPS)	±30 V	900 µV	< -75 dB @ 100 kHz
	±10 V	300 µV	< -80 dB @ 100 kHz
High Resolution (18-bit 350 kSPS)	±30 V	225 µV	< -75 dB @ 10 kHz
	±10 V	80 µV	< -80 dB @ 10 kHz
Audio (18-bit 350 kSPS Differential)	10 V (pk - pk)	50 µV	Better than -96 dB @ 1 kHz
	1 V (pk - pk)	5 µV	Better than -96 dB @ 1 kHz

2.13.7 User Interface

2.13.7.1 Software

All functions of the SPU-100 are programmed using utility function calls from your C test program. These utilities and their syntax and usage are described in The ETS Software Help File.

NOTE: The Master Clock portion of the TCB must be set up and started in order to clock the AWG and/or the digitizer. Please refer to the Test Head Control Board section (Section 2.17 on page 2-113) or the SPU-100 and MCB utility descriptions (in The ETS Software Help File) for further information.

2.13.7.2 Hardware

There are two SPs per card and they are housed in the Floating Resource Card Cage (FR Cage).

SPU-100s may be connected to the Application Board as shown (this example shows SPU-100 connections when using SPU-100 0/1):

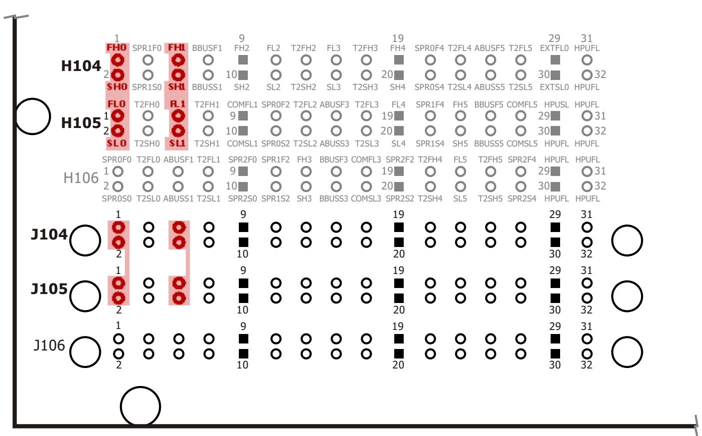


Figure 2-41 – ETS-88™ SPU-100 Application Board Connections

2.13.8 Safety Notes

For safety purposes, the output is inhibited if the DUT board interlock is not satisfied. The user must take great care to assure that potentially hazardous voltages are not accessible to operators or any other personnel who may come in contact with the test apparatus. All electrical surfaces that can be energized to a potential above ±48 VDC must be adequately covered to eliminate possible electrical contact with humans.

SAFETY NOTICE:
This resource is designed to operate in a test system environment that is designed with the following safety features:

- Access to this resource requires the use of a tool to remove a cover
- Access to Input/Output connections to this resource is blocked by mechanical barriers
- An electrical interlock circuit inhibits the output of this resource

CAUTION: RISK OF SHOCK.

****

Hazardous Voltages Present! This resource generates hazardous voltages and must be operated in a properly designed enclosure with safety features in place. Always turn power off prior to handling this resource. Eagle Test Systems, Inc. accepts no responsibility for harm from handling or misuse of this resource.

****

Use high-voltage-insulated wiring when wiring connections from a SPU-100 to points on the application boards. Teflon- and silicone-insulated wire offer dielectric strengths in hundreds to thousands of volts.

2.14 Smart Pin Unit 100 V / 12 A (SPU-112)

2.14.1 Features

- Fully independent dual channel SmartPin™ resource
- Each channel fully floating and stackable
  - 8 A (using two paths, ±1000 VDC from ground max.)
  - 12 A (using three paths, ±200 VDC from ground max.)
- Three voltage ranges, eight current ranges
  - ±100 V @±500mA, 100mA, 10mA, 1mA, 100µA, 10µA, 1µA
  - ±30 V @±12A, 1A, 100mA, 10mA, 1mA, 100µA, 10µA, 1µA
  - ±10 V @±12A, 2A, 200mA, 20mA, 2mA, 200µA, 20µA, 2µA
- Additional 10X and 100X measure gain settings are available in each voltage and current range
- Two 500 KSPS digitizers to capture both voltage and current simultaneously
- Independent high/low programmable voltage/current clamps with alarms
- Kelvin error detect, over temperature, droop, interlock, and measurement full-scale alarms
- Fully backward-compatible with the SPU-100
- Driver/Signal Generator Mode
  - High speed AWG (16 Bit; 25 MSPS; up to 1 MHz Sine)
  - High Resolution AWG (18 Bit; 400 KSPS; up to 50 kHz Sine)
  - Audio Mode (18 Bit; 600 KSPS; better than -96 dB THD @1 kHz)
- Volt meter mode: ±(1000 V, 100 V, 30 V, 10 V)
- Real-time measurement accumulator for instant results averaging
- Change V/I settings under AWG pattern control
  - FV/FI selectable on-the-fly
  - Current ranges selectable on-the-fly
  - ADC gain and filtering selectable on-the-fly
  - ADC sample clock gate on/off on-the-fly
- Results accumulator supports up to 32 sample sets per pattern
- Interlocks provided for operator safety
- Hardware and software designed for multisite applications
- Digitizer self-trigger mode for asynchronous signals (includes pre-trigger sampling)
- Simultaneous voltage and current capture memory: 4k samples
- On-board DSP with robust function library per channel
- Pedestal Mode for precision low voltage DC measurements (<5 V)

NOTE: Pulsed 12 A range requires an optional booster board (each booster board serves up to two SPU-112 boards).

2.14.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]
- Eagle Vision MST [Rev. 2011.1.1002.17 and up]

2.14.3 Theory of Operation

The Smart Pin Unit 100 V / 12 A (SPU-112) is a single slot, dual channel, ±100 V Smart Pin™ with eight current ranges. The SPU-112 spans a wide range of voltage and current combinations, making it an extremely versatile instrument. The Smart Pin architecture incorporates an AWG and a digitizer within a conventional four quadrant V/I. This resource includes all the standard capabilities of full-featured V/Is (programmable clamps, Kelvin detect, alarms, etc.) plus the advanced characteristics that Smart Pin users have come to expect.

The V/I is stable with almost any combination of inductive and/or capacitive loads. The programmable clamps are extremely reliable, crossing over from voltage to current or vice versa with minimal over-shoot or instability. Bandwidth and settling time are optimized for excellent measurement speed. Analog switches are used extensively for excellent reliability and switching speeds.

The 18-bit AWG makes it possible to generate arbitrary voltage and current based signals that are synchronized to all other digital and analog resources in the test system. This enables test engineers to easily create dynamic test conditions that have previously been impossible in ATE without custom application circuitry.

The three special driver modes (18-bit 400 KSPS, 16-bit 25 MSPS and a special audio mode) support waveform generation by providing direct access to the buffered AWG output. This proves extremely useful for general purpose applications requiring AC signals in and above the audio range. These signals may be synchronized to the other analog and digital resources of the system.

In audio mode, a specialized differential line driver is switched into the output force lines to provide a high quality audio signal for THD and noise testing. In this mode, both differential and single-ended configurations are supported.

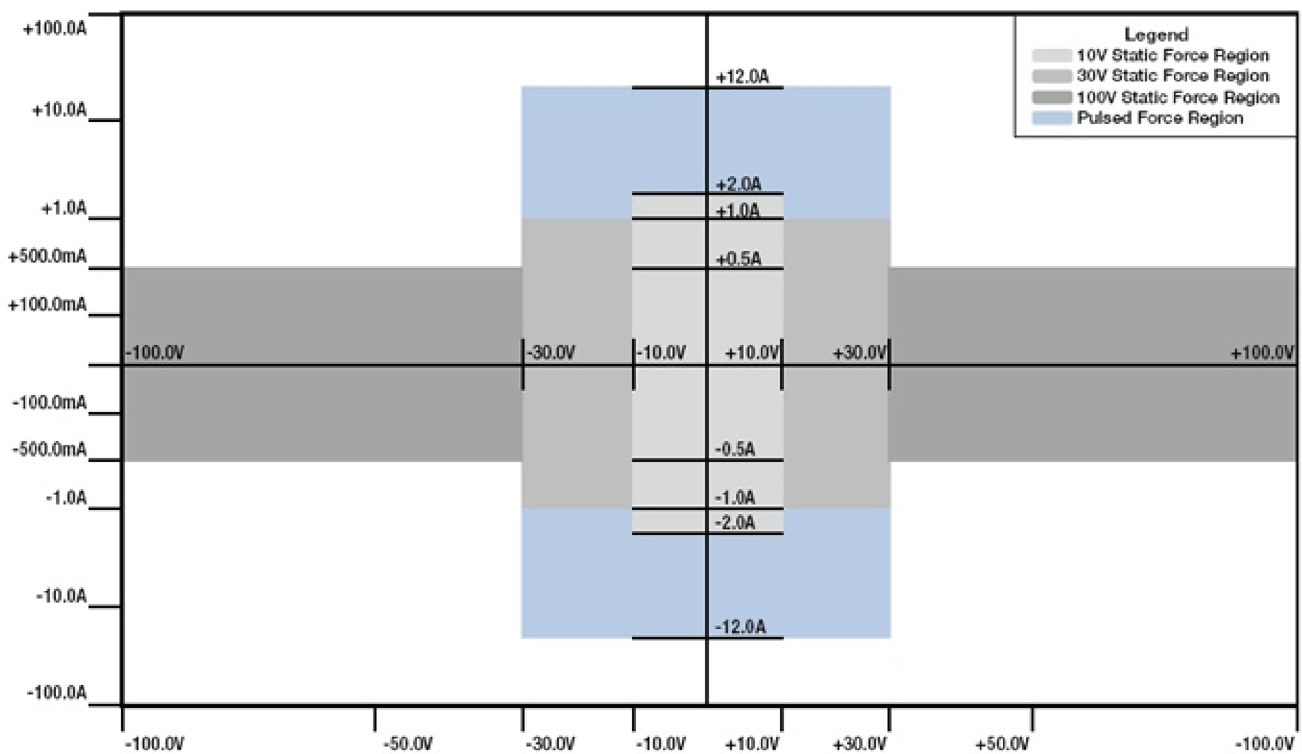


Figure 2-42 – SPU-112 V/I Quadrant Diagram

The high current output stage accommodates pulsed currents on four ranges up to ±12 A. For currents greater than ±2 A, the optional SPU-112 Booster Board can provide ±12 A capability to two SPU-112 boards simultaneously. The high current output stage draws its power from a capacitor bank that is charged continuously. Using pattern-based programming techniques, the SPU-112 can output precise current and/or voltage pulses of any amplitude and duration up to the limits of the selected range. This feature can greatly reduce test time and avoid excessive die heating.

The integrated dual digitizers make parallel measurements a reality. As explained in the following section, the ability to digitize both voltage and current simultaneously simplifies and speeds many applications.

Capturing infrequently occurring asynchronous signals can be a difficult task for a digitizer. The SPU-112 includes a self-trigger mode that greatly simplifies the job. With this mode, it is possible to trigger on the incoming signal, based on a programmable trigger threshold setting. This mode includes the ability to specify a certain number of pre-trigger samples, and a certain number of post-trigger samples. In this way, it is possible to capture and use the entire waveform of interest. The self-trigger mode allows the capture of waveforms with high sample rates, without using great amounts of capture memory.

The pedestal measurement mode of the SPU-112 provides 16-bit resolution in a ±1.1 V measurement range, which can be offset by the following voltages: 0 V, 1 V, 3 V, 5 V. The pedestal measurement mode is operational during the normal V/I forcing mode. This feature provides much higher voltage measurement accuracy for testing voltage regulators, power FETs, and the like.

2.14.4 Applications

The SPU-112 is useful for advanced measurement applications as well as for use as a general purpose V/I. The three voltage ranges and multiple current ranges make it possible to address a wide range of test applications. For static force/measure applications, the real-time measurement accumulator radically reduces measurement times with built-in hardware results averaging. Combining these features with 18-bit force and 16-bit measurement resolution creates a performance standard that is unrivaled in the industry.

Smart Pins are valuable for testing various devices because they make it possible to initiate a wide range of test conditions in rapid sequence. The use of a pattern-based V/I makes it possible to change force conditions on-the-fly. The pattern RAM contains the V/I force values and a number of synchronized control bits that make it possible to change the operating state of the V/I on-the-fly and to enable/disable the on-board digitizer to capture the desired test results at selectable pattern locations. The digitizer is also capable of on-the-fly averaging, such that the average value of each sample set is stored in RAM along with each set of sample values. For pattern-based DC tests, this greatly reduces data transfer times during post processing.

With this type of hardware available, it is possible to string together many test conditions, while simultaneously storing the measured results. After the pattern runs, the system controller will typically read back the results for test limit comparison purposes. Due to the real-time averaging of sample sets (Results Accumulator), very little time is required to read the results from the hardware. If a more complex evaluation method is required, the sampled data can be transferred to controller memory for further mathematical evaluation.

Threshold searches are a common application problem in the industry. Often, test engineers are required to use either successive approximation techniques or design specialized application circuitry to speed up these normally time consuming tests. The SPU-112 makes it possible to locate current-based threshold points and the associated threshold hysteresis levels with a single up/down ramp pattern. This greatly reduces test time and improves measurement repeatability. Because the ramp signal is AWG driven, the resolution and speed of the ramp can easily be optimized for the best tradeoff in test speed versus measurement accuracy and repeatability.

These Smart Pin based test techniques also make it possible to measure parameters such as dynamic PSRR (power supply rejection ratio) or to string together several forcing/loading conditions that are typical in regulator testing. This AWG pattern-based test approach is quickly becoming the method of choice for regulator manufacturers throughout the industry.

For static DC force/measure procedures, which are still heavily used in many applications, the real-time measurement accumulator provides instant averaging of measured results. This means that the samples are summed mathematically in real-time. When sampling is complete, the answer may be read directly from the resource, avoiding any further data transfers; however, the full data is still available for plotting or other engineering purposes, offering the best of both worlds.

Key Uses:

- High-Side and Low-Side Switches
- Battery Management
- Discrete Transistors
- Automotive
- Power Management
- Relay Drivers
- LED Drivers
- General Purpose

2.14.5 Specifications

The tables on the following pages provide the detailed voltage and current force and measure specifications for the SPU-112.

2.14.5.1 10 V Range

Force	Range	Resolution	Accuracy
Voltage	±10 V	80 μV	±(0.8 mV + .025%)
Current	12 A*	90 μA	±(2 mA + .2%)
	2 A	16 μA	±(250 μA + .05%)
	200 mA	1.6 μA	±(25 μA + .05%)
	20 mA	160 nA	±(2.5 μA + .05%)
	2 mA	16 nA	±(250 nA + .05%)
	200 μA	1.6 nA	±(20 nA + .05%)
	20 μA	160 pA	±(7 nA + .1%)
	2 μA	16 pA	±(5 nA + .1%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±10 V	1X	±10 V	300 μV	±(800 μV + .025%)
	±10 V	10X	±1 V	30 μV	±(600 μV + .025%)
	±10 V	100X	±100 mV	3 μV	±(600 μV + .025%)
Current	±12 A*	1X	±12 A	300 μA	±(2 mA + .1%)
	±12 A*	10X	±1.2 A	30 μA	±(1 mA + .1%)
	±12 A*	100X	±120 mA	3 μA	±(500 μA + .1%)
	±2 A	1X	±2 A	60 μA	±(250 μA + .05%)
	±2 A	10X	±200 mA	6 μA	±(125 μA + .05%)
	±2 A	100X	±20 mA	600 nA	±(125 μA + .05%)
	±200 mA	1X	±200 mA	6 μA	±(25 μA + .05%)
	±200 mA	10X	±20 mA	600 nA	±(12 μA + .05%)
	±200 mA	100X	± 2 mA	60 nA	±(12 μA + .05%)
	±20 mA	1X	±20 mA	600 nA	±(2.5 μA + .05%)
	±20 mA	10X	±2 mA	60 nA	±(1.2 μA + .05%)
	±20 mA	100X	±200 μA	6 nA	±(1.2 μA + .05%)
	±2 mA	1X	± 2 mA	60 nA	±(250 nA + .05%)
	±2 mA	10X	±200 μA	6 nA	±(125 nA + .05%)
	±2 mA	100X	±20 μA	600 pA	±(125 nA + .1%)
	±200 μA	1X	±200 μA	6 nA	±(25 nA + .05%)**
	±200 μA	10X	±20 μA	600 pA	±(15 nA + .1%)
	±200 μA	100X	±2 μA	60 pA	±(15 nA + .1%)
	±20 μA	1X	±20 μA	600 pA	±(15 nA + .05%)**
	±20 μA	10X	±2 μA	60 pA	±(10 nA + .1%)
	±20 μA	100X	±200 nA	6 pA	±(10 nA + .1%)
	±2 μA	1X	±2 μA	60pA	±(12 nA + .05%)**
	±2 μA	10X	±200 nA	6 pA	±(6 nA + .1%)
	±2 μV	100X	±20 nA	600 fA	±(6 nA + .1%)

* 12 A range is pulsed only, and requires optional SPU-112 Booster board
** Accuracy improvement with auto-zero tare: ± 2 nA Range = ±(2 nA + .05%)

2.14.5.2 30 V Range

Force	Range	Resolution	Accuracy
Voltage	±30 V	225 µV	±(2.5 mV + .025%)
Current	12 A*	90 µA	±(2 mA + .2%)
	1 A	8 µA	±(125 µA + .05%)
	100 mA	800 nA	±(12.5 µA + .05%)
	10 mA	80 nA	±(1.25 µA + .05%)
	1 mA	8 nA	±(125 nA + .05%)
	100 µA	800 pA	±(25 nA + .05%)
	10 µA	80 pA	±(10 nA + .1%)
	1 µA	8 pA	±(5 nA + .1%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±30 V	1X	±30 V	900 µV	±(2.5 mV + .025%)
	±30 V	10X	±3 V	90 µV	±(2 mV +. 025%)
	±30 V	100X	±300 mV	9 µV	±(2 mV + .025%)
Current	±12 A*	1X	±12 A	300 µA	±(2 mA + .1%)
	±12 A*	10X	±1.2 A	30 µA	±(1 mA + .1%)
	±12 A*	100X	±120 mA	3 µA	±(500 µA + .1%)
	±1 A	1X	± 1 A	30 µA	±(125 µA + .05%)
	±1 A	10X	±100 mA	3 µA	±(60 µA + .05%)
	±1 A	100X	±10 mA	300 nA	±(60 µA + .05%)
	±100 mA	1X	±100 mA	3 µA	±(12.5 µA + .05%)
	±100 mA	10X	±10 mA	300 nA	±(6 µA + .05%)
	±100 mA	100X	±1 mA	30 nA	±(6 µA + .05%)
	±10 mA	1X	±10 mA	300 nA	±(1.25 µA + .05%)
	±10 mA	10X	±1 mA	30 nA	±(600 nA + .05%)
	±10 mA	100X	±100 µA	3 nA	±(600 nA + .05%)
	±1 mA	1X	±1 mA	30 nA	±(125 nA + .05%)
	±1 mA	10X	±100 µA	3 nA	±(60 nA + .05%)
	±1 mA	100X	±10 µA	300 pA	±(60 nA + .1%)
	±100 µA	1X	±100 µA	3 nA	±(20 nA + .05%)
	±100 µA	10X	±10 µA	300 pA	±(12 nA + .1%)
	±100 µA	100X	±1 µA	30 pA	±(12 nA + .1%)
	±10 µA	1X	±10 µA	300 pA	±(12 nA + .05%)
	±10 µA	10X	±1 µA	30 pA	±(8 nA + .1%)
	±10 µA	100X	±100 nA	3 pA	±(8 nA + .1%)
	±1 µA	1X	±1 µA	30 pA	±(8 nA + .05%)*
	±1 µA	10X	±100 nA	3 pA	±(4 nA + .1%)*

* 12 A range is pulsed only, and requires optional SPU-112 Booster board
* Accuracy improvement with auto-zero tare: ±1 µA Range = ±(1.5 nA + .05%)

2.14.5.3 100 V Range

Force	Range	Resolution	Accuracy
Voltage	±100 V	800 µV	±(12.5 mV + .025%)
Current	500 mA	8.0 µA	±(125 µA + .05%)
	100 mA	800 nA	±(12.5 µA + .05%)
	10 mA	80 nA	±(1.25 µA + .05%)
	1 mA	8 nA	±(125 nA + .05%)
	100 µA	800 pA	±(60 nA + .05%)
	10 µA	80 pA	±(30 nA + .1%)
	1 µA	8 pA	±(20 nA + .1%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±100 V	1X	±100 V	3 mV	±(12.5 mV + .025%)
	±100 V	10X	±10 V	300 µV	±(6 mV + .025%)
	±100 V	100X	±1 V	30 µV	±(6 mV + .025%)
Current	±500 mA	1X	±500 mA	15 µA	±(125 µA + .05%)
	±500 mA	10X	±50 mA	1.5 µA	±(60 µA + .05%)
	±500 mA	100X	±5 mA	150 nA	±(60 µA + .05%)
	±100 mA	1X	±100 mA	3 µA	±(12.5 µA + .05%)
	±100 mA	10X	±10 mA	300 nA	±(60 µA + .05%)
	±100 mA	100X	±1 mA	30 nA	±(6 µA + .05%)
	±10 mA	1X	±10 mA	300 nA	±(1.25 µA + .05%)
	±10 mA	10X	±1 mA	30 nA	±(600 nA + .05%)
	±10 mA	100X	±100 µA	3 nA	±(600 nA + .05%)
	±1 mA	1X	±1 mA	30 nA	±(125 nA + .05%)
	±1 mA	10X	±100 µA	3 nA	±(60 nA + .05%)
	±1 mA	100X	±10 µA	300 pA	±(60 nA + .1%)
	±100 µA	1X	±100 µA	3 nA	±(50 nA + .05%)
	±100 µA	10X	±10 µA	300 pA	±(25 nA + .1%)
	±100 µA	100X	±1 µA	30 pA	±(25 nA + .1%)
	±10 µA	1X	±10 µA	300 pA	±(25 nA + .05%)
	±10 µA	10X	±1 µA	30 pA	±(12.5 nA + .1%)
	±1 µA	1X	±1 µA	30 pA	±(12.5 nA + .05%)

2.14.5.4 Volt Meter Mode

Measure	Range	Resolution	Accuracy	Bandwidth	Input R
Voltage	1000 V	30 mV	±(125 mV + 0.05%)	25 kHz	5 Meg
	100 V	3 mV	±(12.5 mV + .025%)	25 kHz	> 200 Meg
	30 V	900 μV	±(4 mV + .025%)	50 kHz	> 200 Meg
	10 V	300 μV	±(1.2 mV + .025%)	50 kHz	> 200 Meg

2.14.5.5 Driver Mode

Force	Range	Resolution	Typical Distortion @ Frequency
High Speed (16-bit 25 MSPS)	±30 V	900 μV	< -75 dB @ 100 kHz
	±10 V	300 μV	< -80 dB @ 100 kHz
High Resolution (18-bit 400 KSPS)	±30 V	225 μV	< -75 dB @ 10 kHz
	±10 V	80 μV	< -80 dB @ 10 kHz
Audio (18-bit 600 KSPS Differential)	10 V (pk - pk)	50 μV	Better than -96 dB @ 1 kHz
	1 V (pk - pk)	5 μV	Better than -96 dB @ 1 kHz

2.14.5.6 Pedestal Voltage Measurement Mode (10 V Range Only)

Measure	Pedestal Voltages	Effective Range	Resolution (16-Bit)	Accuracy
Voltage	0 V	-1.1 V to +1.1 V	35 μV	±(250 μV + .01% of Offset)
	1 V	-0.1 V to +2.1 V	35 μV	±(250 μV + .01% of Offset)
	2 V	-1.9 V to +4.1 V	35 μV	±(250 μV + .01% of Offset)
	5 V	-3.9 V to +6.1 V	35 μV	±(250 μV + .01% of Offset)

2.14.5.7 Pulse Width Curves

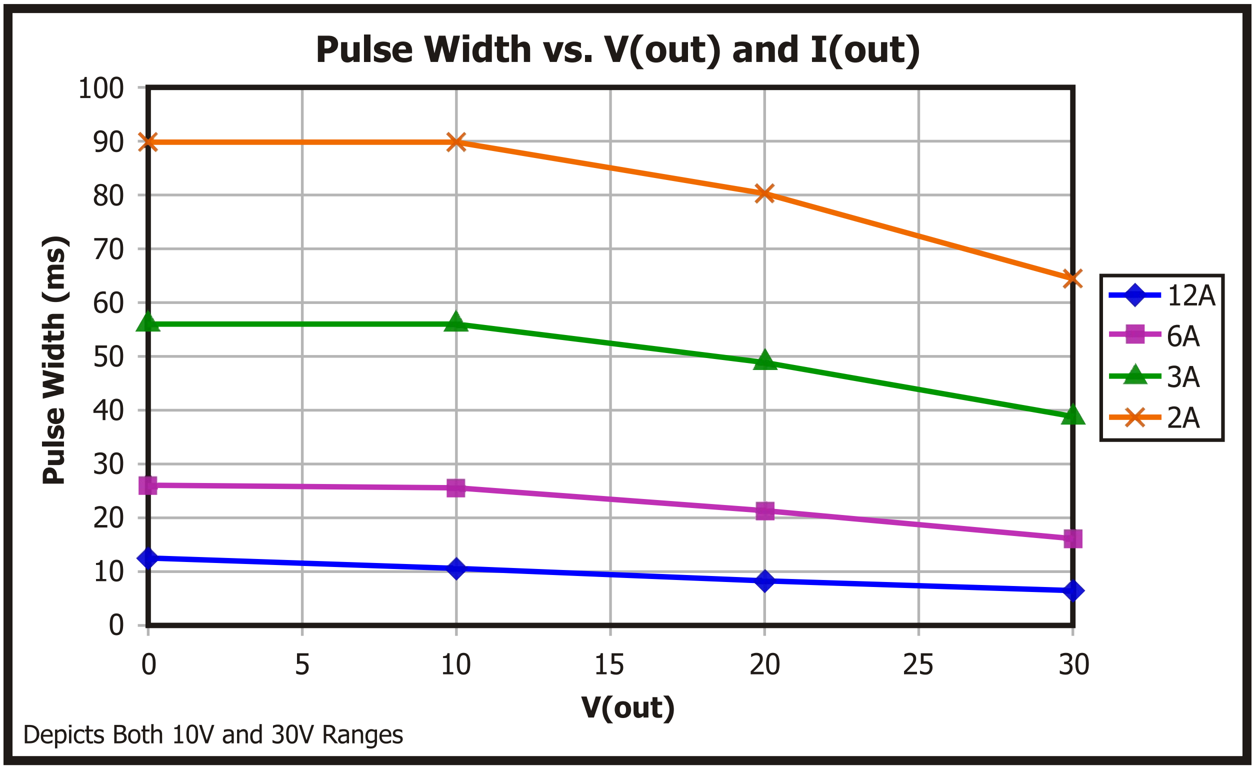


Figure 2-43 – SPU-112 Pulse Width Curves

2.14.6 Block Diagram

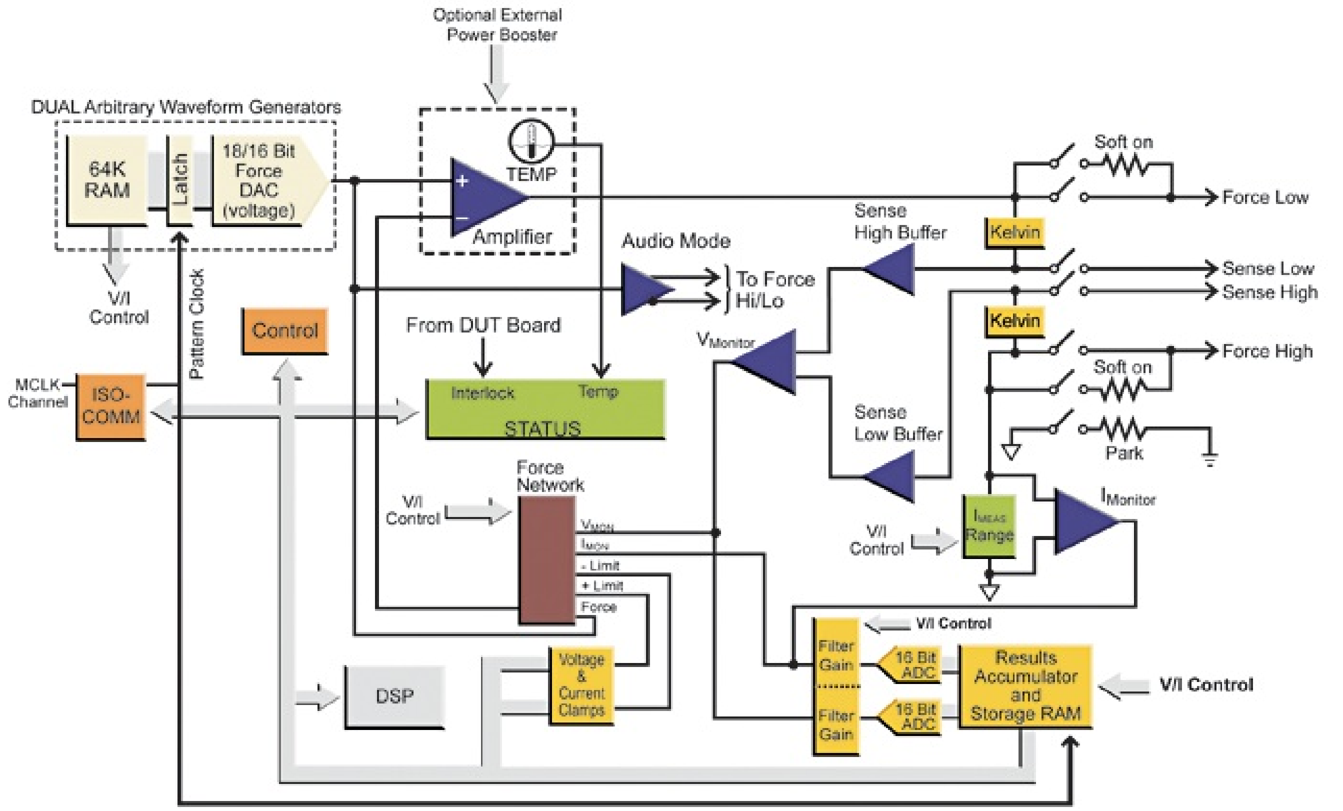


Figure 2-44 – SPU-112 Block Diagram

2.14.7 User Interface

2.14.7.1 Software

All functions of the SPU-112 are programmed using utility function calls from your C test program. These utilities and their syntax and usage are described in The ETS Software Help File.

NOTE: The Master Clock section of the TCB must be set up and started in order to clock the AWG and/or the digitizer. Please refer to the Test Head Control Board section (Section 2.17 on page 2-113) or the SPU-112 and MCB utility descriptions (in The ETS Software Help File) for further information.

2.14.7.2 Hardware

There are two Smart Pins per card, and they are housed in the Floating Resource Card Cage (FR Cage).

Low Current Path

The low current output is designated for DC current less than 2 A, and pulsed current less than 4 A. This connection utilizes a single force high / low pair connection to the DUT.

SPU-112s may be connected to the Application Board as shown in Figure 2-45 (this example shows SPU-112 low current path connections when using SPU-112 0/1):

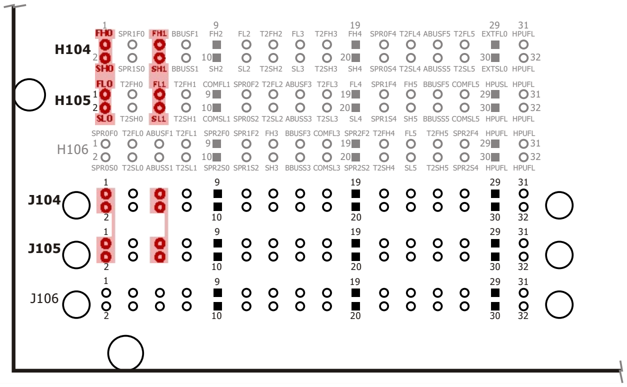


Figure 2-45 – ETS-88™ SPU-112 Application Board Low Current Connections

High Current Path

The second type of output from the SPU-112 is for high current capability. The low current path, plus two additional paths, are utilized for all values up to 12 A.

The SPU-112's high current path may be connected to the application board as shown in Figure 2-46 below (this example shows the SPU-112 high current path connections when using SPU-112 0/1). Please see Chapter 4 for specific connections based on the SPU-112's slot position in the Floating Resource Card Cage.

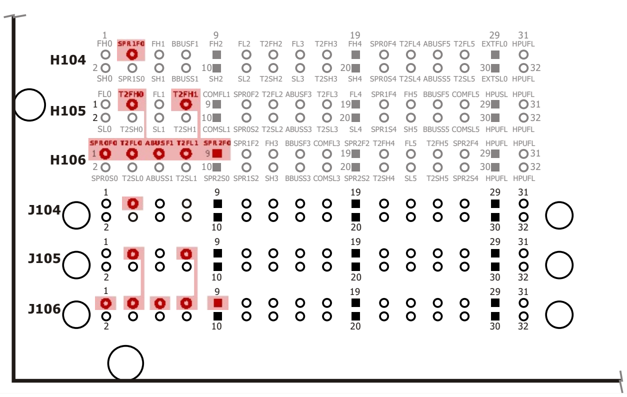


Figure 2-46 – ETS-88™ SPU-112 Application Board High Current Connections

2.14.8 DIB Design Guidelines for Hardware Protection

The DUT Interface Board (DIB) design configuration outlined below must be implemented to prevent potential damage to the backplane when forcing currents >4 A.

Because standard (non-high-current) single trace routing in the system is rated at ±4 A, and the SPU-112 is capable of forcing pulsed currents up to ±12 A, when designing DIBs for use with SPU-112 resources in applications, the three lines from each channel of the SPU-112 must be connected to prepare for high current (>4 A) tests (refer to Figure 1 below). This design is recommended whether or not users have an SPU-112 Booster Board required for the SPU-112's ±12 A capability.

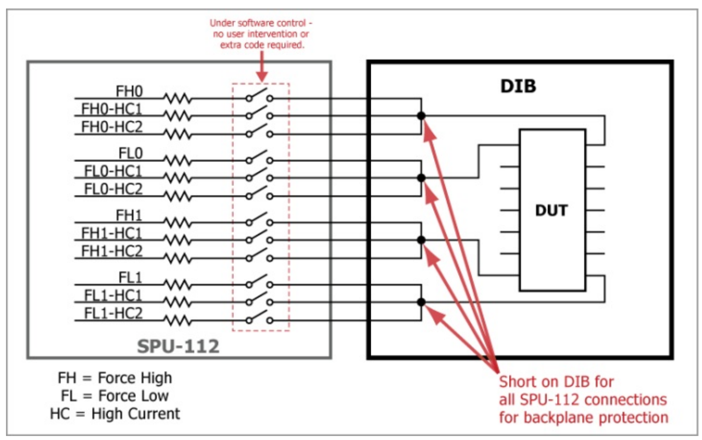


Figure 2-47 – Recommended DIB Connections for SPU-112 Resources

2.14.9 Safety Notes

For safety purposes, the output is inhibited if the application board interlock is not satisfied. The user must take great care to assure that potentially hazardous voltages are not accessible to operators or any other personnel who may come in contact with the test apparatus. All electrical surfaces that can be energized to a potential above ±48 VDC must be adequately covered to eliminate possible electrical contact with humans.

WARNING: RISK OF SHOCK.

****

Hazardous Voltages Present. Due to the nature of this resource and its use, the user must assume the burden of protecting operators and other personnel from possible shock hazard. Eagle Test Systems, Inc. accepts no responsibility for any possible harm this resource may cause to personnel.

****

NOTE: The SPU-112 Booster Board features two green LEDs per channel that indicate the charge state of the positive and negative banks of capacitors (see Figure 2-48), and two red LEDs per channel that light up when the banks are being discharged (see Figure 2-49). These LEDs indicate when hazardous voltage is present. Do not handle the booster board until ALL of these LEDs have turned off!

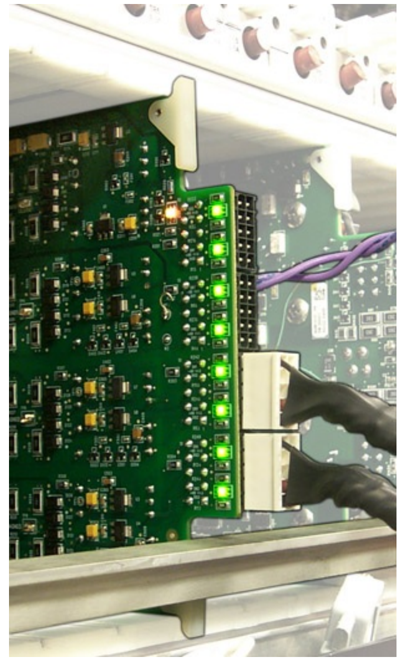


Figure 2-48 – SPU-112 Booster Capacitance Discharge LEDs (Green – System Powered)

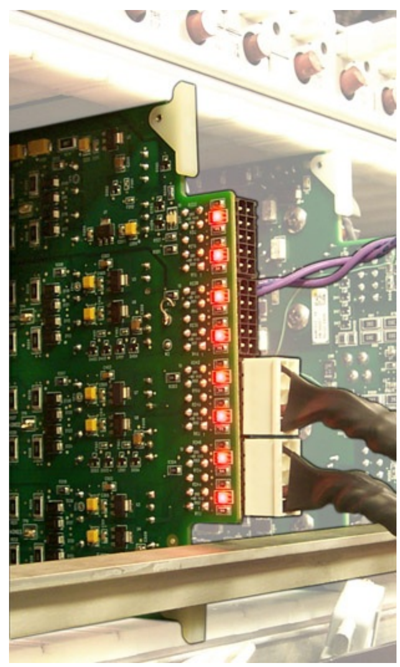


Figure 2-49 – SPU-112 Booster Capacitance Discharge LEDs (Red – Discharging Capacitance)

2.15 Smart Pin Unit 250 V / 100 mA (SPU-250)

2.15.1 Caution to Users

SAFETY NOTICE:

This resource is designed to operate in a test system environment that is designed with the following safety features:

- Access to this resource requires the use of a tool to remove a cover
- Access to Input/Output connections to this resource is blocked by mechanical barriers
- An electrical interlock circuit inhibits the output of this resource

CAUTION: RISK OF SHOCK.

****

Hazardous Voltages Present! This resource generates hazardous voltages and must be operated in a properly designed enclosure with safety features in place. Always turn power off prior to handling this resource. Eagle Test Systems, Inc. accepts no responsibility for harm from handling or misuse of this resource.

****

Use high-voltage-insulated wiring when wiring connections from a SPU-250 to points on the application boards. Teflon- and silicone-insulated wire offer dielectric strengths in hundreds to thousands of volts.

2.15.2 Features

- Fully independent dual channel SmartPin™ resource
- Each channel fully floating and stackable (±1000 VDC from ground max.)
- One (1) force voltage range: ±250 V
- Three (3) measure voltage ranges: ±(250 V; 100 V; 10 V)
- Five (5) force current ranges: ±(100 mA, 10 mA, 1 mA, 100 µA, 10 µA)
- Six (6) measure current ranges: ±(100 mA, 10 mA, 1 mA, 100 µA, 10 µA, 1 µA)
- Real-time measurement accumulator for instant results averaging
- Two (2) digitizers to capture both voltage and current waveforms simultaneously
- Independent high/low programmable voltage/current clamps with alarms
- Measurement alarms are available through software when the resources range limitations are met
- The resource is designed for operator safety by disabling the resource if the application board interlocks are not satisfied
- Hardware and software designed for multisite applications

Each of the SPU-250's two independently programmable channels contains the following components/features:

2.15.2.1 Waveform Digitizer

The 4K RAM Waveform Digitizer provides to the user a powerful measurement tool. The clock coming into the SPU-250 may be divided down to a sampling rate that is between 1 Hz to 100 kHz. This separate clock divider for the digitizer allows the user to measure at one sampling rate and force date with the AWG at a higher frequency.

The digitizer may run concurrently with any forcing function, such as the AWG described above or a DC voltage/current.

See the SPU-250 Specifications for SPU-250 Digitizer capabilities.

2.15.2.2 Arbitrary Waveform Generator (AWG)

The 16 bit AWG provides tremendous forcing capability to this instrument by allowing the user to reproduce any waveform from a sine wave to simulating the output of a digital driver into the DUT. The AWG has a maximum clock rate of 100 kHz with 64K of RAM pattern depth to provide the user the ability to force either voltage or current (depending on the mode selected). The software allows the user to load concatenated patterns into the AWG and then run these patterns individually and/or nonsequentially.

See the SPU-250 Specifications on the following page for SPU-250 AWG capabilities.

2.15.2.3 Voltage and Current Clamps

Each voltage and current range has programmable upper and lower clamps. These clamps essentially define a 'window' of allowable voltage or current. Current clamps are programmed while in the FV mode and voltage clamps are programmed while in the FI mode.

NOTE: Remember that choosing a lower current clamp of 0 mA keeps the resource from sinking current while discharging a load at the end of testing.

2.15.3 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]

2.15.4 Modes

The SPU-250 is a dual-mode resource. You can use the SPU-250 to force voltage (±250 V in power supply mode) and measure current, or as a full V/I with programmable clamps. It is a two-wire resource since it is designed for low current applications.

2.15.4.1 Driver Mode

As stated previously, this is a two-wire resource in all modes and ranges. There are two significant differences in the available DRV mode on the SPU-250:

- The programmable clamps are disabled. Therefore, the current passing through the resource would be limited by the current range selected.
- Faster slew rate capability.

2.15.4.2 V/I Mode

The full V/I mode can force and measure either voltage or current. This mode has the slower slew rate of the two modes (1.5 V/µsec max) to maintain stability with the loop amplifier. The V/I mode also has two 16 Bit limit DACs, which serve as voltage or current clamps depending on whether you're using VFORCE or IFORCE. You set the positive and negative limits, and these clamp limits are independent of each other. The limiting action automatically switches force modes at the limit value.

If using VFORCE with a current limit, the SPU-250 switches to IFORCE once the current reaches the limit value. Because there is no remote sense, a voltage correction circuit corrects the voltage drop of the current measurement circuit to reduce the voltage error.

2.15.5 Block Diagram

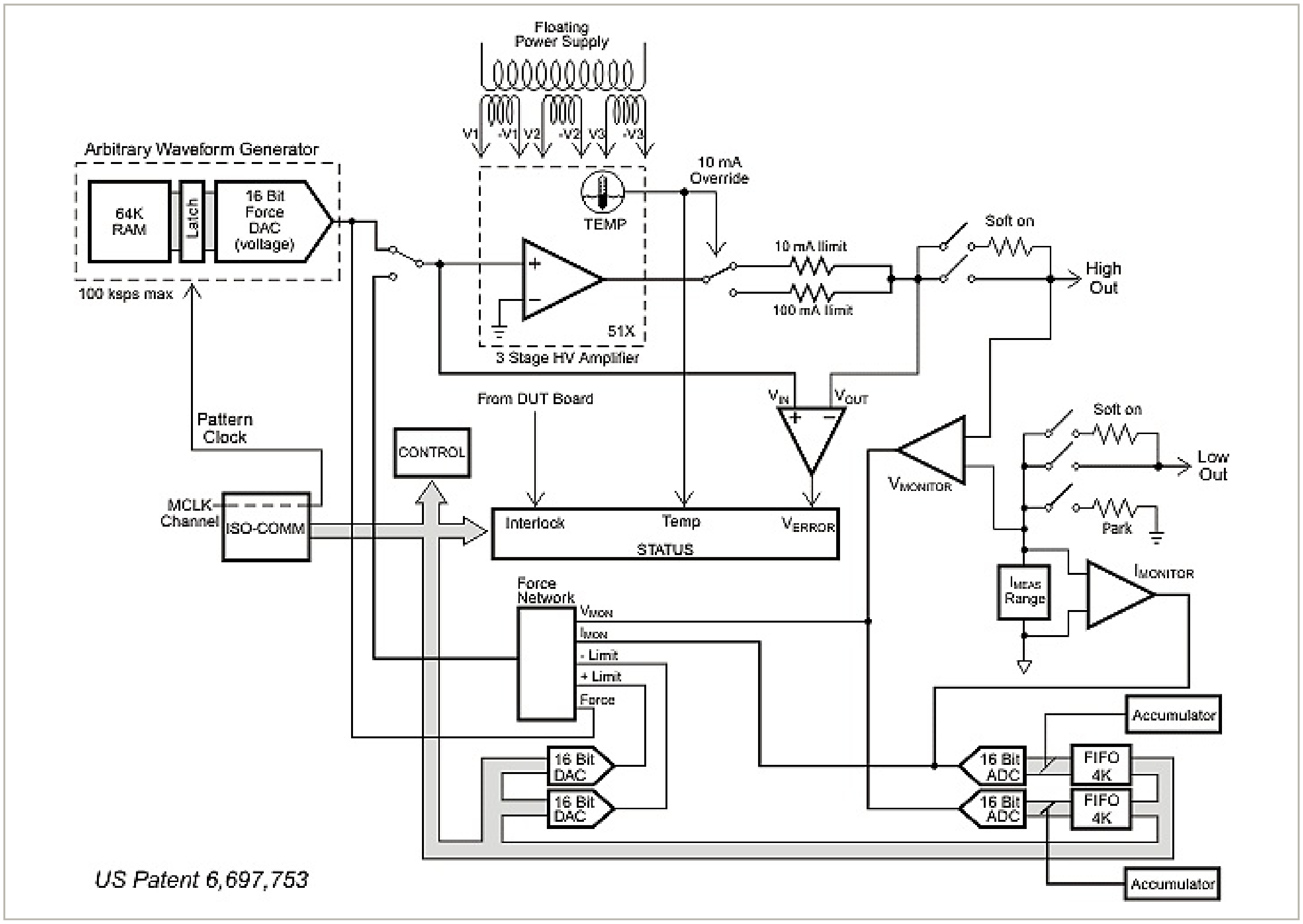


Figure 2-50 – Smart Pin Unit – 250 V / 100 mA – Block Diagram (1 of 2 Channels)

2.15.6 Specifications

Force	Range	Resolution	Accuracy
Driver Mode Voltage	±250 V	7.5 mV	±(60 mV + 0.05% + 100mV/mA)
V/I Mode Voltage	±250 V	7.5 mV	±(60 mV + 0.05%) + 500µV/mA
Current	±100 mA	3 µA	±(12.5 µA + 0.05%)
	±10 mA	300 nA	±(1.25 µA + 0.05%)
	±1 mA	30 nA	±(125 nA + 0.05%)
	±100 µA	3 nA	±(50 nA + 0.05%)
	±10 µA	300 pA	±(30 nA + 0.1%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±250 V	N/A	N/A	7.5 mV	±(60 mV + 0.05%)
	±100 V	N/A	N/A	3 mV	±(18 mV + 0.05%)
	±10 V	N/A	N/A	300 µV	±(4 mV + 0.05%)
Current	±100 mA	1X	100 mA	3 µA	±(12 µA + 0.05%)
	±100 mA	10X	10 mA	300 nA	±(2.5 µA + 0.05%)
	±100 mA	100X	1 mA	30 nA	±(550 nA + 0.1%)
	±10 mA	1X	10 mA	300 nA	±(1.25 µA + 0.05%)
	±10 mA	10X	1 mA	30 nA	±(250 nA + 0.05%)
	±10 mA	100X	100 µA	3 nA	±(125 nA + 0.1%)
	±1 mA	1X	1 mA	30 nA	±(250 nA + 0.05%)
	±1 mA	10X	100 µA	3 nA	±(80 nA + 0.05%)
	±1 mA	100X	10 µA	300 pA	±(60 nA + 0.1%)
	±100 µA	1X	100 µA	3 nA	±(50 nA + 0.05%)
	±100 µA	10X	10 µA	300 pA	±(30 nA + 0.1%)
	±100 µA	100X	1 µA	30 pA	±(20 nA + 0.1%)
	±10 µA	1X	10 µA	300 pA	±(30 nA + 0.1%)
	±10 µA	10X	1 µA	30 pA	±(20 nA + 0.1%)

Driver Mode: Voltage Force with Current Measure *Slew Rate:* 4 V/µsec max.
VI Mode: Voltage Force or Current Force/Measure *Slew Rate:* 2 V/µsec max.

2.15.7 User Interface

2.15.7.1 Software

The SPU-250 is programmed using dedicated software utilities, which are identified by the prefix sp250.

To force a DC voltage: Use the utility `sp250set()` to program the output voltage, select the current range, and set the mode to SPU_DRV or SPU_FV.

To force an AC voltage:

1.) Use the utility `sp250set()` to set the mode to SPU_AWG.

2.) Use the utility `sp250loader()` to load the AWG with a single waveform or multiple waveforms.

3.) Use the MCLK utilities to select, connect, and program a master clock channel that will drive the programmed waveform(s) out the AWG. Program the clock frequency of the selected MCLK channel with the utilities `mclkset()` and `mclkmode()`. Program the timing sequence(s) that will drive the desired AWG pattern(s) with `mclksequence()`. Connect the selected MCLK channel to the SPU-250 with the utility `mclkchannel()`. Refer to The ETS Software Help File in the programming environment, for details on the MCLK utilities.

To read the status of an SPU-250 from the RAIDE debug environment: Call the utility `sp250stat()`. This is easily accomplished by clicking on the "SPU" button. The programmed state and status conditions for all SPU-250s in the tester are displayed graphically.

2.15.7.2 Hardware

The SPU-250 requires two ISO-COMM channels, one for each SPU-250 channel. ISO-COMM channel N, where N = 0 to 255, identifies the SPU-250 address in a call to a SPU-250 software utility. To determine the ISO-COMM addresses assigned to the SPU-250s in the tester, the user can refer to the tester configuration listing, or run LCONFIG from the RAIDE debug environment.

There are two channels per board and they are housed in the Floating Resource Card Cage (FR Cage). The SPU-250 may be connected to the Application Board in two ways:

1.) Direct Connection – Allows for connections up to the DUT area directly.

2.) Matrixed Connection – Allows a channel to be used at multiple locations simultaneously.

On ETS-88™ systems, users can connect the SPU-250 to the 8x8 Matrix via the application board. See Chapter 4 for the connections and pinouts for the SPU-250.

2.16 Smart Pin Unit 500 V / 50 mA (SPU-500)

2.16.1 Caution to Users

SAFETY NOTICE:

This resource is designed to operate in a test system environment that is designed with the following safety features:

- Access to this resource requires the use of a tool to remove a cover
- Access to Input/Output connections to this resource is blocked by mechanical barriers
- An electrical interlock circuit inhibits the output of this resource

CAUTION: RISK OF SHOCK.

****

Hazardous Voltages Present! This resource generates hazardous voltages and must be operated in a properly designed enclosure with safety features in place. Always turn power off prior to handling this resource. Eagle Test Systems, Inc. accepts no responsibility for harm from handling or misuse of this resource.

****

Use high-voltage-insulated wiring when wiring connections from a SPU-500 to points on the application boards. Teflon- and silicone-insulated wire offer dielectric strengths in hundreds to thousands of volts.

2.16.2 Features

- Fully independent dual channel SmartPin™ resource
- Each channel fully floating and stackable (±1000 VDC from ground max.)
- One (1) force voltage range: ±500 V
- Three (3) measure voltage ranges: ±(500 V; 100 V; 10 V)
- Five (5) force current ranges: ±(50 mA, 10 mA, 1 mA, 100 µA, 10 µA)
- Six (6) measure current ranges: ±(50mA, 10mA, 1mA, 100µA, 10µA, 1µA)
- Real-time measurement accumulator for instant results averaging
- Two (2) digitizers to capture both voltage and current waveforms simultaneously
- Independent high/low programmable voltage/current clamps with alarms
- Measurement alarms are available through software when the resources range limitations are met
- The resource is designed for operator safety by disabling the resource if the application board interlocks are not satisfied
- Hardware and software designed for multisite applications

Each of the SPU-500's two independently programmable channels contains the following components/features:

2.16.2.1 Waveform Digitizer

The 4K RAM Waveform Digitizer provides to the user a powerful measurement tool. The clock coming into the SPU-500 may be divided down to a sampling rate that is between 1 Hz to 100 kHz. This separate clock divider for the digitizer allows the user to measure at one sampling rate and force date with the AWG at a higher frequency.

The digitizer may run concurrently with any forcing function, such as the AWG described above or a DC voltage/current.

See the SPU-500 Specifications for SPU-500 Digitizer capabilities.

2.16.2.2 Arbitrary Waveform Generator (AWG)

The 16 bit AWG provides tremendous forcing capability to this instrument by allowing the user to reproduce any waveform from a sine wave to simulating the output of a digital driver into the DUT. The AWG has a maximum clock rate of 100 kHz with 64K of RAM pattern depth to provide the user the ability to force either voltage or current (depending on the mode selected). The software allows the user to load concatenated patterns into the AWG and then run these patterns individually and/or nonsequentially.

See the SPU-500 Specifications on the following page for SPU-500 AWG capabilities.

2.16.2.3 Voltage and Current Clamps

Each voltage and current range has programmable upper and lower clamps. These clamps essentially define a 'window' of allowable voltage or current. Current clamps are programmed while in the FV mode and voltage clamps are programmed while in the FI mode.

NOTE: Remember that choosing a lower current clamp of 0 mA keeps the resource from sinking current while discharging a load at the end of testing.

2.16.3 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]
- Eagle Vision MST [Rev. 2011.1.1002.17 and up]

2.16.4 Modes

The SPU-500 is a dual-mode resource. You can use the SPU-500 to force voltage (±500 V in power supply mode) and measure current, or as a full V/I with programmable clamps. It is a two-wire resource since it is designed for low current applications.

2.16.4.1 Driver Mode

As stated previously this is a two-wire resource in all modes and ranges. There are two significant differences in the available DRV mode on the SPU-500:

- The programmable clamps are disabled. Therefore, the current passing through the resource would be limited by the current range selected.
- Faster slew rate capability.

2.16.4.2 V/I Mode

The full V/I mode can force and measure either voltage or current. This mode has the slower slew rate of the two modes (1.5 V/µsec max) to maintain stability with the loop amplifier. The V/I mode also has two 16 Bit limit DACs, which serve as voltage or current clamps depending on whether you're using V_{FORCE} or I_{FORCE}. You set the positive and negative limits, and these clamp limits are independent of each other. The limiting action automatically switches force modes at the limit value.

If using V_{FORCE} with a current limit, the SPU-500 switches to I_{FORCE} once the current reaches the limit value. Because there is no remote sense, a voltage correction circuit corrects the voltage drop of the current measurement circuit to reduce the voltage error.

2.16.5 Block Diagram

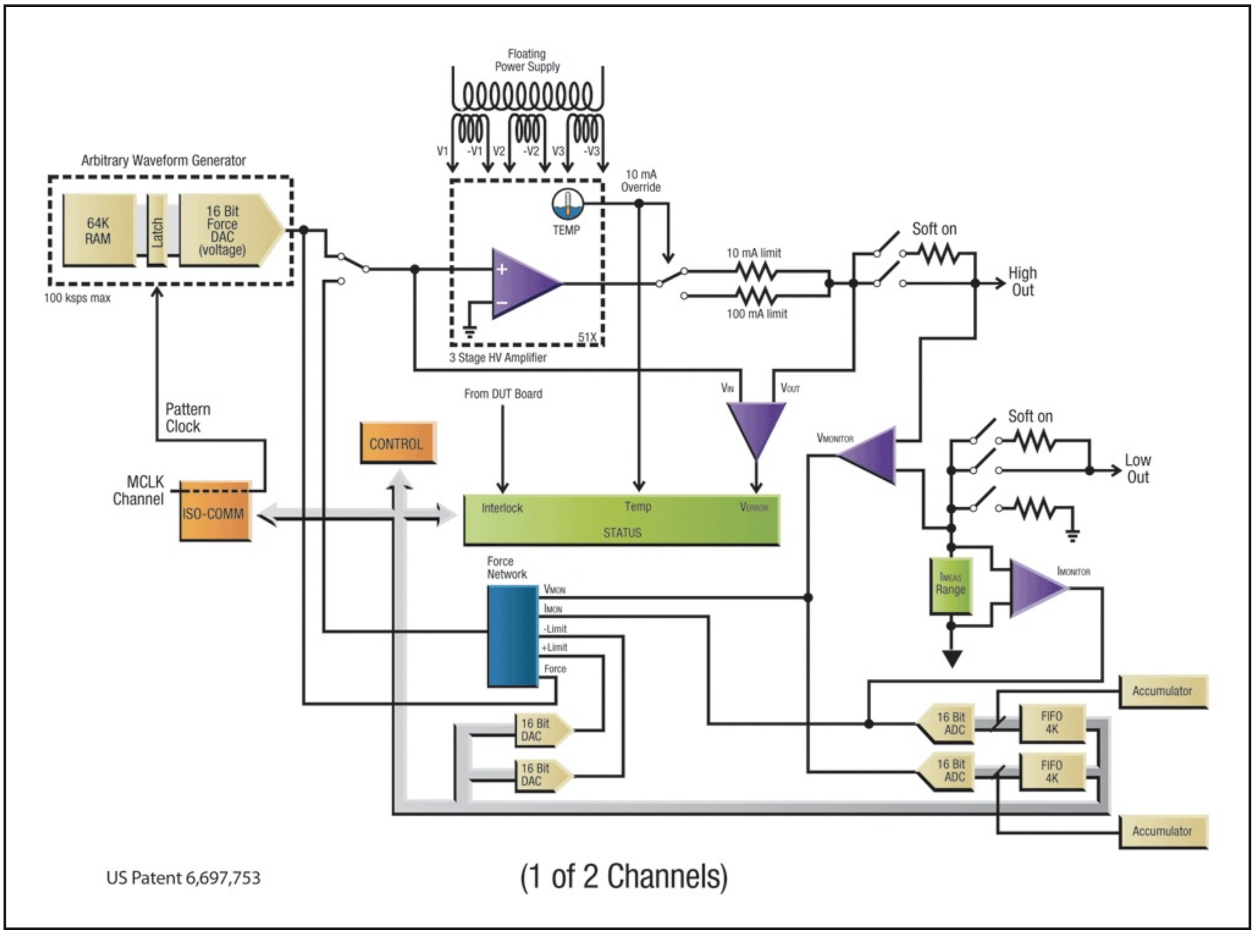


Figure 2-51 – Smart Pin Unit (500 V / 50 mA) Block Diagram (1 of 2 Channels)

2.16.6 Specifications

Force	Range	Resolution	Accuracy
Driver Mode Voltage	±500 V	15 mV	±(60 mV + 0.05% + 100mV/mA)
V/I Mode Voltage	±500 V	15 mV	±(60 mV + 0.05%)
Current	±50 mA	1.5 µA	±(6 µA + 0.05%)
	±10 mA	300 nA	±(1.25 µA + 0.05%)
	±1 mA	30 nA	±(125 nA + 0.05%)
	±100 µA	3 nA	±(50 nA + 0.05%)
	±10 µA	300 pA	±(30 nA + 0.1%)

Measure	Range	Gain	Effective Range	Resolution	Accuracy
Voltage	±500 V	N/A	N/A	15 mV	±(60 mV + 0.05%)
	±100 V	N/A	N/A	3 mV	±(18 mV + 0.05%)
	±10 V	N/A	N/A	300 µV	±(4 mV + 0.05%)
Current	±50 mA	1X	50 mA	1.5 µA	±(6 µA + 0.05%)
	±50 mA	10X	5 mA	150 nA	±(2.5 µA + 0.05%)
	±50 mA	100X	500 µA	15 nA	±(550 nA + 0.1%)
	±10 mA	1X	10 mA	300 nA	±(1.25 µA + 0.05%)
	±10 mA	10X	1 mA	30 nA	±(250 nA + 0.05%)
	±10 mA	100X	100 µA	3 nA	±(125 nA + 0.1%)
	±1 mA	1X	1 mA	30 nA	±(250 nA + 0.05%)
	±1 mA	10X	100 µA	3 nA	±(80 nA + 0.05%)
	±1 mA	100X	10 µA	300 pA	±(60 nA + 0.1%)
	±100 µA	1X	100 µA	3 nA	±(50 nA + 0.05%)
	±100 µA	10X	10 µA	300 pA	±(30 nA + 0.1%)
	±100 µA	100X	1 µA	30 pA	±(20 nA + 0.1%)
	±10 µA	1X	10 µA	300 pA	±(30 nA + 0.1%)
	±10 µA	10X	1 µA	30 pA	±(20 nA + 0.1%)
	±10 µA	100X	100 nA	3 pA	±(20 nA + 0.2%)

Driver Mode: Voltage Force with Current Measure — Slew Rate: 4 V/µsec max.
VI Mode: Voltage Force or Current Force/Measure — Slew Rate: 2 V/µsec max.

2.16.7 User Interface

2.16.7.1 Software

The SPU-500 is programmed using dedicated software utilities, which are identified by the prefix sp500.

To force a DC voltage: Use the utility `sp500set()` to program the output voltage, select the current range, and set the mode to SPU_DRV or SPU_FV.

To force an AC voltage:

1.) Use the utility `sp500set()` to set the mode to SPU_AWG.

2.) Use the utility `sp500loader()` to load the AWG with a single waveform or multiple waveforms.

3.) Use the MCLK utilities to select, connect, and program a master clock channel that will drive the programmed waveform(s) out the AWG. Program the clock frequency of the selected MCLK channel with the utilities `mclkset()` and `mclkmode()`. Program the timing sequence(s) that will drive the desired AWG pattern(s) with `mclksequence()`. Connect the selected MCLK channel to the SPU-500 with the utility `mclkchannel()`. Refer to The ETS Software Help File in the programming environment, for details on the MCLK utilities.

To read the status of an SPU-500 from the RAIDE debug environment: Call the utility `sp500stat()`. This is easily accomplished by clicking on the "SPU" button. The programmed state and status conditions for all SPU-500s in the tester are displayed graphically.

2.16.7.2 Hardware

The SPU-500 requires two ISO-COMM channels, one for each SPU-500 channel. ISO-COMM channel N, where N = 0 to 255, identifies the SPU-500 address in a call to a SPU-500 software utility. To determine the ISO-COMM addresses assigned to the SPU-500s in the tester, the user can refer to the tester configuration listing, or run LCONFIG from the RAIDE debug environment.

There are two channels per board and they are housed in the Floating Resource Card Cage (FR Cage). The SPU-500 may be connected to the Application Board in two ways:

1.) Direct Connection – Allows for connections up to the DUT area.

2.) Matrixed Connections – Allows a channel to be used at multiple locations simultaneously.

On ETS-88™ systems, users can connect the SPU-500 to the 8x8 Matrix via the application board. See Chapter 4 for the connections and pinouts for the SPU-500.

2.17 Test Head Control Board (TCB)

2.17.1 Features

General

- Integrated Multisite Handler Interface
    - 40 I/O lines
    - 8 Start-of-Test (SOT) lines
- System Monitoring
    - On-board temperature and relative humidity monitors
    - DC system voltage monitor

Isolated Communications

- 28 Channels:
    - Floating resource dedicated channels: 12
    - DUT board dedicated channels: 4
    - External channels: 12

Master Clock

- Two (2) selectable DDS-based Master Clocks, programmable from 25 to 66 MHz
- Eight (8) programmable clock sequencer/divider channels (for clocking analog resources)
- True multisite mixed-signal operation
- Dual clock sources facilitate equivalent time sampling
- DDS-based digital DUT clock (1 Hz to 66 MHz – available to all digital channels)
- External reference input/output (frequency locking to external instruments)

Control Bits

- 32 Programmable Control Bits for DUT board relay control
- Operate relays from 5-24 volt coils
- Dynamic pattern-based operation

2.17.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]
- Eagle Vision MST [Rev. 2011.1.1002.17 and up]

2.17.3 Theory of Operation

The TCB is the central control unit for the digital subsystem of the ETS-88. It combines functionality that was formerly divided amongst several resources – system master clocking, isolated communications, test head interfacing, and programmable control bits – into a single board.

2.17.3.1 Isolated Communications

The ISO-COMM portion of the TCB serves as the communication interface with the resources located in the FR Cage (FSSs, APUs, etc.). This communication with the floating resources is composed of two separate parts: the ISO-COMM data, and the Pattern Clock (P-Clock).

The P-Clock is a Master Clock signal that has been connected to a resource using the ETS software `mcuconnect()` utility. Floating resources use these clock signals as the clock input for their on-board AWGs or digitizers. Using the P-Clock signal allows the resources to operate synchronously.

P-Clock signals originate from the Master Clock portion of the TCB. Each of the eight Master Clock signals is connected, through the backplane, to the ISO-COMM section of the TCB. After entering the TCB, these clock signals are routed to any number of ISO-COMM channels. An individual clock channel may be routed to more than one ISO-COMM channel, but an ISO-COMM channel may be connected to only one clock channel. Each Master Clock signal enters the ISO-COMM section of the TCB through only one connection.

ISO-COMM channel assignments are determined by the slot placement of the ISO-COMM modules. Each ISO-COMM channel controls one position of the

FR Cage. ISO-COMM floating resource (Logical ISO-COMM channel) assignments are shown in the table below.

NOTE: Logical assignments are determined by the Config.ets file. For informational purposes, physical ISO-COMM channel assignments are also shown.

Floating Resource Position	Physical ISO-COMM Channel
Pos 0, Pos 1, Pos 4, Pos 5, Pos 8, Pos 9, Application Board Pos 42, Application Board Pos 43	0 – 7 Config.ets "ICOM0"
Pos 2, Pos 3, Pos 6, Pos 7, Pos 10, Pos 11, Pos 14, Pos 15	8 – 15 Config.ets "ICOM1"
Pos 12, Pos 13, Pos 16, Pos 17, Pos 20, Pos 21, Pos 24, Pos 25	16 – 23 Config.ets "ICOM2"
Pos 18, Pos 19, Pos 22, Pos 23, External Pos 44, External Pos 45, External Pos 46, External Pos 47	24 – 31 Config.ets "ICOM3"
Pos 28, Pos 29, Pos 32, Pos 33, Pos 36, Pos 37, Application Board Pos 40, Application Board Pos 41	48 – 55 Config.ets "ICOM6"
Pos 26, Pos 27, Pos 30, Pos 31, Pos 34, Pos 35, Pos 38, Pos 39	56 – 63 Config.ets "ICOM7"

2.17.3.2 Master Clock

The Master Clock portion of the TCB controls the clock timing for its respective dual test head and all associated ETS instruments.

The MCU lets you synchronize analog and/or digital events in the context of a multisite mixed-signal test system. Mixed-signal device testing requires a system architecture that supports frequency- and phase-locked clocking control that is both accurate and reproducible. True parallel multisite testing often requires a system where the sites can be sequenced independently based on the performance of each site's Device Under Test (DUT). The MCU has the capabilities to achieve this level of performance.

The MCUs, two DDS-based timing sources, are available to each of the 16 master clock channels. For most applications, only one clock source is required. Having access to a second frequency locked clock source, makes it possible to perform certain specialized equivalent time sampling operations. This feature facilitates extremely high effective sample rates for digitizing applications (>20 GHz range, limited by bandwidth of appropriate instrument).

The eight master clock channels each include a programmable divide-by-N clock divider and a clocking sequencer. The RAM-based clocking sequencers make it possible to gate the divided clocks on and off as needed to accomplish various test applications. These clocks can be routed to any of the analog resources in the test system for driving AWGs and digitizers as needed.

Timing delays may be programmed for the individual clock channel (they are skewed from the Master Clock signal), and are programmable from 0 nsec to the clock period.

Clock Channel 0

As expressed in Figure 2-52 (on the following page), clock channel 0 must be used in all cases where a clocking signal is desired from the Master Clock. In addition, clock channel 0 must also contain the longest time duration sequence of clock channels 0-7. If multiple sequences are used, the clock channel with the most sequences must again be clock channel 0. Basically, clock channel 0 is used as the primary clock channel to notify the hardware of "end of sequence."

Clock Modes

There are four clock modes available on each clock channel: MCU_OFF, MCU_CLK, MCU_GATE, MCU_FREERUN, MCU_CLKCONT, and MCU_GATECONT. Please refer to the `mcumode()` utility description in the ETS Software help file for details on these modes. The number of clocks, and the way in which the clocks are applied to the instrument, is programmable.

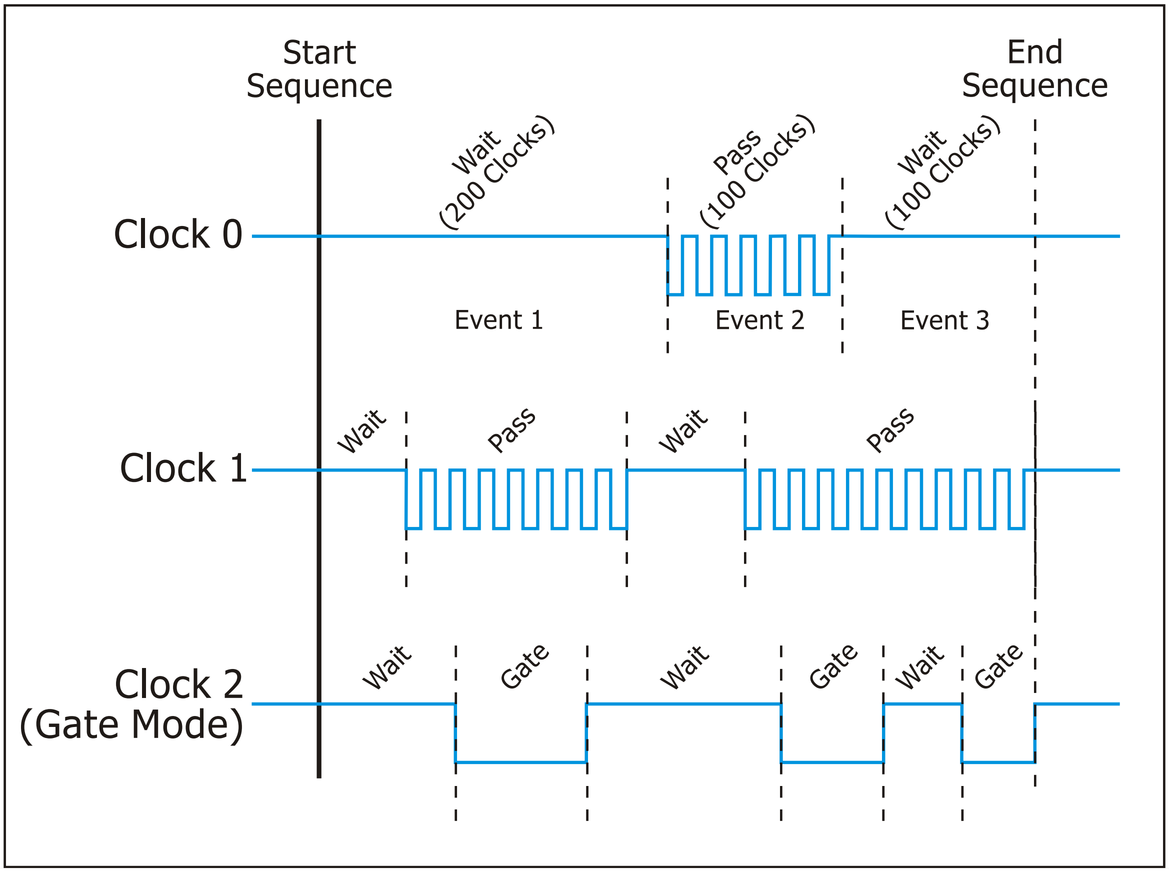


Figure 2-52 – Typical Clocking Sequence

Clock Sequences

In the MCU_CLK mode, the clock sequence describes how many clock pulses to generate. For example…

```
mcusequence(MCU_CH0, "Seq1", "50, ON,
10, OFF, 50, ON, 10, OFF, 50, ON");
```

…will perform the following:

- 50 clock pulses
- Wait 10 pulses (clock hi)
- 50 clock pulses
- Wait 10 pulses (clock hi)
- 50 clock pulses.

The MCU_GATE mode yields a completely different type of signal. Using the same example as above in MCU_GATE mode would do the following:

- 'lo' 50 clock periods
- 'hi' 10 periods
- 'lo' 50 clock periods
- 'hi' 10 periods
- 'lo' 50 clock periods.

For both clock modes, you can declare a <number> up to 65535 "ON/OFF" clocks. In addition, multiple sequences can be loaded into each clock channel up to a cumulative total of 1024 "ON/OFF" events (where "50, ON" is 1 event). Multiple sequences are loaded with additional calls to `mcusequence()` and are selected when calling `mcustart()`. If necessary, you can clear a clock channel by using the keyword "CLEAR" in the `mcusequence()` utility. Figure 2-52 shows a graphical example of a typical clocking sequence.

User Interface

Software

All functions of the Master Clock on the TCB are programmed using utility function calls from your C++ test program. These utilities (`mcuxxx()` and `cbitxxx()`) and their syntax and usage are described in The ETS Software Help File.

Hardware

There is no direct user interfacing with the Master Clock section of the TCB.

2.17.3.3 Programmable Control Bits

The Programmable Control Bits (C-Bits) portion of the TCB – included as a separate module – provides 32 read/write control lines, which are useful for setting and reading the logic state of DUT-board-resident hardware.

The C-Bits are typically used to operate relays and drive logic inputs on such DUT board circuitry. They are designed as low-side switches for direct driving of relay coils of the popular 5 V and 12 V relay families, but can operate with coil voltages up to 24 V. These low-side switches are implemented with CMOS technology to support the low turn-on voltages required for interfacing with logic devices and relays.

The C-Bits also provide a DUT board communication mechanism via their ability to read back hardware logic states. For such read-back applications, the C-Bits can be tied directly to most logic families. Each channel of the C-Bits module includes transient suppression diodes, ESD protection, and built-in 5 V pull-ups; however, user-supplied ESD protection is recommended for read-back applications in order to protect DUT-based logic devices, since these signals are especially vulnerable during handling.

From a software standpoint, the C-Bit Utilities manage the logic state of the control bits, grouping and mapping them by site to simplify multisite coding. The C-Bit Utilities support an "OPEN/CLOSE" syntax to make relay control more intuitive for users, and they also support a "SET:ON/OFF" syntax for logic applications. The C-Bit command set supports C-Bit naming to make user code more readable and intuitive. The RAIDE environment offers a custom C-Bit editor tool to enhance debug efficiency.

User Interface

Software

Most functions of the C-Bits are programmed using C-Bit utility calls from your C test program. These utilities and their syntax and usage are described in The ETS Help File.

The status of the C-Bits can also be checked and changed using RAIDE's C-Bits button and C-Bits Status dialog box. For more information on RAIDE, see the ETS Help File, or the Eagle Vision Software Suite Manual.

Hardware

The C-Bits section of the TCB is installed as a module on the primary side of the TCB board.

2.17.3.4 Test Head Interface

The Test Head Interface portion of the TCB provides a link between the control elements of the resource, the system computer, and the rest of the system.

2.17.4 Block Diagram

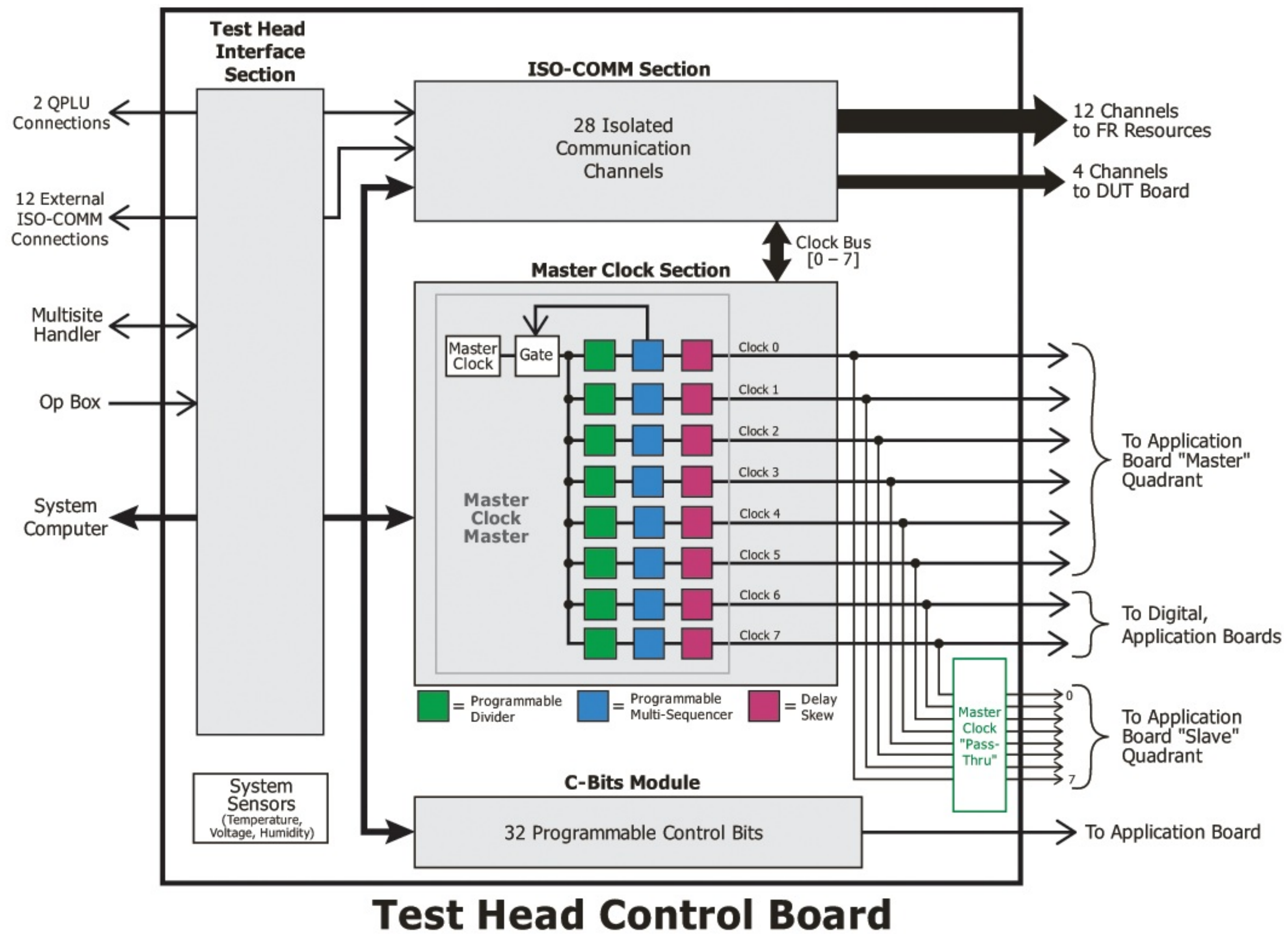


Figure 2-53 – TCB Block Diagram

2.17.5 Specifications

2.17.5.1 General

Feature	Quantity
Programmable Control Bits	32 Bits
Master Clocking	8 Programmable Channels
Isolated Communications	28 Channels
Integrated Multisite Handler Interface	40 I/O Lines, 8 SOT Lines

2.17.5.2 Programmable Control Bits (C-Bits)*

Specification	Condition	Minimum	Maximum
Pull-Up Current, C-Bit Off	V = 0.0 V	800 µA	
Pull-Up Current, C-Bit Off	V = 2.4 V	400 µA	
Pull-Up Current, C-Bit Off	V = 4.8 V	-10 µA	-10 µA
Pull-Down Current, C-Bit On			200 mA
Voltage, C-Bit On	I = 100 mA		0.9 V
Voltage, C-Bit On	I = 200 mA		1.1 V
Clamp Voltage			33 V (Nominal)
Total Current, 32 Outputs	Continuous		4 A
Readback Logic "1" Threshold	Negative Logic		1.3 V
Readback Logic "0" Threshold	Negative Logic	1.5 V	

***NOTE: Eagle Test recommends adding protection diodes to all relay coils used in conjunction with C-Bits.**

2.17.5.3 Master Clocking

Feature	Value or Range
Master Clock	25 – 66 MHz
Digital DUT Clock	1 Hz – 66 MHz
Auxiliary Clock	25 – 66 MHz
Clock Frequency Accuracy	±(250 Hz + 0.1% Rdg)
Channel-to-Channel Skew	10 nsec
Resolution	1 µHz

2.18 Waveform Capture Unit (WCU-2220 or WCU-2220/16K)

2.18.1 Features

- Fully Floating Operation up to 2000 V
- High Speed 8-bit Waveform Digitizer with 8K (WCU-2220) or 16K (WCU-2220/16K) Memory and Variable Sampling Rate up to 250 MHz
- Voltage and Time "Windowing" allows the Digitizer to Zoom in on Portions of a Waveform

2.18.2 Software Support

- Eagle Vision [Rev. 2009.1.2.16 and up]

2.18.3 Theory of Operation

2.18.3.1 Time Zoom

The 8-bit Waveform Capture Unit is designed to digitize up to 16K RAM of input voltage waveforms with sampling frequencies of 1 MHz up to a maximum rate of 250 MHz.

The WCU will digitize at the sampling frequency selected for the duration of a "GATE" signal (from the MCLK Board) and will capture multiple waveforms as the "GATE" signal goes active at user selected intervals.

2.18.3.2 Voltage Zoom

The fully floating high voltage input variable gain attenuator allows maximum flexibility in obtaining Full Scale measurements by enabling the user to center the voltage (window size, as shown in the picture below) of the input attenuator around the anticipated signal in (window midpoint). These voltage ranges are:

- ±0.2 V to ±2000 V

Selected Midpoint	Selectable Windowsize
0 V - 20 V	0.2 V - 20 V
20 V - 200 V	2.0 V - 200 V

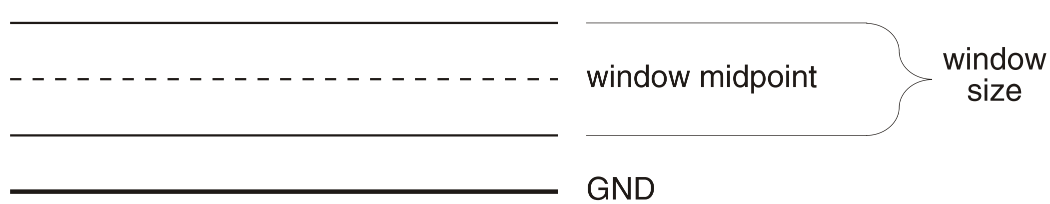


Figure 2-54 – Centering Input Attenuator Voltage

2.18.4 Block Diagram

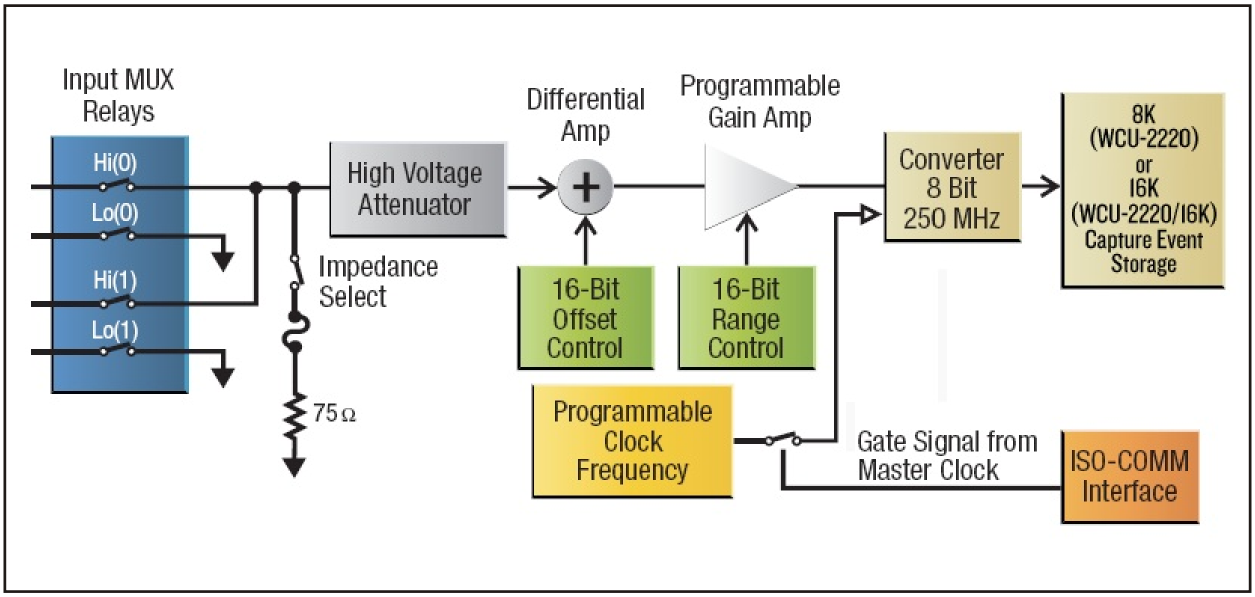


Figure 2-55 – WCU-2220 and WCU-16K Block Diagram

2.18.5 WCU Specifications

Model	WCU-2200	WCU-2220/16K
Number of Digitizers	1	1
Number of Channels	2	2
Bandwidth	DC to 30 MHz	DC to 30 MHz
Captured Signal Memory Depth	8182	16384
Maximum Isolation Voltage (Low to Chassis)	±2000 V	±2000 V
Trigger Source	Master Clock (TCB)	Master Clock (TCB)
Minimum Trigger Certainty	6.3 nsec	6.3 nsec
Amplitude Resolution	8 bits	8 bits

(Cont'd)

WCU Specifications (cont'd)

<table>
<tr><td colspan="2" rowspan="1">Sample Rate</td><td colspan="4">1 MSPS to 250 MSPS Programmable in the Following Steps:
1 MSPS (from 1 MSPS to 10 MSPS)
5 MSPS (from 10 MSPS to 250 MSPS)</td></tr>
<tr><td colspan="2" rowspan="2">Voltage Ranges</td><td>Range</td><td>Resolution</td><td colspan="2">Accuracy</td></tr>
<tr><td>0.2 V-2 V
2 V-20 V
20 V-200 V
200 V-2000 V</td><td>1 mV
10 mV
100 mV
1 V</td><td colspan="2">±1%
±1%
±1%
±1%</td></tr>
<tr><td rowspan="2">Offset</td><td>Range</td><td>Maximum Offset</td><td>Resolution</td><td colspan="2">Accuracy</td></tr>
<tr><td>0.2 V-2 V
2 V-20 V
20 V-200 V
200 V-2000 V</td><td>±20 V
±200 V
±2000 V
±2000 V</td><td>1.25 mV
12.5 mV
125 mV
125 mV</td><td colspan="2">0.2% ±0.01 V
0.2% ±0.1 V
0.2% ±1 V
0.2% ±1 V</td></tr>
<tr><td rowspan="3">Input Impedance</td><td>Range</td><td>|Offset|</td><td colspan="3"></td></tr>
<tr><td>0.2 V-20 V
2 V-200 V
20 V-2000 V</td><td>0 V-20 V
20 V-200 V
200 V-2000 V</td><td colspan="3">1 MΩ ±1% in Parallel with 40 pf ±5%
10 μΩ ±1% in Parallel with 10 pf ±5%
10 MΩ ±1% in Parallel with 1.5 pf ±5%</td></tr>
<tr><td colspan="2">75 Ω (User Selectable)</td><td colspan="3">75 Ω ±1%</td></tr>
<tr><td rowspan="2">Maximum
Input Voltage
(HI to LO)</td><td colspan="2">75 Ω Selected</td><td colspan="3">5 Vrms</td></tr>
<tr><td colspan="2">75 Ω Not Selected</td><td colspan="3">±2000 V (DC + AC pk)</td></tr>
</table>

2.18.6 User Interface

2.18.6.1 Software

All functions of the WCU are programmed using utility function calls from your C test program. These utilities, their syntax, and usage are described in The ETS Software Help File.

2.18.6.2 Hardware

There is one WCU per board with two inputs to the digitizer. Both of the inputs are fully floating high voltage inputs. If using WCU4 – Channel 0 and 1, the connections will be as shown in Figure 2-56.

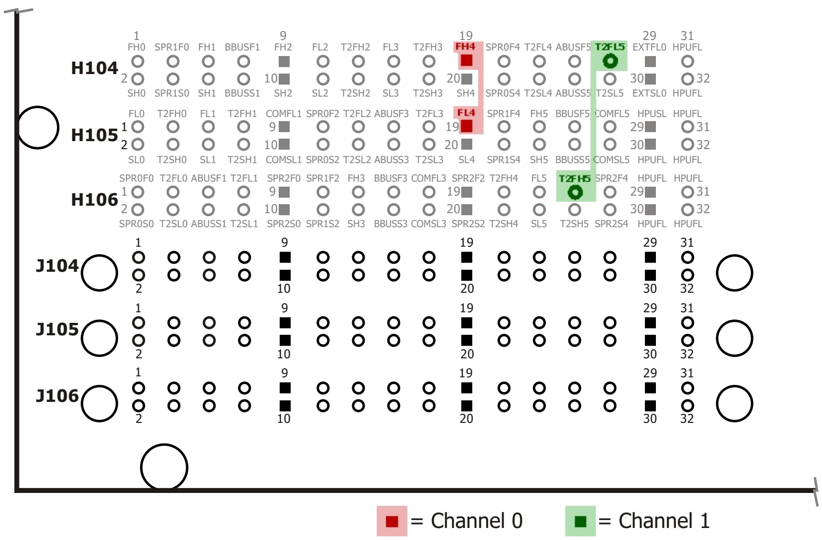


Figure 2-56 – ETS-88™ WCU Application Board Connections

3 System Software

3.1 Introduction

The software environment for ETS-88™ serves as a cohesive interface between the operating system, and the system hardware.

There are two system software options for ETS-88 testers:

- Eagle Vision [Rev. 2009.1.2.16 and up]
- Eagle Vision Multi-Section Technology (MST) [Rev. 2011.1.1002.17 and up]

Different system options are available depending on which software suite you choose. Please refer to Appendix C for a comparision of options and requirements for Eagle Vision operation versus Eagle Vision MST.

This section of the manual provides brief descriptions of the Windows® environment and the ETS software. The operating system, languages, and programmer's tools are each fully documented in their respective manuals. The ETS software suites are described in detail in their respective help files.

3.2 Operating Systems

Eagle Vision:

- Windows® XP Professional 32-Bit (English), Service Pack 2 or Greater

Eagle Vision MST:

- Windows® 7 Ultimate 64-Bit (English), Service Pack 1 or Greater

The Windows® environment provides many sophisticated capabilities. Because of this, it is important that you become as familiar as possible with the operating system and its features. Most of the program development time will be spent using Microsoft Visual Studio®. Knowledge of short cuts and additional features in Visual Studio® will help reduce test development time.

3.3 Development Environment

Eagle Vision:

- Microsoft Visual Studio® 2005 Standard

Eagle Vision MST:

- Microsoft Visual Studio® 2010 Professional

ETS software is based on C++, which comes standard with the ETS-88 system as part of the Microsoft Visual Studio®.

Microsoft Visual Studio® is used in conjunction with the ETS software to perform the following test program development tasks:

1.) Create / modify projects

2.) Build and compile/link test programs/DLLs

3.) Execute test program DLLs

The intermediate step of "linking" the object files greatly enhances the capabilities of the language. Because of the LINK process, reusable modules of code written in C++ or other programming language can be combined to form the final executable file. The linking operation provides a clean interface point for the ETS Family utilities to be integrated into the programming language.

3.3.1 C++ Language

Eagle Test Systems has chosen C++ because of its power, its portability, its structure and its flexibility. As a structured language, it is used to develop programs which are modular, such that a general purpose module can be used in multiple programs. This prevents duplication of development effort. For example, there may be a single module that performs an input leakage test on a pin and reports the results to a file. If properly set up, this module can be used for every test program which performs an input leakage test. General purpose modules can be made into function libraries to which all your programs have access.

Compiler options accommodate direct control over many aspects of the compilation process and provide the programmer access to useful compiler generated listings and interlistings. An extensive error warning and reporting system expedites program development with explicit diagnostic messages.

3.4 Development Environment

Microsoft Visual Studio® is a development environment that integrates the process of creating a program in C++ language. From Visual Studio®, you have access to environments for editing, compiling, linking, building and debugging. This gives you the ability to do all of the following operations from one screen:

- Observe real-time execution of code
- Halt execution of the program at any point
- Examine the contents of variables
- View output from the program
- Edit changes and rebuild the program

Help is available for both Visual Studio® and the Eagle Vision® software suite from within Visual Studio® to assist you with the details of the various processes.

3.5 ETS Utilities

3.5.1 Theory of Operation

The ETS Family utilities provide the software link between the test head hardware and the test program. The utilities are accessed via your C++ program; they are configured as an extension of the operating system. As soon as the test head is booted, the ETS Family utilities are available for use from either a C++ program or the ETS Family Debugger/RAIDE. When you link your test program with the proper util500.lib, the function calls in your program which use the utilities are given the information they need to find the called utility function.

Having the utilities as a DLL provides substantial benefits; e.g. when using the ETS Family Debugger, utilities will execute with the same timing as they would during test execution. From a software maintenance standpoint, the ETS Family utilities may be updated without requiring you to recompile or relink your programs.

The ETS Family utilities themselves provide various programming capabilities. Each aspect of the hardware can be controlled through the use of these utilities. The *ETS software help file* describes the standard utilities in detail and provides examples of their use.

4 Application Board Signal Definitions

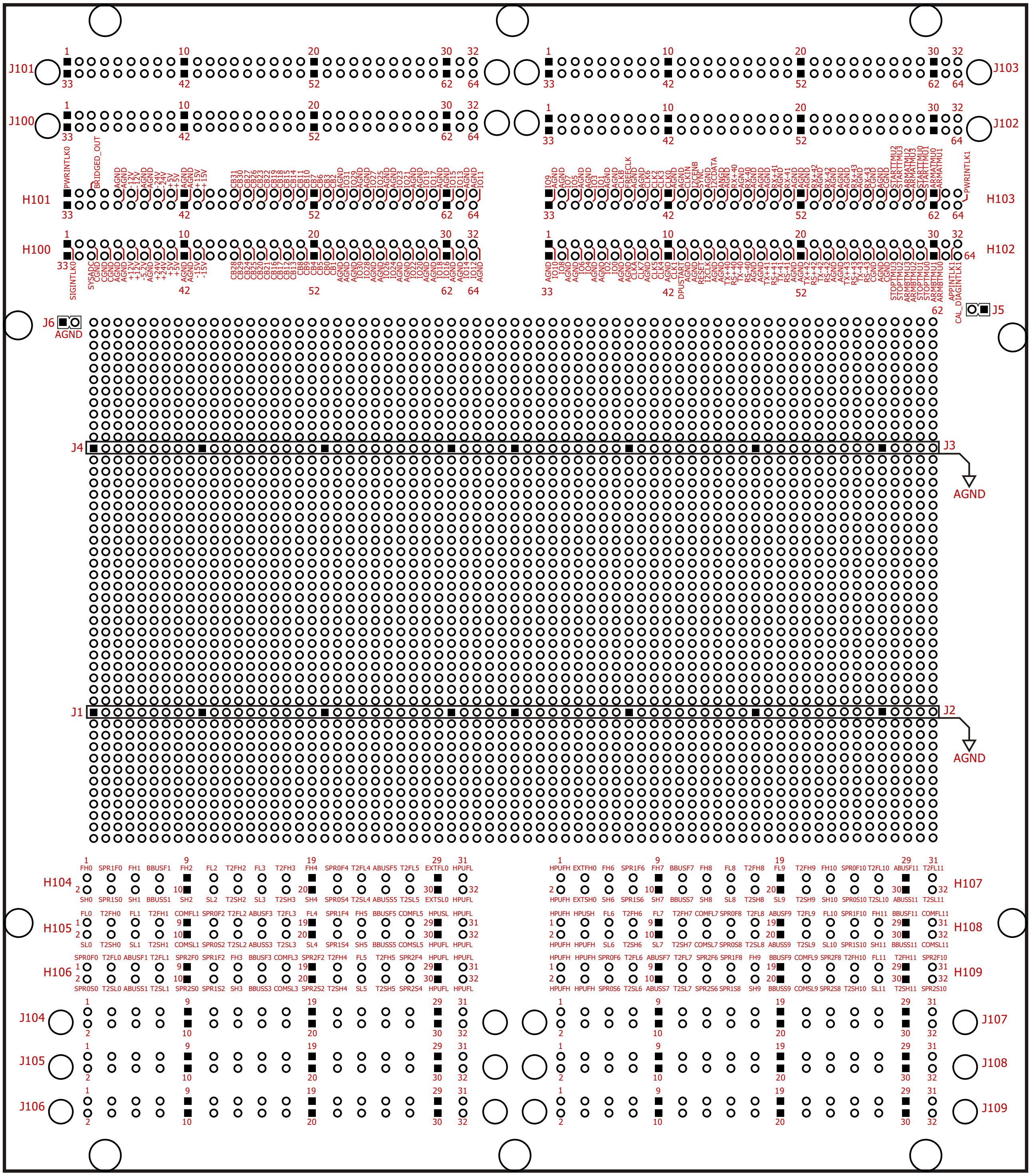


Figure 4-1 – Application Board Layout (Single Quadrant)

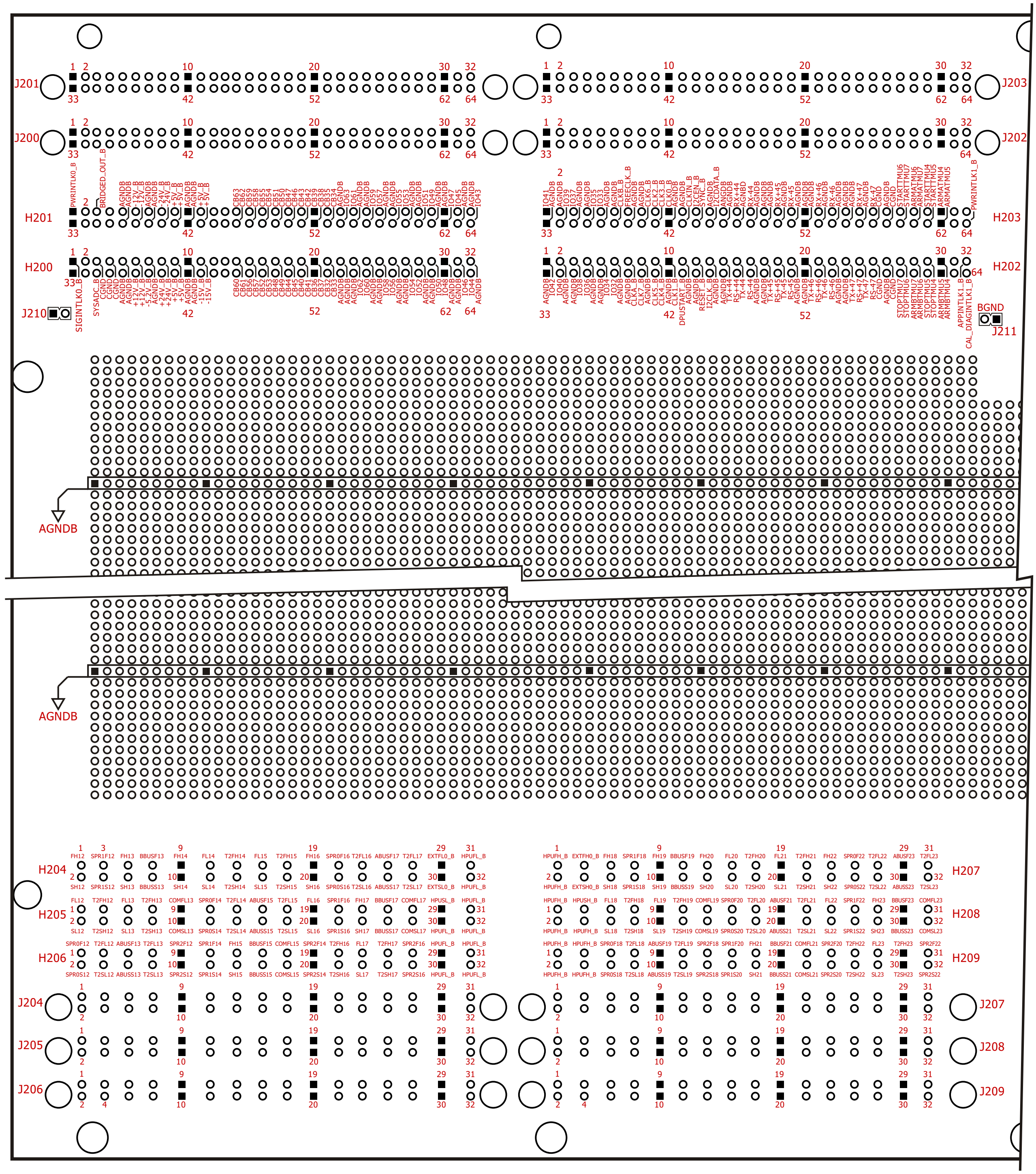


Figure 4-2 – Application Board Layout (Bridged - Left Side)

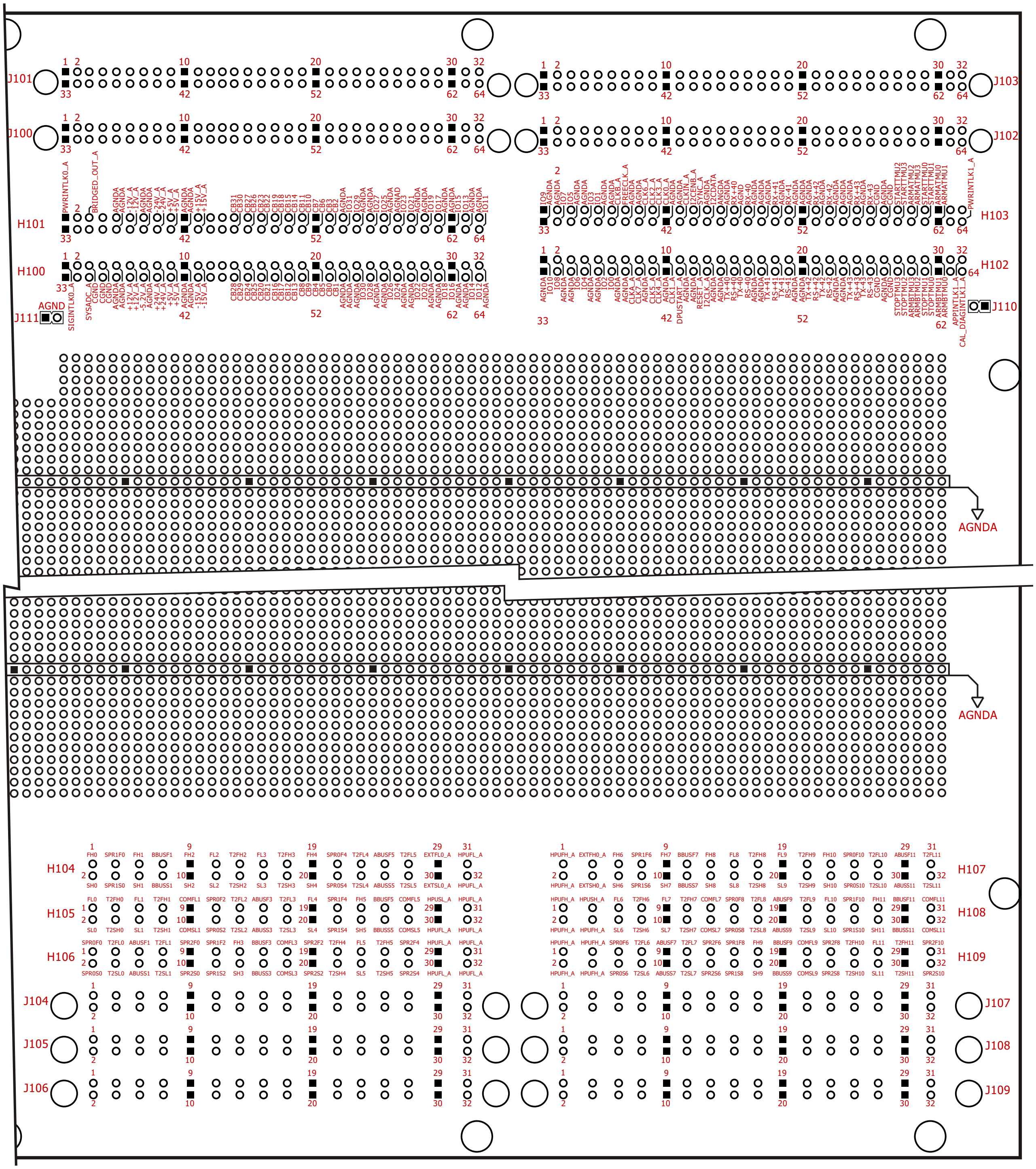


Figure 4-3 – Application Board Layout (Bridged - Right Side)

NOTE: When working with the standard ETS-88 application board, all connections are made at the 'Jxxx' numbered pins. When working with the prototyping version of the application board, connections are made at the 'Hxxx' numbered pins. The signals are the same regardless of 'H' or 'J' designation.

4.1 FR SLOT 0 (ISO-COMM 0/1)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J104-1	(FR)FH0	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 0	Force Hi 0	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 0	Force Hi Input 0
J104-2	(FR)SH0	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 0	Sense Hi 0	AWG 'n'-		Lin Source 'n'-		
J105-1	(FR)FL0	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 0	Force Lo 0	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 0	Force Lo Input 0
J105-2	(FR)SL0	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 0	Sense Lo 0	MEA 'n'-		Ref 1 'n'		
J105-3	(FR)T2FH0	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 0, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J105-4	(FR)T2SH0	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J106-3	(FR)T2FL0	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 0, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J106-4	(FR)T2SL0	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1 'n+1'		
J104-5	(FR)FH1	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 1	Force Hi 1 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 1	Force Hi Input 1
J104-6	(FR)SH1	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 1	Sense Hi 1 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J105-5	(FR)FL1	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 1	Force Lo 1 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 1	Force Lo Input 1
J105-6	(FR)SL1	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 1	Sense Lo 1 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1 'n+2'		
J105-7	(FR)T2FH1	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 1, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J105-8	(FR)T2SH1	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J106-7	(FR)T2FL1	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 1, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J106-8	(FR)T2SL1	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1 'n+3'		
J106-1	(FR)SPR0F0	Input F0	Force Hi Chan 'n+2'			Force Hi 0, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J106-2	(FR)SPR0S0	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J104-3	(FR)SPR1F0	Input F1	Force Hi Chan 'n+3'			Force Lo 0, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J104-4	(FR)SPR1S0	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J105-9	(FR)COMFL1	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J105-10	(FR)COMSL1	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J106-9	(FR)SPR2F0	Input F3	Force Low (6-11)			Force Hi 1, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J106-10	(FR)SPR2S0	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J106-5	(FR)ABUSF1	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 1, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J106-6	(FR)ABUSS1	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J104-7	(FR)BBUSF1	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J104-8	(FR)BBUSS1	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L- HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J104-29	EXTFL0_A	Input F6									
J104-30	EXTSL0_A	Input S6									
J107-3	EXTFH0_A	Input F7									
J107-4	EXTSH0_A	Input S7									

* 1EXT matrix input supported per quadrant

† Future support

4.2 FR SLOT 1 (ISO-COMM 2/3)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J104-9	(FR)FH2	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 2	Force Hi 2	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 2	Force Hi Input 2
J104-10	(FR)SH2	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 2	Sense Hi 2	AWG 'n'-		Lin Source 'n'-		
J104-11	(FR)FL2	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 2	Force Lo 2	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 2	Force Lo Input 2
J104-12	(FR)SL2	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 2	Sense Lo 2	MEA 'n'-		Ref 1 'n'		
J104-13	(FR)T2FH2	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 2, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J104-14	(FR)T2SH2	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J105-13	(FR)T2FL2	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 2, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J105-14	(FR)T2SL2	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1 'n+1'		
J106-13	(FR)FH3	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 3	Force Hi 3 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 3	Force Hi Input 3
J106-14	(FR)SH3	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 3	Sense Hi 3 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J104-15	(FR)FL3	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 3	Force Lo 3 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 3	Force Lo Input 3
J104-16	(FR)SL3	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 3	Sense Lo 3 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1 'n+2'		
J104-17	(FR)T2FH3	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 3, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J104-18	(FR)T2SH3	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J105-17	(FR)T2FL3	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 3, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J105-18	(FR)T2SL3	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1 'n+3'		
J105-11	(FR)SPR0F2	Input F0	Force Hi Chan 'n+2'			Force Hi 2, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J105-12	(FR)SPR0S2	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J106-11	(FR)SPR1F2	Input F1	Force Hi Chan 'n+3'			Force Lo 2, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J106-12	(FR)SPR1S2	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J106-17	(FR)COMFL3	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J106-18	(FR)COMSL3	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J106-19	(FR)SPR2F2	Input F3	Force Low (6-11)			Force Hi 3, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J106-20	(FR)SPR2S2	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J105-15	(FR)ABUSF3	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 3, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J105-16	(FR)ABUSS3	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J106-15	(FR)BBUSF3	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J106-16	(FR)BBUSS3	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L-HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J104-29	EXTFL0_A	Input F6									
J104-30	EXTSL0_A	Input S6									
J107-3	EXTFH0_A	Input F7									
J107-4	EXTSH0_A	Input S7									

* 1 EXT matrix input supported per quadrant

† Future support

4.3 FR SLOT 2 (ISO-COMM 4/5)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J104-19	(FR)FH4	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 4	Force Hi 4	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 4	Force Hi Input 4
J104-20	(FR)SH4	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 4	Sense Hi 4	AWG 'n'-		Lin Source 'n'-		
J105-19	(FR)FL4	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 4	Force Lo 4	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 4	Force Lo Input 4
J105-20	(FR)SL4	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 4	Sense Lo 4	MEA 'n'-		Ref 1 'n'		
J106-21	(FR)T2FH4	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 4, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J106-22	(FR)T2SH4	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J104-23	(FR)T2FL4	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 4, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J104-24	(FR)T2SL4	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1 'n+1'		
J105-23	(FR)FH5	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 5	Force Hi 5 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 5	Force Hi Input 5
J105-24	(FR)SH5	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 5	Sense Hi 5 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J106-23	(FR)FL5	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 5	Force Lo 5 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 5	Force Lo Input 5
J106-24	(FR)SL5	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 5	Sense Lo 5 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1 'n+2'		
J106-25	(FR)T2FH5	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 5, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J106-26	(FR)T2SH5	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J104-27	(FR)T2FL5	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 5, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J104-28	(FR)T2SL5	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1 'n+3'		
J104-21	(FR)SPR0F4	Input F0	Force Hi Chan 'n+2'			Force Hi 4, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J104-22	(FR)SPR0S4	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J105-21	(FR)SPR1F4	Input F1	Force Hi Chan 'n+3'			Force Lo 4, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J105-22	(FR)SPR1S4	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J105-27	(FR)COMFL5	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J105-28	(FR)COMSL5	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J106-27	(FR)SPR2F4	Input F3	Force Low (6-11)			Force Hi 5, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J106-28	(FR)SPR2S4	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J104-25	(FR)ABUSF5	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 5, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J104-26	(FR)ABUSS5	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J105-25	(FR)BBUSF5	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J105-26	(FR)BBUSS5	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L- HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J104-29	EXTFL0_A	Input F6									
J104-30	EXTSL0_A	Input S6									
J107-3	EXTFH0_A	Input F7									
J107-4	EXTSH0_A	Input S7									

* 1 EXT matrix input supported per quadrant

† Future support

4.4 FR SLOT 3 (ISO-COMM 6/7)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J107-5	(FR)FH6	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 6	Force Hi 6	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 6	Force Hi Input 6
J107-6	(FR)SH6	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 6	Sense Hi 6	AWG 'n'-		Lin Source 'n'-		
J108-5	(FR)FL6	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 6	Force Lo 6	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 6	Force Lo Input 6
J108-6	(FR)SL6	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 6	Sense Lo 6	MEA 'n'-		Ref 1 'n'		
J108-7	(FR)T2FH6	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 6, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J108-8	(FR)T2SH6	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J109-7	(FR)T2FL6	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 6, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J109-8	(FR)T2SL6	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1 'n+1'		
J107-9	(FR)FH7	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 7	Force Hi 7 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 7	Force Hi Input 7
J107-10	(FR)SH7	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 7	Sense Hi 7 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J108-9	(FR)FL7	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 7	Force Lo 7 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 7	Force Lo Input 7
J108-10	(FR)SL7	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 7	Sense Lo 7 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1 'n+2'		
J108-11	(FR)T2FH7	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 7, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J108-12	(FR)T2SH7	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J109-11	(FR)T2FL7	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 7, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J109-12	(FR)T2SL7	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1 'n+3'		
J109-5	(FR)SPR0F6	Input F0	Force Hi Chan 'n+2'			Force Hi 6, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J109-6	(FR)SPR0S6	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J107-7	(FR)SPR1F6	Input F1	Force Hi Chan 'n+3'			Force Lo 6, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J107-8	(FR)SPR1S6	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J108-13	(FR)COMFL7	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J108-14	(FR)COMSL7	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J109-13	(FR)SPR2F6	Input F3	Force Low (6-11)			Force Hi 7, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J109-14	(FR)SPR2S6	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J109-9	(FR)ABUSF7	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 7, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J109-10	(FR)ABUSS7	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J107-11	(FR)BBUSF7	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J107-12	(FR)BBUSS7	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L- HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J104-29	EXTFL0_A	Input F6									
J104-30	EXTSL0_A	Input S6									
J107-3	EXTFH0_A	Input F7									
J107-4	EXTSH0_A	Input S7									

* 1EXT matrix input supported per quadrant

† Future support

4.5 FR SLOT 4 (ISO-COMM 8/9)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J107-13	(FR)FH8	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 8	Force Hi 8	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 8	Force Hi Input 8
J107-14	(FR)SH8	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 8	Sense Hi 8	AWG 'n'-		Lin Source 'n'-		
J107-15	(FR)FL8	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 8	Force Lo 8	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 8	Force Lo Input 8
J107-16	(FR)SL8	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 8	Sense Lo 8	MEA 'n'-		Ref 1'n'		
J107-17	(FR)T2FH8	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 8, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J107-18	(FR)T2SH8	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J108-17	(FR)T2FL8	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 8, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J108-18	(FR)T2SL8	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1'n+1'		
J109-17	(FR)FH9	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 9	Force Hi 9 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 9	Force Hi Input 9
J109-18	(FR)SH9	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 9	Sense Hi 9 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J107-19	(FR)FL9	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 9	Force Lo 9 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 9	Force Lo Input 9
J107-20	(FR)SL9	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 9	Sense Lo 9 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1'n+2'		
J107-21	(FR)T2FH9	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 9, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J107-22	(FR)T2SH9	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J108-21	(FR)T2FL9	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 9, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J108-22	(FR)T2SL9	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1'n+3'		
J108-15	(FR)SPR0F8	Input F0	Force Hi Chan 'n+2'			Force Hi 8, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J108-16	(FR)SPR0S8	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J109-15	(FR)SPR1F8	Input F1	Force Hi Chan 'n+3'			Force Lo 8, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J109-16	(FR)SPR1S8	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J109-21	(FR)COMFL9	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J109-22	(FR)COMSL9	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J109-23	(FR)SPR2F8	Input F3	Force Low (6-11)			Force Hi 9, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J109-24	(FR)SPR2S8	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J108-19	(FR)ABUSF9	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 9, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J108-20	(FR)ABUSS9	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J109-19	(FR)BBUSF9	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J109-20	(FR)BBUSS9	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L- HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J104-29	EXTFL0_A	Input F6									
J104-30	EXTSL0_A	Input S6									
J107-3	EXTFH0_A	Input F7									
J107-4	EXTSH0_A	Input S7									

* 1EXT matrix input supported per quadrant
† Future support

4.6 FR SLOT 5 (ISO-COMM 10/11)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J107-23	(FR)FH10	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 10	Force Hi 10	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 10	Force Hi Input 10
J107-24	(FR)SH10	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 10	Sense Hi 10	AWG 'n'-		Lin Source 'n'-		
J108-23	(FR)FL10	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 10	Force Lo 10	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 10	Force Lo Input 10
J108-24	(FR)SL10	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 10	Sense Lo 10	MEA 'n'-		Ref 1 'n'		
J109-25	(FR)T2FH10	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 10, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J109-26	(FR)T2SH10	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J107-27	(FR)T2FL10	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 10, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J107-28	(FR)T2SL10	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1 'n+1'		
J108-27	(FR)FH11	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 11	Force Hi 11 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 11	Force Hi Input 11
J108-28	(FR)SH11	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 11	Sense Hi 11 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J109-27	(FR)FL11	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 11	Force Lo 11 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 11	Force Lo Input 11
J109-28	(FR)SL11	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 11	Sense Lo 11 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1 'n+2'		
J109-29	(FR)T2FH11	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 11, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J109-30	(FR)T2SH11	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J107-31	(FR)T2FL11	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 11, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J107-32	(FR)T2SL11	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1 'n+3'		
J107-25	(FR)SPR0F10	Input F0	Force Hi Chan 'n+2'			Force Hi 10, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J107-26	(FR)SPR0S10	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J108-25	(FR)SPR1F10	Input F1	Force Hi Chan 'n+3'			Force Lo 10, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J108-26	(FR)SPR1S10	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J108-31	(FR)COMFL11	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J108-32	(FR)COMSL11	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J109-31	(FR)SPR2F10	Input F3	Force Low (6-11)			Force Hi 11, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J109-32	(FR)SPR2S10	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J107-29	(FR)ABUSF11	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 11, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J107-30	(FR)ABUSS11	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J108-29	(FR)BBUSF11	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J108-30	(FR)BBUSS11	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L- HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J104-29	EXTFL0_A	Input F6									
J104-30	EXTSL0_A	Input S6									
J107-3	EXTFH0_A	Input F7									
J107-4	EXTSH0_A	Input S7									

* 1EXT matrix input supported per quadrant
† Future support

4.7 FR SLOT 6 (ISO-COMM 12/13 – Bridged Mode Only)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J204-1	(FR)FH12	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 12	Force Hi 12	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 12	Force Hi Input 12
J204-2	(FR)SH12	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 12	Sense Hi 12	AWG 'n'-		Lin Source 'n'-		
J205-1	(FR)FL12	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 12	Force Lo 12	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 12	Force Lo Input 12
J205-2	(FR)SL12	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 12	Sense Lo 12	MEA 'n'-		Ref 1 'n'		
J205-3	(FR)T2FH12	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 12, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J205-4	(FR)T2SH12	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J206-3	(FR)T2FL12	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 12, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J206-4	(FR)T2SL12	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1 'n+1'		
J204-5	(FR)FH13	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 13	Force Hi 13 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 13	Force Hi Input 13
J204-6	(FR)SH13	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 13	Sense Hi 13 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J205-5	(FR)FL13	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 13	Force Lo 13 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 13	Force Lo Input 13
J205-6	(FR)SL13	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 13	Sense Lo 13 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1 'n+2'		
J205-7	(FR)T2FH13	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 13, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J205-8	(FR)T2SH13	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J206-7	(FR)T2FL13	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 13, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J206-8	(FR)T2SL13	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1 'n+3'		
J206-1	(FR)SPR0F12	Input F0	Force Hi Chan 'n+2'			Force Hi 12, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J206-2	(FR)SPR0S12	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J204-3	(FR)SPR1F12	Input F1	Force Hi Chan 'n+3'			Force Lo 12, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J204-4	(FR)SPR1S12	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J205-9	(FR)COMFL13	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J205-10	(FR)COMSL13	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J206-9	(FR)SPR2F12	Input F3	Force Low (6-11)			Force Hi 13, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J206-10	(FR)SPR2S12	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J206-5	(FR)ABUSF13	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 13, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J206-6	(FR)ABUSS13	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J204-7	(FR)BBUSF13	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J204-8	(FR)BBUSS13	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L-HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J204-29	EXTFL0_B	Input F6									
J204-30	EXTSL0_B	Input S6									
J207-3	EXTFH0_B	Input F7									
J207-4	EXTSH0_B	Input S7									

* 1 EXT matrix input supported per quadrant

† Future support

4.8 FR SLOT 7 (ISO-COMM 14/15 – Bridged Mode Only)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J204-9	(FR)FH14	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 14	Force Hi 14	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 14	Force Hi Input 14
J204-10	(FR)SH14	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 14	Sense Hi 14	AWG 'n'-		Lin Source 'n'-		
J204-11	(FR)FL14	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 14	Force Lo 14	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 14	Force Lo Input 14
J204-12	(FR)SL14	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 14	Sense Lo 14	MEA 'n'-		Ref 1 'n'		
J204-13	(FR)T2FH14	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 14, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J204-14	(FR)T2SH14	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J205-13	(FR)T2FL14	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 14, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J205-14	(FR)T2SL14	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1 'n+1'		
J206-13	(FR)FH15	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 15	Force Hi 15 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 15	Force Hi Input 15
J206-14	(FR)SH15	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 15	Sense Hi 15 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J204-15	(FR)FL15	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 15	Force Lo 15 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 15	Force Lo Input 15
J204-16	(FR)SL15	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 15	Sense Lo 15 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1 'n+2'		
J204-17	(FR)T2FH15	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 15, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J204-18	(FR)T2SH15	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J205-17	(FR)T2FL15	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 15, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J205-18	(FR)T2SL15	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1 'n+3'		
J205-11	(FR)SPR0F14	Input F0	Force Hi Chan 'n+2'			Force Hi 14, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J205-12	(FR)SPR0S14	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J206-11	(FR)SPR1F14	Input F1	Force Hi Chan 'n+3'			Force Lo 14, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J206-12	(FR)SPR1S14	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J206-17	(FR)COMFL15	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J206-18	(FR)COMSL15	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J206-19	(FR)SPR2F14	Input F3	Force Low (6-11)			Force Hi 15, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J206-20	(FR)SPR2S14	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J205-15	(FR)ABUSF15	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 15, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J205-16	(FR)ABUSS15	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J206-15	(FR)BBUSF15	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J206-16	(FR)BBUSS15	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L-HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J204-29	EXTFL0_B	Input F6									
J204-30	EXTSL0_B	Input S6									
J207-3	EXTFH0_B	Input F7									
J207-4	EXTSH0_B	Input S7									

* 1EXT matrix input supported per quadrant

† Future support

4.9 FR SLOT 8 (ISO-COMM 16/17 – Bridged Mode Only)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J204-19	(FR)FH16	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 16	Force Hi 16	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 16	Force Hi Input 16
J204-20	(FR)SH16	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 16	Sense Hi 16	AWG 'n'-		Lin Source 'n'-		
J205-19	(FR)FL16	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 16	Force Lo 16	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 16	Force Lo Input 16
J205-20	(FR)SL16	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 16	Sense Lo 16	MEA 'n'-		Ref 1 'n'		
J206-21	(FR)T2FH16	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 16, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J206-22	(FR)T2SH16	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J204-23	(FR)T2FL16	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 16, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J204-24	(FR)T2SL16	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1 'n+1'		
J205-23	(FR)FH17	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 17	Force Hi 17 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 17	Force Hi Input 17
J205-24	(FR)SH17	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 17	Sense Hi 17 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J206-23	(FR)FL17	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 17	Force Lo 17 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 17	Force Lo Input 17
J206-24	(FR)SL17	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 17	Sense Lo 17 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1 'n+2'		
J206-25	(FR)T2FH17	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 17, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J206-26	(FR)T2SH17	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J204-27	(FR)T2FL17	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 17, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J204-28	(FR)T2SL17	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1 'n+3'		
J204-21	(FR)SPR0F16	Input F0	Force Hi Chan 'n+2'			Force Hi 16, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J204-22	(FR)SPR0S16	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J205-21	(FR)SPR1F16	Input F1	Force Hi Chan 'n+3'			Force Lo 16, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J205-22	(FR)SPR1S16	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J205-27	(FR)COMFL17	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J205-28	(FR)COMSL17	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J206-27	(FR)SPR2F16	Input F3	Force Low (6-11)			Force Hi 17, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J206-28	(FR)SPR2S16	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J204-25	(FR)ABUSF17	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 17, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J204-26	(FR)ABUSS17	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J205-25	(FR)BBUSF17	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J205-26	(FR)BBUSS17	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L-HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J204-29	EXTFL0_B	Input F6									
J204-30	EXTSL0_B	Input S6									
J207-3	EXTFH0_B	Input F7									
J207-4	EXTSH0_B	Input S7									

* 1 EXT matrix input supported per quadrant
† Future support

4.10 FR SLOT 9 (ISO-COMM 18/19 – Bridged Mode Only)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J207-5	(FR)FH18	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 18	Force Hi 18	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 18	Force Hi Input 18
J207-6	(FR)SH18	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 18	Sense Hi 18	AWG 'n'-		Lin Source 'n'-		
J208-5	(FR)FL18	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 18	Force Lo 18	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 18	Force Lo Input 18
J208-6	(FR)SL18	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 18	Sense Lo 18	MEA 'n'-		Ref 1 'n'		
J208-7	(FR)T2FH18	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 18, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J208-8	(FR)T2SH18	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J209-7	(FR)T2FL18	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 18, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J209-8	(FR)T2SL18	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1 'n+1'		
J207-9	(FR)FH19	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 19	Force Hi 19 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 19	Force Hi Input 19
J207-10	(FR)SH19	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 19	Sense Hi 19 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J208-9	(FR)FL19	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 19	Force Lo 19 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 19	Force Lo Input 19
J208-10	(FR)SL19	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 19	Sense Lo 19 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1 'n+2'		
J208-11	(FR)T2FH19	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 19, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J208-12	(FR)T2SH19	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J209-11	(FR)T2FL19	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 19, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J209-12	(FR)T2SL19	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1 'n+3'		
J209-5	(FR)SPR0F18	Input F0	Force Hi Chan 'n+2'			Force Hi 18, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J209-6	(FR)SPR0S18	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J207-7	(FR)SPR1F18	Input F1	Force Hi Chan 'n+3'			Force Lo 18, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J207-8	(FR)SPR1S18	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J208-13	(FR)COMFL19	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J208-14	(FR)COMSL19	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J209-13	(FR)SPR2F18	Input F3	Force Low (6-11)			Force Hi 19, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J209-14	(FR)SPR2S18	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J209-9	(FR)ABUSF19	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 19, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J209-10	(FR)ABUSS19	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J207-11	(FR)BBUSF19	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J207-12	(FR)BBUSS19	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L-HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J204-29	EXTFL0_B	Input F6									
J204-30	EXTSL0_B	Input S6									
J207-3	EXTFH0_B	Input F7									
J207-4	EXTSH0_B	Input S7									

* 1 EXT matrix input supported per quadrant

† Future support

4.11 FR SLOT 10 (ISO-COMM 20/21 – Bridged Mode Only)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J207-23	(FR)FH20	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 20	Force Hi 20	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 20	Force Hi Input 20
J207-24	(FR)SH20	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 20	Sense Hi 20	AWG 'n'-		Lin Source 'n'-		
J208-23	(FR)FL20	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 20	Force Lo 20	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 20	Force Lo Input 20
J208-24	(FR)SL20	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 20	Sense Lo 20	MEA 'n'-		Ref 1'n'		
J209-25	(FR)T2FH20	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 20, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J209-26	(FR)T2SH20	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J207-27	(FR)T2FL20	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 20, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J207-28	(FR)T2SL20	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1'n+1'		
J208-27	(FR)FH21	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 21	Force Hi 21 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+	Force Hi 21	Force Hi Input 21
J208-28	(FR)SH21	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 21	Sense Hi 21 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J209-27	(FR)FL21	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 21	Force Lo 21 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'	Force Lo 21	Force Lo Input 21
J209-28	(FR)SL21	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 21	Sense Lo 21 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1'n+2'		
J209-29	(FR)T2FH21	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 21, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+		
J209-30	(FR)T2SH21	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J207-31	(FR)T2FL21	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 21, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'		
J207-32	(FR)T2SL21	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1'n+3'		
J207-25	(FR)SPR0F20	Input F0	Force Hi Chan 'n+2'			Force Hi 20, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J207-26	(FR)SPR0S20	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J208-25	(FR)SPR1F20	Input F1	Force Hi Chan 'n+3'			Force Lo 20, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J208-26	(FR)SPR1S20	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J208-31	(FR)COMFL21	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J208-32	(FR)COMSL21	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J209-31	(FR)SPR2F20	Input F3	Force Low (6-11)			Force Hi 21, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J209-32	(FR)SPR2S20	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J207-29	(FR)ABUSF21	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 21, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J207-30	(FR)ABUSS21	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J208-29	(FR)BBUSF21	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J208-30	(FR)BBUSS21	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L-HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J204-29	EXTFL0_B	Input F6									
J204-30	EXTSL0_B	Input S6									
J207-3	EXTFH0_B	Input F7									
J207-4	EXTSH0_B	Input S7									

* 1EXT matrix input supported per quadrant
† Future support

4.12 FR SLOT 11 (ISO-COMM 22/23 – Bridged Mode Only)

App. Board Pin #	Pin Name	8x8 Matrix*	APU-12	APU or APU-10	FSS / SPU-100	MPU (Low I), HPU, or SPU-112 ‡	QHSU	QMS	QPLU†	SPU-500 or SPU-250	WCU
J207-23	(FR)FH22	Output F0	Force Hi Chan 'n'	Force Hi Chan 'n'	Force Hi 22	Force Hi 22	AWG 'n'+	Hi Side Chan 'n'	Lin Source 'n'+	Force Hi 22	Force Hi Input 22
J207-24	(FR)SH22	Output S0	Sense Hi Chan 'n'	Sense Hi Chan 'n'	Sense Hi 22	Sense Hi 22	AWG 'n'-		Lin Source 'n'-		
J208-23	(FR)FL22	Output F1	Force Hi Chan 'n+1'	Force Hi Chan 'n+1'	Force Lo 22	Force Lo 22	MEA 'n'+	Lo Side Chan 'n'	Ref 0 'n'	Force Lo 22	Force Lo Input 22
J208-24	(FR)SL22	Output S1	Sense Hi Chan 'n+1'	Sense Hi Chan 'n+1'	Sense Lo 22	Sense Lo 22	MEA 'n'-		Ref 1'n'		
J209-25	(FR)T2FH22	Output F2	Force Hi Chan 'n+4'	Force Hi Chan 'n+2'		Force Hi 22, HC2 *(SPU-112 ONLY)*	AWG 'n+1'+	Hi Side Chan 'n+1'	Lin Source 'n+1'+		
J209-26	(FR)T2SH22	Output S2	Sense Hi Chan 'n+4'	Sense Hi Chan 'n+2'			AWG 'n+1'-		Lin Source 'n+1'-		
J207-27	(FR)T2FL22	Output F3	Force Hi Chan 'n+5'	Force Hi Chan 'n+3'		Force Lo 22, HC2 *(SPU-112 ONLY)*	MEA 'n+1'+	Lo Side Chan 'n+1'	Ref 0 'n+1'		
J207-28	(FR)T2SL22	Output S3	Sense Hi Chan 'n+5'	Sense Hi Chan 'n+3'			MEA 'n+1'-		Ref 1'n+1'		
J208-27	(FR)FH23	Output F4	Force Hi Chan 'n+6'	Force Hi Chan 'n+4'	Force Hi 23	Force Hi 23 *(SPU-112 ONLY)*	AWG 'n+2'+	Hi Side Chan 'n+2'	Lin Source 'n+2'+		
J208-28	(FR)SH23	Output S4	Sense Hi Chan 'n+6'	Sense Hi Chan 'n+4'	Sense Hi 23	Sense Hi 23 *(SPU-112 ONLY)*	AWG 'n+2'-		Lin Source 'n+2'-		
J209-27	(FR)FL23	Output F5	Force Hi Chan 'n+7'	Force Hi Chan 'n+5'	Force Lo 23	Force Lo 23 *(SPU-112 ONLY)*	MEA 'n+2'+	Lo Side Chan 'n+2'	Ref 0 'n+2'		
J209-28	(FR)SL23	Output S5	Sense Hi Chan 'n+7'	Sense Hi Chan 'n+5'	Sense Lo 23	Sense Lo 23 *(SPU-112 ONLY)*	MEA 'n+2'-		Ref 1'n+2'		
J209-29	(FR)T2FH23	Output F6	Force Hi Chan 'n+10'	Force Hi Chan 'n+6'		Force Hi 23, HC2 *(SPU-112 ONLY)*	AWG 'n+3'+	Hi Side Chan 'n+3'	Lin Source 'n+3'+	Force Hi 23	Force Hi Input 23
J209-30	(FR)T2SH23	Output S6	Sense Hi Chan 'n+10'	Sense Hi Chan 'n+6'			AWG 'n'+3-		Lin Source 'n+3'-		
J207-31	(FR)T2FL23	Output F7	Force Hi Chan 'n+11'	Force Hi Chan 'n+7'		Force Lo 23, HC2 *(SPU-112 ONLY)*	MEA 'n+3'+	Lo Side Chan 'n'+3	Ref 0 'n+3'	Force Lo 23	Force Lo Input 23
J207-32	(FR)T2SL23	Output S7	Sense Hi Chan 'n+11'	Sense Hi Chan 'n+7'			MEA 'n+3'-		Ref 1'n+3'		
J207-25	(FR)SPR0F22	Input F0	Force Hi Chan 'n+2'			Force Hi 22, HC1 *(SPU-112 ONLY)*			Gnd Force 'n'		
J207-26	(FR)SPR0S22	Input S0	Sense Hi Chan 'n+2'						Gnd Sense 'n'		
J208-25	(FR)SPR1F22	Input F1	Force Hi Chan 'n+3'			Force Lo 22, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+1'		
J208-26	(FR)SPR1S22	Input S1	Sense Hi Chan 'n+3'						Gnd Sense 'n+1'		
J208-31	(FR)COMFL23	Input F2	Force Low (0-5)	Force Low (Bd)					Reserved - Do Not Connect		
J208-32	(FR)COMSL23	Input S2	Sense Low (0-5)	Sense Low (Bd)					Reserved - Do Not Connect		
J209-31	(FR)SPR2F22	Input F3	Force Low (6-11)			Force Hi 23, HC1 *(SPU-112 ONLY)*			Reserved - Do Not Connect		
J209-32	(FR)SPR2S22	Input S3	Sense Low (6-11)						Reserved - Do Not Connect		
J207-29	(FR)ABUSF23	Input F4	Force Hi Chan 'n+8'	Bus A Force		Force Lo 23, HC1 *(SPU-112 ONLY)*			Gnd Force 'n+2'		
J207-30	(FR)ABUSS23	Input S4	Sense Hi Chan 'n+8'	Bus A Sense					Gnd Sense 'n+2'		
J208-29	(FR)BBUSF23	Input F5	Force Hi Chan 'n+9'	Bus B Force					Gnd Force 'n+3'		
J208-30	(FR)BBUSS23	Input S5	Sense Hi Chan 'n+9'	Bus B Sense					Gnd Sense 'n+3'		
8x8 EXT Inputs						‡ Connect FH/L+FH/L-HC1&2 on DIB for each channel for hardware protection in >4A applications. See the SPU-112 DIB Design Guidelines for details.					
J204-29	EXTFL0_B	Input F6									
J204-30	EXTSL0_B	Input S6									
J207-3	EXTFH0_B	Input F7									
J207-4	EXTSH0_B	Input S7									

4.13 High Current Connections

				Bridged Mode Only			
App. Board Pin #	Pin Name	App. Board Pin #	Pin Name	App. Board Pin #	Pin Name	App. Board Pin #	Pin Name
J107.1	HPUFH_A	J104.31	HPUFL_A	J207.1	HPUFH_B	J204.31	HPUFL_B
J107.2	HPUFH_A	J104.32	HPUFL_A	J207.2	HPUFH_B	J204.32	HPUFL_B
J108.1	HPUFH_A	J105.30	HPUFL_A	J208.1	HPUFH_B	J205.30	HPUFL_B
J108.2	HPUFH_A	J105.31	HPUFL_A	J208.2	HPUFH_B	J205.31	HPUFL_B
J108.4	HPUFH_A	J105.32	HPUFL_A	J208.4	HPUFH_B	J205.32	HPUFL_B
J109.1	HPUFH_A	J106.29	HPUFL_A	J209.1	HPUFH_B	J206.29	HPUFL_B
J109.2	HPUFH_A	J106.30	HPUFL_A	J209.2	HPUFH_B	J206.30	HPUFL_B
J109.3	HPUFH_A	J106.31	HPUFL_A	J209.3	HPUFH_B	J206.31	HPUFL_B
J109.4	HPUFH_A	J106.32	HPUFL_A	J209.4	HPUFH_B	J206.32	HPUFL_B
J108.3	HPUSH_A	J105.29	HPUSL_A	J208.3	HPUSH_B	J205.29	HPUSL_B

4.14 Inputs and Outputs

App. Board Pin #	IO#	App. Board Pin #	IO#	Bridged Mode Only: App. Board Pin #	IO#	App. Board Pin #	IO#
J102-39	GND_A	J100-62	IO16	J202-39	GND_B	J200-62	IO48
J102-38	IO0	J100-61	GND_A	J202-38	IO32	J200-61	GND_B
J102-37	GND_A	J101-29	IO17	J202-37	GND_B	J201-29	IO49
J103-6	GND_A	J101-30	GND_A	J203-6	GND_B	J201-30	GND_B
J103-5	IO1	J100-30	GND_A	J203-5	IO33	J200-30	GND_B
J103-4	GND_A	J100-29	IO18	J203-4	GND_B	J200-29	IO50
J102-6	GND_A	J100-28	GND_A	J202-6	GND_B	J200-28	GND_B
J102-5	IO2	J101-60	IO19	J202-5	IO34	J201-60	IO51
J102-4	GND_A	J101-59	GND_A	J202-4	GND_B	J201-59	GND_B
J103-37	GND_A	J100-60	IO20	J203-37	GND_B	J200-60	IO52
J103-36	IO3	J100-59	GND_A	J203-36	IO35	J200-59	GND_B
J103-35	GND_A	J101-27	IO21	J203-35	GND_B	J201-27	IO53
J102-36	IO4	J101-28	GND_A	J202-36	IO36	J201-28	GND_B
J102-35	GND_A	J100-27	IO22	J202-35	GND_B	J200-27	IO54
J103-3	IO5	J100-26	GND_A	J203-3	IO37	J200-26	GND_B
J103-2	GND_A	J101-58	IO23	J203-2	GND_B	J201-58	IO55
J102-3	IO6	J101-57	GND_A	J202-3	IO38	J201-57	GND_B
J102-2	GND_A	J100-58	IO24	J202-2	GND_B	J200-58	IO56
J103-34	IO7	J100-57	GND_A	J203-34	IO39	J200-57	GND_B
J103-33	GND_A	J101-25	IO25	J203-33	GND_B	J201-25	IO57
J102-34	IO8	J101-24	GND_A	J202-34	IO40	J201-24	GND_B
J102-33	GND_A	J100-25	IO26	J202-33	GND_B	J200-25	IO58
J103-1	IO9	J100-24	GND_A	J203-1	IO41	J200-24	GND_B
J102-1	IO10	J101-56	IO27	J202-1	IO42	J201-56	IO59
J101-64	IO11	J101-55	GND_A	J201-64	IO43	J201-55	GND_B
J101-63	GND_A	J100-56	IO28	J201-63	GND_B	J200-56	IO60
J100-64	IO12	J100-55	GND_A	J200-64	IO44	J200-55	GND_B
J100-63	GND_A	J101-23	IO29	J200-63	GND_B	J201-23	IO61
J101-31	IO13	J101-22	GND_A	J201-31	IO45	J201-22	GND_B
J101-32	GND_A	J100-23	IO30	J201-32	GND_B	J200-23	IO62
J100-31	IO14	J100-22	GND_A	J200-31	IO46	J200-22	GND_B
J100-32	GND_A	J101-54	IO31	J200-32	GND_B	J201-54	IO63
J101-62	IO15			J201-62	IO47		
J101-61	GND_A			J201-61	GND_B		

NOTE: When using the Bridged Mode ETS recommends connecting GND_A to GND_B

4.15 Programmable Control Bits (C-Bits)

				Bridged Mode Only			
App. Board Pin #	CBIT#	App. Board Pin #	CBIT#	App. Board Pin #	CBIT#	App. Board Pin #	CBIT#
J100-53	CBIT0	J100-49	CBIT16	J200-53	CBIT32	J200-49	CBIT48
J100-21	CBIT1	J100-17	CBIT17	J200-21	CBIT33	J200-17	CBIT49
J101-53	CBIT2	J101-49	CBIT18	J201-53	CBIT34	J201-49	CBIT50
J101-21	CBIT3	J101-17	CBIT19	J201-21	CBIT35	J201-17	CBIT51
J100-52	CBIT4	J100-48	CBIT20	J200-52	CBIT36	J200-48	CBIT52
J100-20	CBIT5	J100-16	CBIT21	J200-20	CBIT37	J200-16	CBIT53
J101-52	CBIT6	J101-48	CBIT22	J201-52	CBIT38	J201-48	CBIT54
J101-20	CBIT7	J101-16	CBIT23	J201-20	CBIT39	J201-16	CBIT55
J100-51	CBIT8	J100-47	CBIT24	J200-51	CBIT40	J200-47	CBIT56
J100-19	CBIT9	J100-15	CBIT25	J200-19	CBIT41	J200-15	CBIT57
J101-51	CBIT10	J101-47	CBIT26	J201-51	CBIT42	J201-47	CBIT58
J101-19	CBIT11	J101-15	CBIT27	J201-19	CBIT43	J201-15	CBIT59
J100-50	CBIT12	J100-46	CBIT28	J200-50	CBIT44	J200-46	CBIT60
J100-18	CBIT13	J100-14	CBIT29	J200-18	CBIT45	J200-14	CBIT61
J101-50	CBIT14	J101-46	CBIT30	J201-50	CBIT46	J201-46	CBIT62
J101-18	CBIT15	J101-14	CBIT31	J201-18	CBIT47	J201-14	CBIT63

4.16 DC Power

App. Board Pin #	Pin Name	Bridged Mode Only App. Board Pin #	Bridged Mode Only Pin Name
J100-5	GND_A	J200-5	GND_B
J100-6	+12V_A	J200-6	+12V_B
J100-7	GND_A	J200-7	GND_B
J100-8	+24V_A	J200-8	+24V_B
J100-9	+5V_A	J200-9	+5V_B
J100-10	GND_A	J200-10	GND_B
J100-11	-15V_A	J200-11	-15V_B
J100-38	+12V_A	J200-38	+12V_B
J100-39	-5.2V_A	J200-39	-5.2V_B
J100-40	+24V_A	J200-40	+24V_B
J100-41	+5V_A	J200-41	+5V_B
J100-42	GND_A	J200-42	GND_B
J100-43	-15V_A	J200-43	-15V_B
J101-5	GND_A	J201-5	GND_B
J101-6	-12V_A	J201-6	-12V_B
J101-7	GND_A	J201-7	GND_B
J101-8	-24V_A	J201-8	-24V_B
J101-9	+5V_A	J201-9	+5V_B
J101-10	GND_A	J201-10	GND_B
J101-11	+15V_A	J201-11	+15V_B
J101-37	GND_A	J201-37	GND_B
J101-38	-12V_A	J201-38	-12V_B
J101-39	GND_A	J201-39	GND_B
J101-40	-24V_A	J201-40	-24V_B
J101-41	+5V_A	J201-41	+5V_B
J101-42	GND_A	J201-42	GND_B
J101-43	+15V_A	J201-43	+15V_B

NOTE: When using the Bridged Mode, ETS recommends connecting GND_A to GND_B

NOTE: When using the Bridged Mode, ETS recommends connecting A power supplies to B power supplies (Ex. Connect +5V_A to +5V_B)

4.17 QTMU Direct Connections

				Bridged Mode Only			
App. Board Pin #	Pin Name	App. Board Pin #	Pin Name	App. Board Pin #	Pin Name	App. Board Pin #	Pin Name
J103-30	DIRECTARMIN(0)	J102-30	DIRECTARMOUT(0)	J203-30	DIRECTARMIN(4)	J202-30	DIRECTARMOUT(4)
J103-29	DIRECTA(0)	J102-29	DIRECTB(0)	J203-29	DIRECTA(4)	J202-29	DIRECTB(4)
J103-62	DIRECTARMIN(1)	J102-62	DIRECTARMOUT(1)	J203-62	DIRECTARMIN(5)	J202-62	DIRECTARMOUT(5)
J103-61	DIRECTA(1)	J102-61	DIRECTB(1)	J203-61	DIRECTA(5)	J202-61	DIRECTB(5)
J103-28	DIRECTARMIN(2)	J102-28	DIRECTARMOUT(2)	J203-28	DIRECTARMIN(6)	J202-28	DIRECTARMOUT(6)
J103-27	DIRECTA(2)	J102-27	DIRECTB(2)	J203-27	DIRECTA(6)	J202-27	DIRECTB(6)
J103-60	DIRECTARMIN(3)	J102-60	DIRECTARMOUT(3)	J203-60	DIRECTARMIN(7)	J202-60	DIRECTARMOUT(7)
J103-59	DIRECTA(3)	J102-59	DIRECTB(3)	J203-59	DIRECTA(7)	J202-59	DIRECTB(7)

4.18 Clocks and I2C Interlocks

App. Board Pin #	Pin Name	App. Board Pin #	Pin Name	
J103-11	GND_A	J103-12	I2CENB_A	
J103-10	DUTCLK0_A	J103-45	I2CDATA_A	
J102-11	GND_A	J102-45	I2CCLK_A	
J102-10	DUTCLK1_A	J102-12	DUT_RESET_A	
J103-9	DUTCLK2_A	J102-43	DUTDPUSTART_A	
J103-42	GND_A	J103-44	DUTSYNC_A	
J103-41	DUTCLK3_A	J103-43	DUTCLKIN_A	
J102-9	DUTCLK4_A			
J102-42	GND_A	J103-64	PWRINTLK1_A	Must be connected to PWRINTLK0_A for DC power to work
J102-41	DUTCLK5_A	J101-1	PWRINTLK0_A	
J103-40	DUTCLK6_A			
J102-40	DUTCLK7_A	J102-32	APPINTLK1_A	Must be connected to SIGINTLK0_A to satisfy INTERLOCK
J102-8	GND_A	J100-1	SIGINTLK0_A	
J102-7	DUTCLKA_A			
J103-39	GND_A	J102-64	CAL_DIAGINTLK1_A	
J103-38	DUTCLKB_A			
J103-8	GND_A	J101-3	BRIDGED_DUT	**Bridged Mode Only**
J103-7	DUTFREECLK_A	J201-3	BRIDGED_DUT_1	Must be connected to BRIDGED_DUT in Bridged Mode
J203-11	GND_B	J203-12	I2CENB_B	
J203-10	DUTCLK0_B	J203-45	I2CDATA_B	
J202-11	GND_B	J202-45	I2CCLK_B	
J202-10	DUTCLK1_B	J202-12	DUT_RESET_B	
J203-9	DUTCLK2_B	J202-43	DUTDPUSTART_B	
J203-42	GND_B	J203-44	DUTSYNC_B	
J203-41	DUTCLK3_B	J203-43	DUTCLKIN_B	
J202-9	DUTCLK4_B			
J202-42	GND_B	J203-64	PWRINTLK1_B	Must be connected to PWRINTLK0_B for DC power to work
J202-41	DUTCLK5_B	J201-1	PWRINTLK0_B	
J203-40	DUTCLK6_B			
J202-40	DUTCLK7_B	J202-32	APPINTLK1_B	Must be connected to SIGINTLK0_B to satisfy INTERLOCK
J202-8	GND_B	J200-1	SIGINTLK0_B	
J202-7	DUTCLKA_B			
J203-39	GND_B	J202-64	CAL_DIAGINTLK1_B	
J203-38	DUTCLKB_B			
J203-8	GND_B			
J203-7	DUTFREECLK_B			

4.19 Isolated Communications

				Bridged Mode Only			
App. Board Pin #	Pin Name	App. Board Pin #	Pin Name	App. Board Pin #	Pin Name	App. Board Pin #	Pin Name
J102-14	TX+40	J102-20	TX+42	J202-14	TX+44	J202-20	TX+46
J102-15	TX-40	J102-21	TX-42	J202-15	TX-44	J202-21	TX-46
J102-16	GND_A	J102-22	GND_A	J202-16	GND_B	J202-22	GND_B
J102-47	RS+40	J102-53	RS+42	J202-47	RS+44	J202-53	RS+46
J102-48	RS-40	J102-54	RS-42	J202-48	RS-44	J202-54	RS-46
J102-49	GND_A	J102-55	GND_A	J202-49	GND_B	J202-55	GND_B
J103-15	RX+40	J103-21	RX+42	J203-15	RX+44	J203-21	RX+46
J103-16	RX-40	J103-22	RX-42	J203-16	RX-44	J203-22	RX-46
J103-17	GND_A	J103-23	GND_A	J203-17	GND_B	J203-23	GND_B
J102-17	TX+41	J102-23	TX+43	J202-17	TX+45	J202-23	TX+47
J102-18	TX-41	J102-24	TX-43	J202-18	TX-45	J202-24	TX-47
J102-19	GND_A	J102-25	CGND	J202-19	GND_B	J202-25	CGND
J102-50	RS+41	J102-56	RS+43	J202-50	RS+45	J202-56	RS+47
J102-51	RS-41	J102-57	RS-43	J202-51	RS-45	J202-57	RS-47
J102-52	GND_A	J102-58	GND_A	J202-52	GND_B	J202-58	GND_B
J103-18	RX+41	J103-24	RX+43	J203-18	RX+45	J203-24	RX+47
J103-19	RX-41	J103-25	RX-43	J203-19	RX-45	J203-25	RX-47
J103-20	GND_A	J103-26	GND_A	J203-20	GND_B	J203-26	GND_B

4.20 EEPROM Wiring and Use

The EEPROM provides data storage for board ID, serial number, etc. on an ETS application board. These EEPROM devices are included with a purchased application board. The software coding necessary to communicate with EEPROM devices is shown in the program coding example on the following pages.

If you want to place an EEPROM device on the DUT Adapter Board (DAB), Wafer Probe, or on other associated hardware, wire the I2C CLK and I2C DATA lines to a user-supplied EEPROM IC.

The I2C CLK line connects to pin 6 of the EEPROM and the I2C DATA line connects to pin 5 of the EEPROM. On the ASM5111 application board, the I2C CLK signal is at pin H102-45, and the I2C DATA signal is at pin H103-45.

Up to four EEPROMs can be connected to the I2C serial bus. This bus consists of two lines - I2C CLK and I2C DATA. Each EEPROM on the bus must have its own unique hard-wired address per the table below. The ETS application board ID functions use the 'Key' parameter in this table to select the desired EEPROM:

Key	Board Description	A2(pin 3)	A1(pin 2)
0	Application	Ground	Ground
1	DUT Adapter/Wafer Probe	Pull-up	Pull-up
2	MCB (ETS internal use only)	N/A	N/A
3	User Defined	Pull-up	Ground

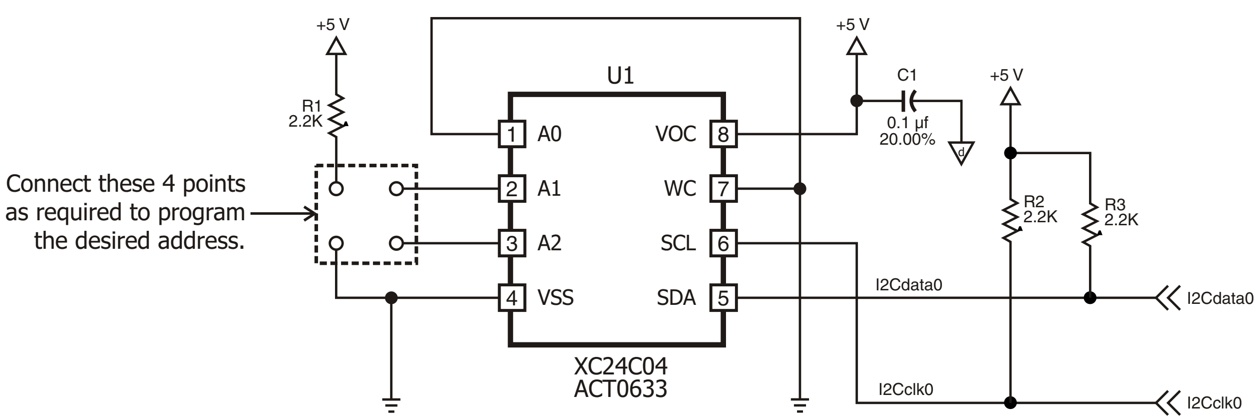


Figure 4-4 – Typical EEPROM Wiring for ETS-88™ Testers

4.20.1 Writing and Reading the EEPROM

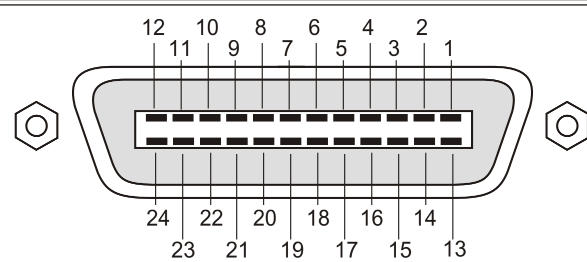
Here is an example of how you can write to and read from an application board EEPROM:

```
// In *.h file

//{{AFX_DEFINE_ACE_GROUP(App Bd ID)
#define PERFORMANCE_BD   0
#define DUT_ADAPTER_BD   1
// Example EEPROM strings
#define PERF_BRD_ID   "ID Rev: 1.0, ETS 300"
#define DAB_ID"ID Rev: 1.0, Voltage Regulator, switcher, Devices: AX2555, Sites: 2"
//}}AFX_DEFINE_ACE_GROUP


// In Test Program

BOOL CAX2555 ::UserInit( void )
{
BOOLEAN status = SUCCESS;
char buff[_MAX_PATH];

    /* Change the '0' to '1' to write to the Performance Board and DUT Adapter Board
    /* EEPROMs, then change it back to '0'.  If this section fails to properly write
    /* to either EEPROM, the test program will abort, and exit the TestExecutive.
     */
#if 0
    if  ( write_board_id( PERFORMANCE_BD, 0, PERF_BRD_ID ) == FAILURE ) return FALSE;
    if  ( write_board_id( DUT_ADAPTER_BD, 0, DAB_ID ) == FAILURE ) return FALSE;
#endif

    /* Check that the correct DUT-boards are installed before bothering
     * with anything else.
     */
    if  ( read_board_id( PERFORMANCE_BD, 0, buff ) == FAILURE ||
        stricmp( buff, PERF_BRD_ID ) != SUCCESS )
    {
    /* If Performance Board EEPROM data is wrong, say so,
     * wait for user acknowledgement then check DAB EEPROM.
     */
        etsMessageBox("\nERROR: App Board ID does not match this test program!\n"
                "\nExpected board ID to contain the string:"
                "\n\"" PERF_BRD_ID "\"\n", MB_OK | MB_ICONSTOP);

        status = FAILURE;
    }
    if  ( read_board_id( DUT_ADAPTER_BD, 0, buff ) == FAILURE ||
        stricmp( buff, DAB_ID ) != SUCCESS )
    {
    /* If DAB EEPROM data is wrong, say so, wait for user acknowledgement then exit.
     */
        etsMessageBox("\nERROR: App Board ID does not match this test program!\n"
                "\nExpected board ID to contain the string:"
                "\n\"" DAB_ID "\"\n", MB_OK | MB_ICONSTOP);
        status = FAILURE;
    }
    if( status == FAILURE )  return FALSE;
    // Abort Run of ax2555.DLL and exit TestExecutive
// ======================================================================
```

Appendix A – Connectors and Pinouts

A.1 User Interface Panel Connectors

A.1.1 IEEE-488 Connector



(Viewed from Back Panel)

PIN NUMBER	FUNCTION	PIN NUMBER	FUNCTION
1	DIO 1	13	DIO 5
2	DIO 2	14	DIO 6
3	DIO 3	15	DIO 7
4	DIO 4	16	DIO 8
5	EOI	17	REN
6	DAV	18	GND
7	NRFD	19	GND
8	NDAC	20	GND
9	JFC	21	GND
10	SRQ	22	GND
11	ATN	23	GND
12	GND	24	GND

Figure A-1 – IEEE-488/GPIB Connector Pinout

A.1.2 Multisite Handler Interface Connections

The MultiSite Handler (MSH) Interface implements a programmable "TTL-style" parallel interface explicitly for the operation of multisite handlers and probers. The Interface supports 32 programmable I/O lines, and eight lines dedicated to the capture of SOT signals.

When configured as outputs, the I/O lines are "open-drain" structured outputs. When configured as inputs, the I/O lines are level-sensitive. The SOT lines are always configured as inputs, but can be programmed for level-sensitive or edge-sensitive operation. All lines are "weakly" pulled-up to +5 V (through ~5 kΩ resistors, see the block diagram in Figure A-3 on the following page).

There are two versions of Eagle Test Systems' MSH Interface. In ETS-88™ systems, the electronics of the MSH Interface are integrated into the Test Head Control Board (TCB), and the 50 pin Centronics port on this board connects to the handling equipment via a handler/prober cable. Figure A-2 shows the pinout of this handler/prober cable.

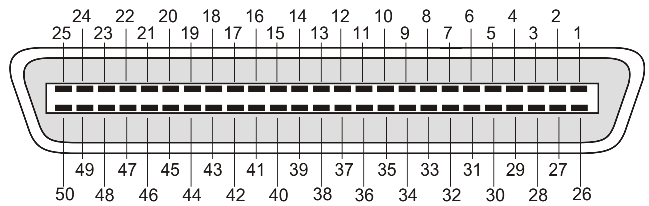


Figure A-2 – The MSH Interface Connector

Pin Number	Function	Pin Number	Function
1	I/O 0	26	I/O 1
2	I/O 2	27	I/O 3
3	I/O 4	28	I/O 5
4	I/O 6	29	I/O 7
5	I/O 8	30	I/O 9
6	I/O 10	31	I/O 11
7	I/O 12	32	I/O 13
8	I/O 14	33	I/O 15
9	SOT 0	34	SOT 1
10	SOT 2	35	SOT 3
11	+5 V (TSTR)	36	+5 V (TSTR)
12	GND	37	GND
13	N/C	38	N/C
14	I/O 16	39	I/O 17
15	I/O 18	40	I/O 19
16	I/O 20	41	I/O 21
17	I/O 22	42	I/O 23
18	I/O 24	43	I/O 25
19	I/O 26	44	I/O 27
20	I/O 28	45	I/O 29
21	I/O 30	46	I/O 31
22	SOT 4	47	SOT 5
23	SOT 6	48	SOT 7
24	+5 V (TSTR)	49	+5 V (TSTR)
25	GND	50	GND

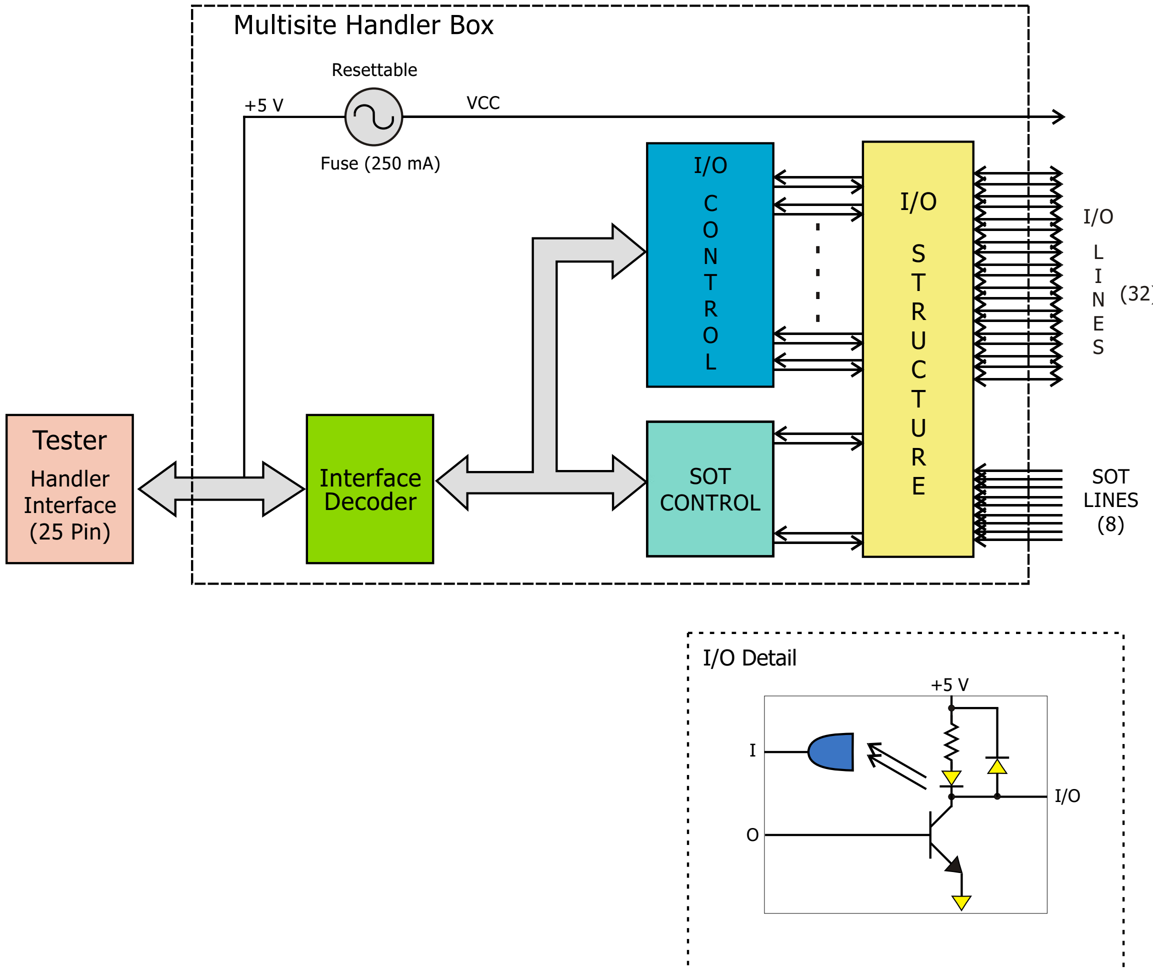


Figure A-3 – MSH Interface Block Diagram

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Appendix B – config.ets

One of the most important configuration files stored on the test system is the \ets\bin\config.ets file. The config.ets file contains the following:

- The mapping of the ISO-COMM boards to the actual FR cage positions being used,
- Resource pin assignments,
- Resource declarations that tell diagnostics where boards should be present in the system,
- Scheduled self-calibration time intervals.

You should only need to edit this file after purchasing a new ETS resource. If you need to change this file, please consult with the Applications department at Eagle before making any modifications.

Below is a sample config.ets file.

```
Test head address                         \0xD000
//////////////////////////////////////////////////////////////////////////////
//          Valid Test Head Types:                                          //
//                 500D, 564, 300, 200, BT2000, 600, 364, 88                //
//////////////////////////////////////////////////////////////////////////////
Test head type:                           \88

Tester Maximum Voltage         \maxv     \1000

//////////////////////////////////////////////////////////////////////////////
// NOTE:                                                                    //
//   All examples use the forward slash '/' instead of the other slash.     //
//   This is so that backward compatibility is maintained.                  //
//                                                                          //
//   If you copy the example, please change the forward slash '/' to the    //
//   other slash.                                                           //
//                                                                          //
//////////////////////////////////////////////////////////////////////////////
//                                                                          //
//                    Iso-comm Position Mapping Syntax:                     //
//                                                                          //
//   Place the logical position number of the floating resource in the      //
//   'pos' field of the line corresponding to the actual Iso-comm channel.  //
//   If a line is missing or the 'pos' field is left blank, the position    //
//   defaults to direct mapping.                                            //
//                                                                          //
//   Example:                                                               //
```

```
//                                                                            //
//        Iso-comm Channel #0            /icom0    /pos<num>                  //
//                                                                            //
//        where: <num> is the logical icom position, 0 to 255                 //
//                                                                            //
////////////////////////////////////////////////////////////////////////////////

////////////////////////////////////////////////////////////////////////////////
//                                                                            //
//                        TH-1A Isocomm Card Cage...                          //
//                                                                            //
////////////////////////////////////////////////////////////////////////////////
 Iso-comm Channel #0            \icom0    \pos0
 Iso-comm Channel #1            \icom1    \pos1
 Iso-comm Channel #2            \icom2    \pos2
 Iso-comm Channel #3            \icom3    \pos3
 Iso-comm Channel #4            \icom4    \pos4
 Iso-comm Channel #5            \icom5    \pos5
 Iso-comm Channel #6            \icom6    \pos6
 Iso-comm Channel #7            \icom7    \pos7
 Iso-comm Channel #8            \icom8    \pos8
 Iso-comm Channel #9            \icom9    \pos9
 Iso-comm Channel #10           \icom10   \pos10
 Iso-comm Channel #11           \icom11   \pos11

 Iso-comm Channel #12           \icom12   \pos80
 Iso-comm Channel #13           \icom13   \pos81
 Iso-comm Channel #14           \icom14   \pos82
 Iso-comm Channel #15           \icom15   \pos83

 Iso-comm Channel #16           \icom16   \pos88
 Iso-comm Channel #17           \icom17   \pos89
 Iso-comm Channel #18           \icom18   \pos90
 Iso-comm Channel #19           \icom19   \pos91

 Iso-comm Channel #20           \icom20   \pos92
 Iso-comm Channel #21           \icom21   \pos93
 Iso-comm Channel #22           \icom22   \pos94
 Iso-comm Channel #23           \icom23   \pos95
 Iso-comm Channel #24           \icom24   \pos96
 Iso-comm Channel #25           \icom25   \pos97
 Iso-comm Channel #26           \icom26   \pos98
 Iso-comm Channel #27           \icom27   \pos99


////////////////////////////////////////////////////////////////////////////////
//                                                                            //
//                        TH-1B Isocomm Card Cage...                          //
```

```
//                                                                          //
//////////////////////////////////////////////////////////////////////////////
 Iso-comm Channel #64           \icom64   \pos12
 Iso-comm Channel #65           \icom65   \pos13
 Iso-comm Channel #66           \icom66   \pos14
 Iso-comm Channel #67           \icom67   \pos15
 Iso-comm Channel #68           \icom68   \pos16
 Iso-comm Channel #69           \icom69   \pos17
 Iso-comm Channel #70           \icom70   \pos18
 Iso-comm Channel #71           \icom71   \pos19
 Iso-comm Channel #72           \icom72   \pos20
 Iso-comm Channel #73           \icom73   \pos21
 Iso-comm Channel #74           \icom74   \pos22
 Iso-comm Channel #75           \icom75   \pos23

 Iso-comm Channel #76           \icom76   \pos84
 Iso-comm Channel #77           \icom77   \pos85
 Iso-comm Channel #78           \icom78   \pos86
 Iso-comm Channel #79           \icom79   \pos87

 Iso-comm Channel #80           \icom80   \pos100
 Iso-comm Channel #81           \icom81   \pos101
 Iso-comm Channel #82           \icom82   \pos102
 Iso-comm Channel #83           \icom83   \pos103

 Iso-comm Channel #84           \icom84   \pos104
 Iso-comm Channel #85           \icom85   \pos105
 Iso-comm Channel #86           \icom86   \pos106
 Iso-comm Channel #87           \icom87   \pos107
 Iso-comm Channel #88           \icom88   \pos108
 Iso-comm Channel #89           \icom89   \pos109
 Iso-comm Channel #90           \icom90   \pos110
 Iso-comm Channel #91           \icom91   \pos111


//////////////////////////////////////////////////////////////////////////////
//                                                                          //
//                         APU Pin mapping syntax:                          //
//                                                                          //
//   Place the logical (mapped) iso-comm position number of the APU in the  //
//   'pos' field of the line corresponding to the pin numbers which that    //
//   APU board represents.  If a line is missing or the 'pos' field is left //
//   blank, auto-pin assignments will occur for each APU present.           //
//                                                                          //
//   Example:                                                               //
//                                                                          //
//        APU Pin Numbers 0-7        /apu0       /pos<num>                  //
//                                                                          //
```

```
//        where: <num> is the logical icom position, 0 to 255              //
//                                                                         //
/////////////////////////////////////////////////////////////////////////////

 APU Pin Numbers 0-7        \apu0       \
 APU Pin Numbers 8-15       \apu8       \
 APU Pin Numbers 16-23      \apu16      \
 APU Pin Numbers 24-31      \apu24      \
 APU Pin Numbers 32-39      \apu32      \
 APU Pin Numbers 40-47      \apu40      \
 APU Pin Numbers 48-55      \apu48      \
 APU Pin Numbers 56-63      \apu56      \
 APU Pin Numbers 64-71      \apu64      \
 APU Pin Numbers 72-79      \apu72      \
 APU Pin Numbers 80-87      \apu80      \
 APU Pin Numbers 88-95      \apu88      \
 APU Pin Numbers 96-103     \apu96      \
 APU Pin Numbers 104-111    \apu104     \
 APU Pin Numbers 112-119    \apu112     \
 APU Pin Numbers 120-127    \apu120     \
 APU Pin Numbers 128-135    \apu128     \
 APU Pin Numbers 136-143    \apu136     \
 APU Pin Numbers 144-151    \apu144     \
 APU Pin Numbers 152-159    \apu152     \
 APU Pin Numbers 160-167    \apu160     \
 APU Pin Numbers 168-175    \apu168     \
 APU Pin Numbers 176-183    \apu176     \
 APU Pin Numbers 184-191    \apu184     \
 APU Pin Numbers 192-199    \apu192     \
 APU Pin Numbers 200-207    \apu200     \
 APU Pin Numbers 208-215    \apu208     \
 APU Pin Numbers 216-223    \apu216     \
 APU Pin Numbers 224-231    \apu224     \
 APU Pin Numbers 232-239    \apu232     \
 APU Pin Numbers 240-247    \apu240     \
 APU Pin Numbers 248-255    \apu248     \


/////////////////////////////////////////////////////////////////////////////
//                                                                         //
//                        APU-12 Pin mapping syntax:                       //
//                                                                         //
//  Place the logical(mapped) odd iso-comm position number of the APU-12 in //
//  the 'pos' field of the line corresponding to the pin numbers which that //
//  APU-12 board represents.  If a line is missing or the 'pos' field is    //
//  left blank, auto-pin assignments will occur for each APU-12 present.    //
//                                                                         //
//   Example:                                                              //
```

```
//                                                                          //
//        APU-12 Pin Numbers   0-11    /apu12-pin0    /pos<num>             //
//        where: <num> is the logical icom position, 0 to 255               //
//                                                                          //
//////////////////////////////////////////////////////////////////////////////

 APU-12 Pin Numbers   0-11    \apu12-pin0    \pos9
 APU-12 Pin Numbers  12-23    \apu12-pin12   \pos11
 APU-12 Pin Numbers  24-35    \apu12-pin24   \pos13
 APU-12 Pin Numbers  36-47    \apu12-pin36   \
 APU-12 Pin Numbers  48-59    \apu12-pin48   \
 APU-12 Pin Numbers  60-71    \apu12-pin60   \
 APU-12 Pin Numbers  72-83    \apu12-pin72   \
 APU-12 Pin Numbers  84-95    \apu12-pin84   \
 APU-12 Pin Numbers  96-107   \apu12-pin96   \
 APU-12 Pin Numbers 108-119   \apu12-pin108  \
 APU-12 Pin Numbers 120-131   \apu12-pin120  \
 APU-12 Pin Numbers 132-143   \apu12-pin132  \
 APU-12 Pin Numbers 144-155   \apu12-pin144  \
 APU-12 Pin Numbers 156-167   \apu12-pin156  \
 APU-12 Pin Numbers 168-179   \apu12-pin168  \
 APU-12 Pin Numbers 180-191   \apu12-pin180  \
 APU-12 Pin Numbers 192-203   \apu12-pin192  \
 APU-12 Pin Numbers 204-215   \apu12-pin204  \
 APU-12 Pin Numbers 216-227   \apu12-pin216  \
 APU-12 Pin Numbers 228-239   \apu12-pin228  \
 APU-12 Pin Numbers 240-251   \apu12-pin240  \
     :      :       :          :        :
 APU-12 Pin Numbers 468-479   \apu12-pin468  \


//////////////////////////////////////////////////////////////////////////////
//                                                                          //
//                        QMS Pin mapping syntax:                           //
//                                                                          //
//  Place the logical (mapped) iso-comm position number of the QMS in the   //
//  'pos' field of the line corresponding to the pin numbers which that QMS //
//  icom num represents.  If a line is missing or the 'pos' field is left   //
//  blank, auto-pin assignments will occur for each QMS pin present.        //
//                                                                          //
//   Example:                                                               //
//                                                                          //
//        QMS Pin Numbers 8-9        /qms8       /pos<num>                  //
//                                                                          //
//        where: <num> is the logical icom position, 0 to 255               //
//                                                                          //
//////////////////////////////////////////////////////////////////////////////
```

```
 QMS Pin Numbers 0-1        \qms0       \pos26
 QMS Pin Numbers 2-3        \qms2       \pos27
 QMS Pin Numbers 4-5        \qms4       \
 QMS Pin Numbers 6-7        \qms6       \
 QMS Pin Numbers 8-9        \qms8       \
 QMS Pin Numbers 10-11      \qms10      \
 QMS Pin Numbers 12-13      \qms12      \
 QMS Pin Numbers 14-15      \qms14      \
 QMS Pin Numbers 16-17      \qms16      \
 QMS Pin Numbers 18-19      \qms18      \
 QMS Pin Numbers 20-21      \qms20      \
 QMS Pin Numbers 22-23      \qms22      \
 QMS Pin Numbers 24-25      \qms24      \
 QMS Pin Numbers 26-27      \qms26      \
 QMS Pin Numbers 28-29      \qms28      \
 QMS Pin Numbers 30-31      \qms30      \
     :      :       :          :        :
 QMS Pin Numbers 124-125    \qms124     \
 QMS Pin Numbers 126-127    \qms126     \


/////////////////////////////////////////////////////////////////////////////
//                                                                         //
//                        QHSU Pin mapping syntax:                         //
//                                                                         //
//  Place the logical (mapped) iso-comm position number of the QHSU in the //
//  'pos' field of the line corresponding to the pin numbers which that QHSU//
//  icom num represents.  If a line is missing or the 'pos' field is left  //
//  blank, auto-pin assignments will occur for each QHSU pin present.      //
//                                                                         //
//   Example:                                                              //
//                                                                         //
//        QHSU Pin Numbers 8-9        /qhsu8       /pos<num>               //
//                                                                         //
//        where: <num> is the logical icom position, 0 to 255              //
//                                                                         //
/////////////////////////////////////////////////////////////////////////////

 QHSU Pin Numbers 0-1        \qhsu0       \
 QHSU Pin Numbers 2-3        \qhsu2       \
 QHSU Pin Numbers 4-5        \qhsu4       \
 QHSU Pin Numbers 6-7        \qhsu6       \
 QHSU Pin Numbers 8-9        \qhsu8       \
 QHSU Pin Numbers 10-11      \qhsu10      \
 QHSU Pin Numbers 12-13      \qhsu12      \
 QHSU Pin Numbers 14-15      \qhsu14      \
 QHSU Pin Numbers 16-17      \qhsu16      \
```

```
 QHSU Pin Numbers 18-19      \qhsu18      \
 QHSU Pin Numbers 20-21      \qhsu20      \
 QHSU Pin Numbers 22-23      \qhsu22      \
 QHSU Pin Numbers 24-25      \qhsu24      \
 QHSU Pin Numbers 26-27      \qhsu26      \
 QHSU Pin Numbers 28-29      \qhsu28      \
 QHSU Pin Numbers 30-31      \qhsu30      \
  :    :     :       :           :         :
  :    :     :       :           :         :
 QHSU Pin Numbers 252-253    \qhsu252     \
 QHSU Pin Numbers 254-255    \qhsu254     \


////////////////////////////////////////////////////////////////////////////////
//                                                                            //
//               8x8 Matrix Resource/Channel mapping syntax:                  //
//                                                                            //
//   Place the logical (mapped) iso-comm position number of the 8x8 Matrix  //
//   in the 'pos' field of the line corresponding to the Resource/Channel   //
//   numbers which that 8x8 Matrix board represents.  If a line is missing  //
//   or the 'pos' field is left blank, resource/channel number assignments  //
//   will occur automatically for each 8x8 Matrix present.                   //
//                                                                            //
//   Example:                                                                 //
//                                                                            //
//        Matrix Res/Chan Numbers 0-7      /mat0       /pos<num>             //
//                                                                            //
//        where: <num> is the logical icom position, 0 to 255                //
//                                                                            //
////////////////////////////////////////////////////////////////////////////////

 Matrix Res/Chan Numbers 0-7     \mat0       \
 Matrix Res/Chan Numbers 8-15    \mat8       \
 Matrix Res/Chan Numbers 16-23   \mat16      \
 Matrix Res/Chan Numbers 24-31   \mat24      \
 Matrix Res/Chan Numbers 32-39   \mat32      \
 Matrix Res/Chan Numbers 40-47   \mat40      \
 Matrix Res/Chan Numbers 48-55   \mat48      \
 Matrix Res/Chan Numbers 56-63   \mat56      \
 Matrix Res/Chan Numbers 64-71   \mat64      \
 Matrix Res/Chan Numbers 72-79   \mat72      \
 Matrix Res/Chan Numbers 80-87   \mat80      \
 Matrix Res/Chan Numbers 88-95   \mat88      \
 Matrix Res/Chan Numbers 96-103  \mat96      \
 Matrix Res/Chan Numbers 104-111 \mat104     \
 Matrix Res/Chan Numbers 112-119 \mat112     \
 Matrix Res/Chan Numbers 120-127 \mat120     \
 Matrix Res/Chan Numbers 128-135 \mat128     \
```

```
 Matrix Res/Chan Numbers 136-143 \mat136     \
 Matrix Res/Chan Numbers 144-151 \mat144     \
 Matrix Res/Chan Numbers 152-159 \mat152     \
 Matrix Res/Chan Numbers 160-167 \mat160     \
 Matrix Res/Chan Numbers 168-175 \mat168     \
 Matrix Res/Chan Numbers 176-183 \mat176     \
 Matrix Res/Chan Numbers 184-191 \mat184     \
 Matrix Res/Chan Numbers 192-199 \mat192     \
 Matrix Res/Chan Numbers 200-207 \mat200     \
 Matrix Res/Chan Numbers 208-215 \mat208     \
 Matrix Res/Chan Numbers 216-223 \mat216     \
 Matrix Res/Chan Numbers 224-231 \mat224     \
 Matrix Res/Chan Numbers 232-239 \mat232     \
 Matrix Res/Chan Numbers 240-247 \mat240     \
 Matrix Res/Chan Numbers 248-255 \mat248     \


//////////////////////////////////////////////////////////////////////////////
//                                                                          //
//                         QPLU Pin mapping syntax:                         //
//                                                                          //
//  Place the logical (mapped) iso-comm position number of the QPLU in the  //
//  'pos' field of the line corresponding to the pin numbers which that     //
//  QPLU icom num represents.  Note that iso-comm positions are shared by   //
//  two consecutive QPLU pins.  Therefore only the even pin of the pair     //
//  should be specified in the pin field.  If a line is missing or the      //
//  'pos' field is left blank, auto-pin assignments will occur for each     //
//  QPLU pin present.                                                       //
//                                                                          //
//   Example:                                                               //
//                                                                          //
//       QPLU Pin Numbers 0-1        /qplu0   /pos<num>                     //
//       QPLU Pin Numbers 2-3        /qplu2   /pos<num>                     //
//                                                                          //
//       where: <num> is the logical icom position, 0 to 255                //
//                                                                          //
//                                                                          //
//                         QPLU Pin-to-CIB mapping syntax:                  //
//                                                                          //
//  Place the logical (mapped) CIB cable position number connected to the   //
//  QPLU pin in the 'cib' field of the line corresponding to the QPLU pin.  //
//  Unmapped pins will have no CIB connection assigned.                     //
//                                                                          //
//  Note that logical CIB connection mapping is valid offline only.  In     //
//  online mode the actual CIB connection is detected automatically.        //
//                                                                          //
//   Example:                                                               //
//                                                                          //
```

```
//        QPLU Pin Number 0           \qplu-pin0      \cib<cibnum>          //
//        QPLU Pin Number 1           \qplu-pin1      \cib<cibnum>          //
//        QPLU Pin Number 2           \qplu-pin2      \cib<cibnum>          //
//        QPLU Pin Number 3           \qplu-pin3      \cib<cibnum>          //
//                                                                          //
//        where: <cibnum> is the cable position on the CIB, 0 to 7          //
//                                                                          //
//////////////////////////////////////////////////////////////////////////////

 QPLU Pin Numbers 0-1        \qplu0      \
 QPLU Pin Numbers 2-3        \qplu2      \
 QPLU Pin Numbers 4-5        \qplu4      \
 QPLU Pin Numbers 6-7        \qplu6      \
 QPLU Pin Numbers 8-9        \qplu8      \
 QPLU Pin Numbers 10-11      \qplu10     \
 QPLU Pin Numbers 12-13      \qplu12     \
 QPLU Pin Numbers 14-15      \qplu14     \
 QPLU Pin Numbers 16-17      \qplu16     \
 QPLU Pin Numbers 18-19      \qplu18     \
 QPLU Pin Numbers 20-21      \qplu20     \
 QPLU Pin Numbers 22-23      \qplu22     \
 QPLU Pin Numbers 24-25      \qplu24     \
 QPLU Pin Numbers 26-27      \qplu26     \
 QPLU Pin Numbers 28-29      \qplu28     \
 QPLU Pin Numbers 30-31      \qplu30     \
  :    :      :      :          :        :
 QPLU Pin Numbers 78-79      \qplu78     \

 QPLU Pin Number 0          \qplu-pin0   \
 QPLU Pin Number 1          \qplu-pin1   \
 QPLU Pin Number 2          \qplu-pin2   \
 QPLU Pin Number 3          \qplu-pin3   \
 QPLU Pin Number 4          \qplu-pin4   \
 QPLU Pin Number 5          \qplu-pin5   \
 QPLU Pin Number 6          \qplu-pin6   \
 QPLU Pin Number 7          \qplu-pin7   \
 QPLU Pin Number 8          \qplu-pin8   \
 QPLU Pin Number 9          \qplu-pin9   \
 QPLU Pin Number 10         \qplu-pin10  \
 QPLU Pin Number 11         \qplu-pin11  \
 QPLU Pin Number 12         \qplu-pin12  \
 QPLU Pin Number 13         \qplu-pin13  \
 QPLU Pin Number 14         \qplu-pin14  \
 QPLU Pin Number 15         \qplu-pin15  \
 QPLU Pin Number 16         \qplu-pin16  \
  :    :      :     :            :       :
 QPLU Pin Number 78         \qplu-pin78  \
 QPLU Pin Number 79         \qplu-pin79  \
```

```
//////////////////////////////////////////////////////////////////////////////
//                                                                          //
//                           GPIB mapping syntax:                           //
//                                                                          //
//   Place here the GPIB addresses versus system resource mapping           //
//                                                                          //
//   Valid GPIB addresses : gpib-0 to gpib-30                               //
//   Valid resource keywords are:                                           //
//         RS0, RS1, RS2, LO                                                //
//                                                                          //
//   Example:                                                               //
//                                                                          //
//        GPIB Address       \gpib-27   \RS0                                //
//        GPIB Address       \gpib-     \RS1   ** not assigned for SMATE**  //
//        GPIB Address       \gpib-28   \LO                                 //
//                                                                          //
//   Please consult ets\inc\gpib500d.h to avoid gpib address conflicts      //
//                                                                          //
//   The following table shows the proper RF connections:                   //
//         RS0 -> SRC#0                                                     //
//         RS1 -> SRC#1                                                     //
//         RS2 -> SRC#2/TS PORT                                             //
//         LO  -> LO                                                        //
//////////////////////////////////////////////////////////////////////////////

 GPIB Address       \gpib-27   \


//////////////////////////////////////////////////////////////////////////////
//                                                                          //
//                   REFCLK GPIB mapping syntax:                            //
//                                                                          //
//   Place here the  type-model and GPIB addresses versus REFCLK system     //
//   resource mapping.                                                      //
//                                                                          //
//   Valid types:  PTS-040, PTS-120, PTS-160, PTS-250, PTS-500, PTS-620     //
//                 PTS1000                                                  //
//   Valid GPIB addresses : gpib-0 to gpib-30                               //
//   Valid resource keywords are:                                           //
//         RCLK0, RCLK1 , RCLK2, RCLK3                                      //
//                                                                          //
//   Example:                                                               //
//                                                                          //
//        GPIB Address       \gpib-24   \RCLK0 PTS-160                      //
//                                                                          //
//   Please consult ets\inc\gpib500d.h to avoid gpib address conflicts      //
```

```
//                                                                          //
//////////////////////////////////////////////////////////////////////////////

 GPIB Address       \gpib-24   \


//////////////////////////////////////////////////////////////////////////////
//                    DPS Emulation syntax:                                 //
//                                                                          //
//  This section allows the user to use either an SPU-100 or an FSS to      //
//  emulate a DPS in an application (without re-compiling the app).         //
//  Place the logical (mapped) iso-comm position number of either the       //
//  SPU-100 or the FSS2000 in the 'pos' field of the line corresponding to  //
//  the specific DPS Power Supply. The utilities will figure out which      //
//  resource is actually at the position number.                            //
//  A translator board does need to be in place in order to physically      //
//  route the FSS or SPU-100 to the DPS Power Supply pins.                  //
//                                                                          //
//  Example: The SPU-100 in position 23 will emulate the DPS Power Supply 1 //
//                                                                          //
//  DPS Power Supply 1         \dpsps1     \pos23                           //
//                                                                          //
//////////////////////////////////////////////////////////////////////////////

 DPS Power Supply 0             \dpsps0     \
 DPS Power Supply 1             \dpsps1     \
 DPS Power Supply 2             \dpsps2     \
 DPS Load Power Supply 0        \dpsldps0   \
 DPS Load Power Supply 1        \dpsldps1   \


//////////////////////////////////////////////////////////////////////////////
//                                                                          //
//              Iso-comm connections to the 'Iso-comm Via FR Bus' Board     //
//                                                                          //
//   Place the logical position numbers of the Iso-comm channels connected  //
//   to the IVFR board in the 'pos' fields below.                           //
//                                                                          //
//   If these lines are missing or ALL 'pos' fields are left blank, the     //
//   IVFR board will not be used even though it may be present.             //
//                                                                          //
//   Example:                                                               //
//                                                                          //
//        IVFR Section #0            \ivfr0    \pos<num>                    //
//                                                                          //
//        where: <num> is the logical icom position, 0 to 255 connected     //
//                to the IVFR board.                                        //
//                                                                          //
```

```
////////////////////////////////////////////////////////////////////////////////

 IVFR Section #0           \ivfr0    \
 IVFR Section #1           \ivfr1    \
 IVFR Section #2           \ivfr2    \
 IVFR Section #3           \ivfr3    \


////////////////////////////////////////////////////////////////////////////////
//                                                                            //
//                  Option 4015/4016 - 3458a Multimeter                       //
//                                                                            //
//  This section is only used if you have the ETS Option 4015 or 4016         //
//  with 3458a Multimeter(s) installed in your tester.                        //
//                                                                            //
//  Place the GPIB address of each 3458a Multimeter(s) that you have in       //
//  your tester in the 'addr' fields below.                                   //
//                                                                            //
//  If these lines are missing or left blank, diagnostics will not run on     //
//  your 3458a meter(s). If these lines are filled in and you do not have     //
//  meter(s) in your tester, a configuration error will occur at the          //
//  beginning of diagnostics.                                                 //
//                                                                            //
//   Example:                                                                 //
//                                                                            //
//       3458a Meter A            \gpibAddr0    \addr<num>                    //
//       3458a Meter B            \gpibAddr1    \addr<num>                    //
//                                                                            //
//       where: <num> is the GPIB address of the 3458a.                       //
//                                                                            //
////////////////////////////////////////////////////////////////////////////////

 3458a Meter A             \gpibAddr0  \
 3458a Meter B             \gpibAddr1  \


////////////////////////////////////////////////////////////////////////////////
//                                                                            //
//                  GiGa Clock GPIB Addresses                                 //
//                                                                            //
//  This section is only used if you have a GiGa Clock installed in your      //
//  tester.                                                                   //
//                                                                            //
//  Place the GPIB address of each GiGa Clock that you have in your tester    //
//  in the 'addr' fields below.                                               //
//                                                                            //
//  If these lines are missing or left blank, diagnostics will not run on     //
//  your GiGa Clock. If these lines are filled in and you do not have a       //
```

```
//  GiGa Clock in your tester, a configuration error will occur at the      //
//  beginning of diagnostics.                                               //
//                                                                          //
//  Note: An Embedded Instrument Diagnostic board must be in place in order //
//        to run GiGa Clock diagnostics.                                    //
//                                                                          //
//   Example:                                                               //
//                                                                          //
//    GiGa Clock 0               \gigaGpibAddr0  \    \addr<num>            //
//    GiGa Clock 1               \gigaGpibAddr1  \    \addr<num>            //
//                                                                          //
//        where: <num> is the GPIB address of the GiGa Clock.               //
//                                                                          //
//////////////////////////////////////////////////////////////////////////////

 GiGa Clock 0               \gigaGpibAddr0  \
 GiGa Clock 1               \gigaGpibAddr1  \


//////////////////////////////////////////////////////////////////////////////
//                                                                          //
//              Scheduled Self-Calibration Time Intervals                   //
//                                                                          //
//  Place the resource's self-calibration keyword and the number of days    //
//  that you would like to set its self-calibration time interval to.       //
//                                                                          //
//  A Scheduled Self-Calibration will not run, if these lines are missing   //
//  or left blank.                                                          //
//                                                                          //
//  Valid scheduled self-calibration keywords are:                          //
//          \selfcalqms                                                     //
//                                                                          //
//  Also, enter the max time in minutes that you will allow ETS software to //
//  consume on any given self-calibration event. This allows you to         //
//  minimize the time consumed by self-calibrating various resources. ETS   //
//  software will manage each resource's interval and the time it consumes  //
//  self-calibrating all resources.                                         //
//                                                                          //
//  If no time is entered, a scheduled self-cal will consume all the time   //
//  it needs to calibrate any and all resources that have reached their     //
//  elapsed time interval.                                                  //
//                                                                          //
//  Valid time allowed keyword is:                                          //
//          \maxselfcaltime                                                 //
//                                                                          //
//  Example:                                                                //
//                                                                          //
//  Resource Self Cal Time Interval  \selfcalqms      \days-<numdays>       //
```

```
//  Max Self-Cal Time Allowed        \maxselfcaltime  \mins-<nummins>       //
//                                                                          //
//  where <numdays> is the number of days in between running QMS            //
//  self-calibrations.  Valid values are 1 to 180 days.                     //
//  and <nummins> is the number of minutes that the ETS software is allowed //
//  to consume per a self-cal event. Valid values are 5 to 1200 minutes.    //
//                                                                          //
//////////////////////////////////////////////////////////////////////////////

 Resource Self-Cal Time Interval    \selfcalqms         \
 Max Self-Cal Time Allowed          \maxselfcaltime     \


//////////////////////////////////////////////////////////////////////////////
//                                                                          //
//                         Resource List syntax:                            //
//                                                                          //
//   Place the keyword for ALL resources in the 'resource' field. Blank     //
//   entries in any or all of the 'resource' fields are ignored.            //
//                                                                          //
//   Valid resource keywords are:                                           //
//                                                                          //
//   PCIB-PCI-T, TCB-Q1, TCB-Q2,                                            //
//   QTMU0, QTMU1                                                           //
//   DPU16-0, DPU16-1, DPU16-2, DPU16-3                                     //
//   DPU16-8M-0, DPU16-8M-1, DPU16-8M-2, DPU16-8M-3                         //
//   NOTE: DPU16-8M and DPU16 resources cannot be mixed.                    //
//                                                                          //
//   HSDU0 (channels 0-3), HSDU1 (channels 4-7)                             //
//   CIB0                                                                   //
//   SPEC-II-CAGE       TH-1A and TH-1B each controls half of 12 icom chnls //
//   SPEC-II-CAGE-A     TH-1A controls all 12 icom channels                 //
//   SPEC-II-CAGE-B     TH-1B controls all 12 icom channels                 //
//                                                                          //
//   For an ONLINE tester, this list represents all resources which exist   //
//   in the tester.                                                         //
//                                                                          //
//   For an OFFLINE tester, this list represents resources to be simulated  //
//   in a tester.                                                           //
//                                                                          //
//   Example:                                                               //
//                                                                          //
//        Resource                  /res        /MCB                        //
//        Resource                  /res        /SPEC-II-CAGE               //
//                                                                          //
//////////////////////////////////////////////////////////////////////////////

//                                        Resource Field
```

```
 Resource                   \res        \
 Resource                   \res        \
 Resource                   \res        \
 Resource                   \res        \
 Resource                   \res        \
 Resource                   \res        \
 Resource                   \res        \
 Resource                   \res        \
 Resource                   \res        \


//////////////////////////////////////////////////////////////////////////////
//                                                                          //
//                  Isocomm based Resource List syntax:                     //
//                                                                          //
//   Place the keyword for ALL iso-comm based resources in the 'resource'   //
//   field. Blank entries in any or all of the 'resource' fields are        //
//   ignored.                                                               //
//                                                                          //
//   Valid iso-comm based resource keywords are:                            //
//                                                                          //
//   FSS (FSS2000), FSS-64K (FSS2000, 64k AWG), FSS-2010 (FSS2010),         //
//   FSS-2010-64K (FSS2010, 64k AWG), APU, MPU, MPU-64K, WCU (WCU2220),     //
//   WCU-2000 (WCU2000), WCU-200 (WCU200), WCU-16K, VI2K, HCM, HSVS, GD,    //
//   QMS, DHVR, DHVR-2, MAT8X8, MAT11X16, DUALMAT11X16, RF3000, RFSC,       //
//   HOAL, USER_BRD, QHSU, QHSU-RF (when RF-6000 cables are attached),      //
//   RTP ( when no Dual Loops are connected ),                              //
//   RTP-DL-NONE ( when Dual Loop chans 0 & 1 are only connected ),         //
//   RTP-NONE-DL ( when Dual Loop chans 2 & 3 are only connected ),         //
//   RTP-DL-DL ( when Dual Loop chans 0, 1, 2, & 3 are all connected )      //
//   SPU-500, SPU-100 (wo/DSP option), SPU-100-DSP (w/DSP option), SPU-250, //
//   APU-10, APU-12, IVFR, QPLU, HPU (wo/DSP option), HPU-DSP (w/DSP option)//
//   HPU-BOOSTER (HPU connected to a booster),                              //
//   HPU-DSP-BOOSTER (HPU w/DSP option & connected to a booster),           //
//   HPUBOOSTER, HCMUX, QMS-T (for Turbo QMS),                              //
//   SPU-112, SPU-112-BOOSTER (for SPU-112 connected to a booster),         //
//   SPU-112BOOSTER (SPU-112 Booster board), PSU-6030                       //
//                                                                          //
//  NOTE: When you are replacing a FSS with a SPU-100 and you are           //
//  expecting the SPU-100 to emulate the FSS in an application (i.e. you do //
//  not want to change the application's FSS source code), you must modify  //
//  the Resource List as follows:                                           //
//                                                                          //
// The previous entries,                                                    //
//     Iso-comm base Resource     \ires0             \fss                   //
//     Iso-comm base Resource     \ires1             \fss-2010              //
//     Iso-comm base Resource     \ires2             \fss-64k               //
//     Iso-comm base Resource     \ires3             \fss-2010-64k          //
```

```
// should be changed to:                                                  //
//                     For an SPU with DSP option,                        //
//     Iso-comm base Resource     \ires0            \spu-100-dsp:fss      //
//                     For an SPU without DSP option,                     //
//     Iso-comm base Resource     \ires1            \spu-100:fss-2010     //
//     Iso-comm base Resource     \ires2            \spu-100:fss-64k      //
//     Iso-comm base Resource     \ires3            \spu-100:fss-2010-64k //
//                                                                        //
// Use a normal entry if you are not trying to emulate a FSS.             //
//                     For an SPU with DSP option,                        //
//     Iso-comm base Resource     \ires0            \spu-100-dsp          //
//                                                                        //
//                     For an SPU-112                                     //
//     Iso-comm base Resource     \ires0            \spu-112              //
//                     For an SPU-112 connected to a booster              //
//     Iso-comm base Resource     \ires0            \spu-112-booster      //
//                     For an SPU-112 booster board                       //
//     Iso-comm base Resource     \ires0            \spu-112booster       //
//                                                                        //
//   For an ONLINE tester, this list represents all resources which exist //
//   in the tester.                                                       //
//                                                                        //
//   For an OFFLINE tester, this list represents resources to be simulated //
//   in a tester.                                                         //
//                                                                        //
//   Example:                                                             //
//                                                                        //
//        Iso-comm base Resource     /ires5      /FSS                     //
//                                                                        //
//////////////////////////////////////////////////////////////////////////////

//                          Mapped Icom Pos    Resource Field
// TH-1A
 Iso-comm base Resource     \ires0      \
 Iso-comm base Resource     \ires1      \
 Iso-comm base Resource     \ires2      \
 Iso-comm base Resource     \ires3      \
 Iso-comm base Resource     \ires4      \
 Iso-comm base Resource     \ires5      \
 Iso-comm base Resource     \ires6      \
 Iso-comm base Resource     \ires7      \
 Iso-comm base Resource     \ires8      \
 Iso-comm base Resource     \ires9      \
 Iso-comm base Resource     \ires10     \
 Iso-comm base Resource     \ires11     \

// TH-1B
 Iso-comm base Resource     \ires12     \
```

```
 Iso-comm base Resource     \ires13     \
 Iso-comm base Resource     \ires14     \
 Iso-comm base Resource     \ires15     \
 Iso-comm base Resource     \ires16     \
 Iso-comm base Resource     \ires17     \
 Iso-comm base Resource     \ires18     \
 Iso-comm base Resource     \ires19     \
 Iso-comm base Resource     \ires20     \
 Iso-comm base Resource     \ires21     \
 Iso-comm base Resource     \ires22     \
 Iso-comm base Resource     \ires23     \

// ICOM Resources Hardwired to DUT Area
 Iso-comm base Resource     \ires80     \
 Iso-comm base Resource     \ires81     \
 Iso-comm base Resource     \ires82     \
 Iso-comm base Resource     \ires83     \
 Iso-comm base Resource     \ires84     \
 Iso-comm base Resource     \ires85     \
 Iso-comm base Resource     \ires86     \
 Iso-comm base Resource     \ires87     \

// Remote Cabled ICOM Resources
 Iso-comm base Resource     \ires88     \
 Iso-comm base Resource     \ires89     \
 Iso-comm base Resource     \ires90     \
 Iso-comm base Resource     \ires91     \
 Iso-comm base Resource     \ires92     \
 Iso-comm base Resource     \ires93     \
 Iso-comm base Resource     \ires94     \
 Iso-comm base Resource     \ires95     \
```

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Appendix C – Instrument Software Support Matrix

C.1 Resource Support

System Software:	Eagle Vision	Eagle Vision MST
Analog Pin Unit, 10 µA (APU-10)	X	
Analog Pin Unit, 12 Channel (APU-12)	X	X
Digital Pin Unit, 16 Channel (DPU-16, DPU-16/8M)	X	X
High Power Unit (HPU-25/100)	X	X
Matrix, 8x8	X	
Medium Power Unit (MPU)	X	
Quad High Speed Signal Unit (QHSU)	X	
Quad Precision Linearity Unit (QPLU)	X	
Quad Time Measurement Unit (QTMU)	X	X
Smart Pin Unit, 100 V / 2 A (SPU-100)	X	X
Smart Pin Unit, 100 V / 12 A (SPU-112)	X	X
Smart Pin Unit, 250 V / 100 mA (SPU-250)	X	
Smart Pin Unit, 500 V / 50 mA (SPU-500)	X	X
Test Head Control Board (TCB)	X	X
Waveform Capture Unit (WCU-2220 or WCU-2220/16K)	X	

X = Supported
(see resource sections for the specific software revision in which support was added for each resource)

C.2 Computer Requirements

C.2.1 Hardware

System Software:	Eagle Vision	Eagle Vision MST
Online Operation	Intel® Pentium® III Processor, 700MHz, 1GB of RAM, DVD-ROM drive, USB port, 10GB of available hard drive space	Dual Intel® Xeon® E5620 Processor, 2.40 GHz, Quad Core, 32GB of RAM, DVD-ROM drive, USB port, 10GB of available hard drive space
Offline Operation	Intel® Pentium® III Processor, 700MHz, 1GB of RAM, DVD-ROM drive, USB port, 10GB of available hard drive space	Intel® Core™2 Duo Processor, 2.50 GHz, 4GB of RAM (8GB preferred), DVD-ROM drive, USB port, 10GB of available hard drive space

C.2.2 Software

System Software:	Eagle Vision	Eagle Vision MST
Operating System	Windows® XP Professional 32-Bit (English), Service Pack 2 or Greater	Windows® 7 Ultimate 64-Bit (English), Service Pack 1 or Greater
Development Environment	Microsoft Visual Studio® 2005 Standard	Microsoft Visual Studio® 2010 Professional
Miscellaneous	Internet Explorer® 6.0 or greater	Internet Explorer® 8.0 or greater
Video Settings	Screen Resolution: 1024 x 768 or Greater	Screen Resolution: 1440 x 900 or Greater

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